

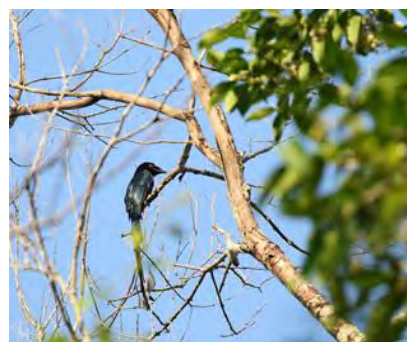
Intended for  
Public Utilities Board

Document type  
TAC/21009/Draft EIA/RT02

Date  
31 May 2022

## RECONSTRUCTION OF CHOA CHU KANG WATERWORKS

### Environmental Impact Assessment (EIA) Report



Submitted by:



**TEMBUSU**  
Asia

Description Draft Environmental Impact Assessment Report for Reconstruction of Choa Chu Kang Waterworks (CCKWW)

Project Ref TAC / 21009

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| 05       | 24/01/2022 | Revised based on comments from PUB         | Holly Siow,<br>Jefferson                                                   | Komal<br>Pujara | S C Tan     |
| 06       | 22/02/2022 | Revised based on comments from NParks      | Holly Siow,<br>Jefferson                                                   | Komal<br>Pujara | S C Tan     |
| 07       | 11/03/2022 | Revised based on comments from NParks      | Holly Siow,<br>Jefferson                                                   | Komal<br>Pujara | S C Tan     |
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| 10       | 04/05/2022 | Revised based on comments from PUB         | Jefferson                                                                  | Holly Siow      | S C Tan     |
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# CONTENTS

|                                                                   |           |
|-------------------------------------------------------------------|-----------|
| <b>ABBREVIATIONS .....</b>                                        | <b>11</b> |
| <b>GLOSSARY.....</b>                                              | <b>14</b> |
| <b>EXECUTIVE SUMMARY .....</b>                                    | <b>16</b> |
| <b>1 INTRODUCTION .....</b>                                       | <b>20</b> |
| 1.1 Project Background .....                                      | 20        |
| 1.2 Objectives of the Environmental Impact Assessment Report..... | 20        |
| 1.3 Report Structure .....                                        | 21        |
| 1.4 Limitations of the EIA Report .....                           | 21        |
| <b>2 PROJECT OVERVIEW.....</b>                                    | <b>22</b> |
| 2.1 Project Background .....                                      | 22        |
| 2.2 Project Scope .....                                           | 22        |
| 2.3 Project Location .....                                        | 22        |
| 2.4 Overview of Construction Work Areas .....                     | 24        |
| 2.5 Proposed Plant Layout Plan .....                              | 26        |
| 2.6 Analysis of Alternatives .....                                | 29        |
| 2.7 Construction Approach and Implementation Schedule.....        | 31        |
| 2.8 Implementation Schedule.....                                  | 35        |
| 2.9 Operational Activities Associated with Project .....          | 36        |
| 2.10 Concurrent Project .....                                     | 36        |
| <b>3 EIA APPROACH .....</b>                                       | <b>37</b> |
| 3.1 Singapore's EIA Context.....                                  | 37        |
| 3.2 Scope of EIA .....                                            | 37        |
| 3.3 EIA Study Area and Environmental Aspects.....                 | 39        |
| 3.4 Identification of Sensitive Receptors .....                   | 41        |
| 3.5 Impact Assessment Methodology .....                           | 42        |
| 3.6 Project Team .....                                            | 45        |
| <b>4 BIODIVERSITY .....</b>                                       | <b>46</b> |
| 4.1 Introduction .....                                            | 46        |
| 4.2 Baseline Methodology .....                                    | 46        |
| 4.3 Desktop Review .....                                          | 54        |
| 4.4 Current Baseline Results and Discussion.....                  | 57        |
| 4.5 Impact Assessment .....                                       | 84        |
| 4.6 Mitigation Measures .....                                     | 93        |
| 4.7 Residual Impacts.....                                         | 102       |

|          |                                                                    |            |
|----------|--------------------------------------------------------------------|------------|
| <b>5</b> | <b>HYDROLOGY AND SURFACE WATER QUALITY.....</b>                    | <b>110</b> |
| 5.1      | Introduction .....                                                 | 110        |
| 5.2      | Relevant Environmental Legislation, Guidelines and Standards ..... | 110        |
| 5.3      | Hydrology Profile of Project Study Area.....                       | 110        |
| 5.4      | Natural Stream Characterisation.....                               | 113        |
| 5.5      | Surface Water Quality Baseline Methodology .....                   | 125        |
| 5.6      | Surface Water Quality Baseline Results and Discussion .....        | 127        |
| 5.7      | Impact Assessment.....                                             | 131        |
| 5.8      | Mitigation Measures .....                                          | 135        |
| 5.9      | Residual Impacts.....                                              | 140        |
| <b>6</b> | <b>NOISE .....</b>                                                 | <b>147</b> |
| 6.1      | Introduction .....                                                 | 147        |
| 6.2      | Relevant Environmental Legislation, Guidelines and Standards ..... | 147        |
| 6.3      | Baseline Methodology.....                                          | 148        |
| 6.4      | Baseline Results and Discussion.....                               | 151        |
| 6.5      | Impact Assessment.....                                             | 157        |
| 6.6      | Residual Impacts.....                                              | 169        |
| <b>7</b> | <b>AMBIENT AIR QUALITY.....</b>                                    | <b>173</b> |
| 7.1      | Introduction .....                                                 | 173        |
| 7.2      | Relevant Environmental Legislation, Guidelines and Standards ..... | 173        |
| 7.3      | Baseline Methodology.....                                          | 173        |
| 7.4      | Baseline Results and Discussion.....                               | 175        |
| 7.5      | Impact Assessment.....                                             | 179        |
| 7.6      | Mitigation Measures .....                                          | 183        |
| 7.7      | Residual Impacts.....                                              | 184        |
| <b>8</b> | <b>GROUND-BORNE VIBRATION .....</b>                                | <b>187</b> |
| 8.1      | Introduction .....                                                 | 187        |
| 8.2      | Relevant Environmental Legislation, Guidelines and Standards ..... | 187        |
| 8.3      | Baseline Methodology.....                                          | 188        |
| 8.4      | Baseline Results and Discussion.....                               | 190        |
| 8.5      | Impact Assessment.....                                             | 190        |
| 8.6      | Mitigation Measures .....                                          | 193        |
| 8.7      | Residual Impacts.....                                              | 194        |
| <b>9</b> | <b>LIGHT.....</b>                                                  | <b>196</b> |
| 9.1      | Introduction .....                                                 | 196        |
| 9.2      | Impact Assessment.....                                             | 197        |

|                                                                                   |                                                                       |            |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------|------------|
| 9.3                                                                               | Mitigation Measures .....                                             | 199        |
| 9.4                                                                               | Residual Impacts.....                                                 | 202        |
| <b>10</b>                                                                         | <b>WASTE MANAGEMENT.....</b>                                          | <b>204</b> |
| 10.1                                                                              | Introduction.....                                                     | 204        |
| 10.2                                                                              | Relevant Environmental Legislation, Guidelines and Standards .....    | 204        |
| 10.3                                                                              | Impact Assessment.....                                                | 205        |
| 10.4                                                                              | Mitigation Measures .....                                             | 207        |
| 10.5                                                                              | Residual Impacts.....                                                 | 208        |
| <b>11</b>                                                                         | <b>VECTOR CONTROL.....</b>                                            | <b>210</b> |
| 11.1                                                                              | Introduction.....                                                     | 210        |
| 11.2                                                                              | Relevant Environmental Legislation, Guidelines and Standards .....    | 210        |
| 11.3                                                                              | Impact Assessment.....                                                | 211        |
| 11.4                                                                              | Mitigation Measures .....                                             | 212        |
| 11.5                                                                              | Residual Impacts.....                                                 | 214        |
| <b>12</b>                                                                         | <b>SUMMARY OF IMPACT ASSESSMENT &amp; MITIGATION MEASURES.....</b>    | <b>216</b> |
| <b>13</b>                                                                         | <b>ENVIRONMENTAL MANAGEMENT &amp; MONITORING PLAN FRAMEWORK .....</b> | <b>233</b> |
| 13.1                                                                              | Overview .....                                                        | 233        |
| 13.2                                                                              | Construction EMMP .....                                               | 233        |
| 13.3                                                                              | Wildlife Management Plan .....                                        | 238        |
| 13.4                                                                              | Flora Management Plan.....                                            | 250        |
| 13.5                                                                              | Lighting Management Plan (LMP) .....                                  | 259        |
| 13.6                                                                              | Biodiversity Monitoring Requirements .....                            | 260        |
| 13.7                                                                              | Environmental Monitoring Plan .....                                   | 261        |
| <b>14</b>                                                                         | <b>Bibliography .....</b>                                             | <b>267</b> |
| <b>APPENDIX A Project Layout Plan .....</b>                                       |                                                                       | <b>270</b> |
| <b>APPENDIX B List of Plant Species Recorded at Project Study Area.....</b>       |                                                                       | <b>271</b> |
| <b>APPENDIX C Photographs of Plants Observed at Project Study Area .....</b>      |                                                                       | <b>272</b> |
| <b>APPENDIX D Flora Maps Depicting Species of Conservation Significance .....</b> |                                                                       | <b>273</b> |
| <b>APPENDIX E List of Fauna Species Recorded at Project Study Area .....</b>      |                                                                       | <b>274</b> |
| <b>APPENDIX F Photographs of Animals Observed at Project Study Area.....</b>      |                                                                       | <b>275</b> |
| <b>APPENDIX G Baseline Surface Water Quality Report .....</b>                     |                                                                       | <b>276</b> |
| <b>APPENDIX H Recommended ECM Plan Approach for Area 2.....</b>                   |                                                                       | <b>277</b> |
| <b>APPENDIX I Baseline Ambient Noise Level Report.....</b>                        |                                                                       | <b>278</b> |
| <b>APPENDIX J Baseline Ambient Air Quality Study Report .....</b>                 |                                                                       | <b>279</b> |
| <b>APPENDIX K Baseline Vibration Study Report .....</b>                           |                                                                       | <b>280</b> |

|                                                               |            |
|---------------------------------------------------------------|------------|
| <b>APPENDIX L Preliminary Vibration Analysis Report .....</b> | <b>281</b> |
| <b>APPENDIX M Wildlife Incident Form.....</b>                 | <b>282</b> |

## FIGURES

|                                                                                                               |    |
|---------------------------------------------------------------------------------------------------------------|----|
| Figure 2.1. Location plan of CCKWW.....                                                                       | 23 |
| Figure 2.2. Construction work areas of CCKWW reconstruction project .....                                     | 24 |
| Figure 2.3. Soil investigation borehole layout plan for Area 2 .....                                          | 25 |
| Figure 2.4. Soil investigation borehole layout plan for Area 3 .....                                          | 26 |
| Figure 2.5. Proposed layout plan of Area 1 .....                                                              | 27 |
| Figure 2.6. Proposed layout plan of Area 2 .....                                                              | 28 |
| Figure 2.7. 3D representation of layout plan of Area 1 and Area 2 .....                                       | 28 |
| Figure 2.8. Proposed layout plan of sewerage alignment at Area 3.....                                         | 29 |
| Figure 2.9. Construction approach at Area 1 .....                                                             | 31 |
| Figure 2.10. Proposed temporary construction access road in Area 2 (circled light green)..                    | 32 |
| Figure 2.11. Adjacent project development .....                                                               | 36 |
| Figure 3.1. EIA study area .....                                                                              | 40 |
| Figure 4.1. Layout of MGPs used to survey vascular plants within project study area .....                     | 48 |
| Figure 4.2. Fauna transects in the project study area.....                                                    | 51 |
| Figure 4.3. Location of camera traps within project study area.....                                           | 52 |
| Figure 4.4. Location of aquatic surveys within project study area.....                                        | 54 |
| Figure 4.5. Study area of the 2016 EIS shown with the present study area.....                                 | 55 |
| Figure 4.6. Summary of fauna results from previous baseline study.....                                        | 56 |
| Figure 4.7. Conservation status of recorded CCKWW flora species.....                                          | 61 |
| Figure 4.8. Habitat map of the project study area.....                                                        | 63 |
| Figure 4.9. Vegetation map of surveyed flora with girth $\geq 1$ metre .....                                  | 64 |
| Figure 4.10. Vegetation map of surveyed Critically Endangered flora .....                                     | 65 |
| Figure 4.11. Vegetation map of surveyed Endangered flora.....                                                 | 65 |
| Figure 4.12. Vegetation map of surveyed Vulnerable flora .....                                                | 66 |
| Figure 4.13. Distribution of number of threatened bird species in project study area .....                    | 67 |
| Figure 4.14. Barred eagle owl (L) and chestnut-bellied malkoha (R) in project study area ..                   | 68 |
| Figure 4.15. Location of mammal species of conservation value within project study area..                     | 69 |
| Figure 4.16. Spectrograms showing acoustic call characteristics of detected bat species ...                   | 70 |
| Figure 4.17. Adult wild pig with two piglets along Dunbar Walk within project study area ....                 | 71 |
| Figure 4.18. Wild pig feeding activities observed within project study area.....                              | 71 |
| Figure 4.19. Location of reptile species of conservation value within project study area.....                 | 72 |
| Figure 4.20. Puff-faced water snake found at Stream 2.....                                                    | 73 |
| Figure 4.21. Malesian frog in stream within project site .....                                                | 74 |
| Figure 4.22. Location of amphibian species of conservation value within project study area                    | 74 |
| Figure 4.23. Malayan snow flat within forested part of project study area.....                                | 75 |
| Figure 4.24. Location of butterfly species of conservation value within project study area...                 | 76 |
| Figure 4.25. Location of odonates species of conservation value within project study area.                    | 77 |
| Figure 4.26. Elf and variable featherlegs dragonflies within forested area of project site. ....              | 77 |
| Figure 4.27. Number of species sightings captured by each camera.....                                         | 80 |
| Figure 4.28. Wild pig ( <i>Sus scrofa</i> ) activity as recorded by the camera trap over 24-hour period ..... | 81 |
| Figure 4.29. Local conservation status overview of species recorded within project study area .....           | 81 |

|                                                                                                          |     |
|----------------------------------------------------------------------------------------------------------|-----|
| Figure 4.30. Global conservation status overview of species recorded within project study area .....     | 82  |
| Figure 4.31. Location of species of local and global conservation value vis-à-vis habitat types .....    | 82  |
| Figure 4.32. Conservation significant flora affected by planned borehole works.....                      | 98  |
| Figure 4.33. Conservation significant flora affected by planned borehole, manhole and trench works ..... | 101 |
| Figure 5.1. Catchment areas of Singapore (Public Utilities Board, 2019) .....                            | 111 |
| Figure 5.2. Annual monthly rainfall at Jurong West weather station during 2011–2021 .....                | 111 |
| Figure 5.3. Location of natural streams in the project study area .....                                  | 112 |
| Figure 5.4. Photographs showing various streams within project study area .....                          | 113 |
| Figure 5.5. Location of stream characterisation survey points .....                                      | 114 |
| Figure 5.6. Location of surface water quality sampling points in project study area.....                 | 127 |
| Figure 5.7. Surface water sampling activities .....                                                      | 127 |
| Figure 5.8. Project boundary, including original footprint (magenta) and revised footprint (cyan) .....  | 137 |
| Figure 6.1. Setup of the baseline noise monitoring equipment .....                                       | 150 |
| Figure 6.2. Location of baseline noise monitoring stations within project study area .....               | 151 |
| Figure 6.3. Baseline Leq 5 mins monitoring results at Station N1 over one-week period ...                | 152 |
| Figure 6.4. Baseline Leq 5 mins monitoring results at Station N2 over one-week period ...                | 153 |
| Figure 6.5. Baseline Leq 5 mins monitoring results at Station N3 over one-week period ...                | 154 |
| Figure 6.6. Baseline Leq 5 mins monitoring results at Station N4 over one-week period ...                | 155 |
| Figure 6.7. Baseline Leq 5 mins monitoring results at Station N5 over one-week period ...                | 156 |
| Figure 6.8. Predicted daytime noise level from Area 1 & 2 activities without mitigation measures.....    | 161 |
| Figure 6.9. Predicted night-time noise level from Area 1 activities without mitigation measures .....    | 162 |
| Figure 6.10. Predicted daytime noise level from Area 3 activities without mitigation measures .....      | 163 |
| Figure 6.11. Predicted night-time noise level from Area 3 activities without mitigation measures .....   | 163 |
| Figure 6.12. Predicted day-time noise level at Area 1 and 2 with mitigation measures.....                | 165 |
| Figure 6.13. Predicted night-time noise level at Area 1 with mitigation measures.....                    | 165 |
| Figure 6.14. Predicted day-time noise level at Area 3 with mitigation measures.....                      | 166 |
| Figure 6.15. Predicted night-time noise level at Area 3 with mitigation measures.....                    | 167 |
| Figure 7.1. Setup of the baseline ambient air quality monitoring equipment .....                         | 175 |
| Figure 7.2. Location of baseline ambient air quality monitoring stations within project area .....       | 175 |
| Figure 8.1. Setup of the baseline ground-borne vibration monitoring equipment .....                      | 189 |
| Figure 8.2. Location of ambient vibration monitoring stations within project area .....                  | 189 |
| Figure 9.1. Artificial light illuminating from CCKWW facility observed during night surveys              | 196 |
| Figure 9.2. Artificial lighting observed along Dunbar Walk.....                                          | 197 |
| Figure 9.3. Example of light shielding.....                                                              | 199 |
| Figure 9.4. Example of hedge planting using <i>Murraya paniculata</i> .....                              | 201 |
| Figure 13.1. Environmental incidence reporting flowchart .....                                           | 237 |
| Figure 13.2. Directional Clearance 1 .....                                                               | 243 |
| Figure 13.3. Directional Clearance 2 .....                                                               | 244 |
| Figure 13.4. Directional Clearance 3 .....                                                               | 245 |

|                                                       |     |
|-------------------------------------------------------|-----|
| Figure 13.5. Directional Clearance 4 .....            | 246 |
| Figure 13.6. Directional Clearance 5 .....            | 247 |
| Figure 13.7. Example of a tagged native sapling ..... | 252 |
| Figure 13.8. Example of plants in a nursery .....     | 252 |
| Figure 13.9. Tree Protection Zone Diagram .....       | 256 |
| Figure 13.10. Illustration of a notch cut .....       | 258 |
| Figure 13.11. Illustration of edge effect .....       | 258 |

## TABLES

|                                                                                                                                                                                            |     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Table 2.1. Sewerage alignment route analysis .....                                                                                                                                         | 30  |
| Table 2.2. Sewerage alignment construction method analysis .....                                                                                                                           | 30  |
| Table 2.3. List of planned night work activities in Area 2 .....                                                                                                                           | 33  |
| Table 2.4. Implementation schedule of entire project .....                                                                                                                                 | 35  |
| Table 3.1. List of applicable Singapore legislations, regulations, and guidelines .....                                                                                                    | 38  |
| Table 3.2. Overview of identified sensitive receptors .....                                                                                                                                | 41  |
| Table 3.3. List of parameters and respective scores assigned in RIAM method .....                                                                                                          | 43  |
| Table 3.4. Description of the value of magnitudes in RIAM method .....                                                                                                                     | 44  |
| Table 3.5. List of ES range along with the degree of impact associated with each range ....                                                                                                | 44  |
| Table 3.6. List of EIA project team members .....                                                                                                                                          | 45  |
| Table 4.1. Conservation status for flora & fauna species and their respective definitions, adapted from IUCN Red List (2021) and Singapore Red Data Book (Davison, Ng, & Chew, 2008) ..... | 47  |
| Table 4.2. Static parameters collected for surveyed flora .....                                                                                                                            | 49  |
| Table 4.3. Habitat types found within Singapore and general description of each habitat, modified from the BIA Guidelines (National Parks Board, 2020) .....                               | 49  |
| Table 4.4. Survey timings, frequency, and methodology for each fauna group .....                                                                                                           | 50  |
| Table 4.5. Coordinates of the camera traps within project study area .....                                                                                                                 | 52  |
| Table 4.6. Conservation significant flora encountered within the project study area .....                                                                                                  | 59  |
| Table 4.7. Distribution of conservation significant flora species by numbers & percentages                                                                                                 | 61  |
| Table 4.8. List of bird species of conservation value currently found in project study area ..                                                                                             | 67  |
| Table 4.9. List of mammal species of conservation value currently found in project study area .....                                                                                        | 68  |
| Table 4.10. List of reptile species of conservation value recorded by the present study .....                                                                                              | 72  |
| Table 4.11. List of amphibian species of conservation value within project study area .....                                                                                                | 73  |
| Table 4.12. List of butterfly species of conservation value within project study area .....                                                                                                | 75  |
| Table 4.13. List of odonates species of conservation value within project study area .....                                                                                                 | 76  |
| Table 4.14. Recorded aquatic fauna composition in the streams within project study area ..                                                                                                 | 78  |
| Table 4.15. Number of sightings from each camera by species (*=locally threatened species) .....                                                                                           | 79  |
| Table 4.16. List of threatened flora species in the Area 2 .....                                                                                                                           | 87  |
| Table 4.17. List of threatened fauna species in the Area 2 .....                                                                                                                           | 87  |
| Table 4.18. Environmental Scores of the identified impacts on site's biodiversity with corresponding mitigation measures for Area 2 activities .....                                       | 104 |
| Table 4.19. Environmental Scores of the identified impacts on site's biodiversity with corresponding mitigation measures for Area 3 activities .....                                       | 107 |
| Table 5.1. Classification of substrate types (Thein, 1979) .....                                                                                                                           | 115 |
| Table 5.2. Classification of sand particle sizes (Wentworth, 1922) .....                                                                                                                   | 115 |
| Table 5.3. Results of stream 1 characterization survey .....                                                                                                                               | 116 |

|                                                                                                                                                     |     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Table 5.4. Results of stream 2 characterization survey .....                                                                                        | 119 |
| Table 5.5. Results of stream 3 characterization survey .....                                                                                        | 122 |
| Table 5.6. Surface water quality parameters to be surveyed .....                                                                                    | 125 |
| Table 5.7. Coordinate of surface water quality sampling locations .....                                                                             | 126 |
| Table 5.8. Result of first dry weather period surface water quality samples.....                                                                    | 128 |
| Table 5.9. Result of second dry weather period surface water quality samples .....                                                                  | 128 |
| Table 5.10. Result of wet weather period surface water quality samples.....                                                                         | 129 |
| Table 5.11. Environmental Scores of the identified impacts on site's hydrology & water quality with corresponding mitigation measures (Area 1)..... | 141 |
| Table 5.12. Environmental Scores of the identified impacts on site's hydrology & water quality with corresponding mitigation measures (Area 2)..... | 142 |
| Table 5.13. Environmental Scores of the identified impacts on site's hydrology & water quality with corresponding mitigation measures (Area 3)..... | 145 |
| Table 6.1. Maximum permissible noise levels from construction sites (source: SSO).....                                                              | 147 |
| Table 6.2. Value of correction factor accounting for baseline noise level exceedance.....                                                           | 148 |
| Table 6.3. List of identified noise-sensitive receptors .....                                                                                       | 149 |
| Table 6.4. Location of baseline noise monitoring stations .....                                                                                     | 149 |
| Table 6.5. Summary of the baseline Leq 5 mins noise monitoring results at Station N1 ....                                                           | 151 |
| Table 6.6. Summary of the baseline Leq 5 mins noise monitoring results at Station N2 ....                                                           | 152 |
| Table 6.7. Summary of the baseline Leq 5 mins noise monitoring results at Station N3 ....                                                           | 153 |
| Table 6.8. Summary of the baseline Leq 5 mins noise monitoring results at Station N4 ....                                                           | 154 |
| Table 6.9. Summary of the baseline Leq 5 mins noise monitoring results at Station N5 ....                                                           | 156 |
| Table 6.10. Adjusted permissible noise limit at respective receptors on Monday–Saturday .....                                                       | 158 |
| Table 6.11. Typical sound power levels of construction equipment .....                                                                              | 160 |
| Table 6.12. Predicted noise levels at Area 1 and Area 2 without mitigation measures .....                                                           | 161 |
| Table 6.13. Predicted noise levels at Area 3 without mitigation measures.....                                                                       | 162 |
| Table 6.14. Predicted noise levels at Area 1 and Area 2 with mitigation measures .....                                                              | 164 |
| Table 6.15. Predicted noise levels at Area 3 with mitigation measures .....                                                                         | 166 |
| Table 6.16. Environmental Scores of the identified noise impacts with corresponding mitigation measures for Area 2 .....                            | 170 |
| Table 6.17. Environmental Scores of the identified noise impacts with corresponding mitigation measures for Area 3 .....                            | 171 |
| Table 7.1. Applicable Singapore Ambient Air Quality Targets (NEA, 2021) .....                                                                       | 173 |
| Table 7.2. List of identified air-sensitive receptors .....                                                                                         | 174 |
| Table 7.3. Location of ambient air quality monitoring stations .....                                                                                | 174 |
| Table 7.4. 24-hour average of PM <sub>10</sub> , PM <sub>2.5</sub> , and SO <sub>2</sub> levels at Station A1.....                                  | 176 |
| Table 7.5. 8-hour and maximum daily hourly average of CO level at Station A1 .....                                                                  | 176 |
| Table 7.6. 8-hour average of O <sub>3</sub> and maximum daily hourly average of NO <sub>2</sub> at Station A1                                       | 177 |
| Table 7.7. 24-hour average of PM <sub>10</sub> and PM <sub>2.5</sub> levels at Station A2.....                                                      | 177 |
| Table 7.8. 8-hour and maximum daily hourly average of CO level at Station A2 .....                                                                  | 177 |
| Table 7.9. 8-hour average of O <sub>3</sub> and maximum daily hourly average of NO <sub>2</sub> at Station A2                                       | 178 |
| Table 7.10. Environmental Scores of the identified impacts on site's ambient air quality with corresponding mitigation measures.....                | 185 |
| Table 8.1. BS 5228 Part 2 Guidance on Human Response to Vibration Levels.....                                                                       | 187 |
| Table 8.2. List of identified vibration sensitive receptor .....                                                                                    | 188 |
| Table 8.3. Location of vibration monitoring stations.....                                                                                           | 189 |
| Table 8.4. Ground-borne vibration level at Station V2 .....                                                                                         | 190 |

|                                                                                                                          |     |
|--------------------------------------------------------------------------------------------------------------------------|-----|
| Table 8.5. Perceptible vibration levels as per BS 5228-2 .....                                                           | 191 |
| Table 8.6. Environmental Scores of the identified vibration impacts with corresponding mitigation measures .....         | 195 |
| Table 9.1. Environmental Scores of the identified light impacts with corresponding mitigation measures .....             | 203 |
| Table 10.1. Types of waste likely to be generated during construction phase .....                                        | 204 |
| Table 10.2. Environmental Scores of the identified waste management impacts with corresponding mitigation measures ..... | 209 |
| Table 11.1. Common vectors in Singapore .....                                                                            | 210 |
| Table 11.2. Environmental Scores of the identified vector control impacts with corresponding mitigation measures .....   | 215 |
| Table 12.1. Summary of Environmental Scores of the identified impacts with corresponding mitigation measures .....       | 216 |
| Table 13.1. Roles and responsibilities of EMMP team members .....                                                        | 234 |
| Table 13.2. Training Programme for Site Personnel .....                                                                  | 236 |
| Table 13.3. Monthly Environmental Performance Report framework .....                                                     | 238 |
| Table 13.4. List of identified target species .....                                                                      | 240 |
| Table 13.5. Wildlife response and rescue plan .....                                                                      | 248 |
| Table 13.6. Maintenance regimes for salvaged plants at nursery .....                                                     | 253 |
| Table 13.7. TPZ size required for different girth range .....                                                            | 255 |
| Table 13.8. Parameters to take note during monthly tree inspections .....                                                | 261 |
| Table 13.9. Recommended Environmental Monitoring Plan for construction phase .....                                       | 262 |

## ABBREVIATIONS

| Abbreviation      | Definition                                                |
|-------------------|-----------------------------------------------------------|
| µg/m <sup>3</sup> | microgram per cubic meter                                 |
| ACC               | Alternative Control Center                                |
| ASR               | Air Sensitive Receptor                                    |
| BMP               | Best Management Practices                                 |
| CCKWW             | Choa Chu Kang Waterworks                                  |
| CEMMP             | Construction Environmental Management and Monitoring Plan |
| CCT               | Chlorine Contact Tank                                     |
| CWT               | Clear Water Tank                                          |
| CF                | constant frequency                                        |
| CO                | carbon monoxide                                           |
| COP               | Code of Practice                                          |
| Cult.             | cultivated                                                |
| dB(A)             | A-weighted decibel                                        |
| DTSS              | Deep Tunnel Sewerage System                               |
| ECM               | Earth Control Measure                                     |
| ECMO              | Earth Control Measure Officer                             |
| ECO               | Environmental Control Officer                             |
| EIA/S             | Environmental Impact Assessment/Study                     |
| EMMP              | Environmental Management and Monitoring Plan              |
| EPH               | Environmental Public Health                               |
| EPHA              | Environmental Public Health Act                           |
| EPM               | Environmental Protection and Management                   |
| EPMA              | Environmental Protection and Management Act               |
| ERSS              | Earth Retaining Stabilizing Systems                       |
| ES                | Environmental Score                                       |
| FM                | Frequency Modulated                                       |
| GPS               | Global Positioning System                                 |
| HDB               | Housing Development Board                                 |
| kVA               | Kilovolt-amps                                             |
| ha                | Hectors                                                   |
| hp                | Horsepower                                                |
| hr                | hour                                                      |
| ISA               | International Society of Arboriculture                    |
| ITT               | Invitation to Tender                                      |
| IUCN              | International Union for the Conservation of Nature        |
| JRL               | Jurong Region Line                                        |
| kHz               | kilohertz                                                 |
| km                | kilometre                                                 |

| <b>Abbreviation</b> | <b>Definition</b>                                  |
|---------------------|----------------------------------------------------|
| LA <sub>eq</sub>    | equivalent continuous A-weighted sound power level |
| Leq                 | equivalent continuous noise level                  |
| LTA                 | Land Transport Authority                           |
| m                   | meter                                              |
| m <sup>2</sup>      | meter square                                       |
| mg/L                | milligram per litre                                |
| mg/m <sup>3</sup>   | milligram per cubic meter                          |
| mgd                 | million gallon per day                             |
| mins                | minutes                                            |
| ms                  | milliseconds                                       |
| NSR                 | Noise Sensitive Receptor                           |
| MOB                 | Membrane-Ozone-BAC                                 |
| MND                 | Ministry of National Development                   |
| MPA                 | Maritime and Port Authority of Singapore           |
| NEA                 | National Environment Agency                        |
| n.d.                | no date                                            |
| NO <sub>2</sub>     | nitrogen dioxide                                   |
| NParks              | National Parks Board                               |
| NSFD                | Near Sulphur-Free Diesel                           |
| NTU                 | Nanyang Technological University                   |
| PAR                 | Photosynthetic Active Radiation                    |
| PCN                 | Park Connector Network                             |
| PM <sub>10</sub>    | particulate matter with diameter ≤ 10 micrometres  |
| PM <sub>2.5</sub>   | particulate matter with diameter ≤ 2.5 micrometres |
| ppb                 | parts per billion                                  |
| PPE                 | Personal Protective Equipment                      |
| ppm                 | parts per million                                  |
| PSI                 | Pollutant Standards Index                          |
| PUB                 | Public Utilities Board                             |
| QECF                | Qualified Erosion Control Professional             |
| RFP                 | Request for Proposal                               |
| RIAM                | Rapid Impact Assessment Matrix                     |
| RC                  | Reinforced Concrete                                |
| s                   | seconds                                            |
| SAFTI               | Singapore Armed Forces Training Institute          |
| SCDF                | Singapore Civil Defence Force                      |
| SFA                 | Singapore Food Agency                              |
| SHD                 | Singapore Height Datum                             |
| SS                  | Singapore Standards                                |
| SSO                 | Singapore Statutes Online                          |

| <b>Abbreviation</b> | <b>Definition</b>                             |
|---------------------|-----------------------------------------------|
| t                   | Tonnes                                        |
| TAC                 | TEMBUSU Asia Consulting Pte Ltd               |
| TIW                 | Toxic Industrial Waste                        |
| TPZ                 | Tree Protection Zone                          |
| TSS                 | Total Suspended Solid                         |
| TWPS                | Treated Water Pumping Station                 |
| URA                 | Urban Redevelopment Authority                 |
| US EPA              | United States Environmental Protection Agency |
| VSR                 | Vibration Sensitive Receptor                  |
| .wav                | Waveform Audio File format                    |
| WHO                 | World Health Organisation                     |

## GLOSSARY

**Abundance:** The number of a single species recorded at any given time period or location.

**Biodiversity:** The variety of plant and animal life in the world, habitat or location, a high level of which is usually considered to be important and desirable. Biodiversity can be assessed at more focused taxonomic groups such as “bird biodiversity”, in which case it is interchangeably with “diversity”.

**Conservation Status:** A status given to a species that is threatened with becoming extinct either locally or globally. These species may be restricted to only a small area, show noticeable decline in abundance over time, or have a historically low global population size. Assessments can be made either at global level under the IUCN’s Red List of Threatened Species or at national level (e.g., Singapore’s Red Data Book of Threatened Plants and Animals).

**Constant Frequency (CF):** The portion of an ultrasonic bat call that has a constant (or near constant) frequency; in a sonogram display, this will appear as horizontal line.

**Ecology:** The pattern of relations between organisms and their environment.

**Edge Effect:** The effect of an abrupt transition between two quite different adjoining ecological communities on the numbers and kinds of organisms in the marginal habitat

**Fauna:** Referring to all animal life present in an area. Animals are defined as any species from the Kingdom Animalia.

**Flora:** Referring to all plant life present in an area. Plants are defined as any species from the Kingdom Plantae.

**Frequency Modulated (FM):** Portion of an ultrasonic bat call that has a constantly changing frequency; this will typically appear in a sonogram as a steeply dipping line which, depending on the species, may be straight, slightly curved, or may have varying steepness.

**Genus:** A taxonomic group above species. A genus consists of closely related species. For example, Grey Heron and Purple Heron are closely related species in the same genus *Ardea*, hence their scientific names are *Ardea cinerea* and *Ardea purpurea* respectively.

**Habitat:** The natural home or environment of an animal, plant, or other organisms.

**Herpetofauna:** A taxonomic sub-group that includes amphibians and reptiles.

**Impact:** Any positive or negative alteration of existing conditions caused directly or indirectly by the project.

**kHz (kilohertz):** A measure of frequency equivalent to 1,000 cycles per second. Human hearing may extend up to 20 kHz. Most bat calls are beyond 20 kHz, extending locally up to 245 kHz in the case of *Kerivoula hardwickii*.

**Leachate:** Any liquid that, in the course of passing through matter, extracts soluble or suspended solids, or any other component of the material through which it has passed.

**Microclimate:** Local atmospheric zone where the climate differs from the surrounding area.

**Mitigation Measure:** Means to prevent, reduce, or control negative environmental effects of a project, and repair any damage to the environment caused by those effects through replacement, restoration, compensation, or any other means.

**ms (milliseconds):** 1/1000 of a second. Duration of individual bat pulses typically range from 2 ms, in some species of *Myotis*, to more than 50 ms, in some local emballonurids such as *Saccolaimus saccolaimus*.

**Odonates:** A taxonomic sub-group of Insects that includes dragonflies and damselflies.

**Population:** The term population can be in reference to the total number of a species found in a given area (e.g., global population, or Singapore population). It is also used as a term to define distinct sub-sets of a species based on the level of inter-mixing. For example, an island may hold two populations of a species if there are two groups of the same species present and those groups are sufficiently prevented (geographically or behaviourally) from mixing, forming separate breeding populations.

**Regenerated Forest:** a forest that has its trees and plants regrown through various means, e.g., seed dispersion by wind and animals or through human assistance.

**Species:** The standard classification of living organisms. It is defined as a group of living organisms consisting of similar individuals capable of exchanging genes or interbreeding. It is represented by the second word of the scientific name of an organism. For example, the scientific name of a long-tailed macaque is *Macaca fascicularis*, where *fascicularis* is its species name.

**Taxa:** In reference to a specific taxonomic group. In order of specificity, the taxonomic groups are Kingdom, Phylum, Class, Order, Family, Genus, Species.

**Transect:** A predefined line or belt along which observations and/or measurements are taken.

**.wav:** Denotes a WAV (Waveform Audio) file format, which is used for raw and typically uncompressed audio recordings, which in this EIA is used for bat acoustic recordings.

## EXECUTIVE SUMMARY

Long term planning and timely renewal of water infrastructure are key to safeguard Singapore's water security. PUB, Singapore's National Water Agency intends to reconstruct the Choa Chu Kang Waterworks (CCKWW) to ensure our water security. CCKWW, the second largest waterworks in Singapore, was constructed in phases between 1975 and 1981 to supply 80 million gallons per day (MGD) of good and safe drinking water to the western part of Singapore. Drawing water from Kranji, Pandan and Tengah reservoirs, CCKWW is the sole waterworks in the western part of Singapore and a key pillar in our water security. It is paramount to renew and re-construct the aged infrastructure at CCKWW to ensure a continued supply of good and safe drinking water well into the future.

During the reconstruction of CCKWW, which is expected to be completed by 2026, CCKWW must remain in operation to continue serving the western region of Singapore. Taking into account these operational considerations and space constraints within the existing plant, additional land is therefore needed for the new water infrastructure.

PUB has appointed Binnies Singapore Pte Ltd to provide professional engineering services for the detailed design and construction supervision of the project. TEMBUSU Asia Consulting Pte Ltd is commissioned by Binnies Singapore Pte Ltd to provide the consultancy services to conduct an Environmental Impact Assessment (EIA) study for this project and prepare an EIA report for developing the project in an environmentally sensitive manner.

The EIA study was undertaken at the preliminary design stage of the project and it assesses the proposed development's activities and impacts during construction and operation phase. The EIA report has been prepared in accordance with scope of work provided by PUB and outcome of technical agency consultation processes.

This EIA report aims to provide a baseline assessment of the project site, identify the sensitive receptors, assess the potential impacts, recommend appropriate mitigation measures to reduce the residual impacts to acceptable levels. The potential impacts on the following environmental aspects have been considered in this assessment:

- Biodiversity
- Noise
- Hydrology & Water Quality
- Ambient Air Quality
- Vibration
- Light
- Waste Management
- Vector Control

Overall, it is concluded that environmental impacts from this development project are expected to be of slight to major negative levels, which can be reduced to moderate negative levels and lower by diligently implementing the environmental mitigation measures recommended in this report.

The main findings of each environmental aspect studied has been summarised as follows.

### **Biodiversity**

The project site is located within the Western Water Catchment Area. Baseline surveys were conducted within project study area comprising various field survey methodologies including

modified gentry plots for flora, visual fauna transects, camera trapping, netting, and trapping of aquatic fauna, and acoustic bat recording. The site consists of various habitat types, of which the old native-dominated secondary forest and one of the stream habitats is identified to be of high conservation priority.

Impacts from project activities, including vegetation clearance, increased noise, and increased artificial light, may have undesirable impacts to biodiversity, especially to species of conservation value found within and around the project site. For plant species of conservation significance that are likely to be directly impacted, transplanting will be carried out, where feasible. The footprint for the CCKWW plant extension will also affect two freshwater streams, with the backfilling of the northern section of Stream 2 and the eastern branch of Stream 3. Stream 3, located on the west of the project boundary, is considered to be of higher conservation value of the two streams. As part of the pre-design mitigation measures, the project footprint has been adjusted to create a vegetated buffer between the project boundary and the main tributary of Stream 3. Care will be taken to avoid human-wildlife conflict during the construction phase due to presence of wild pigs, feral dogs, long-tailed macaques, and smooth-coated otters. With the implementation of recommended mitigation measures – e.g., training the construction staff on wildlife encounters, erecting hoarding to prevent fauna from running onto roads, and designating areas for food and waste disposal – it is expected that any such impacts will be reduced to moderate levels and below.

### **Hydrology & Water Quality**

The project site is a part of the Jurong Lake and Poyan-Tengeh catchment area. There are three natural streams within project study area. In general, the surface water quality of these streams complied with the applicable standards except for the pH and the Dissolved Oxygen. Likely impacts from the construction phase of proposed project includes disturbance to local hydrological pattern due to backfilling of two natural streams within project footprint, impact to water quality due to sediment runoff and trade effluent discharge. The significant impacts to Stream 3, which is of higher conservation value, were avoided by setting back the project footprint to safeguard the main tributary of the stream. Other recommended mitigation measures – such as properly implementing Earth Control Measures (ECMs) and managing trade effluent/sewage discharge – will help to reduce the impact on water quality of project area.

### **Noise**

As the project site is vegetated, it is inhabited by various fauna species which may be sensitive to noise. There are sensitive human receptors situated within the project study area at NTU premises. A seven-day continuous (24x7) noise monitoring was carried out at five locations to establish baseline noise levels. The baseline noise levels generally complied with Singapore's noise regulations. It is anticipated that the main noise sources during the construction phase will be associated with high-impact stationary and mobile equipment. The overall impact of noise on the sensitive receptors has been assessed as minor and can be further controlled if recommended noise mitigation measures – such as avoiding night works, erecting noise barriers, using quieter construction methods, and using construction equipment and vehicles with low noise levels – are suitably designed and implemented.

### **Ambient Air Quality**

Currently there are no stationary sources of air pollution within project study area as well as its surrounding area. A seven-day continuous (24x7) air monitoring was carried out at two locations to establish baseline air quality levels. The baseline air quality levels are within Singapore's air quality targets and project study area enjoys good air quality. There are sensitive human receptors within project study area at NTU premises. The air quality will potentially be affected by activities associated with construction of treatment units, especially from an increase in airborne particulates and heavy-vehicle exhaust emissions. These may create a moderate and direct impact on local air quality and affect sensitive receptors located in the proximity of dust generating construction activities, including biodiversity of project area. These impacts can be brought down to an acceptable level with the implementation of various mitigation measures – e.g., implementing dust suppression plan, installation of dust screens, covering stockpiles of dusty materials with impervious sheeting, and regular watering of construction site.

### **Vibration**

Currently there are no major sources of vibration within project study area as well as its surrounding area. A baseline monitoring was carried out at two locations to establish baseline vibration levels. There are sensitive receptors located adjacent to project study area, including animals in forested areas and humans in NTU premises. The overall impact of vibration has been assessed as minor and can be further controlled if recommended mitigation measures – such as avoiding night works, use of equipment or method which generate lower vibration levels, implementation of vibration monitoring plan – are suitably designed and implemented. Please see Appendix K for the Preliminary Vibration Analysis Report.

### **Light**

Impacts of light pollution during the construction period may take place, but such impacts are expected to be minor considering nightworks within forested areas will be avoided as far as practicable. Additionally, the baseline light levels are high, so any nightworks within the existing plant should not significantly contribute to light pollution. Nonetheless, mitigation measures such as adjustment of working hours and appropriate positioning of artificial lights during the construction phase and appropriate design of lights for operational phase are recommended to further reduce the impacts of light pollution.

### **Waste Management**

It is anticipated that the main sources of waste during construction phase will be from cut vegetation, excavated material, general construction waste, personal waste and/or hazardous waste. With the implementation of the mitigation measures proposed – such as implementing a waste management plan, reusing timber wastes for wood industry, and properly disposing the waste through licensed collectors – any negative impact is expected to be reduced to an acceptable level.

### **Vector Control**

The primary impact of the construction phase of the project is potential increase in the immediate vector population. Secondary to this, an increase in the number of vectors has the potential to increase the likelihood of vector-borne diseases affecting humans. Key mitigation measures include source reduction and effective drainage through implementation of vector control plan. With its implementation, it is expected that these impacts can be reduced to an

insignificant level. Considering the sensitive nature of forested area, fogging is not recommended to be carried out as part of vector control measures.

An Environmental Management and Monitoring Plan (EMMP) has been proposed to manage the identified environmental impacts during construction phase. It also includes environmental monitoring requirements containing on-site visual compliance monitoring and physical monitoring which will help to verify the effective implementation of mitigation measures during construction stage. It is recommended that the appointed Contractor develops a Construction EMMP (CEMMP) based on the recommended EMMP framework in this EIA report before initiating construction work and to strictly implement the CEMMP requirements throughout construction phase to ensure the development of this project in environmentally sensitive manner.

# 1 INTRODUCTION

## 1.1 Project Background

PUB, Singapore's National Water Agency intends to reconstruct the Choa Chu Kang Waterworks (CCKWW) to ensure our water security. PUB has appointed Binnies Singapore Pte Ltd to provide professional engineering services for the detailed design and construction supervision of the project. TEMBUSU Asia Consulting Pte Ltd (TAC) is commissioned by Binnies Singapore Pte Ltd to provide the consultancy services to conduct an Environmental Impact Assessment (EIA) for this project and prepare an EIA report for developing the project in an environmentally sensitive manner.

An EIA Inception Report covering the environmental baseline survey methodology and EIA study area was prepared and submitted to all relevant stakeholders in July 2021. Upon confirmation of the Inception Report, the environmental baseline surveys were carried out in the period between August 2021 and October 2021.

## 1.2 Objectives of the Environmental Impact Assessment Report

The EIA study was undertaken at the preliminary design stage of the project and assesses the proposed development's activities and impacts during construction and operation phase.

The principal objective of the EIA study is to provide clear and concise technical information for decision-making on potential environmental impacts associated with proposed development.

The key objectives of this EIA study are to:

- Understand the environmental baseline through the collection of primary and secondary data.
- Assess the impacts of the CCKWW development during the construction and operational phases of the project on the environment.
- Present appropriate mitigation measures to reduce the level of impact for each activity assessed that has a moderate to major impact.
- Recommend an environmental management framework to monitor the mitigation measures implementation.

It is understood that the information presented in the EIA report will contribute to decisions on:

- The overall acceptability of any adverse environmental consequences that are likely to arise as a result of the proposed re-construction of CCKWW.
- The conditions and requirements for the re-construction of CCKWW project to mitigate adverse environmental consequences whenever practicable; and
- The acceptability of residual impacts after the proposed mitigation measures are implemented.

The undertaking of the EIA study will therefore promote environmentally sound and sustainable development.

### **1.3 Report Structure**

The EIA report is structured as follows:

- Chapter 1 introduces the project background and the general information of the EIA, covering its objectives, report structure, and limitations.
- Chapter 2 provides the description of the proposed project, and activities associated with the proposed project.
- Chapter 3 outlines the EIA approach, including its scope and impact assessment methodology that is applied in the preparation of this report.
- Chapter 4 presents assessment of environmental impacts on biodiversity and its proposed mitigation measures.
- Chapter 5 discusses the noise impacts and proposes the mitigation measures.
- Chapter 6 describes the hydrology and water quality of the project study area and provides the impacts and mitigation measures.
- Chapter 7 provides a description of the ambient air quality impacts of the project and proposes the mitigation measures.
- Chapter 8 discusses the impacts from ground-borne vibration and proposes the mitigation measures.
- Chapter 9 discusses the impacts from light pollution and proposes the mitigation measures.
- Chapter 10 discusses the waste impacts and proposes the mitigation measures.
- Chapter 11 details the vector control assessment and mitigation requirements.
- Chapter 12 summarises the impact assessment and mitigation measures of each environmental aspect.
- Chapter 13 outlines the proposed the Environmental Management and Monitoring Plan (EMMP) framework.
- Chapter 14 lists the references used in the report.

### **1.4 Limitations of the EIA Report**

The EIA study is conducted as per the requirements specified by PUB in the Invitation to Tender (ITT), Request for Proposal (RFP), and outcome of consultancy with Technical Agencies. The extent of collection of baseline data is guided by these requirements.

In preparing this report, we relied, in whole or in part, on data and information provided by PUB, Binnies, and third parties, which have been assumed to be accurate, complete, and reliable as of the time of writing.

## 2 PROJECT OVERVIEW

### 2.1 Project Background

Long term planning and timely renewal of water infrastructure are key to safeguard Singapore's water security. PUB, Singapore's National Water Agency intends to reconstruct the Choa Chu Kang Waterworks (CCKWW) to ensure our water security. CCKWW, the second largest waterworks in Singapore, was constructed in phases between 1975 and 1981 to supply 80 million gallons per day (MGD) of good and safe drinking water to the western part of Singapore. Drawing water from Kranji, Pandan and Tengah reservoirs, CCKWW is the sole waterworks in the western part of Singapore and a key pillar in our water security. It is paramount to renew and re-construct the aged infrastructure at CCKWW to ensure a continued supply of good and safe drinking water well into the future.

During the reconstruction of CCKWW, which is expected to be completed by 2026, CCKWW must remain in operation to continue serving the western region of Singapore. Taking into account these operational considerations and space constraints within the existing plant, additional land is therefore needed for the new water infrastructure.

### 2.2 Project Scope

The project comprises of construction of the following new water treatment units and facilities not limited to:

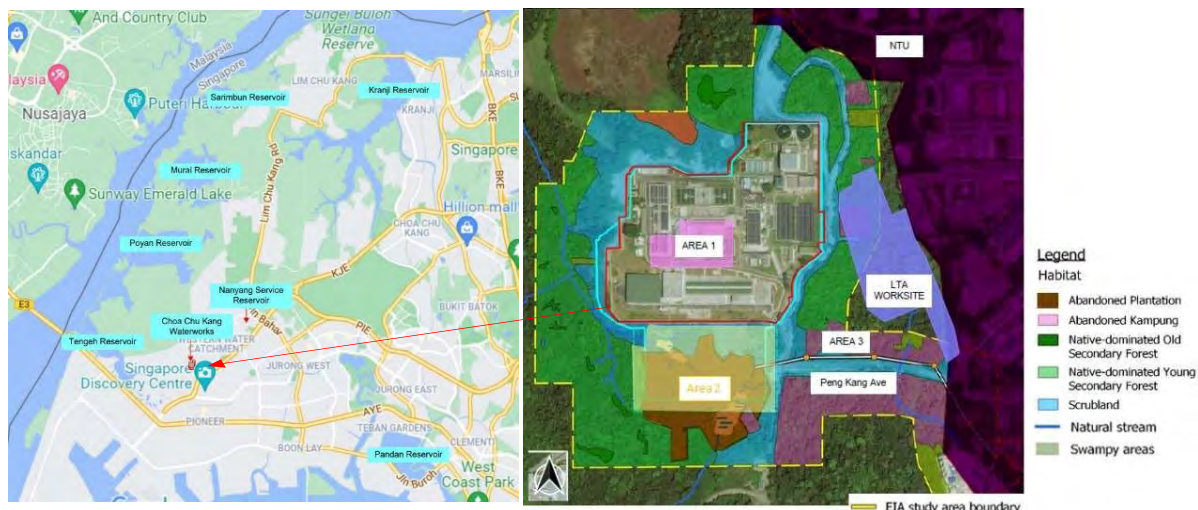
- New sedimentation tanks.
- Treated water tanks and pumping stations.
- Pre-treatment and post-treatment chemical buildings, Membrane Filtration building, administration building.
- Electrical buildings, fuel storage facilities, underground piping and services, perimeter fence, and earth-retaining structures, sewer pipeline alignment along Peng Kang Ave and Nanyang Drive and demolition of old treatment facilities upon completion of new facilities.

The works will be carried out in the following three stages:

- Stage I – Investigation and preliminary engineering design
- Stage II – Detailed engineering design and tendering
- Stage III – Supervision, construction, and commissioning of the Works

### 2.3 Project Location

CCKWW is located within the Western Catchment Area, with Nanyang Technological University (NTU) to its east, forested area to its south, and the rest of the Western Catchment Area to its north and west (see Figure 2.1). The proposed site for reconstruction of CCKWW is adjacent to existing CCKWW.



**Figure 2.1.** Location plan of CCKWW

As existing CCKWW treatment infrastructure must remain operational until the new water infrastructure is completed, options for the additional land required to house the new infrastructure were studied. The plot of green-field land to the south of the existing CCKWW plant is proposed for the following reasons:

- Northwest of CCKWW: There is a sludge lagoon that is currently not in use, the soil condition is soft and poor in quality. This is unsuitable for new building construction and for new pipework, which conveys drinking water, to be buried. Any soil improvement works will take ~2 years and adversely impact the expected project completion by 2026. Construction of new facilities to the north would require an additional relift pump system to convey the water from the existing plant to the new land parcel as the north plot is on higher elevation. This results in additional pumping energy and cost.
- Northeast of CCKWW: The land plot is more elevated. This will incur more energy usage and cost due to the need to pump water to a higher ground. 99 flora species were identified during our environmental survey, out of which 68 are native.
- South of CCKWW: The good soil condition is suitable for new construction without the need for further ground treatment / improvement works, which would then enable the project to be completed by 2026. Faster completion results in shorter period of disturbance to the surrounding environment. The greenfield parcel is also at comparatively lower ground level compared with the plot at the north and therefore relift pumping is not needed, hence improving energy efficiency and carbon footprint during the operation of the plant. The design of new facilities is based on compact design to minimise landtake. PUB also altered the project footprint away from stream 3 to protect biodiversity. This area consists of Abandoned Plantation and Scrubland type vegetation, 100 flora species were identified during our environmental survey, out of which 68 are native.
- West and Southwest of CCKWW: The land parcel to the west will encroach into areas of good forest, freshwater streams and biodiversity.
- East of CCKWW: The land parcel has been safeguarded by LTA for the construction of the Jurong Region Line Station.

Additionally, an external Sewer pipeline (700 mm diameter) will be laid from Area 2 to the DTSS chamber along PIE/Nanyang Drive through pipe jacking method.

## 2.4 Overview of Construction Work Areas

The project sites are divided into 3 main construction work areas (Area 1, Area 2 and Area 3) as shown in Figure 2.2 and described below. **Appendix A** provides the overall layout plan of the project area.



**Figure 2.2.** Construction work areas of CCKWW reconstruction project

### 2.4.1 Construction of New Facilities & Demolition of Old Facilities within Existing CCKWW Plant Area (Area 1)

All Area 1 activities are within the existing CCKWW plant. Area 1, also known as pre-treatment area, covers approximately 1.2 hectares and is north of the existing MOB (Membrane-Ozone-BAC) Building within CCKWW. Activities in this area involves first stage of demolition of existing buildings and facilities (see Figure 2.9), to give space for new construction of sedimentation tank, membrane filtration building and pre-treatment chemical building. Area 1 has highest platform level, ranging from RL 41 m to 44 m.

Second stage of demolition at Area 1 can only be proceeded upon completion of new treatment

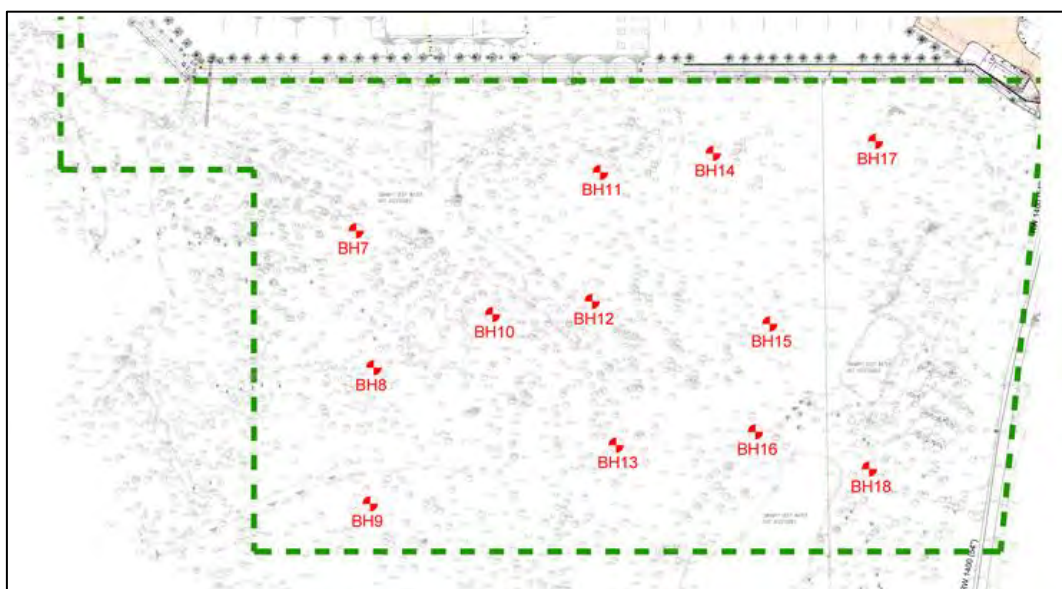
buildings and facilities at both Area 1 & 2. Buildings to be demolished during this stage are shown in Figure 2.9.

#### **2.4.2 Construction of New Facilities within CCKWW Extension in Green Field Area (Area 2)**

All Area 2 activities are in new land parcel of CCKWW Extension Area (green field area). Area 2 or also known as post-treatment area, is to the south of existing CCKWW plant. This area includes the existing forested area, and in general an undulating terrain, ranging from RL 15m to RL 38m with swampy deep-water areas. Site clearance and earthworks are to be carried out to remove unsuitable soil and form the required building/ road level. Post treatment units such as Chlorine Contact Tank (CCT), Clear Water Tank (CWT), Treated Water Pumping Station (TWPS), electrical buildings, administration building, Post Treatment Chemical buildings, and other ancillary buildings will be constructed here.

##### Site Investigation Work

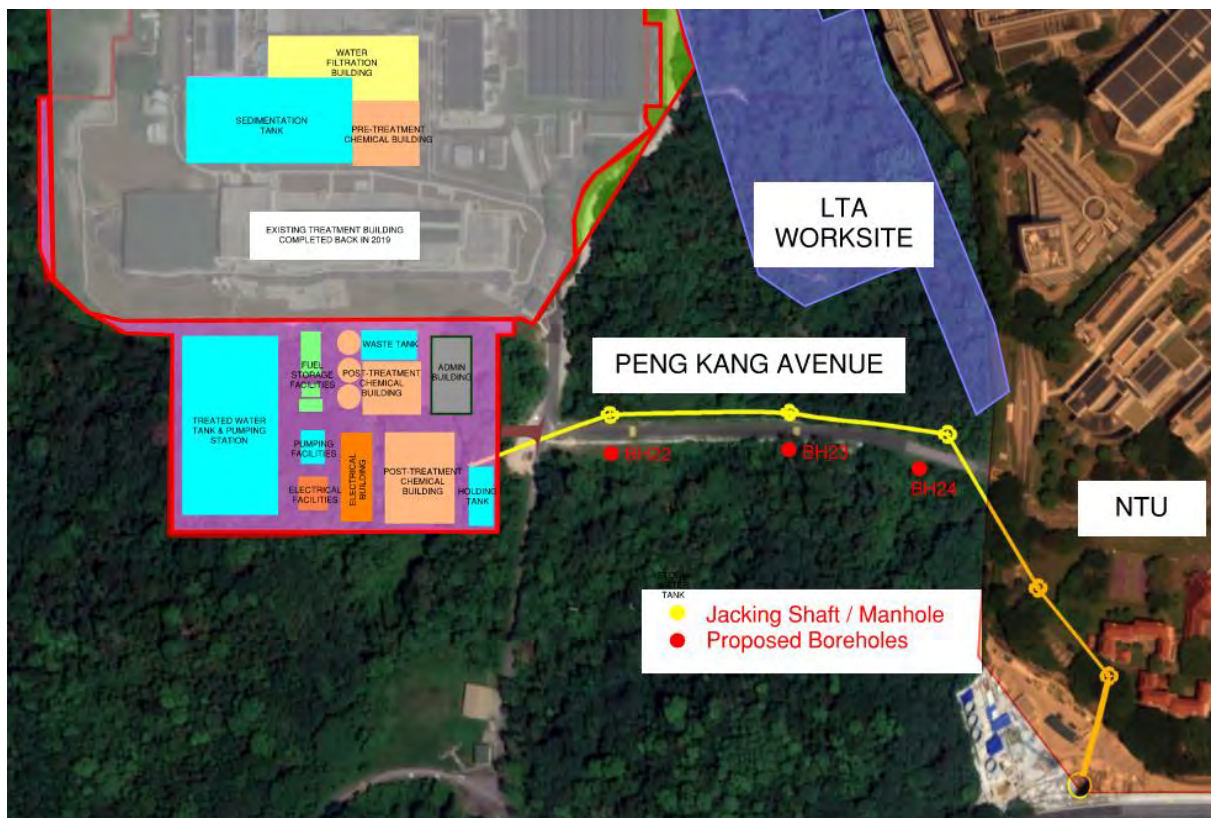
A total of 12 nos. of boreholes are proposed to be carried out within this Area 2 as part of the soil investigation programme for the proposed building foundation design (Figure 2.3).



**Figure 2.3.** Soil investigation borehole layout plan for Area 2

#### **2.4.3 Construction of New Sewerage Alignment from CCKWW Plant to DTSS Chamber (Area 3)**

Approximately 600 m long external sewerage main will be constructed outside the boundary of CCKWW plant by pipe jacking method for connection to the DTSS chamber off PIE/Nanyang Drive. Total 3 nos. of soil investigation boreholes are proposed along the new sewerage alignment at Area 3 off Peng Kang Avenue (refer to Figure 2.4).



**Figure 2.4.** Soil investigation borehole layout plan for Area 3

## 2.5 Proposed Plant Layout Plan

The overall plant layout is planned to meet design, operation and maintenance, safety and constructability, and local permitting requirements. The followings considerations are incorporated in the plant layout during preliminary design:

- Min. 6 m wide fire engine accessway & its turning radius complying with Fire Code 2018
- New alternate access road from Area 2 to Area 1 to facilitate delivery of heavy plant and equipment
- Siting of building footprint to ensure sufficient working space for manoeuvring of heavy plant and machinery for demolition and construction works
- Min. 2 m wide green buffer along Area 2 perimeter, complying with NParks' requirement
- Maintain safe setback from building footprint to existing critical underground services i.e., incoming raw water pipes
- Siting of building footprint to optimize the length of new underground yard piping, cabling and chemical trenching works to be constructed interconnecting among buildings
- Siting of building openings adjacent to road access to ease loading / unloading and other operation and maintenance works

### 2.5.1 Layout Plan within Existing CCKWW Plant Area (Area 1)

Area 1 is within the existing CCKWW, towards the north of existing Membrane-Ozone-BAC (MOB) Building. An internal new access road will be provided on the north and west perimeter of the building connecting to existing roads on the south and east, thereby providing access to the entire perimeter of the building. The proposed layout plan of new treatment units of Area 1 is presented in Figure 2.5.



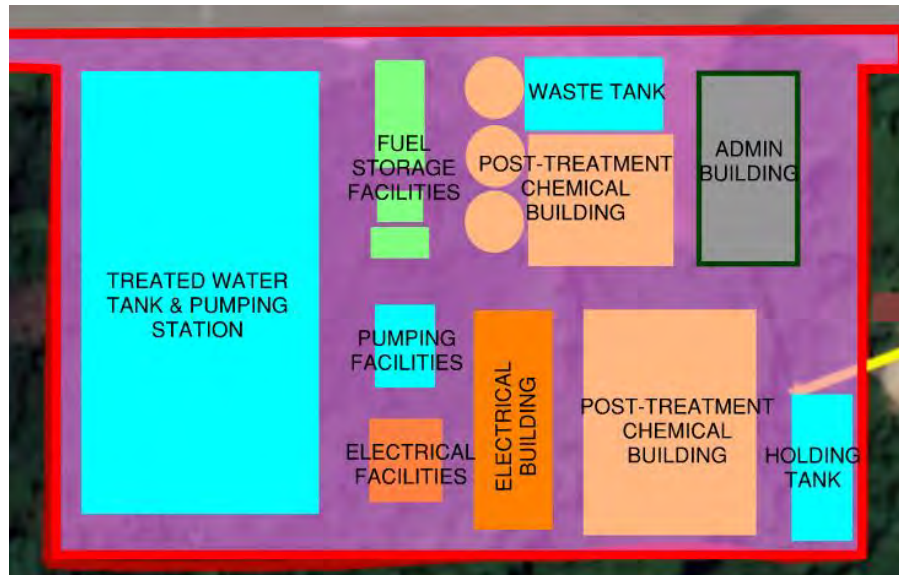
**Figure 2.5.** Proposed layout plan of Area 1

### **2.5.2 Layout Plan within CCKWW Extension (Green-Field Area) (Area 2)**

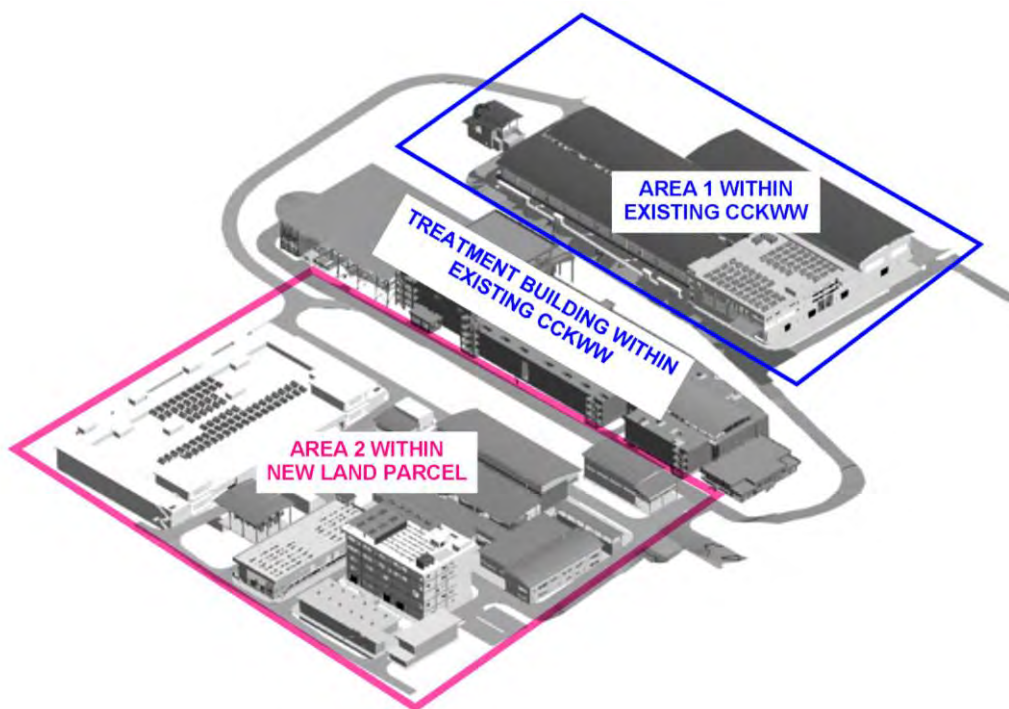
The proposed Area 2 layout has taken into consideration the following approaches to reduce impact to the environment:

- The project footprint has been adjusted to create a vegetated buffer between the project boundary and the main tributary of Stream 3.
- Minimised earthwork/excavation works by designing building levels to suit existing terrain.
- Waste tanks to store and contain all trade effluents from treatment process for discharge to external sewer line along Peng Kang Avenue.
- Watertight construction and containment measures for all water/waste tanks to prevent leaks from tank structure to nearby environment.
- Provision of containment for all chemicals complying with both SCDF's Fire Code and NEA's Code of Practice for Pollution Control. Design plans will be submitted to both SCDF and NEA for approval before construction.
- Storage of fuel oil complying with SS532: Code of practice for the storage of flammable liquids. Design plans will be submitted to SCDF and NEA for approval before construction.
- All chemical pipes will be contained within concrete trench with containment kerb provided around chemical charging areas with collection sump. Any leak and wash water are to be conveyed to waste tanks for discharge to external sewer.
- Containment kerb provided around fuel oil charging area with oil interceptor to contain any oil spills and to separate oil and rainwater as per NEA's requirement.
- Stormwater run-off from Area 2 will be collected at stormwater detention tank with controlled discharge back to Stream 2, hence preventing stream bank erosion and flooding downstream without changing the catchment area of Poyan-Tengeh reservoirs.

The proposed layout plan of various treatment units in Area 2 is shown in Figure 2.6. The road / building levels are set to reduce earthwork/excavation works required within the green-field area for the construction of future building structures. Figure 2.7 provides the 3D representation of layout plan of Area 1 and Area 2.



**Figure 2.6.** Proposed layout plan of Area 2



**Figure 2.7.** 3D representation of layout plan of Area 1 and Area 2

### 2.5.3 Layout Plan of Proposed Sewerage Alignment from CCKWW Plant to DTSS Chamber (Area 3)

The sewer, 700 mm in diameter, will be laid using pipe jacking method which is a trenchless method to reduce amount of earth to be excavated along the proposed pipe alignment. The total length of sewer is approximately 600 m with three jacking shafts/manholes along Peng Kang Avenue and two other jacking shafts / manholes along Nanyang Drive. The proposed number of jacking shafts required is according to PUB's Code of Practice on Sewerage and Sanitary Works (i.e., distance between manholes shall not be more than 120 m). This is to ensure that future inspection, cleaning, and repair works can be carried out safely.

Figure 2.8 shows the proposed sewerage alignment layout plan including proposed work boundary around pipe jacking shaft locations in Area 3.

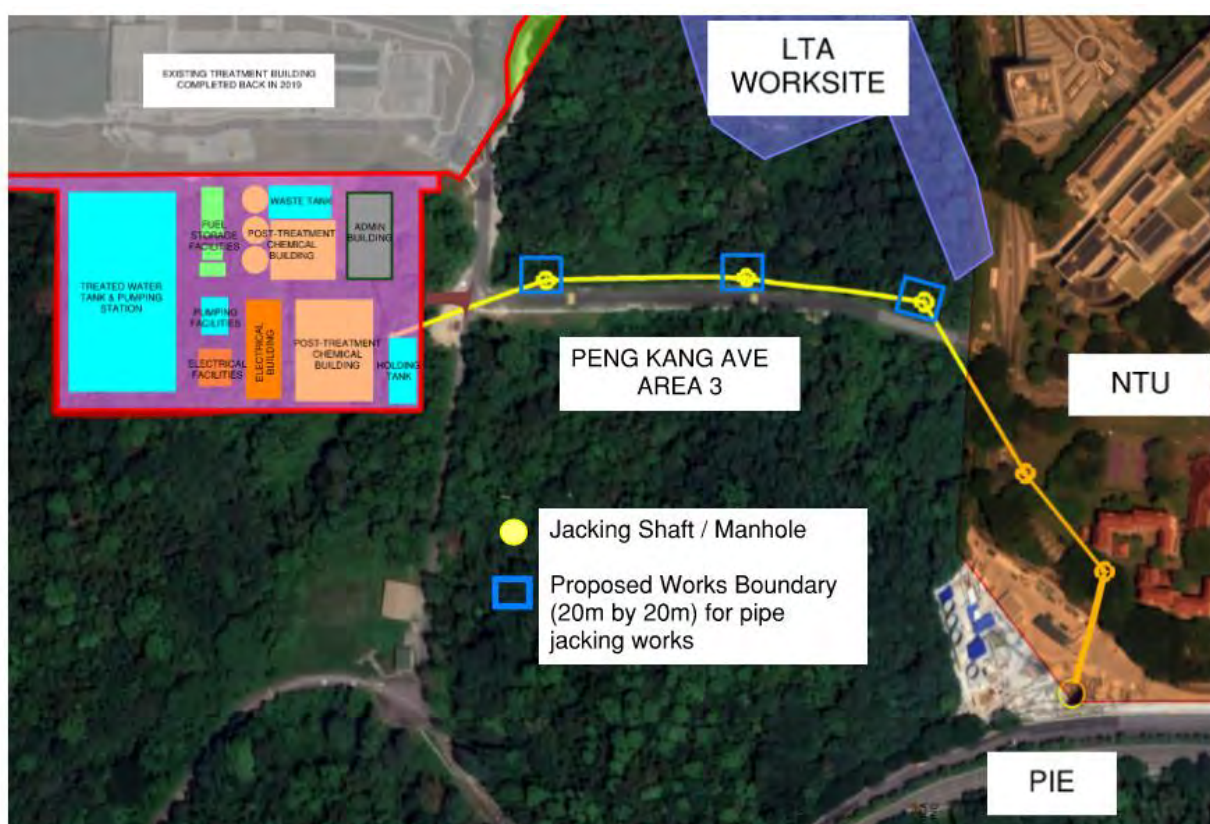


Figure 2.8. Proposed layout plan of sewerage alignment at Area 3

## 2.6 Analysis of Alternatives

During the project conception, alternatives were considered for sewerage alignment for two aspects – route of the alignment and the construction methodology. Table 2.1 and Table 2.2 outline the results of the analysis. Based on various considerations for route alignment, **Option 1 (North of Peng Kang Avenue) is the chosen alignment**. For construction methodology, **the pipejacking method (Option 1) is selected** as it would minimise impacts compared to open-cut method.

**Table 2.1.** Sewerage alignment route analysis

| Option 1 – Current alignment (north of Peng Kang Avenue) [SELECTED]                                                                                                                                                                     | Option 2 – Initial alignment (south of Peng Kang Avenue)                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>a. Route map</b></p>                                                                                                                             | <p><b>a. Route map</b></p>                                                                                                           |
| <p><b>b. Stakeholder considerations</b></p> <ul style="list-style-type: none"> <li>Stakeholder agencies are agreeable to this alignment.</li> </ul>                                                                                     | <p><b>b. Stakeholder considerations</b></p> <ul style="list-style-type: none"> <li>The south of Peng Kang Avenue is safeguarded for other agency's use only</li> </ul>                                                 |
| <p><b>c. Environmental considerations</b></p> <ul style="list-style-type: none"> <li>Scrubland habitat</li> <li>Few flora species of conservation significance within localised areas</li> </ul>                                        | <p><b>c. Environmental considerations</b></p> <ul style="list-style-type: none"> <li>Scrubland habitat</li> <li>Flora of conservation significance distributed along vegetated areas south of Peng Kang Ave</li> </ul> |
| <p><b>Selection Rationale for Option 1</b></p> <ul style="list-style-type: none"> <li>Land use constraint to the south of Peng Kang Avenue</li> <li>Lower environmental impact.</li> <li>Preferred by relevant stakeholders.</li> </ul> |                                                                                                                                                                                                                        |

**Table 2.2.** Sewerage alignment construction method analysis

| Option 1: Pipejacking method [SELECTED]                                                                                                                                                                                                                                                                        | Option 2: Open-cut method                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>a. Stakeholder considerations</b></p> <ul style="list-style-type: none"> <li>Stakeholder agencies are both agreeable to the proposed pipejacking method as only localized excavation at jacking / receiving shafts are required.</li> </ul>                                                              | <p><b>a. Stakeholder considerations</b></p> <ul style="list-style-type: none"> <li>Open-cut methodology will impact a substantial area of stakeholders' land.</li> </ul>                                                                                                                                                  |
| <p><b>b. Construction consideration</b></p> <ul style="list-style-type: none"> <li>Construction working area – approximately 20m x 20m working space and 10m diameter working shafts</li> <li>Equipment used – Jacking machine, Crawler crane, Excavator, Generator, Discharge pump, Settling tanks</li> </ul> | <p><b>b. Construction consideration</b></p> <ul style="list-style-type: none"> <li>Construction working area – Along the full stretch of the proposed pipeline, including working areas left and right of the pipeline</li> <li>Equipment used – Vibro hammer, Crawler crane, Excavator, Power pack, Generator</li> </ul> |
| <p><b>c. Environmental considerations</b></p> <ul style="list-style-type: none"> <li>Minimal vegetation clearance</li> <li>Flexibility in placement of working shafts</li> <li>Ability to avoid important habitats</li> <li>Lower impact construction methodology</li> </ul>                                   | <p><b>c. Environmental considerations</b></p> <ul style="list-style-type: none"> <li>Complete vegetation clearance for alignment</li> <li>Habitat fragmentation</li> <li>Major Impact to Stream 1</li> <li>Higher impact construction methodology</li> </ul>                                                              |

#### **Selection Rationale for Option 1**

- Reduced amount of ground disturbance and earth handling
- Reduced project footprint
- Minimal environmental impact.
- Preferred by relevant stakeholders

## **2.7 Construction Approach and Implementation Schedule**

### **2.7.1 Area 1 Construction Approach**

#### Demolition Works to cater space for future advance treatment processes requirements

Area 1 involves demolition of existing buildings and tank structures before the space can be freed up for the new buildings and tank structures to be constructed. The existing structures highlighted in Figure 2.9 will be demolished prior to construction of new buildings/tank structures. Existing structures highlighted in orange will be demolished upon full completion and commissioning of all new treatment building and facilities at both Area 1 & Area 2 to cater space for future advance treatment processes requirements.



**Figure 2.9.** Construction approach at Area 1

Typical method of demolition include:

- Cut about 0.5 m beyond existing column/beam-floor slab interface
- Remaining areas to demolish by controlled hacking and crushing
- All underground structures, such as footings and column stumps are to be crushed and removed
- Install temporary earth retaining stabilizing structures and carry out localised excavation by open excavation to expose underground structure for removal

### Construction Hoarding

The proposed works at Area 1 involves diversion, demolition, and reconstruction. The work site area will be properly hoarded, secured, and segregated from the Plant operations.

### Foundation

Bored piles would be suitable foundation to support the structures and multi-storey buildings.

### Prefabricated Steelworks

A prefabricated steel roof system is proposed, where possible. A steel roof system improves construction productivity and safety, using less time and manpower and eliminates the need for temporary staging. Each roof truss will be prefabricated offsite in segments no longer than 12 m to facilitate ease of transport and manoeuvrability. The roof truss will be assembled only via bolting.

### Lifting and Tower Cranes

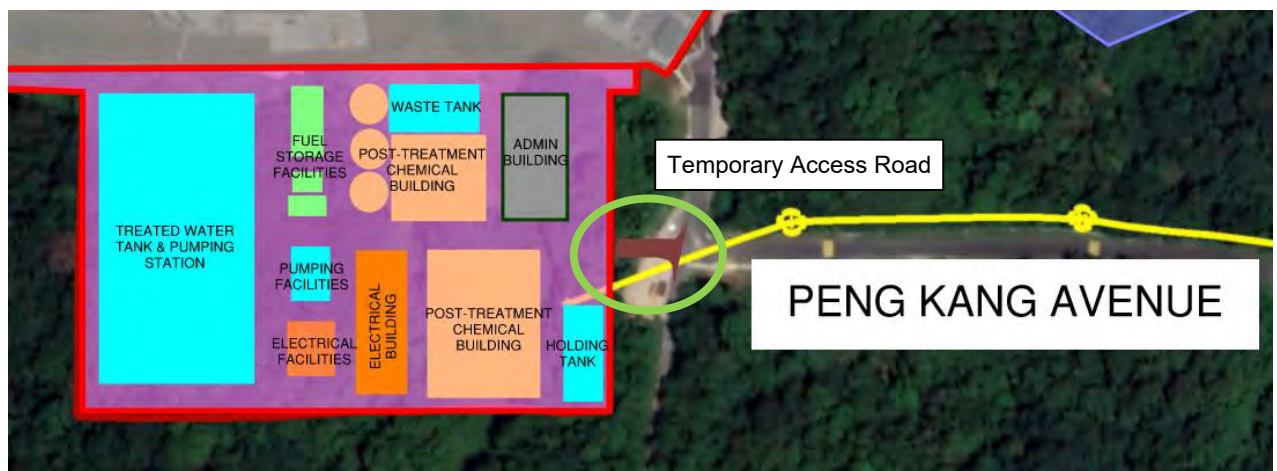
As there is limited working space available at Area 1, the mobile cranes and tower cranes will be used.

## **2.7.2 Area 2 Construction Approach**

Area 2 is a new plot with a land area of approximately 3.2 ha. It is located immediate south of the existing CCKWW. The proposed treatment building and facilities within the new land comprise the CCT, CWT, TWPS, Post Treatment Chemical Buildings, Administration Building, Waste Balancing Tank, Electrical buildings. A hoarding will be erected along the new boundary, integrated with the existing CCKWW fence line.

### Temporary Construction Access Road

A temporary construction access road is proposed to create a separate entry into Area 2. This allows construction traffic to be redirected away from the main CCKWW plant, avoiding heavy traffic flow along Peng Kang Avenue during construction, hence minimizing inconvenience to other road users.



**Figure 2.10.** Proposed temporary construction access road in Area 2 (circled light green)

### Site Clearance and Earth Works

The terrain at Area 2 is undulating. The proposed building levels will be designed to minimise cut and fill works as well as earth retaining structures required. This approach not only eases construction but also minimises impact to the nearby environment.

### Soil Investigation

In view of the undulating terrain, it is anticipated that excavator and crawler drilling rig will be deployed to access these boreholes to ensure the investigation works can be carried out safely. In rotary borehole drilling, the borehole is advanced by rotary action of a drill bit attached at the end of hollow drill rods. The drilling fluid which is pumped down to the bit through the hollow drill rods lubricates the bit and flushes the drill debris up the borehole. The drilling fluid used is bentonite.

### Temporary Works for Underground Tank Structures/Basement

The CCT and CWT are partially buried below the proposed platform level. To facilitate the safe construction of the tank, and due to the undulating terrain, a combination of open excavation and Earth Retaining Stabilizing Systems (ERSS) in the form of cantilevered sheet pile walls will be considered.

### Foundation

Bored piles will be suitable foundation to support the heavier tank structures and multi-storey buildings. Due to the varying depths of the hard ground across Area 2, some of the lightly loaded tanks and valve chambers could be supported directly on rigid ground. These will be further reviewed once the soil investigation is completed. Due to depth of the CCT and CWT, the bored piles can be drilled prior to the full excavation to the base of tank level.

### Yard Piping

The pipes will be laid at its designed bedding level before further filling of the ground to the platform level wherever possible. At areas where the existing ground level is higher than the platform level, the ground will be trimmed to the platform level prior to any trench or open excavation to install the yard piping. It is likely that the yard piping will be supported by compacting the ground without the need for piled foundation.

### Planned Night works

No major night work will be carried out within Area 2 except in the unavoidable situations due to safety and structural integrity considerations, or in the case where works are to be contained in already built-up superstructure, as described below.

**Table 2.3.** List of planned night work activities in Area 2

| <b>Planned Night Work Activity</b> | <b>Duration</b>                                                                                                                                                                   | <b>Rationale</b>                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Casting of bored concrete piles    | <ul style="list-style-type: none"><li>• Estimated Timing: 8am to 12 midnight</li><li>• Frequency: 5 times per week for 12 months</li><li>• Timeline: Q3 2023 to Q2 2024</li></ul> | <u>Safeguard building's safety and structural integrity</u><br>A single pile typically takes ~15 hours to complete, within the same working day, to reduce risk of soil collapse into boreholes or poor quality of bentonite slurry which affects soil strength and pile working capacity. Driving of long casing to prevent soil collapse in boreholes is not feasible considering soil properties at this site. |

| Planned Night Work Activity                            | Duration                                                                                                                                                                               | Rationale                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Thick slab/wall casting for water retaining structures | <ul style="list-style-type: none"> <li>Estimated Timing: 4pm to 12 midnight</li> <li>Frequency: Intermittent, maximum 2 days per week</li> <li>Timeline: Q4 2023 to Q2 2025</li> </ul> | <u>Safeguard structural integrity</u><br>Continued wall/slab panel casting is required to limit thermal stresses and cold joints which could otherwise affect the water tightness of water retaining structures (e.g. CWT/CCT, clarifier etc). Thick raft slabs are cast at night to limit the rise of concrete core temperature which would otherwise result in cracking and undermine the tank water tightness. |
| Work within completed superstructure                   | <ul style="list-style-type: none"> <li>Estimated Timing: 8am to 12 midnight</li> <li>Frequency: Continuous for 12 months</li> <li>Timeline: Q2 2025 to Q1 2026</li> </ul>              | Mechanical and electrical installation works, and architectural works within completed superstructure which will not generate much noise and vibration.                                                                                                                                                                                                                                                           |
| Delivery under traffic police escort                   | <ul style="list-style-type: none"> <li>Estimated Timing: 11pm to 4am</li> <li>Frequency: Maximum 1 day per week</li> <li>Timeline: Q2 2025 to Q4 2025</li> </ul>                       | Road safety                                                                                                                                                                                                                                                                                                                                                                                                       |

### 2.7.3 Area 3 Sewerage Alignment

The external sewerage alignment (700 mm diameter) will be laid from Area 2 to the DTSS chamber along PIE/Nanyang Drive through pipe jacking method. The total length of sewer is approximately 600 m and will be laid at the gradient of 1:300 as a gravity main. Three (3) sewerage manholes are located along Peng Kang Avenue and another two along Nanyang Drive. The distance between manholes shall not exceed 120 m complying with PUB's requirement.

Figure 2.8 shows the proposed sewerage alignment layout plan including proposed work boundary around pipe jacking shaft locations in Area 3.

There are two types of pipe jacking machine, namely Earth Pressure Balancing (EPB) machine and Slurry machines. Choosing between EPB and Slurry machines depends on ground conditions at Area 3 which is to be verified by soil investigation. To ensure excavation safety, use of EPB machine is suitable for cohesive soil while slurry machine works best for granular soil. In terms of site utilization, slurry machines require relatively large construction yard for slurry and separation plants. As the excavated material is too fluid, solid-liquid separation unit is required to allow excavated material to settle in settling tank before disposal off to approved dumping ground. Irrespective of selection of either EPB or Slurry machines, a comprehensive waste management plan will be implemented on site to minimize wastage, ensuring proper disposal and prevent pollution to the environment, complying with all applicable statutory regulations, including the Sewerage and Drainage Regulations and Environmental Protection and Management Regulations.

The pipe jacking method include following:

- a) Equipment used – Settling tanks, water tanks, jacking machine, crawler crane, excavator, power pack, generator, discharge pump
- b) Construction working area – approximately 20 x 20 m working space and 10 m diameter working shafts
- c) Construction activities:
  - Construction of concrete working shafts (i.e., jacking/ receiving shafts)
  - Installation of guide rails at jacking shaft and lower jacking machine unto guide rail, connect slurry lines to the machine. Fill slurry tanks and charge tank with water.
  - Commence jacking and install jacking pipes progressively behind the jacking machine by extending the main jack at the jacking shaft while the jacking machine advances along the proposed pipe alignment.
  - The pipe jacking operation is completed when the jacking machine reaches the receiving shaft
  - Retrieve jacking machine from the receiving shaft and construct permanent manhole at shafts before backfilling and reinstatement works

In summary, the likely construction activities expected to take place at all work areas during execution of this project are:

- Land clearing, earth works, and general construction activities (e.g., soil mixing clearing and preparation, trench excavation, backfill, compaction, spoil handling and transport, building of permanent structures);
- Vehicle movements and emissions (as a result of equipment, materials, and personnel movement in and out of construction work areas);
- Stockpiling and materials handling (delivery, unloading, and use of construction aggregates, etc.);
- Use of high impact stationary and mobile equipment such as bored pile drivers, vibro hammers, rotary drilling, mobile cranes, haul trucks, excavators, dozers, etc. leading to variable and sporadic noise levels, typically repeating over time;
- Minor equipment/ plant maintenance activities;
- Storage and disposal of solid and liquid waste; and
- Workforce activities leading to generation of general waste.

## 2.8 Implementation Schedule

Although Area 1 and Area 2 are distinct locations within and outside of CCKWW plant, concurrent progress of works at both areas is required to ensure timely completion of the reconstruction and proposed works. The total duration for execution of this project is expected to last around 4 years The implementation schedule is provided in Table 2.4.

**Table 2.4.** Implementation schedule of entire project

| Sr No. | Name of Area                                                                                                | Period             |
|--------|-------------------------------------------------------------------------------------------------------------|--------------------|
| 1      | Existing CCKWW Plant area – Area 1                                                                          | Q1 2023 to Q1 2026 |
| 2      | CCKWW extension area (Greenfield area) - Area 2                                                             | Q1 2023 to Q1 2026 |
| 3      | Sewerage alignment from CCKWW plant to DTSS Chamber – Area 3                                                | Q4 2023 to Q4 2025 |
| 4      | Demolition of existing buildings within Area 1 upon full completion of new treatment buildings & facilities | Q2 2026 to Q4 2026 |

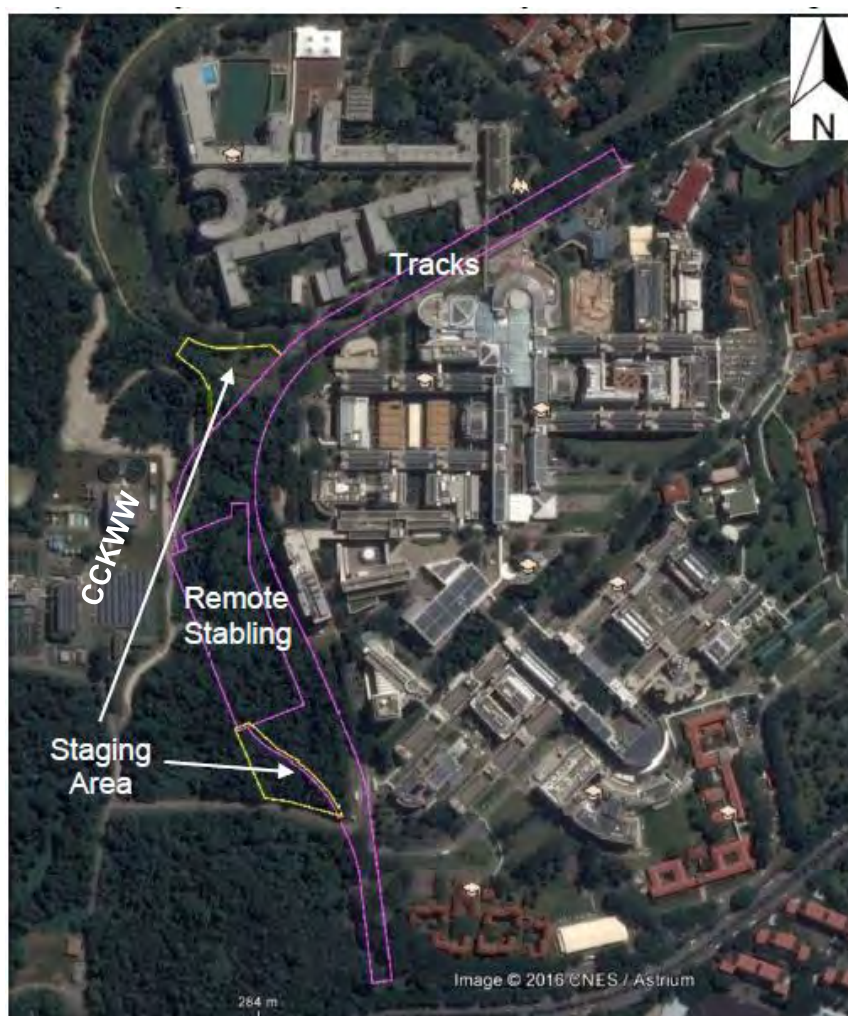
## 2.9 Operational Activities Associated with Project

Operational phase activities will mainly include operation and maintenance of the CCKWW plant including:

- 24-hour manning of control room to monitor the plant process
- Regular servicing of pumps and other mechanical equipment
- Material delivery

## 2.10 Concurrent Project

There is one concurrent project identified adjacent to the project site. The project includes construction of Remote Stabling Facility as part of Jurong Region Line (JRL) development (T. Y. Lin, 2018). This project is in close proximity to the proposed sewer pipeline alignment at Area 3 of the current Project (Figure 2.11).



**Figure 2.11.** Adjacent project development

## 3 EIA APPROACH

### 3.1 Singapore's EIA Context

Singapore adopts a systematic framework to determine and mitigate the potential impact of any new development on the environment. In general, development projects are required to undergo a thorough evaluation process that addresses the development's potential impact on traffic, public health, heritage, and the environment. In addition, proposed development projects near sensitive areas, such as Nature Reserves, Nature Areas, marine and coastal areas, other areas of significant biodiversity or with potential trans-boundary impact, are subject to greater scrutiny.

For such projects, Technical Agencies (e.g., the National Parks Board, National Environment Agency, Maritime and Port Authority of Singapore, and Singapore Food Agency) are consulted more extensively, in which the developer sets out the relevant locational and environmental factors, make a considered statement on the potential impacts of the project based on these factors, and also indicate the measures that will be taken to minimize negative impact on the surrounding environment.

This is intended to help set the grounds (on a reasonable basis) on the potential impacts (whether it will cause substantial pollution of or significant and harmful change) and is also consistent with principles of the screening stage in international EIA practices. Government agencies will assess the impact of the project and recommend whether further environmental studies are required.

Due to the potential impacts, it is deemed that a comprehensive study comprising impact analysis, assessment, and mitigation management is required for this proposed development. For this EIA, an Inception Report illustrating the EIA scope was submitted to PUB and relevant Technical Agencies in advance of the commencement of the EIA to confirm the scope.

### 3.2 Scope of EIA

This EIA study is conducted as per the requirements specified by PUB in the Invitation to Tender (ITT), Request for Proposal (RFP), and scoping consultation process with Technical Agencies. As noted earlier, an Inception Report describing the scope of EIA was submitted to the relevant Technical Agencies prior to the commencement of the EIA study.

The salient points of EIA scope as follows:

- Establishment of baseline data within EIA study area including physical site surveys.
- Impact assessment of the proposed development to include:
  - Scale and effects of vegetation removal from site clearance, and construction;
  - Introduction, loss or changes to native plant or animal species such as trees, understory plants, and wildlife as well as habitats which play a role in supporting resident, seasonal or migrant species;
  - Wildlife displacement;
  - Wildlife connectivity;
  - Surface hydrology (loss of natural stream, disturbance to drainage pattern, water quality, impact to reservoirs/ catchment area);
  - Noise impact on sensitive receptors;

- Air quality impact on sensitive receptors;
- Waste management (e.g., toxic and general waste); and
- Other impacts of the proposed development on the biodiversity, identified upon completion of baseline study.
- Recommend mitigation measures for impacts due to the construction and operational activities of the proposed development.
- Develop an Environmental Monitoring and Management Plan (EMMP) framework for construction phase.
- Stakeholder consultations and engagement with relevant stakeholders (i.e., government, non-government groups and general public) to be carried out during the EIA process.

Applicable Singapore and International Legislations (Table 3.1) lists relevant legislation, regulations and guidelines that govern the various environmental parameters within Singapore. The latest legislations and relevant subsidiary regulations can be accessed from the website of Singapore Statutes Online (SSO) (n.d.) at <https://sso.agc.gov.sg/>.

**Table 3.1.** List of applicable Singapore legislations, regulations, and guidelines

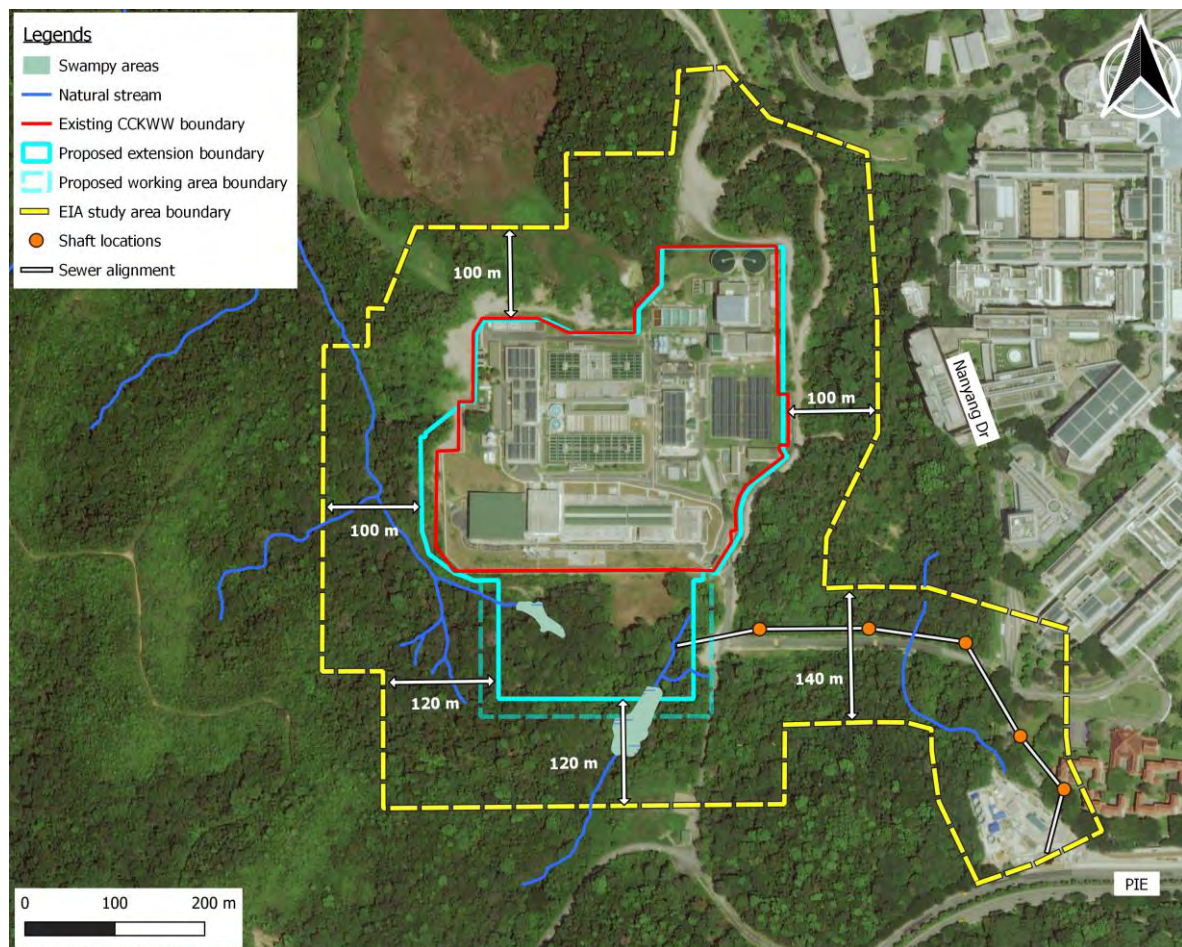
| Parameter             | Legislation, Regulations and Guidelines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General               | <ul style="list-style-type: none"> <li>• Environmental Protection and Management Act, 2002</li> <li>• Environmental Public Health Act, rev. 2002</li> <li>• Singapore Code of Practice on Pollution Control (SS593: 2013)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Biodiversity          | <ul style="list-style-type: none"> <li>• The Wildlife Act 2020</li> <li>• The Parks and Trees Act 2006</li> <li>• The Parks &amp; Trees Regulations 2006</li> <li>• The Parks &amp; Trees Preservation Order 1998</li> <li>• Parks &amp; Trees (Composition of Offences Regulations) 2006</li> <li>• Parks &amp; Trees (Planning Areas) Notifications 2006</li> <li>• Parks &amp; Trees (Heritage Road Green Buffers) Order 2006</li> <li>• Singapore Red Data Book, Second Edition, 2008</li> <li>• Singapore Red Data Book, Third Edition (online), 2021</li> <li>• IUCN Red List of Threatened Species to assess species vulnerability (2021)</li> <li>• CITES (the Convention on International Trade in Endangered Species of Wild Fauna and Flora, also known as the Washington Convention) 1983</li> </ul> |
| Noise                 | <ul style="list-style-type: none"> <li>• Environmental Protection and Management Act 2002, Part VIII Noise Control</li> <li>• Environmental Protection and Management (Control of Noise at Construction Sites) Regulations 2008</li> <li>• Environmental Protection and Management (Boundary Noise Limits for Factory Premises) Regulations 2008</li> <li>• NEA Code of Practice on Pollution Control SS 593 (2013)</li> <li>• Code of Practice for Noise Control on Construction and Demolition Sites SS602 (2014)</li> </ul>                                                                                                                                                                                                                                                                                   |
| Surface Water Quality | <ul style="list-style-type: none"> <li>• Sewerage and Drainage Act 2001</li> <li>• Sewerage and Drainage (Surface Water Drainage) Regulations 2007</li> <li>• Sewerage and Drainage (Trade Effluent) Regulations revised 2007</li> <li>• Environmental Protection and Management Act 2002, Part V on water pollution</li> <li>• Environmental Protection and Management Act (Trade Effluent) Regulations 2008</li> <li>• PUB Code of Practice on Surface Water Drainage (2018)</li> </ul>                                                                                                                                                                                                                                                                                                                        |

| Parameter           | Legislation, Regulations and Guidelines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <ul style="list-style-type: none"> <li>• PUB Handbook on Managing Urban Runoff (2013)</li> <li>• NEA Code of Practice on Pollution Control SS 593 (2013)</li> <li>• PUB Guidebook on Erosion and Sediment Control at Construction Sites (2018)</li> </ul>                                                                                                                                                                                                                                                                                                           |
| Ambient Air Quality | <ul style="list-style-type: none"> <li>• Environmental Protection and Management Act 2002, Part IV on Air Pollution Control</li> <li>• Environmental Protection and Management (Vehicle Emissions) Regulations 2008</li> <li>• Environmental Protection and Management (Prohibition on Use of Open Fires) Order 2008</li> <li>• Environmental Protection and Management (Air Impurities) Regulations 2008</li> <li>• NEA Singapore Ambient Air Quality Targets (2011)</li> <li>• NEA Code of Practice on Pollution Control SS 593 (2013)</li> </ul>                 |
| Waste Management    | <ul style="list-style-type: none"> <li>• Environmental Protection and Management Act 2002, Part VII on Hazardous Substances</li> <li>• Environmental Protection and Management (Hazardous Substances) Regulations 2008</li> <li>• Environmental Public Health (General Waste Collection) Regulations 2000</li> <li>• Environmental Public Health (Toxic Industrial Waste) Regulations 2000</li> <li>• Best Environmental Practices: Construction Waste Management at LTA Sites (2009)</li> <li>• NEA Code of Practice on Pollution Control SS 593 (2013)</li> </ul> |
| Vector Control      | <ul style="list-style-type: none"> <li>• NEA guidelines on "Rainwater Collection System and Mosquito Prevention"</li> <li>• Control of Vectors and Pesticides Act 2002</li> <li>• Environmental Public Health Act (EPHA) 2002</li> </ul>                                                                                                                                                                                                                                                                                                                            |

### 3.3 EIA Study Area and Environmental Aspects

The EIA study area denotes the area where construction of the proposed project is predicted to have impacts on various environmental aspects within the site. Figure 3.1 below shows the extent of the EIA study area.

The EIA study area mainly covered a 100-m distance around the existing CCKWW plant. This is the typical distance beyond the working boundaries where TAs require Developers to study and to take into account any potential impacts on flora and fauna, and habitats within the vicinity of the working boundary. Along the additional greenfield area which will form the reconstructed CCKWW, the study area extended to 120 m to take into account the working area requirements (20m) during construction stage. The study area covered 140 m along the sewerage alignment.



**Figure 3.1.** EIA study area

The environmental aspects that were studied for this project, along with a brief description of each aspect and explanation of its relevance in the study, are presented below. The identified sensitive receptors are listed down in Table 3.2.

### ***Biodiversity***

This aspect covers flora and fauna groups inhabiting the project site that may be impacted by future construction and operation of the project. The main groups studied are vegetation on site, birds, mammals, amphibians, reptiles, butterflies, and odonates. Aquatic surveys of natural streams were also conducted, covering freshwater fish, molluscs, and decapod crustaceans. Impacts to biodiversity during the project implementation were assessed, and mitigation measures to address the potential impacts were recommended.

### ***Hydrology and Water Quality***

This aspect concerns natural waterbodies within project site that may be impacted during the construction stage. Free-flowing streams were mapped, and water quality analysis and stream characteristics were carried out at points along each stream to establish the baseline condition. Direct and indirect impacts to streams during the project implementation were studied.

### ***Noise***

The construction and operation of the CCKWW and associated infrastructure will likely increase noise levels beyond current ambient level. Many fauna species are sensitive to noise. Noise

levels were measured at sensitive receptors and analysed to assess the baseline condition. Impacts of noise from construction activities were assessed to recommend mitigation measures.

### ***Ambient Air Quality***

Construction activities are known to release dust and other particulate matters that may harm the health of sensitive receptors in surrounding area, in this case flora and fauna species and human receptors within project study area. The baseline ambient air quality was established. The impacts generated from construction activities were assessed and recommendations to mitigate the impacts are provided.

### ***Vibration***

Construction activities involves use of heavy equipment generating vibration that may harm the health of sensitive receptors in surrounding area. The baseline vibration levels were established. The impacts generated from construction activities were assessed and recommendations to mitigate the impacts are provided.

### ***Light***

Increased artificial light during night-time—either from construction activities or from the operation of the project—could disrupt circadian cycles of animals and distort the day–night cycle of plants. Impacts from increased artificial light were studied, and mitigation measures proposed to reduce adverse impacts to the surrounding environment.

### ***Waste Management***

The main impacts in relation to the storage, handling, transport and disposal of waste include deterioration of the environment and health & safety risks with regards to hazardous waste, if they are not managed properly. These impacts were assessed and mitigation measures to overcome these impacts were proposed.

### ***Vector Control***

Possibility of vector borne diseases in Singapore's warm tropical climate is always present. As the majority of vector-borne diseases in Singapore are transmitted by mosquitoes, these impacts were assessed and mitigation measures to overcome these impacts were proposed.

## **3.4 Identification of Sensitive Receptors**

Table 3.2 provides the overview of the identified sensitive environmental receptors that may be affected during the development of the proposed project.

**Table 3.2.** Overview of identified sensitive receptors

| <b>Environmental Aspects</b> | <b>Sensitive receptors</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Biodiversity                 | <ul style="list-style-type: none"> <li>Native flora and fauna of international conservation significance (i.e., classified as Critically Endangered, Endangered, or Vulnerable according to IUCN classification system) within project study area</li> <li>Native flora and fauna of national conservation significance (i.e., classified as Critically Endangered, Endangered, and/or Vulnerable according to Singapore Red Data Book (Davison et al., 2008) or other relevant local status publications)</li> </ul> |

| <b>Environmental Aspects</b> | <b>Sensitive receptors</b>                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              | within project study area <ul style="list-style-type: none"> <li>• Native flora and fauna endemic to Singapore within project study area</li> <li>• Habitats with high ecological value (i.e., environments that support species of conservation significance)</li> </ul>                                                                                                                                                    |
| Surface Water Quality        | <ul style="list-style-type: none"> <li>• Aquatic flora and fauna within project study area</li> <li>• Surface waterways that support habitats of conservation significance</li> <li>• Tengeh Reservoir, Poyan Reservoir, and Jurong Lake</li> </ul>                                                                                                                                                                          |
| Noise                        | <ul style="list-style-type: none"> <li>• Species that are susceptible to noise pollution (e.g., species that require a quiet environment to find prey and species with acute hearing)</li> <li>• People working on the site (e.g., construction workers)</li> <li>• NTU's Experimental Medicine Building, Animal House, Wee Kim Wee School of Communication and Information, and residents of Hall of Residence 7</li> </ul> |
| Ambient Air Quality          | <ul style="list-style-type: none"> <li>• Flora and fauna living within project study area</li> <li>• People working on the site (e.g., construction workers, consultants)</li> <li>• NTU's Experimental Medicine Building and Animal House</li> </ul>                                                                                                                                                                        |
| Vibration                    | <ul style="list-style-type: none"> <li>• Fauna living within project study area</li> <li>• NTU's Experimental Medicine Building, Animal House, Wee Kim Wee School of Communication and Information, and residents of Hall of Residence 7</li> </ul>                                                                                                                                                                          |
| Light                        | <ul style="list-style-type: none"> <li>• Species that are susceptible to light disturbance (e.g., nocturnal fauna) within project study area</li> </ul>                                                                                                                                                                                                                                                                      |

### 3.5 Impact Assessment Methodology

#### 3.5.1 Identification of Impacts

The proposed project involves the re-construction of CCKWW and construction of sewerage alignment within the project study area. This will involve clearance of vegetation and earthworks for working spaces (e.g., project footprint, access routes, storage space for construction equipment and materials), which has several potential impacts on surrounding environment. These impacts include those that are direct in nature, such as loss of species, habitat destruction due to vegetation clearance, and indirect impacts such as fragmentation of habitats, isolation of population due to reduced connectivity and fauna movement, increased edge effect<sup>1</sup>, and noise, light and air pollution during construction and operation phases of the project.

Assessment of the impacts of the proposed project will include:

- Scale and effects of vegetation removal from site clearance, and construction;
- Introduction, loss, or changes to native species such as trees, understory plants, and wildlife as well as habitats which support resident, seasonal, or migrant species;
- Wildlife displacement;
- Wildlife connectivity;
- Surface hydrology (loss of natural stream, disturbance to drainage pattern, water quality, impact to reservoirs/ catchment area);

<sup>1</sup> i.e., higher temperature, light, noise, and pollution levels on the edges compared to the interior of a forest resulting in retraction or loss of species sensitive to these disturbances.

- Noise impact on sensitive receptors;
- Air quality impact on sensitive receptor;
- Waste management (e.g., toxic and general waste); and
- Other impacts of the proposed development on biodiversity, identified upon completion of baseline study.

### 3.5.2 Assessment of Impacts

Based on the impact analysis of construction and operational activities of the proposed development, suitable mitigation measures are recommended to lower the magnitude of negative impacts on the environment to within acceptable levels as much as practically possible.

Potential impacts were quantified using the Rapid Impact Assessment Matrix (RIAM) assessment, a scoring system in which impacts of each project activity are evaluated against environmental receptors (adapted from Pastakia & Barber, 1998). The RIAM method attributes values to each condition based on its importance (I), magnitude (M), permanence (P), reversibility (R), and cumulativeness (C).

The parameters are tabulated in Table 3.3.

**Table 3.3.** List of parameters and respective scores assigned in RIAM method

| Parameter                | Description                                                                     | Score |
|--------------------------|---------------------------------------------------------------------------------|-------|
| <b>Importance (I)</b>    | Important to national/international interests                                   | 5     |
|                          | Important to regional/national interests                                        | 4     |
|                          | Important to areas immediately outside the local condition                      | 3     |
|                          | Important to the local condition (within a large direct impact area)            | 2     |
|                          | Important only to the local condition (within a small direct impact area)       | 1     |
| <b>Magnitude (M)</b>     | Major positive benefit or change                                                | +4    |
|                          | Moderate positive benefit or change                                             | +3    |
|                          | Minor positive benefit or change                                                | +2    |
|                          | Slight positive benefit or change                                               | +1    |
|                          | No change/status quo                                                            | 0     |
|                          | Slight negative disadvantage or change                                          | -1    |
|                          | Minor negative disadvantage or change                                           | -2    |
|                          | Moderate negative disadvantage or change                                        | -3    |
|                          | Major negative disadvantage or change                                           | -4    |
| <b>Permanence (P)</b>    | No change/Not applicable                                                        | 1     |
|                          | Temporary                                                                       | 2     |
|                          | Permanent                                                                       | 3     |
| <b>Reversibility (R)</b> | No change/Not applicable                                                        | 1     |
|                          | Reversible or controllable through Environmental Management and Monitoring Plan | 2     |
|                          | Irreversible                                                                    | 3     |
| <b>Cumulativity (C)</b>  | No change / Not applicable                                                      | 1     |
|                          | Non-cumulative/single                                                           | 2     |
|                          | Cumulative/synergistic                                                          | 3     |

Given the ambiguity in the nature of assessing the “magnitude” component, we use the following criteria, tabulated in Table 3.4, to aid the assessment.

**Table 3.4.** Description of the value of magnitudes in RIAM method

| <b>Magnitude</b>                         | <b>Description</b>                                                                                                                                                                                                                                                                                          |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Major positive benefit or change         | Refers to significant improvements in baseline conditions and a significant reduction of stress or improvement in the baseline states of sensitive receptors.                                                                                                                                               |
| Moderate positive benefit or change      | Refers to significant improvements in local baseline conditions, which may lead to a moderate reduction of stress to sensitive receptors or improvement in their baseline state.                                                                                                                            |
| Minor positive benefit or change         | Refers to minor positive benefit or change implies that positive changes to baseline conditions are discernible but local. These changes may lead to local and limited reduction of stress to sensitive receptors.                                                                                          |
| Slight positive benefit or change        | Refers to slight positive benefit or change implies that changes in baseline conditions are unlikely to be detectable on-site, and thus are unlikely to cause discernible reduction of stress to sensitive receptors.                                                                                       |
| No change/status quo                     | Refers to no change/status quo implies that changes in the baseline conditions are not expected, and unlikely to cause any stress to sensitive receptors.                                                                                                                                                   |
| Slight negative disadvantage or change   | Refers to changes in baseline conditions that are unlikely to be detectable in the field, and thus are unlikely to cause discernible stress to sensitive receptors.                                                                                                                                         |
| Minor negative disadvantage or change    | Refers to negative changes to baseline conditions that are discernible but local. These may also refer to changes that are approaching thresholds for established standards or guidelines. These changes may lead to a local and limited increase in stress to sensitive receptors.                         |
| Moderate negative disadvantage or change | Refers to significant adverse changes in local baseline conditions. These may also refer to changes that are very close to exceeding established standards or guidelines or causing significant ecological impacts. These changes may lead to a moderate increase of stress to sensitive receptors.         |
| Major negative disadvantage or change    | Refers to significant adverse changes in baseline conditions. These may also refer to changes that exceed established standards or guidelines or causing a complete loss of certain habitats or ecological components. These changes may lead to an unacceptable increase of stress to sensitive receptors. |

These values will then contribute to the condition’s environmental score, where:

$$Environmental\ Score\ (ES) = I * M * (P + R + C).$$

The ES attained for each condition correlates to a measure of its impact, tabulated in Table 3.5.

**Table 3.5.** List of ES range along with the degree of impact associated with each range

| <b>Range</b> | <b>Impact</b>                           |
|--------------|-----------------------------------------|
| 116 to 180   | Major positive change/impact            |
| 81 to 115    | Moderate positive change/impact         |
| 37 to 80     | Minor positive change/impact            |
| 7 to 36      | Slight positive impact                  |
| -6 to +6     | No impact / Status quo / Not applicable |

| Range        | Impact                          |
|--------------|---------------------------------|
| -7 to -36    | Slight negative change/impact   |
| -37 to -80   | Minor negative change/impact    |
| -81 to -115  | Moderate negative change/impact |
| -116 to -180 | Major negative change/impact    |

### 3.5.3 Mitigation Measures

In general, mitigation measures follow two concepts:

- ALARP : “As Low as Reasonably Practical”
- BATNEEC : “Best Available Technology Not Entailing Excessive Costs”

The first concept is a hierarchy of actions that aims to find anything that can be done to avoid, minimise, or reduce the predicted/ potential adverse (negative) environmental impacts, as practically feasible and reasonable. The second concept comes in when discussing whether or not to adopt certain available technology that could address/reduce environmental impacts.

This EIA utilises both concepts for the development of mitigation measures for this project.

## 3.6 Project Team

The team members involved in the fieldwork, analysis, impact assessment, and report preparation for this project are listed in Table 3.6.

**Table 3.6.** List of EIA project team members

| No | Name                       | Designation                               |
|----|----------------------------|-------------------------------------------|
| 1  | Mr Tan Seng Chuan          | Project Director                          |
| 2  | Mr Komal Pujara            | Project Manager / EIA Lead                |
| 3  | Ms Holly Siow              | Lead Ecologist                            |
| 4  | Ms Henrietta Woo           | Fauna Expert (Birds)                      |
| 5  | Mr Robin Ngiam             | Fauna Expert (Invertebrates)              |
| 6  | Dr Alex Figueroa           | Fauna Expert (Bats)                       |
| 7  | Mr Ivan Kwan Wei Ming      | Fauna Expert (Aquatic Fauna)              |
| 8  | Ms Sarah Pascoe            | Biodiversity Consultant                   |
| 9  | Mr Jefferson               | Environmental Consultant                  |
| 10 | Ms Lisa Lim                | Environmental Consultant                  |
| 11 | Mr Martin Tay              | ISA-certified Arborist                    |
| 12 | Ms Jong Ying Wei           | Flora Specialist / ISA-certified Arborist |
| 13 | Mr Rushan Bin Abdul Rahman | Ecologist/ Flora Surveyor                 |
| 14 | Mr Dominic Ng              | Ecologist/ Flora Surveyor                 |

## 4 BIODIVERSITY

### 4.1 Introduction

The project site covers a small area around the existing CCKWW plant, in the south of the Western Water Catchment Area. The site consists of a range of different habitat and vegetation types, including vegetation from abandoned kampung and plantation, native-dominated old secondary forest, native dominated young secondary forest, natural stream, and scrubland habitat. Parts of the area south of the existing CCKWW also consists of swampy ground, especially close to the streams.

Vegetation has been allowed to mature in the larger site of the Western Water Catchment Area. This has facilitated the formation of many important habitat types, including secondary forest, mangroves, freshwater marsh, and manmade reservoirs.

The habitats in the Western Catchment area prove critical for many animals found there. For example, this area is possibly one of the most important habitats on mainland Singapore for native terrestrial mammals, including the Critically Endangered leopard cat (*Prionailurus bengalensis*) and Sunda pangolin (*Manis javanica*). Other threatened fauna, including the elusive Endangered king cobra (*Ophiophagus hannah*) and Critically Endangered Asian leaf terrapin (*Cyclemys dentata*), are also found there.

This following section describes the methodology of field surveys that were carried out to update the baseline for biodiversity and identify the existing sensitive receptors within project study area. The results of these surveys are then analysed and discussed to assess the likely impacts of the proposed development to the receptors. Measures to mitigate these impacts are recommended for implementation during project implementation.

### 4.2 Baseline Methodology

#### 4.2.1 Desktop Review

A desktop review to collect secondary data was conducted. This included a thorough review of publicly available literature on the ecology and biodiversity of the site, as well as other publicly available material, which may include land use and other maps, photographs, and environmental data found on government websites.

#### **Conventions**

The species' local conservation status was mainly based on *Singapore Red Data Book* (Davison, Ng, & Ho, 2008). However, the National Parks Board (NParks) is in the process of updating local conservation statuses of native species, and as the statuses of mammals, reptiles, and amphibians have been updated, these were taken from the lists uploaded on NParks' Website (National Parks Board, 2021).

For flora, *The Checklist of the Total Vascular Plant Flora of Singapore* (Chong, Tan, & Corlett, 2009) was the main reference, with the Flora of Singapore as a supplementary resource. Conservation status for odonates was derived from *The Dragonflies of Singapore: An updated checklist and revision of the national conservation statuses* (Ngiam & Cheong, 2016). For butterflies, Singapore does not have conservation statuses per se, but conservation status is

listed according to the species' rarity based on Khew, 2015. Global conservation statuses were derived from the IUCN Red List of Threatened Species (IUCN, 2021).

**Table 4.1.** Conservation status for flora & fauna species and their respective definitions, adapted from IUCN Red List (2021) and Singapore Red Data Book (Davison, Ng, & Chew, 2008)

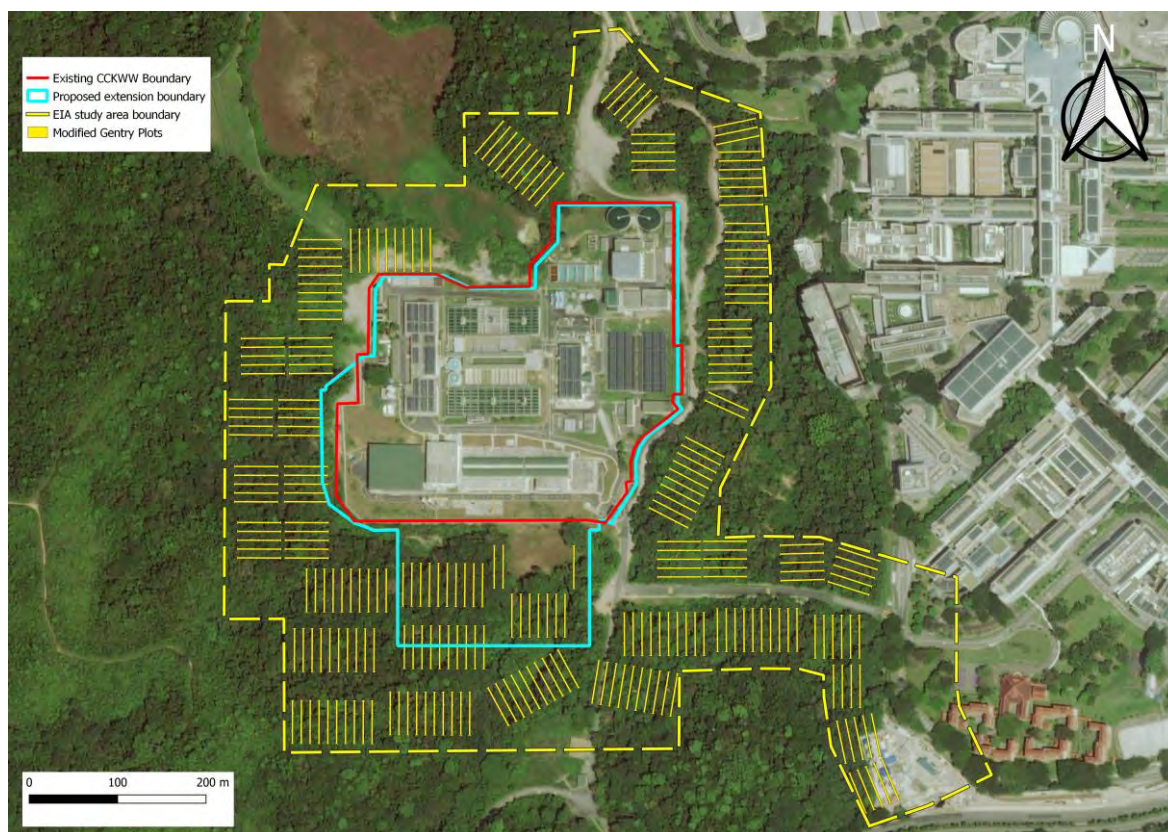
| Conservation Status              | Definition                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Global</b>                    |                                                                                                                                                                                                                                                                                                                                                                        |
| Extinct (EX)                     | There is no reasonable doubt that the last individual has died. Exhaustive surveys in known and/or expected habitat, at appropriate times, throughout its historic range have failed to record an individual. Surveys should be over a time frame appropriate to the taxon's life cycle and life form.                                                                 |
| Extinct in the Wild (EW)         | Known only to survive in cultivation, in captivity or as a naturalized population (or populations) well outside the past range. Exhaustive surveys in known and/or expected habitat, at appropriate times, throughout its historic range have failed to record an individual. Surveys should be over a time frame appropriate to the taxon's life cycle and life form. |
| Critically Endangered (CR)       | Considered to be facing an extremely high risk of extinction in the wild.                                                                                                                                                                                                                                                                                              |
| Endangered (EN)                  | Considered to be facing a very high risk of extinction in the wild.                                                                                                                                                                                                                                                                                                    |
| Vulnerable (VU)                  | Considered to be facing a high risk of extinction in the wild.                                                                                                                                                                                                                                                                                                         |
| Near Threatened (NT)             | Does not qualify as Critically Endangered, Endangered or Vulnerable now, but is close to qualifying for or is likely to qualify for a threatened category in the near future.                                                                                                                                                                                          |
| Least Concern (LC)               | Does not qualify for Critically Endangered, Endangered, Vulnerable or Near Threatened. Widespread and abundant taxa are included in this category.                                                                                                                                                                                                                     |
| Data Deficient (DD)              | Inadequate information to make a direct, or indirect, assessment of its risk of extinction based on its distribution and/or population status. A taxon in this category may be well studied, and its biology well known, but appropriate data on abundance and/or distribution are lacking. Data Deficient is therefore not a category of threat.                      |
| Not Evaluated (NE)               | Not yet been evaluated against the criteria.                                                                                                                                                                                                                                                                                                                           |
| <b>Local</b>                     |                                                                                                                                                                                                                                                                                                                                                                        |
| Presumed Nationally Extinct (NE) | This species is extinct in Singapore but still survives outside Singapore. It has not been recorded with the last 30 years (plants) and 50 years (animals).                                                                                                                                                                                                            |
| Critically Endangered (CR)       | There are fewer than 50 mature individuals, or if more than 50 mature individuals but less than 250, with some evidence of decline or fragmentation.                                                                                                                                                                                                                   |
| Endangered (EN)                  | There are fewer than 250 mature individuals, and no other evidence of decline or fragmentation.                                                                                                                                                                                                                                                                        |
| Vulnerable (VU)                  | There are fewer than 1000 mature individuals, but more than 250 and there may or may not be any other evidence of decline, small range size, or fragmentation.                                                                                                                                                                                                         |

#### 4.2.2 Flora

##### *Modified Gentry Plots*

Twenty-seven Modified Gentry Plots (MGPs) were plotted within vegetated areas of the study site in order to cover the requisite 10% of the vegetated area stated in the Biodiversity Impact

Assessment (BIA) Guidelines (National Parks Board, 2020). Each MGP consists of ten 2 x 50 m subplots, within which woody flora with a diameter at breast height (DBH) of 2.5cm or more would be recorded, and each subplot also has six 1 x 1m quadrats to record herbaceous flora only (“Core Standardized Methods for Rapid Biological Field Assessment”, 2016; National Parks Board, 2020). Figure 4.1 provides layout map of MGPs surveyed within project study area. Flora including non-woody shrubs, climbing plants, epiphytes, and herbs of conservation significance that were encountered outside of the MGPs were recorded opportunistically.



**Figure 4.1.** Layout of MGPs used to survey vascular plants within project study area  
Each MGP has ten subplots, with each subplot having six quadrats.

Survey teams utilized the Trimble® R2 integrated Global Network Satellite System (GNSS) units. The Trimble® R2 GNSS is a receiver that is capable of connecting to multiple satellites in orbit if they are flying over the unit. In open areas with no cloud cover, as many as 20 satellites can connect to the Trimble® R2 GNSS unit, making submeter positioning accuracy possible. Accuracy underneath closed canopies is up to 1 m (Trimble, 2021). Data is collected on a 6<sup>th</sup> Generation iPad iOS 13.4.1, with locations from the Trimble® R2 GNSS unit linked to the iPad via the GNSS iPad application version 3.0.8.673 by Trimble® (Trimble, 2020). The geolocations and details of the surveyed flora is recorded using the TreePlotter™ Software Suite by PlanIT Geo™ (PlanIT Geo, 2021). TreePlotter™ is a tree inventory and asset management software that allows users to geolocate and add trees in real time. This allows multiple users to work on a specific site and view trees as they are located and recorded, thus ensuring that the same tree is not recorded twice. The locations of the trees surveyed were verified during the topography survey.

Static parameters collected for recorded flora can be found in Table 4.2. Plant samples were collected to facilitate species identification where required.

**Table 4.2.** Static parameters collected for surveyed flora

| Static Parameter                        | Description                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------|
| <i>Coordinates (Eastings/Northings)</i> | In SVY21 (EPSG:3414) Coordinate Reference System                                   |
| <i>Collection date</i>                  | Date when the tree was geolocated                                                  |
| <i>Flora Identification Number</i>      | Unique identification number for each tree encountered on site                     |
| <i>Flora Species</i>                    | Latin and common name*                                                             |
| <i>Form</i>                             | Is the flora a tree, shrub, palm, herb, fern, climber/vine?                        |
| <i>Height</i>                           | Height of the tree (m)                                                             |
| <i>Girth</i>                            | Circumference of the tree (cm) at 1.2 m height                                     |
| <i>Assessor's remarks</i>               | Any remarks about the tree, if applicable                                          |
| <i>Image</i>                            | At least 1 colour image of each tree surveyed (provided as supplementary material) |

Note: Some flora species within the Southeast Asian region have no common name

### Vegetation and Habitat Mapping

Vegetation and habitat maps were produced based on the flora recorded in the MGPs and opportunistic encounters. Vegetation maps were produced for woody flora with a girth of one metre or greater, and for flora of Conservation Significance. Habitat types within the site identified during vegetation surveys were demarcated based on species compositions within various areas of the site (Table 4.3).

**Table 4.3.** Habitat types found within Singapore and general description of each habitat, modified from the BIA Guidelines (National Parks Board, 2020)

| Habitat                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Source(s)                   |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Primary forest                          | Contains an emergent layer that has dipterocarp trees such as <i>Shorea</i> and <i>Dipterocarpus</i> . Has a continuous layer of tall native trees, a sub-canopy consisting of smaller trees, and an understorey dominated by saplings of big tree species interspersed with other shrubs and treelets                                                                                                                                                                                                                         | (Tan et al., 2007)          |
| Native-dominated young secondary forest | Developed on land cleared not long before the 1960s, or on degraded soils and not near other native-dominated forests. Dominated by native pioneer trees such as <i>Adinandra</i> , <i>Macaranga</i> , <i>Mallotus</i> and <i>Trema</i> .                                                                                                                                                                                                                                                                                      | (A. T. K. Yee et al., 2016) |
| Native-dominated old secondary forest   | Developed on land cleared much earlier than the 1950s, often on less degraded soil and with higher species richness than early successional native dominated secondary forest. Common species found in the canopy layer include <i>Alstonia</i> spp., <i>Calophyllum</i> spp., <i>Camptosperma</i> spp., <i>Elaeocarpus</i> spp., <i>Garcinia</i> spp., <i>Litsea</i> spp., <i>Rhodamnia</i> spp. and <i>Syzygium</i> spp. Common understorey plants include <i>Anisophyllea disticha</i> and <i>Agrostistachys borneensis</i> | (A. T. K. Yee et al., 2016) |
| Abandoned kampung                       | Developed from an abandoned kampung or orchard, usually dominated by fruit trees such as Durian ( <i>Durio zibethinus</i> ) or Rambutan ( <i>Nephelium lappaceum</i> ), or ornamental plants such as <i>Spathodea campanulata</i> ,                                                                                                                                                                                                                                                                                            | (A. T. K. Yee et al., 2016) |

| Habitat                 | Description                                                                                                                                                                                                                                                                                                                                                                    | Source(s)                                                          |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
|                         | <i>Aglaonema commutatum</i> , <i>Dieffenbachia seguine</i> and <i>Heliconia</i> spp.                                                                                                                                                                                                                                                                                           |                                                                    |
| Abandoned plantation    | Developed from an abandoned plantation, usually dominated by Para Rubber ( <i>Hevea brasiliensis</i> )                                                                                                                                                                                                                                                                         | (A. T. K. Yee et al., 2016)                                        |
| Scrubland/Grassland     | Exposed areas with very little tree cover, typically dominated by grasses, shrubs and herbs.                                                                                                                                                                                                                                                                                   | (Lee Kong Chian Natural History Museum, 2017; A. Yee et al., 2011) |
| Freshwater swamp forest | Formed where slow-flowing streams drain into shallow valleys. The swamp is flooded periodically or semi-permanently, resulting in waterlogged soils that are anaerobic and unstable. Dominated by plants with special adaptations such as stilt roots, plank-like buttresses and pneumatophores. Examples include <i>Xylocarpus fusca</i> and <i>Palaquium xanthochyllum</i> . | (Tan et al., 2007)                                                 |

#### 4.2.3 Fauna

The fauna field assessment covered the various fauna groups: birds, mammals, herpetofauna (reptiles and amphibians), odonates (dragonflies and damselflies), and butterflies. For the natural, free-flowing streams within the site, aquatic surveys covering fish, molluscs, and decapod crustaceans are also conducted. Table 4.4 summarizes the survey methods and appropriate survey timings for each taxonomic group. Fauna species encountered outside their dedicated survey timings are also recorded.

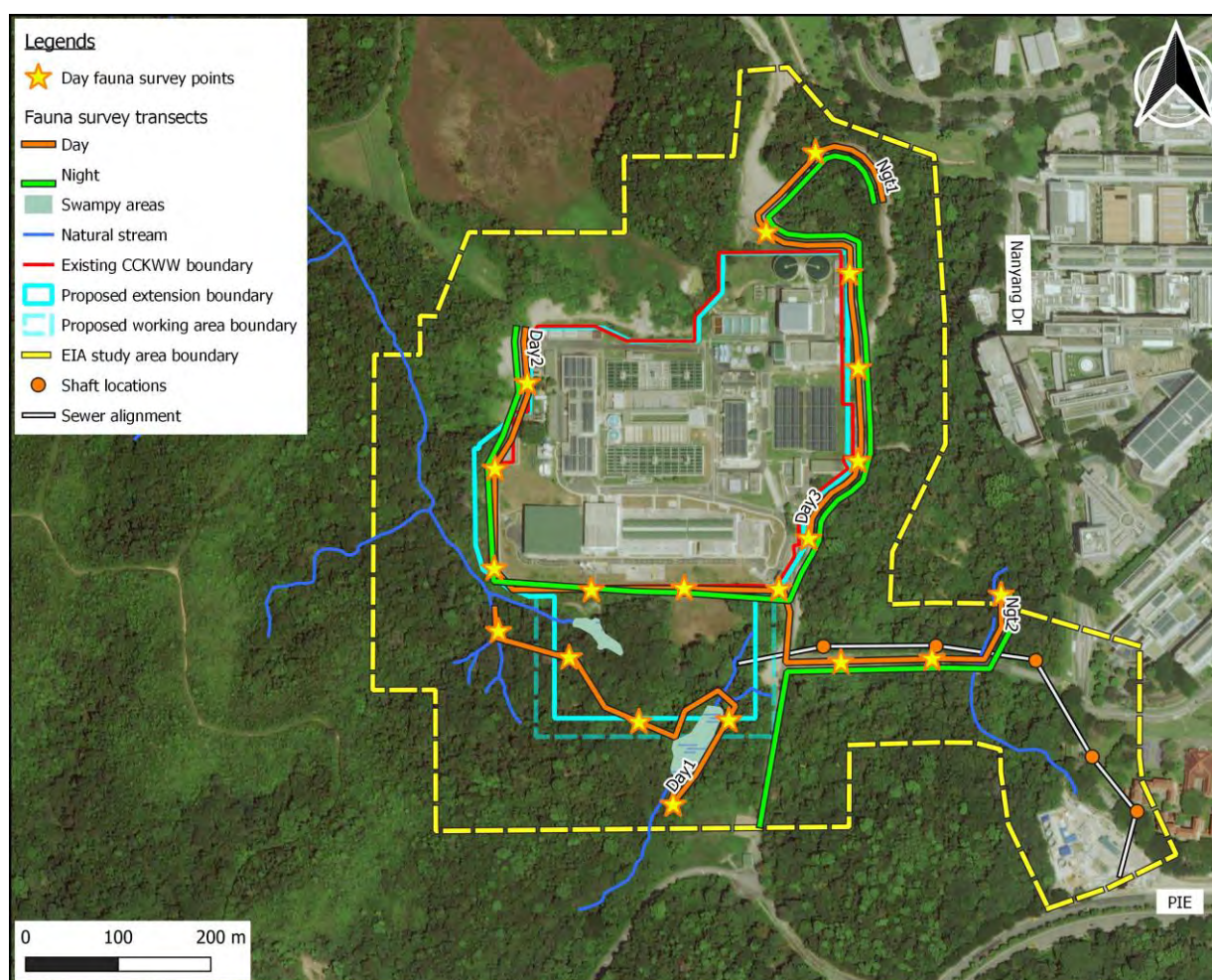
**Table 4.4.** Survey timings, frequency, and methodology for each fauna group.

| Taxonomic Group                                     | Survey Timings                                   | No. of surveys            | Survey Methodology                                                                                                 |
|-----------------------------------------------------|--------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------|
| Birds                                               | 0700 – 1100                                      | 5                         | Visual encounter survey; call recognition along transects                                                          |
| Mammals (non-volant)                                | 2000 – 0000                                      | 4                         | Visual encounter survey; call recognition along transects along transects                                          |
|                                                     | 24 hrs                                           | 10 weeks                  | Camera traps attached on tree trunks 30 cm above ground level                                                      |
| Mammals (bats)                                      | 1900 – 0700                                      | 5                         | Acoustic recording                                                                                                 |
| Herpetofauna                                        | 0700 – 1100<br>2000 – 0000                       | 4                         | Visual encounter survey; call recognition along transects                                                          |
| Butterflies                                         | 0900 – 1200                                      | 5                         | Visual encounter survey along transects                                                                            |
| Odonates                                            | 0900 – 1200                                      | 5                         | Visual encounter survey along transects                                                                            |
| Aquatic Fauna (fish, molluscs, decapod crustaceans) | Diurnal (0900 – 1500)<br>Nocturnal (2000 – 0000) | 2 Diurnal and 1 Nocturnal | Sweep sampling with dip or hand nets for diurnal. Visual detection for nocturnal. Baited traps for both, overnight |

### Visual Encounter Surveys

A baseline survey of fauna found in the study area were conducted along systematic transects, following available trails and roads that span the project site (Figure 4.2). A total of 5 diurnal surveys and 4 nocturnal surveys for dedicated groups were conducted. Visual observations were conducted by at least two observers through point count locations (day) and walking at a steady pace along each transect (night). These surveys focused on birds, mammals, reptiles, amphibians, odonates, and butterflies.

Both diurnal and nocturnal surveys were conducted, since a large proportion of the target groups, particularly mammals and herpetofauna (reptiles and amphibians), are predominantly nocturnal. Locations of fauna species of conservation interest were indicated on maps.



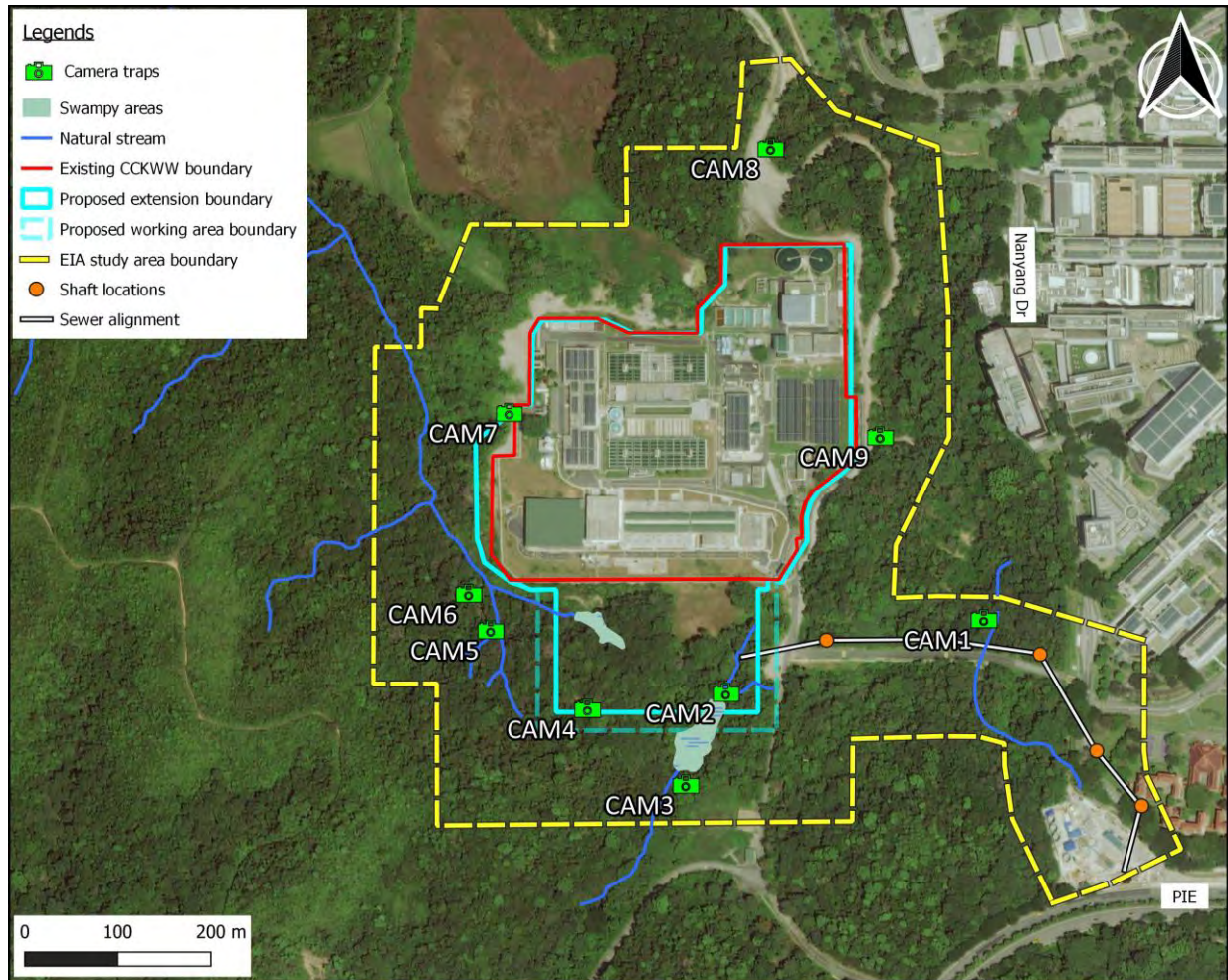
**Figure 4.2.** Fauna transects in the project study area

### Camera Trapping

To supplement the visual encounter surveys, camera trapping was conducted within project study area (Table 4.5, Figure 4.3). 9 camera traps were deployed over a period of at least 60 days, and were secured on tree trunks at 30 cm above ground level. This method is particularly useful for elusive or rare animals that are not often observed during visual encounter surveys.

**Table 4.5.** Coordinates of the camera traps within project study area

| ID   | Latitude | Longitude  | ID   | Latitude | Longitude  |
|------|----------|------------|------|----------|------------|
| CAM1 | 1.342186 | 103.678152 | CAM6 | 1.342432 | 103.673174 |
| CAM2 | 1.341474 | 103.675660 | CAM7 | 1.344180 | 103.673566 |
| CAM3 | 1.340589 | 103.675271 | CAM8 | 1.346731 | 103.676093 |
| CAM4 | 1.341320 | 103.674324 | CAM9 | 1.343951 | 103.677148 |
| CAM5 | 1.342082 | 103.673387 |      |          |            |



**Figure 4.3.** Location of camera traps within project study area

### Acoustic Bat Surveys

The echolocation calls of bats can be used to identify bats to species as they are unique to each species (Fenton and Bell, 1981). However, call structure varies depending on habitat type and foraging mode (Schnitzler *et al.*, 2003). Microchiropteran bats use echolocation to navigate through their environment and locate food (Schnitzler *et al.*, 2003), whereas megachiropteran bats rely on their vision and do not echolocate. Thus, visual and/or trapping surveys are necessary to detect megachiropteran bats.

Acoustic sampling was performed using a handheld Echo Meter Touch 2 Pro (Wildlife Acoustics, Inc.) attached to an iPad (Apple, Inc.) during nocturnal terrestrial and aquatic transects. The Echo Meter functions by detecting ultrasonic sounds in real-time and converting them into

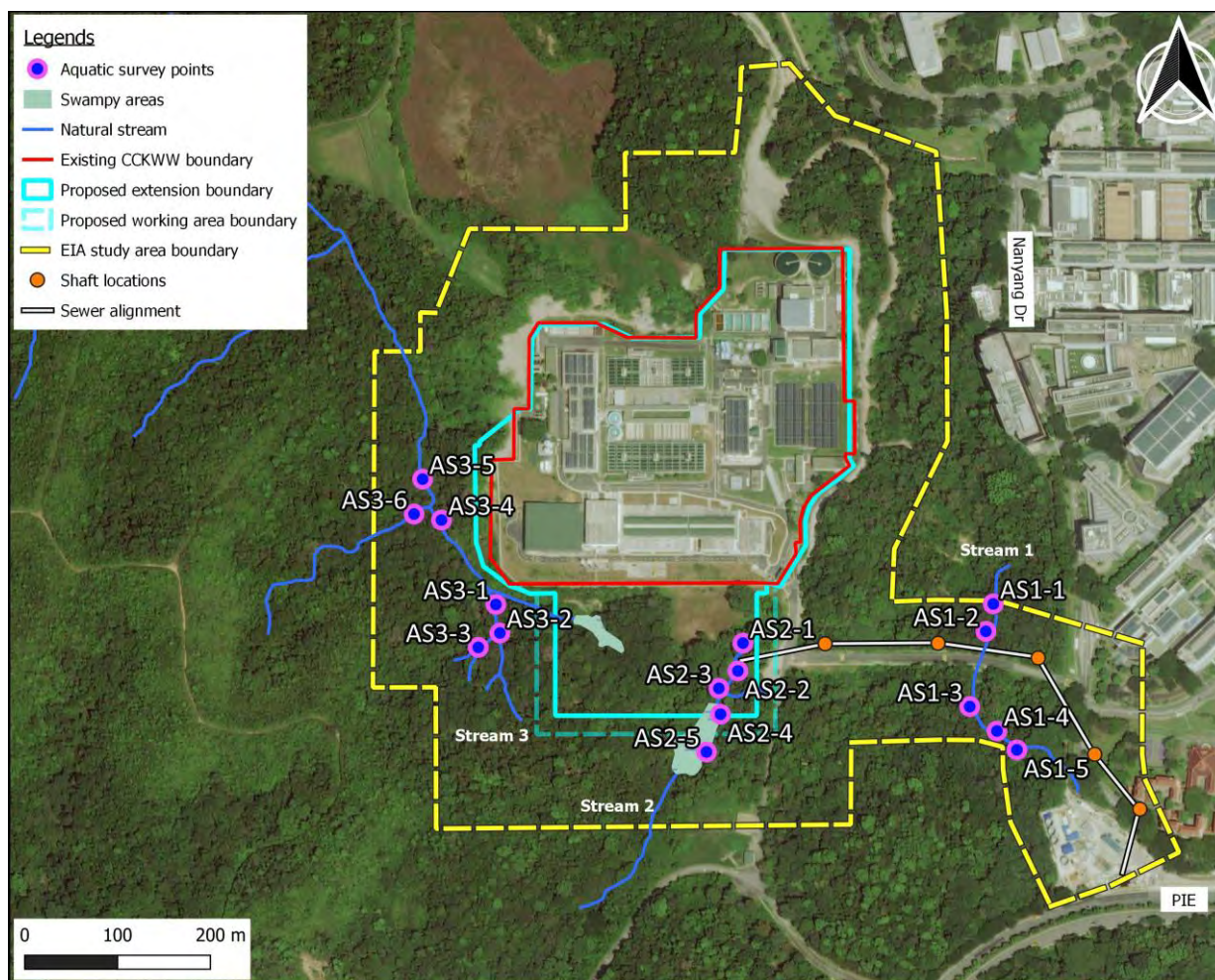
audible digital signals that can be heard and visualized using the Echo Meter Touch App on the iPad. Each time sounds resembling a bat call is detected, it is automatically recorded and saved onto the iPad as a 16-bit WAV file.

Bat call structure was then analysed using Kaleidoscope v.5.1.9 (Wildlife Acoustics, Inc.). Key call structure parameters include call shape, frequency (kHz) and duration (ms). Each echolocation recording was identified to species level based on call shape, frequency (minimum, maximum, and peak) and call duration (Pottie *et al.*, 2005). Once these parameters were inspected, the results were compared to Pottie *et al.* (2005), who provided bat echolocation signatures for several species in Singapore.

### **Aquatic Surveys**

Surveys were conducted along earth streams within project study area. A total of 1 night and 2 diurnal surveys, were conducted at 16 points, focusing on decapod crustaceans, freshwater fishes, and molluscs (e.g., gastropods and bivalves) but also included other vertebrate species (mammals, reptiles, and amphibians) found along the streams. Ten-minute point count surveys were conducted at these points. At the swampy area east of Stream 3, one round of ten-minute point count survey was conducted. The survey locations are indicated in Figure 4.4.

Surveys relied on visual detection, as well as hand-netting and overnight baited cage traps. Baited traps were left on-site for at least 12–24 hours before retrieval. A portion of the traps remained above the water surface, to reduce the risk of drowning of air-breathing species. All aquatic fauna caught by hand netting or baited traps were sorted, photographed, and identified to species or family level where possible, before being released. Locations of species of conservation significance encountered during the aquatic surveys, if any, were recorded.



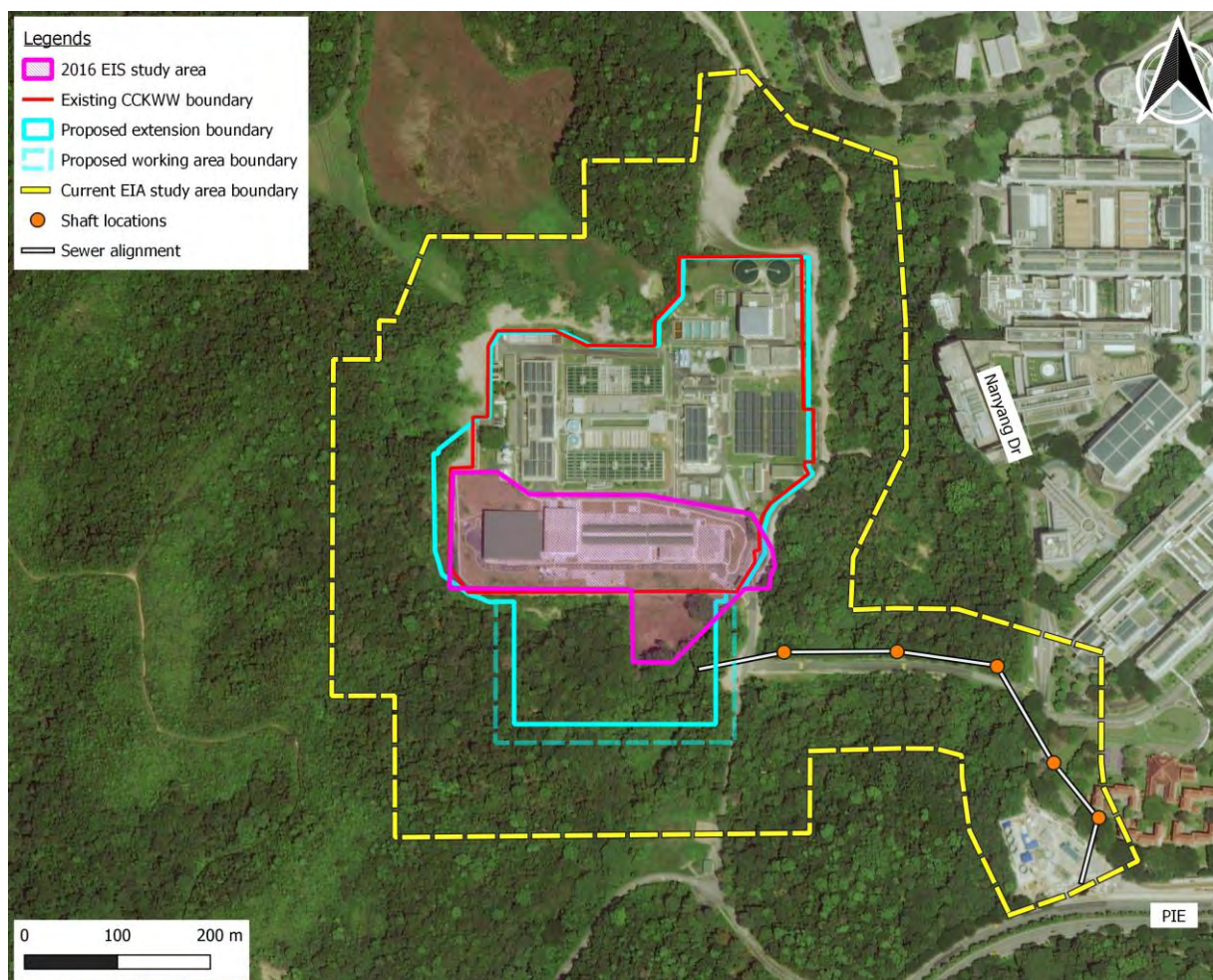
**Figure 4.4.** Location of aquatic surveys within project study area

### 4.3 Desktop Review

This section presents the summary of findings from desktop study carried out for project study area. It covers the review of previous environmental baseline studies conducted in the project area, field studies undertaken by experts, and other publicly available literature.

#### 4.3.1 Review of the Previous Environmental Baseline Studies

An Environmental Impact Study was carried out by DHI in 2016 for the earlier expansion of the CCKWW facility. The previous baseline study area, as shown in Figure 4.5 below, is now part of the current existing CCKWW boundary.



**Figure 4.5.** Study area of the 2016 EIS shown with the present study area

### Habitat Types

At the time the previous study was conducted, the project area studied consisted of 5 main fauna habitat types including, secondary forest, secondary scrub, secondary swamp forest and grassy marshland. The study area was dominated by secondary forest similar to the adjacent Western Catchment Area with young forest patches. Native colonizing plants such as fishtail palms (*Caryota mitis*), wild cinnamon (*Cinnamomum iners*), Tembusu (*Cyrtophyllum fragrans*) and tree ferns dominated the young forest patches.

On the northern periphery wooded area adjacent of the CCKWW facility, distributed secondary scrub with native shrubs was found. Prior to tree clearance for agriculture and other uses, the forest had supported freshwater swamp habitats, which contained a series of shallow streams channels over waterlogged mud. Freshwater streams currently present are remnants from the past. Past human usage may have modified soil composition in the area, leading to the presence of grassy habitats along drainage depressions.

### Flora

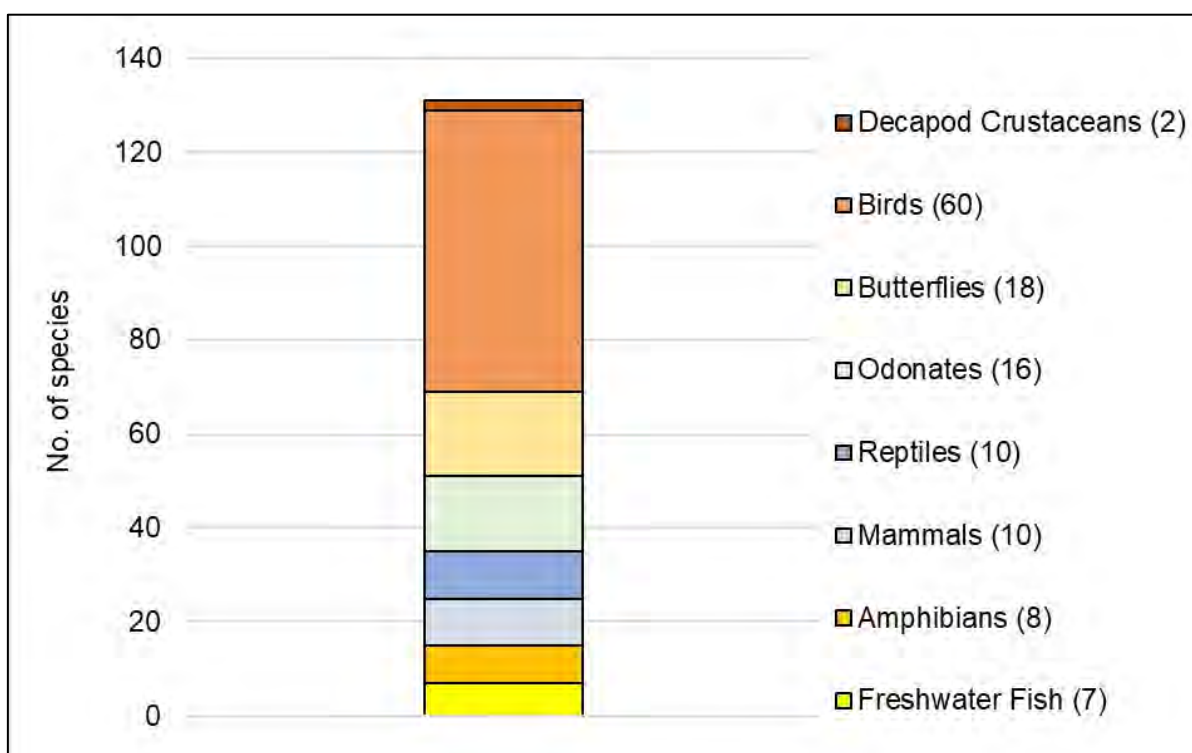
The arboriculture study conducted in 2016 recorded 1,915 trees in the site, with 6 tree species of conservation interest which included the Presumed Nationally Extinct *kelat paya* (*Syzygium myrtifolium*), the Critically Endangered rambutan (*Nephelium lappaceum*), rambai (*Baccaurea motleyana*), yellow-eyed pong pong tree (*Cerbera odollam*), tree fern (*Cyathea latebrosa*) and

the Indonesian bay leaf (*Syzygium polyanthum*). With the exception of the tree fern, the status of the 5 other recorded species were considered irrelevant in the study as they were cultivated remnants of historic human-dependent planting.

Other plants of conservation interest included 4 understory plants such as the Endangered grass fern (*Schizaea dichotoma*), the Vulnerable scorpion's tail (*Pentaphragma ellipticum*), pitcher (*Nepenthes ampullaria*) and the golden hairy fig (*Ficus aurata*). Multiple stands of *Pandanus* species that were potentially the Endangered *Pandanus atrocarpus* were also observed as riparian and low valley components of vegetation downstream beyond study area boundary.

## Fauna

A total of 129 fauna species (see Figure 4.6 for summary of results) from selected vertebrate and invertebrate groups were recorded in the baseline study. Various species of conservation significance were observed in the study area. Threatened species detected included the internationally Critically Endangered straw-headed bulbul (*Pycnonotus zeylanicus*), the locally Endangered changeable hawk-eagle (*Nisaetus cirrhatus*), blue-crowned hanging parrot (*Loriculus galgulus*), red junglefowl (*Gallus gallus*) and the Vulnerable variable featherlegs (*Copra vittata*). Aquatic fauna surveys revealed that the Nationally Endangered Johnson's freshwater crab (*Irmengardia johnsoni*) was present in the vicinity of Stream 3 of current study area. This species of freshwater crab is one of the three endemic species found in Singapore.



**Figure 4.6.** Summary of fauna results from previous baseline study

## 4.3.2 Review of Other Biodiversity Literature

### Flora

As Singapore is a small-city state which has undergone rapid urbanization and faces land use

pressure due to its limited land resources much of the island's primeval land conditions have changed. The island has sustained deforestation resulting in the loss of original forest and secondary forest cover (Tan, Wang, & Sia, 2013). Though much of it has been urbanised, large remnant tracts of secondary forest and regenerated abandoned plantations notably on the western edge of the island. Reserved for water catchment purposes, the Western Catchment Area is a significant extensive nature area (Yong, et al., 2017).

The swamp forests present within the Western Catchment Area act as a refuge for Singapore's native swamp forest associate palm species which are Endangered due to the loss of freshwater swamp habitats on the island. A small population of *Cyrtostachys renda* was rediscovered in within the catchment area offering hope for the native palm flora of Singapore (Loo et al., 2015).

## **Fauna**

A study on the ecology of leopard cats in Singapore provided the first confirmed observation of the locally Critically Endangered leopard cat (*Prionailurus bengalensis*) in Western Catchment Area. This study also provided the first confirmed observations for the Critically Endangered Malayan porcupine (*Hystrix brachyura*) and greater mousedeer (*Tragulus napu*) (Chua, 2015).

Due to the area's inaccessibility, the secondary forest and freshwater swamp forests in the Western Catchment Area provide habitat and refuge for the straw-headed bulbul. Singapore is a stronghold for the globally threatened species as populations elsewhere in South-east Asia have declined due to the regional bird trade. Protected nature areas in Singapore such as nature reserves, parks and the restricted Western Catchment Area collectively provide habitat to support the species and contribute to its protection in Singapore. (Yong, et al., 2017).

The freshwater forest streams and marshlands found within the western catchment also support various species of native freshwater fauna such as species like the Malayan pygmy halfbeak (*Dermogenys collettei*), Malayan forest betta (*Betta pugnax*), saddle barb (*Barbodes banksi*) and brilliant rasbora (*Rasbora einthovenii*) (Ho, et al., 2016). In 2016 NParks conducted a survey of the streams found within the study area and found 5 fish species, and 4 crustacean species. Two of the crustacean species, Johnson's freshwater crab (*Irmengardia johnsoni*) and Peracca's land crab (*Geosesarma peraccae*), are locally threatened. Several of the species recorded are forest-dependent species, indicating that the stream was relatively undisturbed when the survey was conducted (National Biodiversity Centre, National Parks Board, 2016).

## **4.4 Current Baseline Results and Discussion**

This section outlines the results of the current biodiversity baseline study conducted for this EIA.

### **4.4.1 Flora**

#### **Flora Inventory**

An inventory of the surveyed flora was produced, containing the information described in Table 4.2. *Dillenia suffruticosa* was the most commonly encountered species of vascular flora throughout the project study area, which occurred often in both the native-dominated young secondary forest and the abandoned plantation habitat types. *Hevea brasiliensis*, *Spathodea campanulata*, and *Cinnamomum iners* were the most commonly recorded trees in the abandoned plantation and abandoned kampung habitat types, as well as occurring occasionally

in the native-dominated young secondary forest areas.

The prevalence of *Dillenia suffruticosa* throughout all habitat types was to be expected, as it is known to be an early-successional species that has low nutrient requirements, as well as being able to spread quickly in moist soils (Lim & Turner, 1996; Wickramathilake et al., 2013). The propensity of *Dillenia suffruticosa* to shade out undergrowth when it is the dominant crown cover was also observed within large *Dillenia* stands, such as the area to the north of the CCKWW, as there was no recorded herbaceous flora found under the canopy cover of the *Dillenia* (Wickramathilake et al., 2013).

The number of cultivated flora species found within the study area, especially to the south and southeast of the CCKWW, was also expected as the area was labelled as being used for sundry tree cultivation in historical maps of the area (Mapping Unit Ministry of Defence, 1975, 1987, 1998). These species include flora likely planted for commercial value, such as para rubber trees (*Hevea brasiliensis*), cocoa trees (*Theobroma cacao*) and oil palm (*Elaeis guineensis*), as well as flora cultivated for their fruit or ornamental value, such as coconut (*Cocos nucifera*), durian (*Durio zibenthinus*) and African tulip (*Spathodea campanulata*). Some species with herbal medicinal values were also recorded, such as Sabah snake grass (*Clinacanthus nutans*), black face general (*Strobilanthes crispus*) and climbing ginseng (*Eleutherococcus trifolius*). Yee et al. (2016) stated that these species are common in the abandoned land habitat type, which consists of the abandoned plantation and abandoned kampung habitat subtypes, and that these subtypes tend to be spatially heterogeneous when found.

Most of these abandoned land areas show signs of early succession, as the pioneer species *Macaranga bancana* was commonly found in these areas, along with *Dillenia suffruticosa* (Yee et al., 2016). *Adinandra dumosa* is another commonly recorded species that is known to be an indicator of early secondary forest succession, although it tends to occur in depleted soil (Yee et al., 2016). However, multiple genera suggestive of later stages of succession were also recorded, usually in close proximity to *Adinandra dumosa* or *Macaranga bancana*, such as flora from the *Litsea*, *Rhodamnia*, *Elaeocarpus*, *Alstonia*, *Calophyllum* and *Syzygium* genera (Yee et al., 2016). Areas with these species were usually labelled as native-dominated old secondary forests, although several individuals belonging to these genera were also found along forest edges, where their habitat type was labelled as scrubland due to the low canopy cover and high undergrowth density, as well as in areas labelled as native-dominated young secondary forest due to the prevalence of early successional species. These early succession areas are also essential as buffer zones for the native dominated old secondary forest areas, which have been shown to have a rather high biodiversity, and could also act as stepping stones for the dispersion of the Conservation Significant species found onsite, as well as the late succession secondary forest species (Tan, Feng, & Hwang, 2016).

*Acacia auriculiformis* and *Falcataria moluccana*, both of which are usually found as successional species in former scrubland, were also found within the study area, typically at or near forest edges, although they were also found near a naturalized stream likely formed by water discharged from the CCKWW (Yee et al., 2016).

A total of 307 species had been recorded for the study area. Out of the 307 species, 89 are of conservation significance, which comprises of 20 Critically Endangered, 22 Endangered, 44 Vulnerable, two rediscovered, one newly discovered, and one Nationally Extinct flora species, which is higher than the six conservation significant species found in the 2016 study. Table 4.6

provides the list of flora species of conservation significance recorded within study area. Table 4.7 presents distribution of conservation significant flora species by numbers and percentages. Figure 4.7 depict the status of recorded flora species in graphical form.

Several conservation significant floras – e.g., the Critically Endangered rambutan (*Nephelium lappaceum*), rambai (*Baccaurea motleyana*), and Borneo mahogany (*Calophyllum inophyllum*) as well as the presumed Nationally Extinct kelat oil (*Syzygium myrtifolium*) – are likely of cultivated stock and thus do not have any conservation value (Neo et al., 2013).

**Table 4.6.** Conservation significant flora encountered within the project study area

| Species Name                   | Common Name                                    | Local Conservation Status <sup>1</sup> |
|--------------------------------|------------------------------------------------|----------------------------------------|
| <i>Ampelocissus elegans</i>    | Elegant ivy vine                               | EN                                     |
| <i>Ampelocissus gracilis</i>   | NA                                             | EN                                     |
| <i>Aphanamixis polystachya</i> | Pithraj tree                                   | EN                                     |
| <i>Archidendron contortum</i>  | Lesser grasshopper tree, <i>Petai belalang</i> | VU                                     |
| <i>Archidendron jiringa</i>    | Jiring                                         | VU                                     |
| <i>Ardisia elliptica</i>       | Sea-shore ardisia                              | EN                                     |
| <i>Artabotrys maingayi</i>     | NA                                             | CR                                     |
| <i>Artabotrys suaveolens</i>   | NA                                             | EN                                     |
| <i>Artocarpus lacucha</i>      | Green <i>tampang</i>                           | EN                                     |
| <i>Baccaurea motleyana</i>     | <i>Rambai</i>                                  | CR (cultivated)                        |
| <i>Bridelia stipularis</i>     | <i>Kenidai</i>                                 | VU                                     |
| <i>Callicarpa longifolia</i>   | Long leaved beauty berry                       | EN                                     |
| <i>Calophyllum inophyllum</i>  | <i>Penaga laut</i>                             | CR                                     |
| <i>Causonis trifolia</i>       | Bush grape                                     | VU                                     |
| <i>Chassalia curviflora</i>    | <i>Wan guan hua</i>                            | EN                                     |
| <i>Cissus repens</i>           | Malayan wild vine                              | CR                                     |
| <i>Clerodendrum villosum</i>   | <i>Capah, cempening, labu-labu</i>             | VU                                     |
| <i>Cratogeomys maingayi</i>    | Derum                                          | CR                                     |
| <i>Cyathea latebrosa</i>       | Tree fern                                      | VU                                     |
| <i>Cyclosorus opulentus</i>    | NA                                             | EN                                     |
| <i>Cyclosorus polycarpus</i>   | NA                                             | VU                                     |
| <i>Cyrtococcum accrescens</i>  | NA                                             | VU                                     |
| <i>Cyrtococcum patens</i>      | Bowgrass                                       | VU                                     |
| <i>Dalbergia junghuhnii</i>    | NA                                             | CR                                     |
| <i>Dendrotrophe varians</i>    | NA                                             | VU                                     |
| <i>Dioscorea cf polyclados</i> | NA                                             | CR                                     |
| <i>Dissochaeta sp.</i>         | NA                                             | VU                                     |
| <i>Dracaena cantleyi</i>       | Cantley's dracaena                             | VU                                     |
| <i>Embelia canescens</i>       | NA                                             | EN                                     |
| <i>Enkleia malaccensis</i>     | NA                                             | CR                                     |
| <i>Eurycoma longifolia</i>     | <i>Tongkat ali</i>                             | CR                                     |
| <i>Ficus apiocarpa</i>         | <i>Ara akar</i>                                | EN                                     |
| <i>Ficus aurata</i>            | Yellow hairy fig                               | VU                                     |
| <i>Ficus sagittata</i>         | <i>Akar beringing, sepedih</i>                 | CR                                     |
| <i>Ficus superba</i>           | Sea fig                                        | EN                                     |

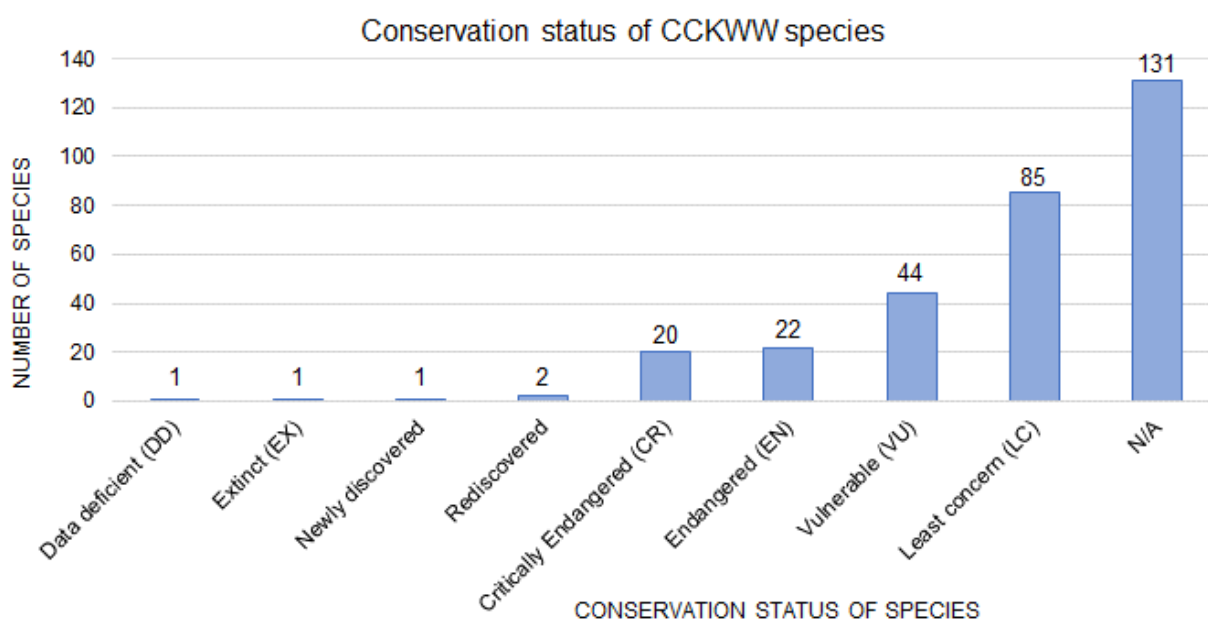
| Species Name                                        | Common Name                    | Local Conservation Status <sup>1</sup> |
|-----------------------------------------------------|--------------------------------|----------------------------------------|
| <i>Ficus vasculosa</i>                              | White fig                      | EN                                     |
| <i>Glochidion zeylanicum</i> var. <i>zeylanicum</i> | NA                             | VU                                     |
| <i>Gnetum latifolium</i>                            | NA                             | CR                                     |
| <i>Goniophlebium percussum</i>                      | NA                             | VU                                     |
| <i>Grenacheria amentacea</i>                        | NA                             | Rediscovered                           |
| <i>Guioa pubescens</i>                              | NA                             | VU                                     |
| <i>Gynochthodes coriacea</i>                        | Akar lampai hitam              | VU                                     |
| <i>Gynochthodes praetermissa</i>                    | NA                             | VU                                     |
| <i>Gynochthodes ridleyi</i>                         | NA                             | CR                                     |
| <i>Gynochthodes rigida</i>                          | NA                             | VU                                     |
| <i>Helminthostachys zeylanica</i>                   | Kamraj, Tunjuk-langin          | Newly discovered                       |
| <i>Homalomena griffithii</i>                        | NA                             | VU                                     |
| <i>Jasminum elongatum</i>                           | Common Malayan jasmine         | EN                                     |
| <i>Labisia pumila</i>                               | Kacip fatimah                  | VU                                     |
| <i>Lygodium circinnatum</i>                         | Red finger fern                | VU                                     |
| <i>Lygodium longifolium</i>                         | NA                             | VU                                     |
| <i>Macaranga recurvata</i>                          | Swamp mahang                   | CR                                     |
| <i>Mangifera foetida</i>                            | Horse mango, <i>bacang</i>     | VU                                     |
| <i>Melicope lunu-ankenda</i>                        | NA                             | CR                                     |
| <i>Memecylon</i> sp.                                | NA                             | EN                                     |
| <i>Nepenthes ampullaria</i>                         | Narrow-lid pitcher plant       | VU                                     |
| <i>Nepenthes rafflesiana</i>                        | Raffles' pitcher plant         | VU                                     |
| <i>Nephelium lappaceum</i>                          | Rambutan                       | CR (cultivated)                        |
| <i>Oncosperma tigillarum</i>                        | Nibung                         | VU                                     |
| <i>Oxyceros longiflorus</i>                         | Akar berdara laut              | VU                                     |
| <i>Pandanus atrocarpus</i>                          | Bengkuang, Pandan duri         | EN                                     |
| <i>Pellacalyx axillaris</i>                         | Buloh-buloh                    | EN                                     |
| <i>Pentaphragma ellipticum</i>                      | Kubis hutan, Scorpion's tail   | VU                                     |
| <i>Phytocrene bracteata</i>                         | Nanas batu                     | VU                                     |
| <i>Planchonella obovata</i>                         | Menasi, Sea gutta              | VU                                     |
| <i>Porterandia anisophylla</i>                      | Wild randa                     | VU                                     |
| <i>Pternandra coerulescens</i>                      | Cursed shade                   | VU                                     |
| <i>Pternandra echinata</i>                          | Cursed shade                   | VU                                     |
| <i>Rourea asplenifolia</i>                          | NA                             | CR                                     |
| <i>Rubus moluccanus</i>                             | Tempu rengat                   | VU                                     |
| <i>Schizaea dichotoma</i>                           | Grass fern, branched comb fern | EN                                     |
| <i>Selaginella intermedia</i>                       | Spikemoss                      | VU                                     |
| <i>Sloetia elongata</i>                             | Tempinis                       | VU                                     |
| <i>Strophanthus caudatus</i>                        | Tandok-tandok                  | EN                                     |
| <i>Suregada multiflora</i>                          | False lime                     | CR                                     |
| <i>Symplocos fasciculata</i>                        | Kerenang, menasi               | VU                                     |
| <i>Syzygium cf rugosum</i>                          | NA                             | CR                                     |
| <i>Syzygium myrtifolium</i>                         | Kelat oil                      | NE                                     |
| <i>Syzygium polyanthum</i>                          | Indonesian bayleaf             | VU                                     |

| Species Name                    | Common Name         | Local Conservation Status <sup>1</sup> |
|---------------------------------|---------------------|----------------------------------------|
| <i>Tetracera cf arborescens</i> | NA                  | EN                                     |
| <i>Tetracera cf fagifolia</i>   | NA                  | VU                                     |
| <i>Thunbergia laurifolia</i>    | Akar ketuau         | Rediscovered (cultivated)              |
| <i>Timonius flavescens</i>      | Medang suid         | CR                                     |
| <i>Uncaria cordata</i>          | Gambir-gambir hutan | EN                                     |
| <i>Urceola cf torulosa</i>      | NA                  | EN                                     |
| <i>Urophyllum arboreum</i>      | NA                  | VU                                     |

<sup>1</sup> (Chong, Tan, & Corlett, 2009; Davison, Ng, & Ho, 2008)

**Table 4.7.** Distribution of conservation significant flora species by numbers & percentages

| Status                          | Number of species | Percentage of species (%) |
|---------------------------------|-------------------|---------------------------|
| DD                              | 1                 | 0.3                       |
| EX                              | 1                 | 0.3                       |
| Newly discovered                | 1                 | 0.3                       |
| Rediscovered                    | 2                 | 0.6                       |
| CR                              | 20                | 6.5                       |
| EN                              | 22                | 7.2                       |
| VU                              | 44                | 14.3                      |
| LC                              | 85                | 27.8                      |
| NA                              | 131               | 42.7                      |
| <b>Total number of species</b>  | <b>307</b>        | <b>–</b>                  |
| <b>Number of consig species</b> | <b>89</b>         | <b>–</b>                  |



**Figure 4.7.** Conservation status of recorded CCKWW flora species

The complete list of flora species recorded within project study area is provided in **Appendix B**. Photographs of some of dominant plant species are presented in **Appendix C**.

Conservation significant flora that were encountered within the proposed extension area, and

are thus likely to be affected by the future development, include *Aphanamixis polystachya* (EN), *Baccaurea motleyana* (CR), *Callicarpa longifolia* (EN), *Cissus repens* (CR), *Cyrtococcum patens* (VU), *Dioscorea cf polyclados* (CR), *Dracaena cantleyi* (VU), *Eurycoma longifolia* (CR), *Ficus aurata* (VU), *Goniophlebium percussum* (VU), *Jasminum elongatum* (EN), *Lygodium circinnatum* (VU), *Melicope lunu-ankenda* (CR), *Pellacalyx axillaris* (EN), *Porterandia anisophylla* (VU), *Symplocos fasciculata* (VU) and *Urceola cf torulosa* (EN), as well as a newly discovered fern species in Singapore, *Helminthostachys zeylanica*, which was previously known only to occur in other areas of South-east Asia, and is a species listed as Critically Endangered or Endangered by several countries including India, China, and Philippines (Balasubramanian et al., 2013; Yang et al., 2011; Coritico & Amoroso, 2020). *Thunbergia laurifolia*, while listed as a rediscovery, is persistent from cultivation. Thus, individuals of these species on the project site could be from cultivated stock.

The proposed sewerage alignment running from the CCKWW to the southeast may affect a number of additional conservation significant flora, including *Archidendron contortum* (VU), *Artocarpus lacucha* (EN), *Cyathea latebrosa* (VU), *Ficus sagittata* (CR), *Ficus vasculosa* (EN), *Gynochthodes coriacea* (VU), *Suregada multiflora* (CR), *Tetracera fagifolia* (VU), *Oxyceros longiflorus* (VU), and *Uncaria cordata* (EN) (refer to Figure 4.33).

To summarise, the current development plan may affect a total of 12 Vulnerable, 7 Endangered, 6 Critically Endangered and 1 newly discovered species of flora. It is therefore recommended to salvage any conservation significant flora found within these development areas where possible, especially those considered to be Critically Endangered and those with few known localities.

The study area contains a rather large number of species of conservation significance, which are distributed throughout the site, including in abandoned land areas. Some of the species may have been formerly cultivated for domestic or commercial purposes, such as Borneo mahogany (*Calophyllum inophyllum*), whose timber is prized in the woodworking industry, and *tongkat ali* (*Eurycoma longifolia*), a tree formerly used for medicinal purposes, which could possibly account for their presence within the study area (Friday & Okano, 2006; Turner I. , 2021).

Due to the lack of connection between Bukit Timah Nature Reserve and the Western Catchment area, the Western Catchment is a significant source of native flora germplasm as well as possibly being the most biodiverse area in the west of Singapore. This is evident from the unique set of flora diversity not found in other areas. For example, the Critically Endangered climber *Gynochthodes ridleyi* has only been recorded at Western Catchment, Jurong and Kranji only so far, and two Critically Endangered species, *Melicope lunu-ankenda* and *Ficus sagittata* have been found relatively frequently throughout the study. Species like *Syzygium rugosum*, *Grenacheria amentacea*, *Artabotrys maingayi*, *Ampelocissus gracilis*, *Macaranga recurvata*, *Pentaphragma ellipticum*, *Dioscorea polyclados* and *Enkleia malaccensis* had mostly been recorded near Mandai, Chestnut and Bukit Timah previously based on herbarium records, and this is likely the only known locality for these species at west of Singapore.

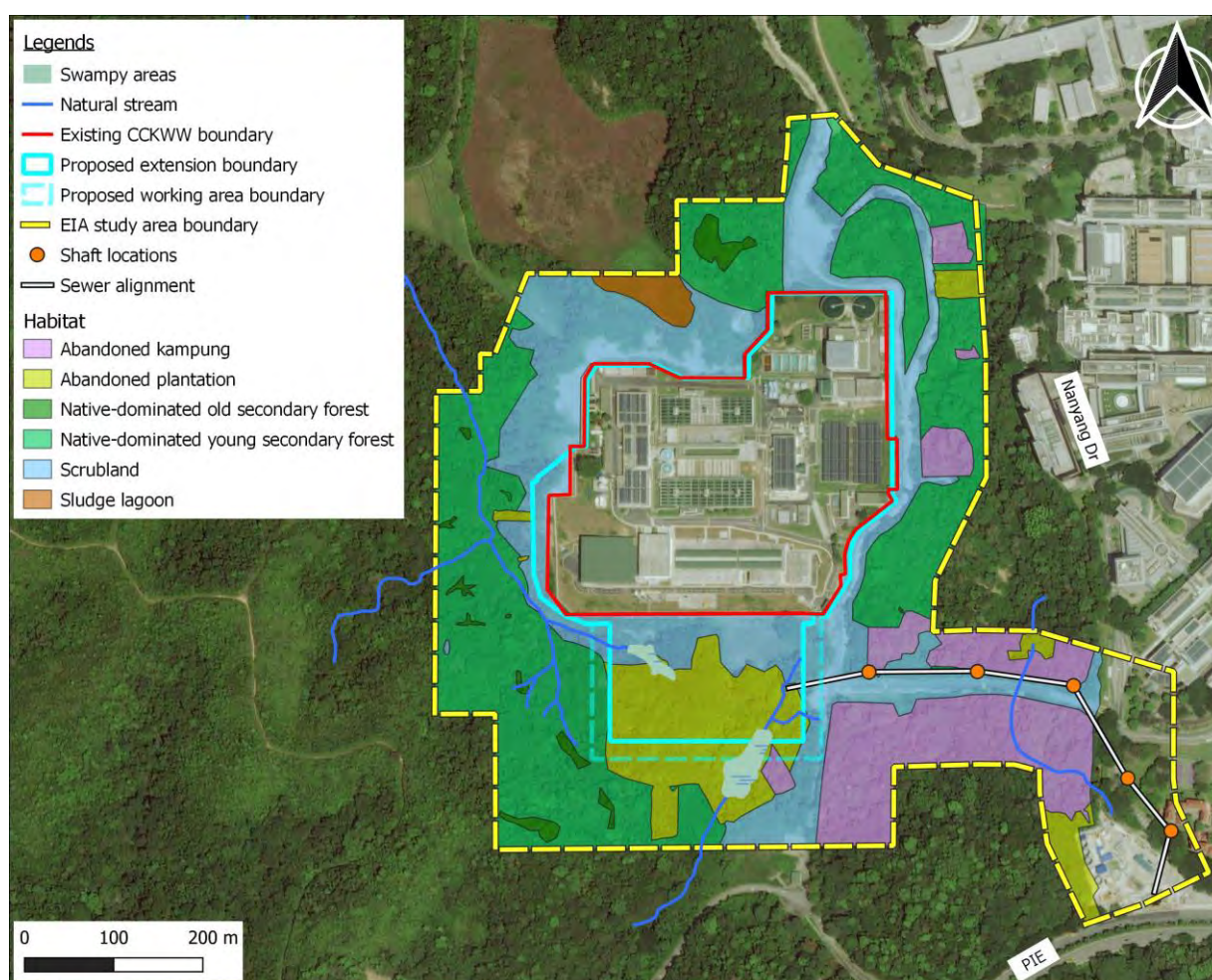
### **Vegetation and Habitat Mapping**

Native-dominated young secondary forest, defined as land not cleared long before the 1960s, and comprising of native pioneer species including *Adinandra* and *Macaranga* (Yee et al.,

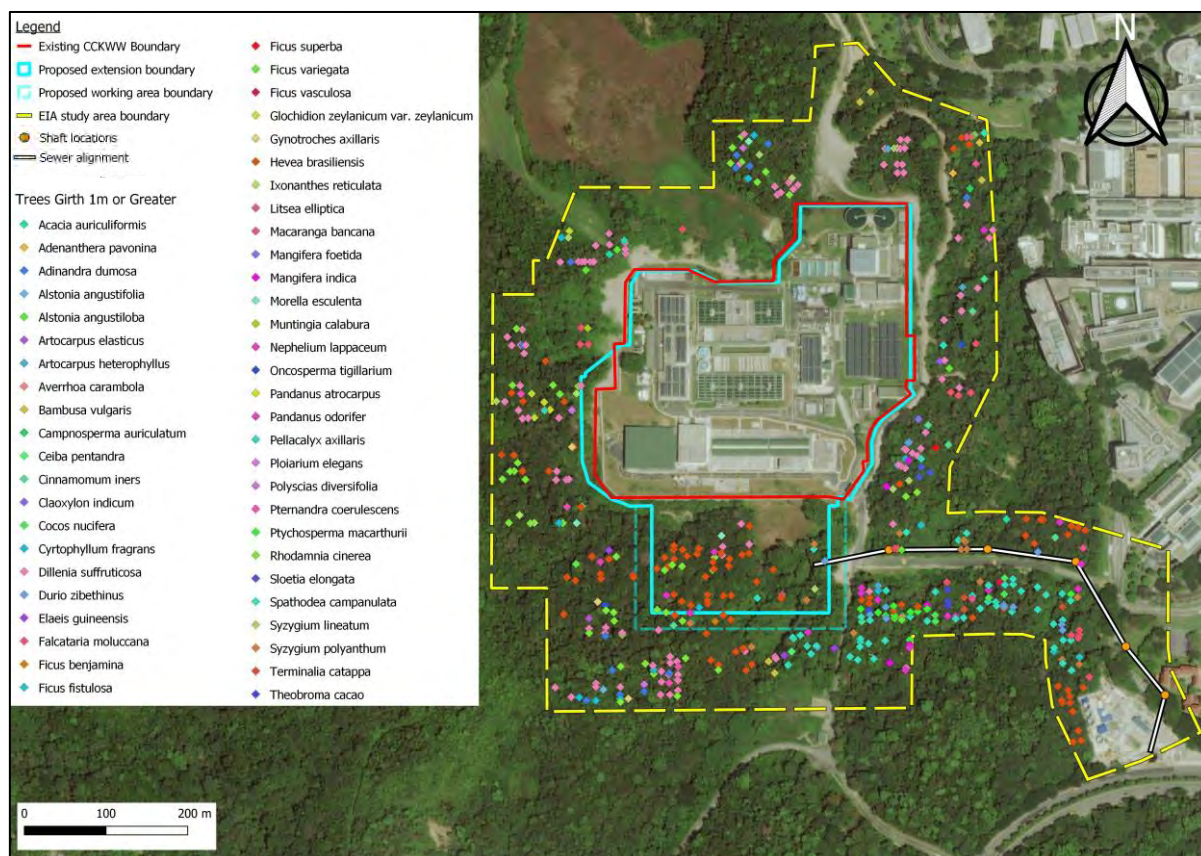
2016), was the habitat type that was most encountered throughout the forested areas of the study site. This habitat type comprised of a total area of approximately 12.16 hectares, with the majority of the area occurring to the southwest of the CCKWW. Small patches of native-dominated old secondary forest were also found distributed within the young secondary forest areas, with a total area of about 0.44 hectares. Scrubland was another fairly commonly encountered habitat type, occurring around the perimeter of the CCKWW as well as along all the roads in the area, and accounting for about 9.93 hectares in total.

The abandoned kampung habitat type occurred mainly to the southeast of the CCKWW and made up about 4.74 hectares of the study site, while abandoned plantation was found mainly in the south of the study area, covering a total of about 3.53 hectares. A sludge lagoon created by discharge from the CCKWW to the north of the study area, which was not surveyed, covered approximately 0.39 hectares.

Figure 4.8 provides the habitat map of the project study area. Figure 4.9 presents vegetation map of surveyed flora with girth  $\geq 1$  meter. Figure 4.10 provides vegetation map of surveyed Critically Endangered flora. Figure 4.11 presents vegetation map of surveyed Endangered flora and Figure 4.12 provides vegetation map of surveyed Vulnerable flora. Larger flora maps depicting species of conservation significance have been included in **Appendix D**.

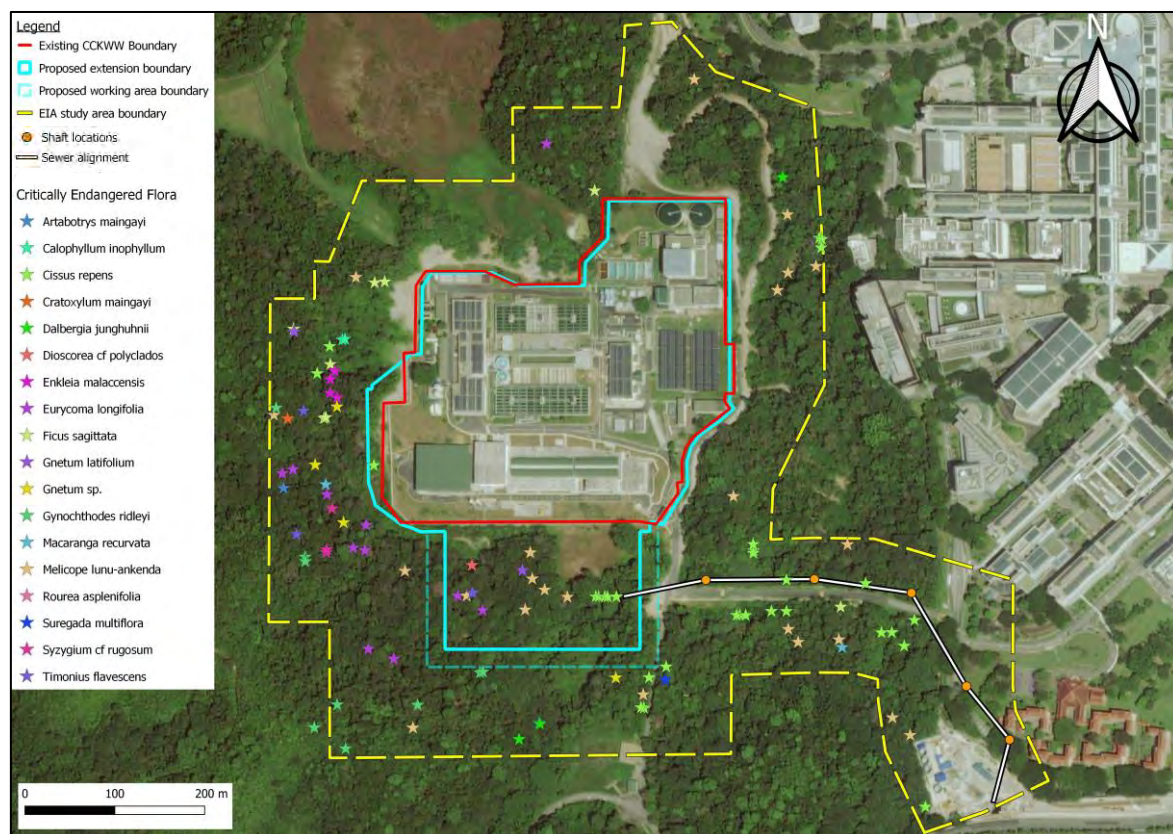


**Figure 4.8.** Habitat map of the project study area

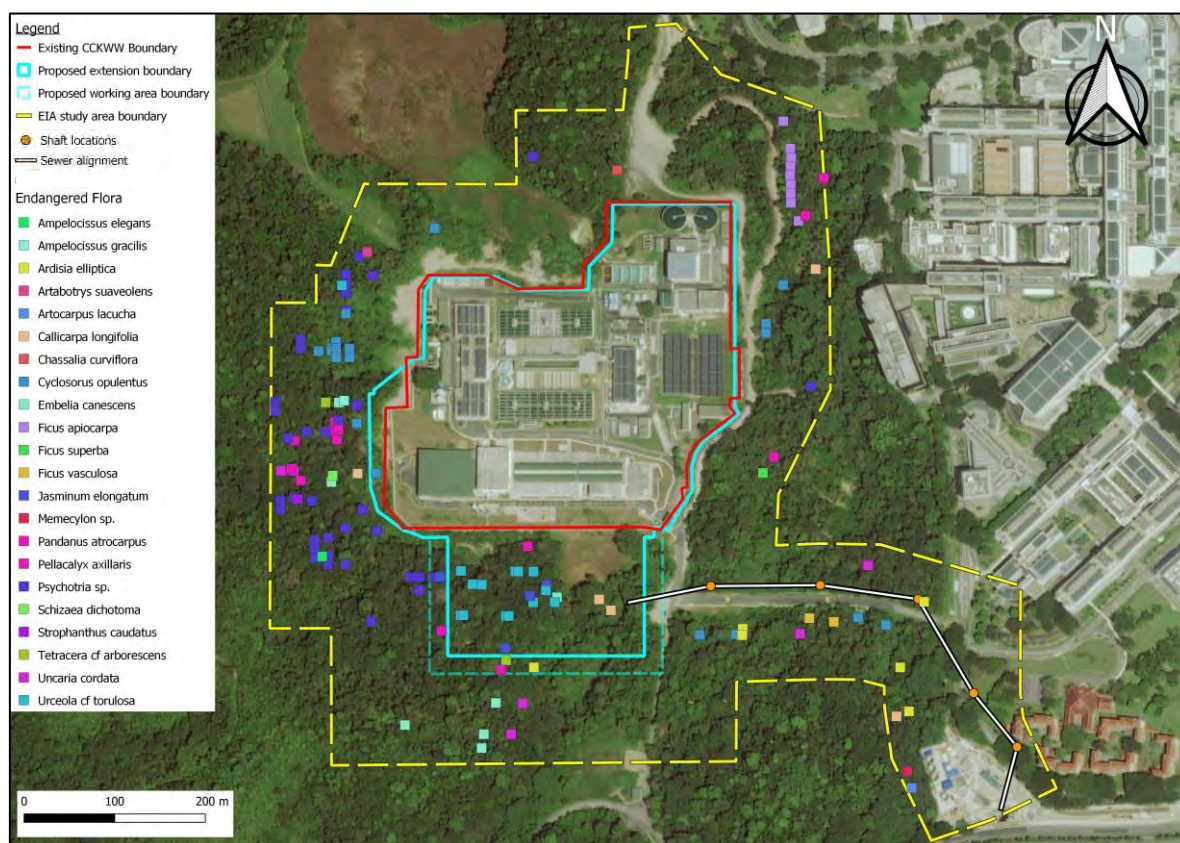


**Figure 4.9.** Vegetation map of surveyed flora with girth  $\geq 1$  metre

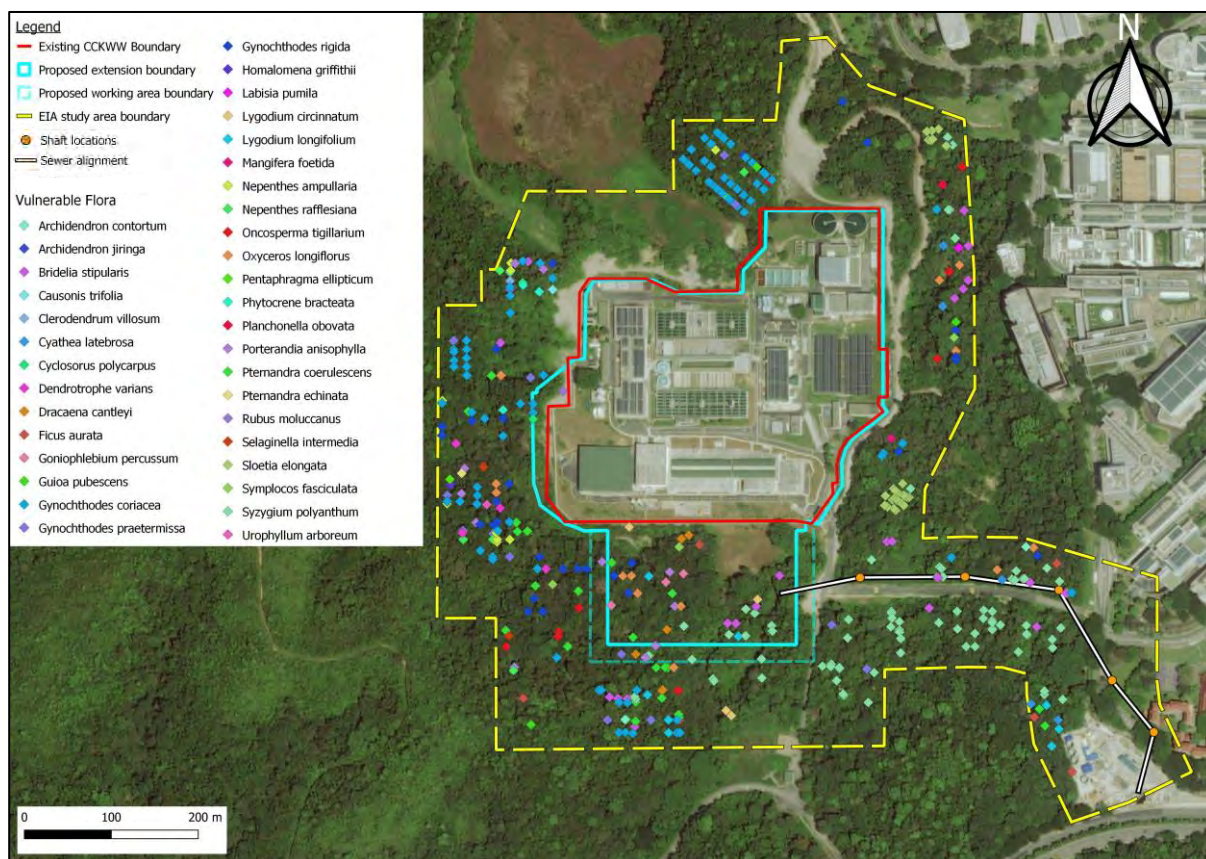
While the area marked for future development is relatively small compared to the rest of the study site, the clearing of the entire area of early succession forest remains an irreversible loss of habitat for both flora and fauna and contributes to the continuing deforestation in Singapore (Tan, Feng, & Hwang, 2016). The impact of the loss of this habitat is even greater considering the variety of Conservation Significant flora that was found within the future development area and the study area as a whole. The proximity of the development site to areas marked as native-dominated old secondary forest is also of concern, as it could affect the regeneration of these areas which have potential to develop into primary forest given time, which is a habitat type that currently only exists within the Central Catchment Nature Reserve and Bukit Timah (Turner & Corlett, 1996; Turner et al., 1997).



**Figure 4.10.** Vegetation map of surveyed Critically Endangered flora



**Figure 4.11.** Vegetation map of surveyed Endangered flora



**Figure 4.12.** Vegetation map of surveyed Vulnerable flora

#### 4.4.2 Fauna

The complete list of fauna sightings and photographs of observed animals can be found respectively in **Appendix E** and **Appendix F**.

##### Birds

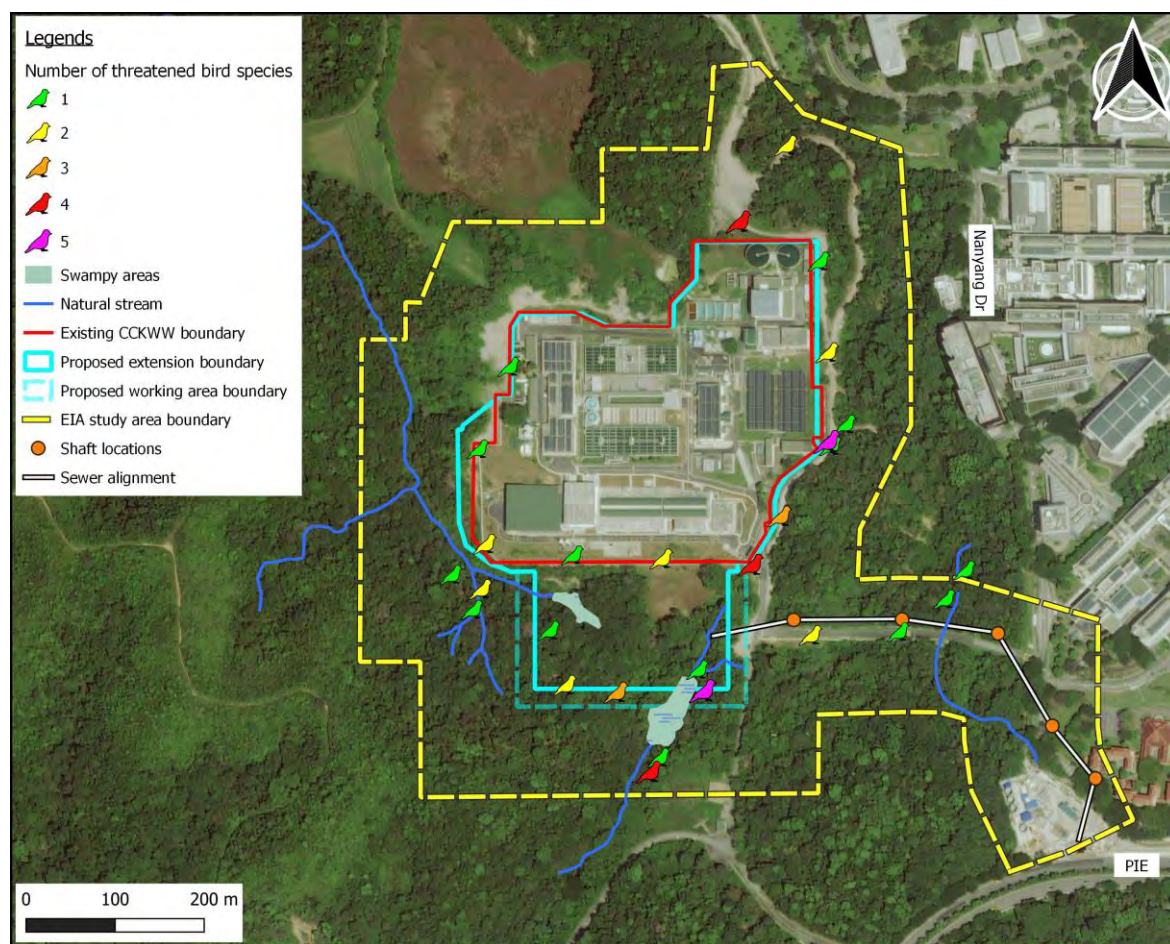
The surveys yielded 69 species of birds within the project site, of which 6 are locally threatened and 6 are globally threatened or near threatened (refer to Table 4.8 for the list of species of conservation value and Figure 4.13 for distribution of number of species of the threatened bird species at each point count location). Together with the data from DHI EIS 2016, this brings the number of birds observed on the project site to at least 82. Locally threatened species include the Endangered blue-crowned hanging parrot (*Loriculus galgulus*), changeable hawk-eagle (*Nisaetus cirrhatus*), oriental magpie-robin (*Copsychus saularis*), red junglefowl (*Gallus gallus*), red-legged crane (*Rallina fasciata*), and straw-headed bulbul (*Pycnonotus zeylanicus*). The straw-headed bulbul is also listed as globally Critically Endangered. Other globally threatened or Near-threatened species include the Vulnerable Javan myna (*Acridotheres javanicus*) and long-tailed parakeet (*Psittacula longicauda*), and the Near Threatened chestnut-bellied malkoha (*Phaenicophaeus sumatranus*) and red-breasted parakeet (*Psittacula alexandri*). However, both Javan myna and red-breasted parakeet are introduced to Singapore and are not considered to be of conservation significance locally.

There was the notable record of two barred eagle owls (*Bubo sumatranus*), which are rare in Singapore (Lim et al., 2020), within the project site. The barred eagle owl wasn't assessed in the previous Singapore Red Data Book (2008) due to insufficient data on the status of this

species in Singapore. However, the review committee for the upcoming Red Data list for birds has recommended that this species be listed as Critically Endangered. The current bird list is being updated and is due to be released in the near future (Lim Kim Seng, pers. comms.).

**Table 4.8.** List of bird species of conservation value currently found in project study area

| No. | Species name                 | Common Name                 | Local Status | Global Status |
|-----|------------------------------|-----------------------------|--------------|---------------|
| 1   | <i>Nisaetus cirrhatus</i>    | Changeable hawk-eagle       | EN           | LC            |
| 2   | <i>Copsychus saularis</i>    | Oriental magpie-robin       | EN           | LC            |
| 3   | <i>Gallus gallus</i>         | Red junglefowl              | EN           | LC            |
| 4   | <i>Loriculus galgulus</i>    | Blue-crowned hanging parrot | EN           | LC            |
| 5   | <i>Psittacula longicauda</i> | Long-tailed parakeet        | –            | VU            |
| 6   | <i>Pycnonotus zeylanicus</i> | Straw-headed bulbul         | EN           | CR            |
| 7   | <i>Rallina fasciata</i>      | Red-legged crane            | VU           | LC            |



**Figure 4.13.** Distribution of number of threatened bird species in project study area

As the surveys overlapped with the annual peak bird migration period, several migratory birds were thus recorded within the site. These included the yellow-rumped flycatcher (*Ficedula zanthopygia*), Arctic warbler (*Phylloscopus borealis*), and blue-throated bee-eater (*Merops viridis*) which were observed during September and October 2021.

Several species found within the site are forest species that are rarely found outside forests of good quality. Examples of such species are the chestnut-bellied malkoha, barred eagle owl, and

straw-headed bulbul. This indicates the site's importance as habitat and foraging ground for such species.



**Figure 4.14.** Barred eagle owl (L) and chestnut-bellied malkoha (R) in project study area

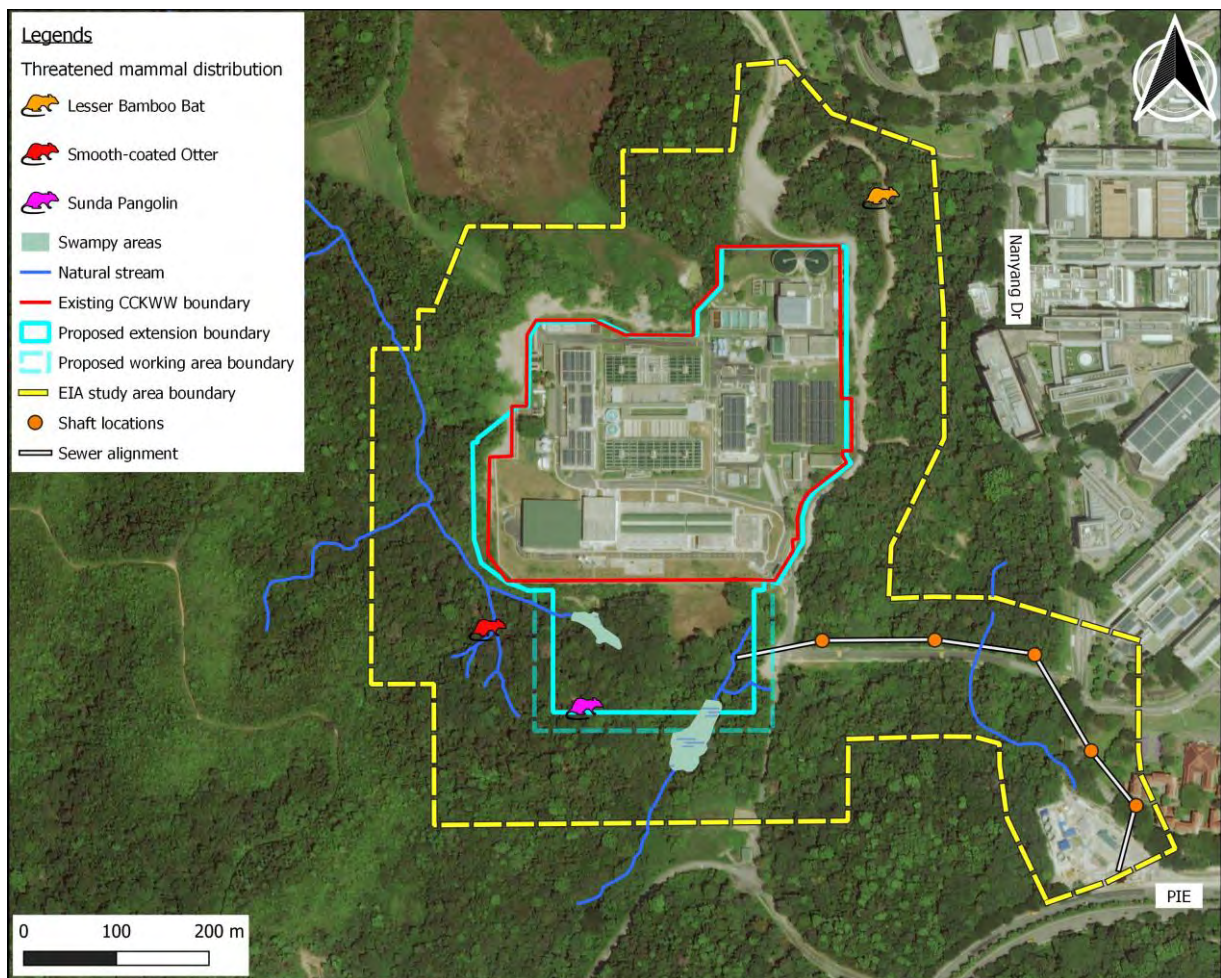
The straw-headed bulbul is also a noteworthy species given its globally Critically Endangered status. While populations straw-headed bulbuls on Singapore's mainland have been stable over the past few years, this species' numbers in the rest of Southeast Asia have been declining (Yong et al, 2018). Singapore is an important stronghold of this species, and its habitat need to be preserved for its continued persistence.

### **Mammals**

There are currently 74 species of mammals extant in Singapore. A total of 16 mammals were encountered within the project site during recent surveys, including three that were of conservation significance (Table 4.9, Figure 4.15). These mammals were detected through bat acoustic surveys, walking transects, and camera trapping. Together with the data from previous DHI EIS 2016, the number of mammals observed on the project site is at least 17.

**Table 4.9.** List of mammal species of conservation value currently found in project study area

| No. | Species name                   | Common Name         | Local Status | Global Status |
|-----|--------------------------------|---------------------|--------------|---------------|
| 1   | <i>Manis javanica</i>          | Sunda pangolin      | CR           | CR            |
| 2   | <i>Lutrogale perspicillata</i> | Smooth-coated otter | EN           | VU            |
| 3   | <i>Tylonycteris fulvida</i>    | Lesser bamboo bat   | VU           | LC            |



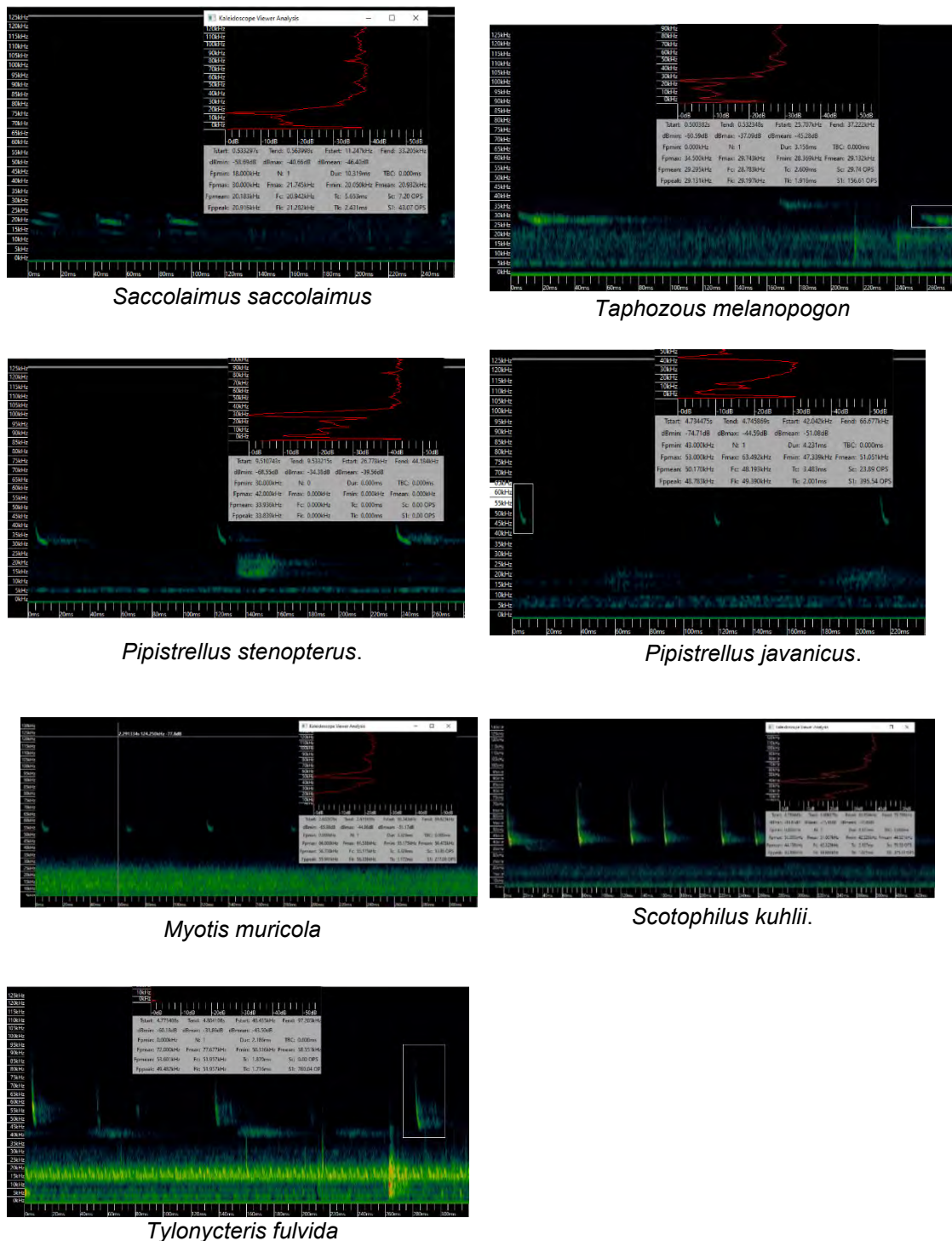
**Figure 4.15.** Location of mammal species of conservation value within project study area

Singapore's bat species diversity is ambiguous. In an analysis of Singapore's bat fauna, Lane *et al.* (2006) concluded that 30 species of bats were documented in Singapore (six Megachiroptera and 24 Microchiroptera), with 10 species considered as having gone extinct. However, they inferred that prior to 1819 between 60 to 72 species occurred in Singapore (Lane *et al.*, 2006). At the time of their publication, species diversity had declined to 20 (five Megachiroptera and 15 Microchiroptera), five fewer than what Baker and Lim (2012) had listed in their book. In the updated Singapore Red Data Book, NParks (2021) lists Singapore's bat species richness at 31 (five Megachiroptera and 26 Microchiroptera). Two fairly recently recorded species (long-winged tomb bat [*Taphozous longimanus*] and big-eared pipistrelle [*Hypsugo macrotis*]) represent new records for Singapore and are only known from Pulau Ubin (Yang, 2018).

Of the non-bat species encountered, most are common in Singapore and found within a range of vegetated sites around Singapore. However, there were two species of conservation significance within the project site: the globally and locally Critically Endangered Sunda pangolin (*Manis javanica*) and locally Endangered and globally Vulnerable smooth-coated otter (*Lutrogale perspicillata*).

Five rounds of bat acoustic surveys were conducted, split between two nights covering terrestrial transects (10 Aug and 7 Sep) and three nights covering stream transects (14 & 28 Sep and 5 Oct). Of the mammal species found on site, 8 of these species are bats. Acoustic sampling at

study area resulted in seven species, while the common fruit bat (*Cynopterus brachyotis*) was recorded during nocturnal visual encounter surveys. The spectrograms for the seven bat species recorded during acoustic sampling are presented in Figure 4.16.



**Figure 4.16.** Spectrograms showing acoustic call characteristics of detected bat species

Of the eight bat species detected, only the lesser bamboo bat (*Tylonycteris fulvida*) is of conservation significance. This species was recorded on 14 September at 19:47 hrs. The lesser bamboo bat is recognized as locally Vulnerable. Apparently, recent research has shown that population numbers of this highly specialised species are increasing as it was previously

regarded as Critically Endangered. As this bat species (as well as the greater bamboo bat detected from DHI's previous surveys) is dependent upon bamboo groves for roosting, its longevity in Singapore depends on access to suitable bamboo stands. Previously, the lesser bamboo bat was considered extirpated from Singapore until it was rediscovered in 1997 in Bukit Timah (Teo, 2018). Bamboo occurs throughout Singapore, but only with increased field research are we beginning to understand the extent of bamboo bat populations throughout Singapore.

Wild pigs (*Sus scrofa*) sightings were frequent throughout the site (Figure 4.17). Feeding of wild pigs was also observed along Dunbar Walk (Figure 4.18). Analyses of wild pig activities and locations can be found in Figure 4.28. There were also several encounters of domestic dogs (*Canis lupus familiaris*) along the road, outside the CCKWW main entrance. The presence of these animals, particularly wild pigs, could be a potential human wildlife-conflict issue.



**Figure 4.17.** Adult wild pig with two piglets along Dunbar Walk within project study area



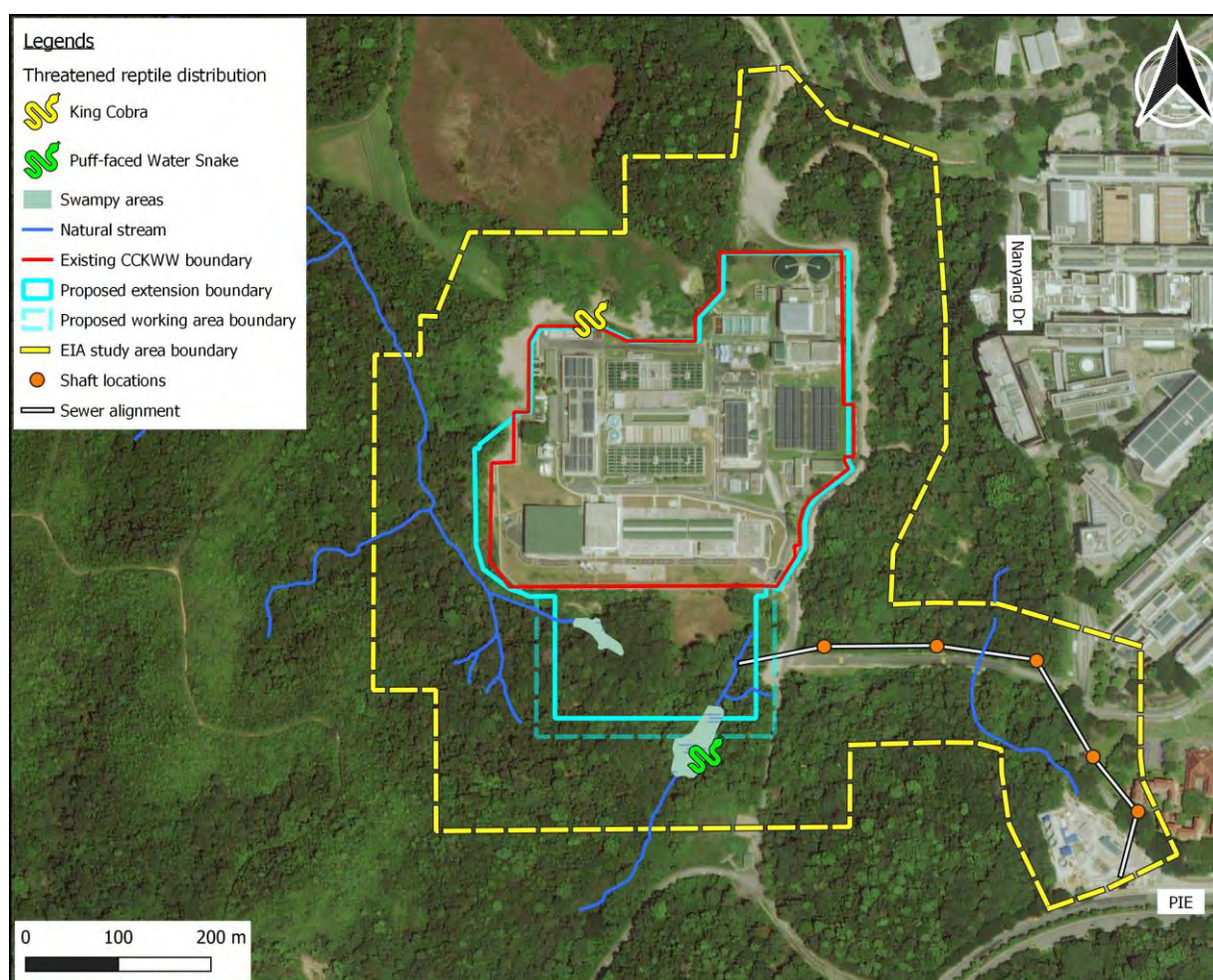
**Figure 4.18.** Wild pig feeding activities observed within project study area

## Reptiles

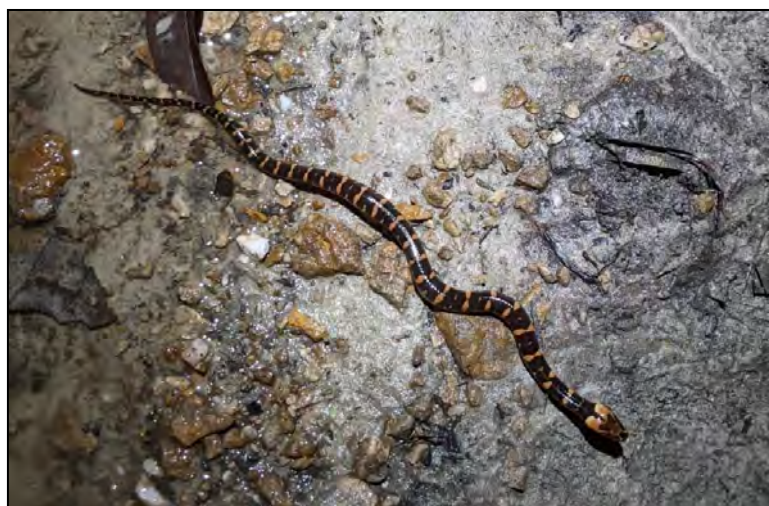
Singapore holds a relatively high diversity of reptiles, with at least 109 species being found in Singapore. A total of 12 reptile species were recorded on the project site during the surveys. The species observed included 2 locally threatened species (Table 4.10 and Figure 4.19, one of which is also globally threatened) – the locally Vulnerable puff-faced water snake (*Homalopsis buccata*) and the locally and globally Vulnerable king cobra (*Ophiophagus hannah*). One sighting of a *Draco* sp. was not identified to species level. Of the three species of *Draco* found in Singapore, two are considered locally threatened. Previous surveys from DHI EIS 2016 also found the locally Vulnerable Malayan blue coral snake (*Calliophis bivirgatus*) and the locally Near Threatened gold-ringed cat snake (*Boiga dendrophila*). With additional data from DHI EIS 2016, this brings the number of reptiles observed on the project site to at least 15. Other species are considered widespread and common except the green crested lizard (*Bronchocela cristatella*), which is widespread but uncommon in Singapore.

**Table 4.10.** List of reptile species of conservation value recorded by the present study

| No. | Species name              | Common Name            | Local Status | Global Status |
|-----|---------------------------|------------------------|--------------|---------------|
| 1   | <i>Ophiophagus hannah</i> | King cobra             | VU           | VU            |
| 2   | <i>Homalopsis buccata</i> | Puff-faced water snake | VU           | LC            |



**Figure 4.19.** Location of reptile species of conservation value within project study area



**Figure 4.20.** Puff-faced water snake found at Stream 2

The changeable lizard (*Calotes versicolor*) was the only introduced reptile species found on the project site.

It is likely that the number of reptile species observed on the site is under-represented due to the nature of most reptile species. Reptiles are generally elusive and are not observed frequently in surveys. However, the presence of several species of conservation value, particularly the puff-faced water snake, which relies on freshwater habitats, indicates the importance of the site for reptile species.

### Amphibians

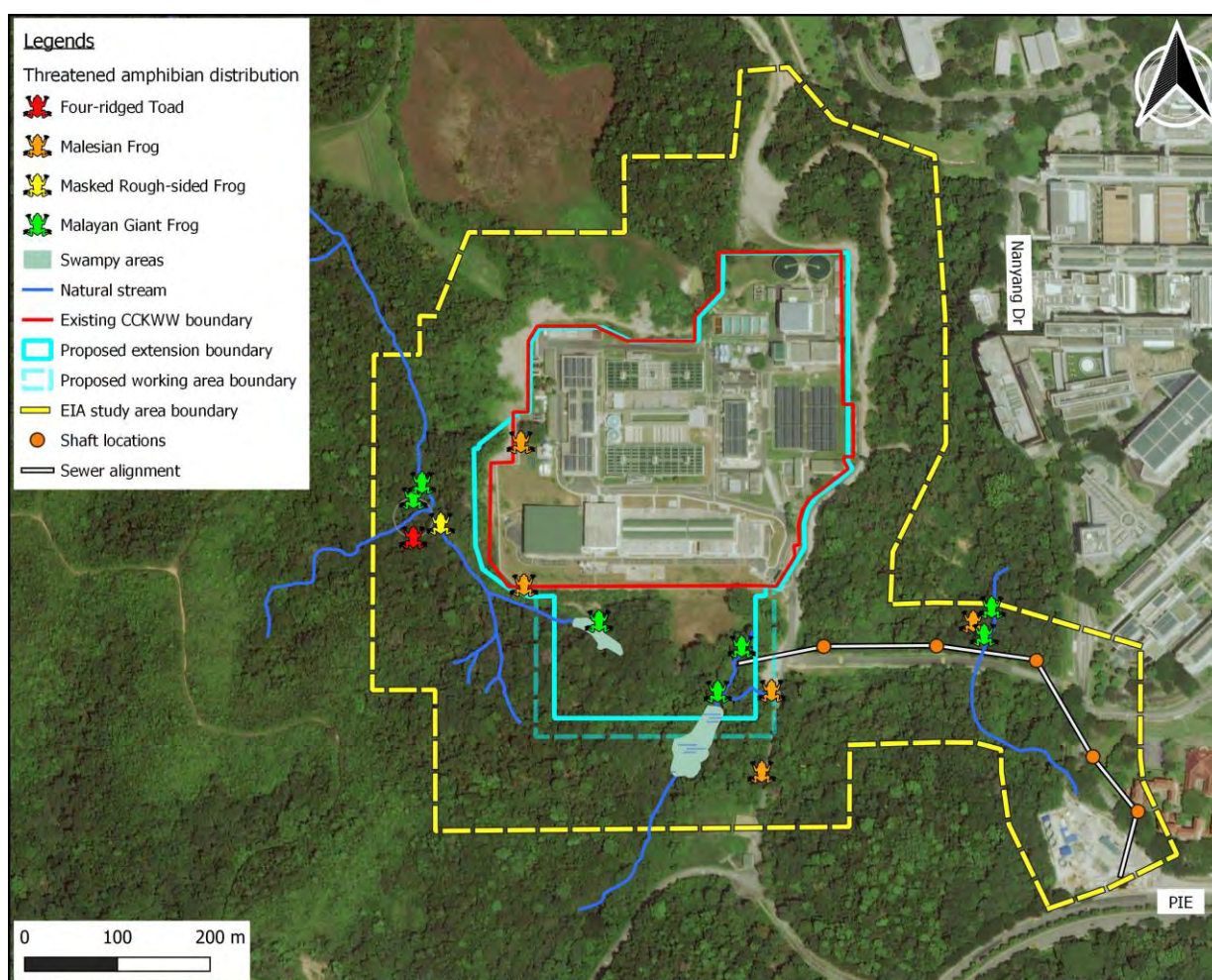
There are currently 30 species of amphibians known from Singapore. Of these, 14 amphibian species were found on the project site during the recent surveys. The species observed included 3 locally (near-)threatened and 2 globally (near-)threatened species (Table 4.11 and Figure 4.22). These include the locally Vulnerable four-ridged toad (*Ingerophrynus quadriporcatus*) Near-threatened Malesian frog (*Limnonectes malesianus*) and masked rough-sided frog (*Pulchrana laterimaculata*), and the globally Near Threatened Malayan giant frog (*Limnonectes blythii*). Together with the data from DHI EIS 2016, this brings the number of amphibians observed on the project site to at least 15.

**Table 4.11.** List of amphibian species of conservation value within project study area

| No. | Species name                        | Common Name             | Local Status | Global Status |
|-----|-------------------------------------|-------------------------|--------------|---------------|
| 1   | <i>Ingerophrynus quadriporcatus</i> | Four-ridged toad        | VU           | LC            |
| 2   | <i>Limnonectes blythii</i>          | Malayan giant frog      | LC           | NT            |
| 3   | <i>Limnonectes malesianus</i>       | Malesian frog           | NT           | NT            |
| 4   | <i>Pulchrana laterimaculata</i>     | Masked rough-sided frog | NT           | LC            |



**Figure 4.21.** Malesian frog in stream within project site



**Figure 4.22.** Location of amphibian species of conservation value within project study area

Several of the species detected are forest dependent. In particular, the masked rough-sided frog and four-ridged toad (both detected through calls) are swamp forest specialists and are uncommon outside the Central Nature Reserves of Singapore. The Malesian frog and copper

cheeked frog (*Chalcorana labialis*) are dependent on forest streams or waterbodies, and also rarely found outside the Central Nature Reserves and buffer nature parks.

Three of the species recorded have been introduced to Singapore: banded bullfrog (*Kaloula pulchra*), Mukhlesur's narrow-mouthed frog (*Microhyla mukhlesuri*), and Gunther's frog (*Sylvirana guentheri*).

The presence of several forest-dependent species is significant, and indicative of the quality of the habitats present in project study area. Most amphibians have specific requirements for waterbodies, where they live out the larval stage of their lives. The habitats these species are reliant on, in particular swamp forests and forest streams, are rare in Singapore due to the country's history of largescale deforestation. The habitats present in the project site are therefore important for the conservation of such species.

### Terrestrial Invertebrates

#### Butterflies

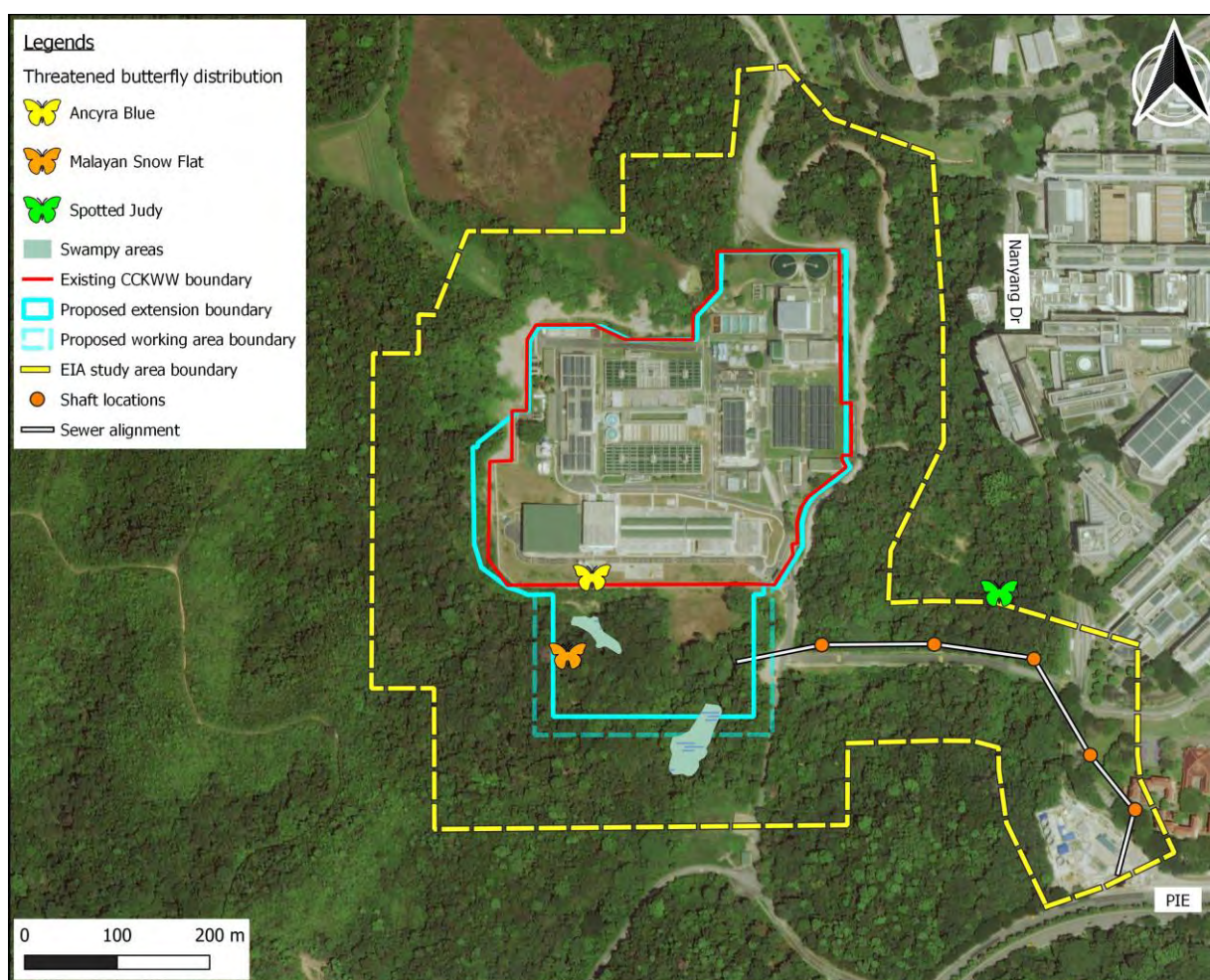
Singapore is home to about 338 species of butterflies. At least 34 butterfly species were recorded within the project site. These include the moderately rare (see Table 4.12 and Figure 4.24) Malayan snow flat (*Tagiades calligana*), ancyra blue (*Catopyrops ancyra*), and spotted judy (*Abisara geza niya*). With additional data from DHI EIS 2016, the total number of butterflies observed on the site amounts to 38.

**Table 4.12.** List of butterfly species of conservation value within project study area

| No. | Species name              | Common Name       | Local Status ( <i>Butterflies of Singapore</i> , 2nd ed) |
|-----|---------------------------|-------------------|----------------------------------------------------------|
| 1   | <i>Tagiades calligana</i> | Malayan snow flat | Moderately Rare                                          |
| 2   | <i>Catopyrops ancyra</i>  | Ancyra blue       | Moderately Rare                                          |
| 3   | <i>Abisara geza niya</i>  | Spotted judy      | Moderately Rare                                          |



**Figure 4.23.** Malayan snow flat within forested part of project study area



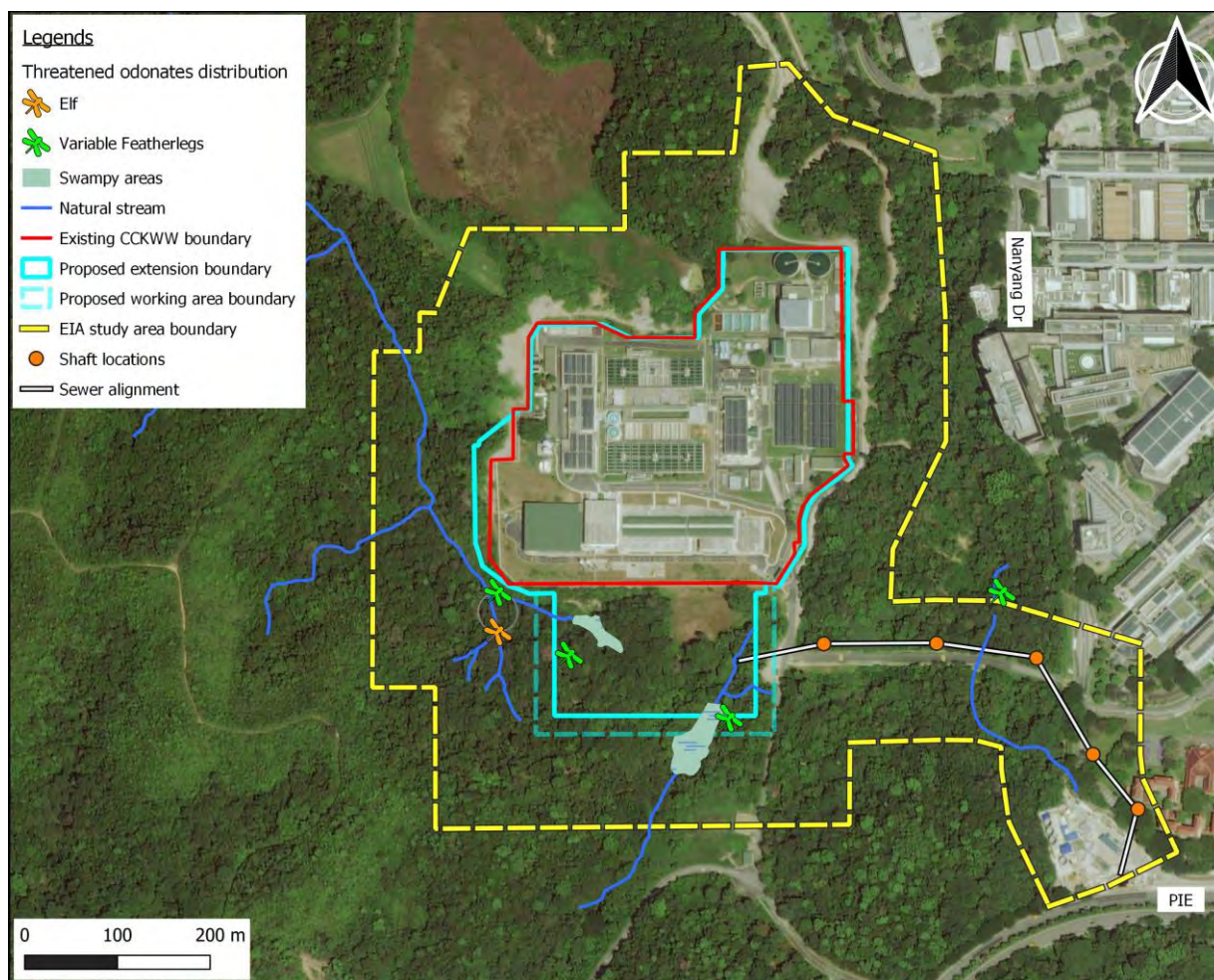
**Figure 4.24.** Location of butterfly species of conservation value within project study area

### Odonates

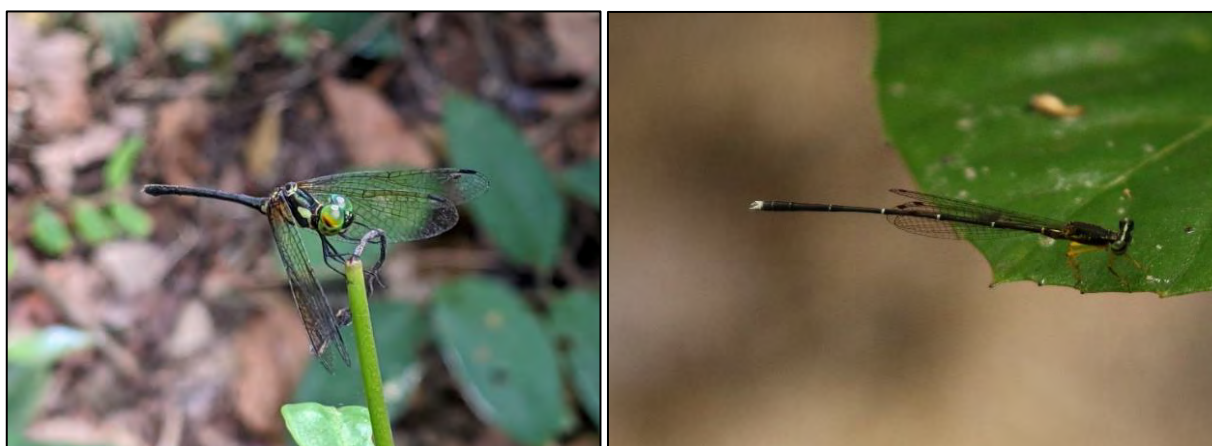
At least 133 species of odonates are currently known from Singapore. Of these, 25 odonates (dragonflies and damselflies) were found within the project site. These include the locally Critically Endangered species, the elf (previously known as the subspecies *Tetrathemis irregularis hyalina* but recently revised taxonomically as a separate species, *Tetrathemis hyalina*), which is considered very rare and found only in Central Catchment Nature Reserve (CCNR), and the Vulnerable variable featherlegs (*Copera vittata*) (Table 4.13 and Figure 4.25). The observation of the elf is particularly significant given that there have been few sightings of this species within Singapore, and this is the first record outside CCNR (Tang, Wang, & Hämäläinen, 2010). Several uncommon species were also observed during the surveys, namely sapphire flutterer (*Rhyothemis triangularis*), bronze flutterer (*Rhyothemis obsolescens*), variable sprite (*Argioctenidia rubescens*), and banded skimmer (*Pseudothemis jorina*). A dragonfly, *Gynacantha* sp., was not identified to species level. However, this genus represents four swamp forest-dependent species known locally, two of which are Vulnerable. Together with data from DHI EIS 2016, there are at least 28 odonates species observed on the project site.

**Table 4.13.** List of odonates species of conservation value within project study area

| No. | Species name               | Common Name          | Local Status | Global Status |
|-----|----------------------------|----------------------|--------------|---------------|
| 1   | <i>Copera vittata</i>      | Variable featherlegs | VU           | LC            |
| 2   | <i>Tetrathemis hyalina</i> | Elf                  | CR           | LC            |



**Figure 4.25.** Location of odonates species of conservation value within project study area



**Figure 4.26.** Elf and variable featherlegs dragonflies within forested area of project site.

In Singapore, all species of odonates that are no longer found here had faced extirpation due to the loss of key forest habitats they dependent on for their survival (Tang et al., 2010). The presence of forest swamp/stream species, including the elf, variable featherlegs, and treehugger (*Tyriobapta torrida*), once again illustrates the importance of these habitats within the project site for the conservation of rare forest-dependent species.

## Aquatic Fauna

21 aquatic fauna species were detected in the project study area (see Figure 4.4 for survey locations), covering 9 freshwater fish species, 9 molluscs, and 3 crustaceans. None of the species recorded in the present study are threatened. Together with the DHI EIS 2016 data, 22 aquatic fauna species are known in the study area, with Johnson's freshwater crab as the only species from the report not detected in the present study.

As reported by Table 4.14, species composition differs significantly between the three streams. Three species were recorded in Stream 1, 15 in Stream 2 and eight species in Stream 3 (Table 4.14). This is not surprising considering that the water quality in Stream 1 and Stream 3 (represented in Section 5.6 by WS1 and WS2 for Stream 1 and by WS6 and WS7 for Stream 3) were more acidic than Stream 2. Moreover, low overall species diversity in Stream 1 may be attributed to water discharge from stormwater drain and CCKWW discharge channelled via roadside drain. Moreover, two species in Stream 1, guppy, and Malaysian trumpet snail, are known to be able to live in polluted waters with strong current due to their high tolerance of a wide range of environmental conditions (Ivan Kwan Wei Ming, pers. comms.).

**Table 4.14.** Recorded aquatic fauna composition in the streams within project study area

| Species Name                    | Common Name             | Stream 1 | Stream 2 | Stream 3 | Swamp* |
|---------------------------------|-------------------------|----------|----------|----------|--------|
| <b>Freshwater Fish</b>          |                         |          |          |          |        |
| <i>Aplocheilichthys panchax</i> | Whitespot               |          |          | v        | v      |
| <i>Channa striata</i>           | Common snakehead        |          | v        | v        |        |
| <i>Clarias</i> sp.              | Walking catfish species |          | v        | v        |        |
| <i>Rasbora einthovenii</i>      | Brilliant rasbora       |          |          | v        |        |
| <i>Barbodes banksi</i>          | Saddle barb             |          |          | v        |        |
| <i>Trichopodus trichopterus</i> | Three-spot gourami      |          | v        |          | v      |
| <i>Betta pugnax</i>             | Malayan forest betta    |          |          | v        |        |
| <i>Poecilia reticulata</i>      | Guppy                   | v        | v        |          |        |
| <i>Monopterus javanensis</i>    | Sunda swamp-eel         | v        |          |          |        |
| <b>Molluscs</b>                 |                         |          |          |          |        |
| <i>Bithyniidae</i> sp.          |                         |          | v        |          |        |
| <i>Melanoides tuberculata</i>   | Malaysian trumpet snail | v        | v        | v        |        |
| <i>Tarebia granifera</i>        | Quilted melania         |          | v        |          |        |
| <i>Mieniplotia scabra</i>       |                         |          | v        |          |        |
| <i>Indoplanorbis exustus</i>    | Ramshorn snail (large)  |          | v        |          |        |
| <i>Gyraulus convexiusculus</i>  | Ramshorn snail (small)  |          | v        |          | v      |
| <i>Amerianna carinata</i>       |                         |          | v        |          | v      |
| <i>Physa acuta</i>              | Bladder snail           |          | v        |          | v      |
| <i>Radix rubiginosa</i>         | Pond snail              |          | v        |          |        |
| <b>Crustaceans</b>              |                         |          |          |          |        |
| <i>Caridina</i> sp.             |                         |          | v        |          |        |
| <i>Macrobrachium malayanum</i>  |                         |          |          | v        |        |
| <i>Macrobrachium</i> sp.        |                         |          | v        |          |        |
| <b>Total</b>                    |                         | 3        | 15       | 8        | 5      |

\* The swamp was surveyed for only one round of hand-netting.

## Camera Trapping

The camera traps were deployed for a total of at least 60 days, eight of which captured animal pictures from 3 August to 12 October 2021. CAM9 was found to be malfunctioning during the first round of SD card changes and was thus redeployed from 25 August to 26 October 2021. As the camera traps took a sequence of three photos each time they were triggered, the photos were separated according to their respective sequences. Sequences separated by more than

60s were assumed to be independent sightings.

A total of 1841 independent sightings (Table 4.15) identified from 6176 photos of 24 identified fauna species were collected from the 9 camera traps.

**Table 4.15.** Number of sightings from each camera by species (\*=locally threatened species)

| Species                                   | CAM1       | CAM2       | CAM3       | CAM4       | CAM5       | CAM6      | CAM7     | CAM8      | CAM9      |
|-------------------------------------------|------------|------------|------------|------------|------------|-----------|----------|-----------|-----------|
| Blue-winged pitta                         | 3          | 0          | 0          | 0          | 0          | 0         | 0        | 0         | 0         |
| Changeable hawk-eagle*                    | 0          | 0          | 0          | 0          | 0          | 1         | 0        | 0         | 0         |
| Common emerald dove                       | 4          | 0          | 0          | 0          | 0          | 0         | 0        | 0         | 0         |
| Common palm civet                         | 0          | 1          | 0          | 1          | 0          | 0         | 0        | 0         | 0         |
| Common treeshrew                          | 1          | 0          | 0          | 9          | 0          | 0         | 0        | 0         | 1         |
| Greater racket-tailed drongo              | 0          | 0          | 0          | 1          | 0          | 0         | 0        | 0         | 0         |
| Javan myna                                | 2          | 1          | 0          | 1          | 0          | 0         | 0        | 0         | 0         |
| Laced woodpecker                          | 0          | 1          | 1          | 3          | 0          | 0         | 0        | 0         | 0         |
| Long-tailed macaque                       | 0          | 3          | 21         | 6          | 2          | 12        | 0        | 0         | 0         |
| Malayan water monitor                     | 10         | 0          | 0          | 0          | 0          | 0         | 0        | 0         | 0         |
| Olive-winged bulbul                       | 0          | 0          | 0          | 1          | 0          | 0         | 0        | 0         | 0         |
| Oriental magpie-robin*                    | 12         | 0          | 0          | 2          | 0          | 0         | 0        | 0         | 0         |
| Pink-necked green pigeon                  | 1          | 0          | 0          | 0          | 0          | 0         | 0        | 0         | 0         |
| Plantain squirrel                         | 2          | 45         | 15         | 7          | 0          | 21        | 0        | 0         | 0         |
| Red junglefowl*                           | 0          | 18         | 4          | 12         | 2          | 0         | 0        | 0         | 2         |
| Smooth-coated otter*                      | 0          | 0          | 0          | 0          | 1          | 0         | 0        | 0         | 0         |
| Spine-tufted skimmer                      | 1          | 0          | 0          | 0          | 0          | 0         | 0        | 0         | 0         |
| Sunda pangolin*                           | 0          | 0          | 0          | 1          | 0          | 0         | 0        | 0         | 0         |
| White-breasted waterhen                   | 3          | 2          | 4          | 0          | 3          | 0         | 0        | 0         | 0         |
| White-crested laughingthrush              | 0          | 7          | 3          | 7          | 2          | 0         | 0        | 1         | 0         |
| White-throated kingfisher                 | 1          | 0          | 0          | 0          | 0          | 0         | 0        | 0         | 0         |
| Wild pig                                  | 127        | 277        | 678        | 178        | 136        | 17        | 2        | 65        | 16        |
| Yellow-vented bulbul                      | 0          | 2          | 0          | 4          | 0          | 0         | 0        | 2         | 0         |
| Zebra dove                                | 0          | 0          | 0          | 1          | 0          | 0         | 0        | 0         | 0         |
| Unidentified                              | 2          | 2          | 4          | 2          | 9          | 13        | 0        | 5         | 2         |
| Unidentified bird                         | 0          | 0          | 0          | 2          | 0          | 0         | 0        | 0         | 0         |
| Unidentified mammal                       | 0          | 18         | 5          | 4          | 0          | 0         | 0        | 0         | 3         |
| <b>Total Number of Detections</b>         | <b>169</b> | <b>377</b> | <b>735</b> | <b>242</b> | <b>155</b> | <b>64</b> | <b>2</b> | <b>73</b> | <b>24</b> |
| <b>Total Number of Identified Species</b> | <b>12</b>  | <b>10</b>  | <b>7</b>   | <b>15</b>  | <b>6</b>   | <b>4</b>  | <b>1</b> | <b>3</b>  | <b>3</b>  |

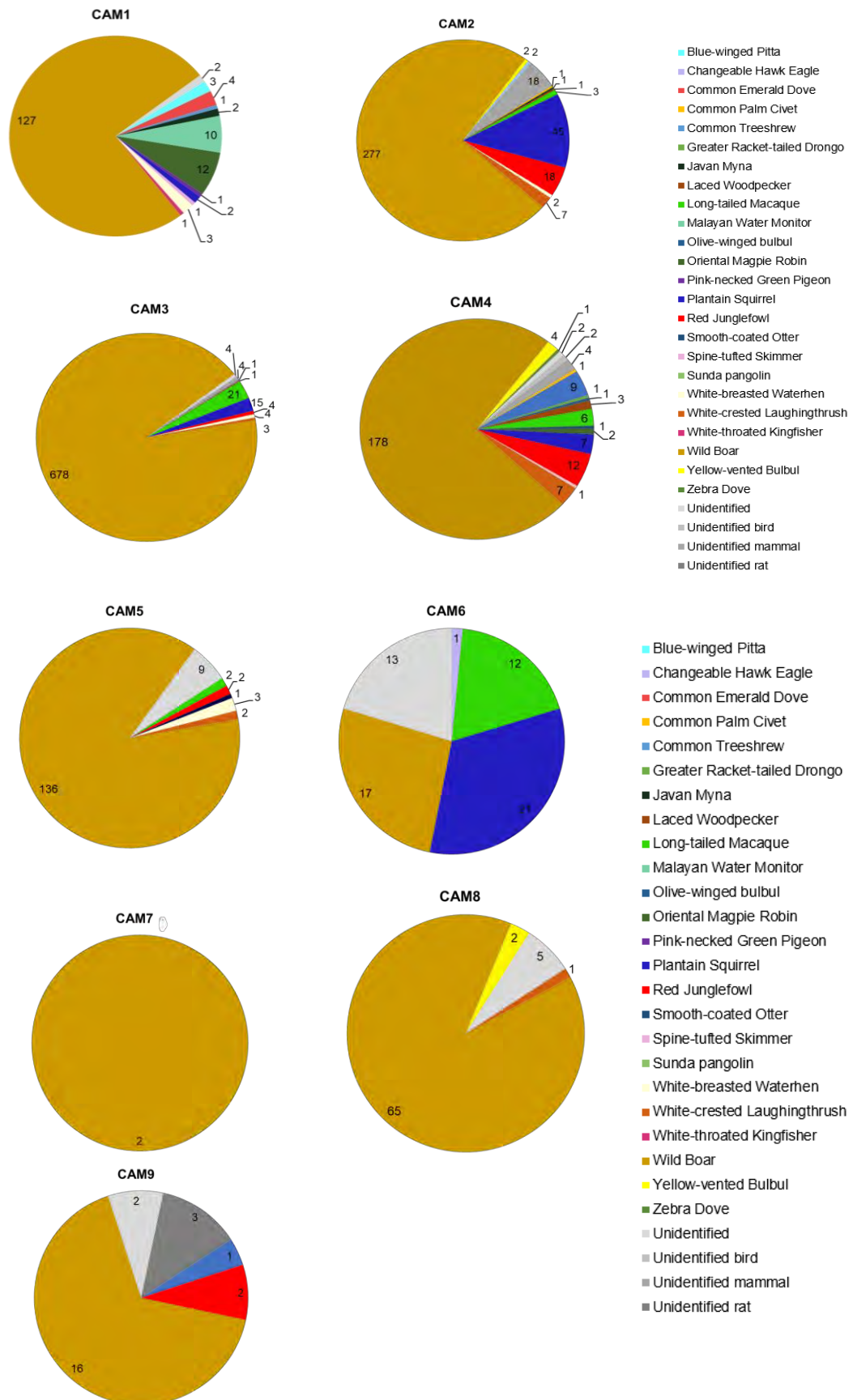
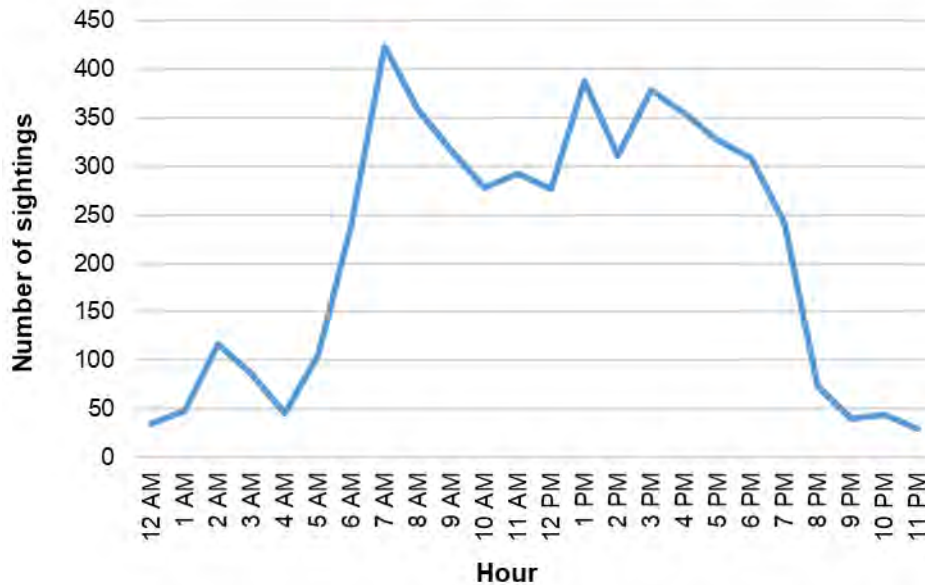


Figure 4.27. Number of species sightings captured by each camera

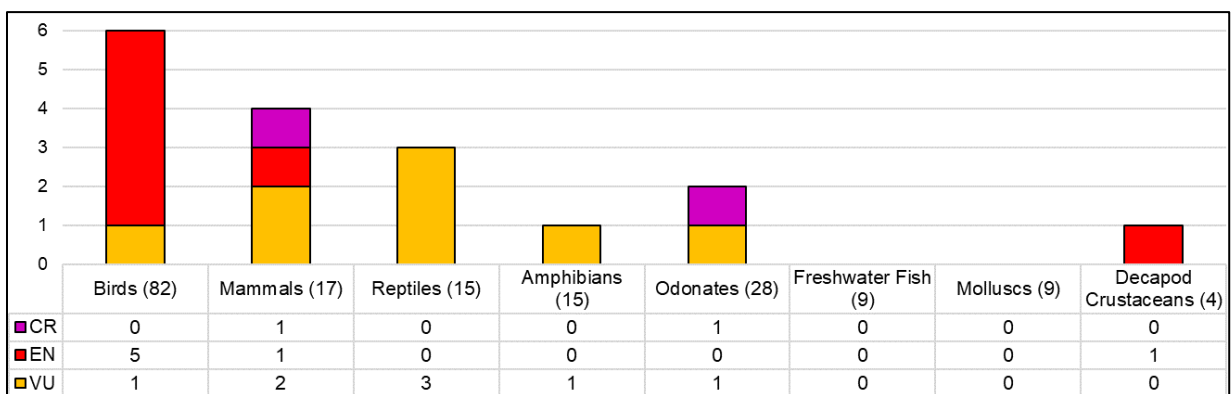
The most frequently recorded species was the wild pig (*Sus scrofa*), of which 1,496 individual sightings were detected by all camera traps. Six of the nine camera traps also showed groups of wild pigs with piglets. Wild pig activity as recorded by the camera traps is presented in Figure 4.28.



**Figure 4.28.** Wild pig (*Sus scrofa*) activity as recorded by the camera trap over 24-hour period

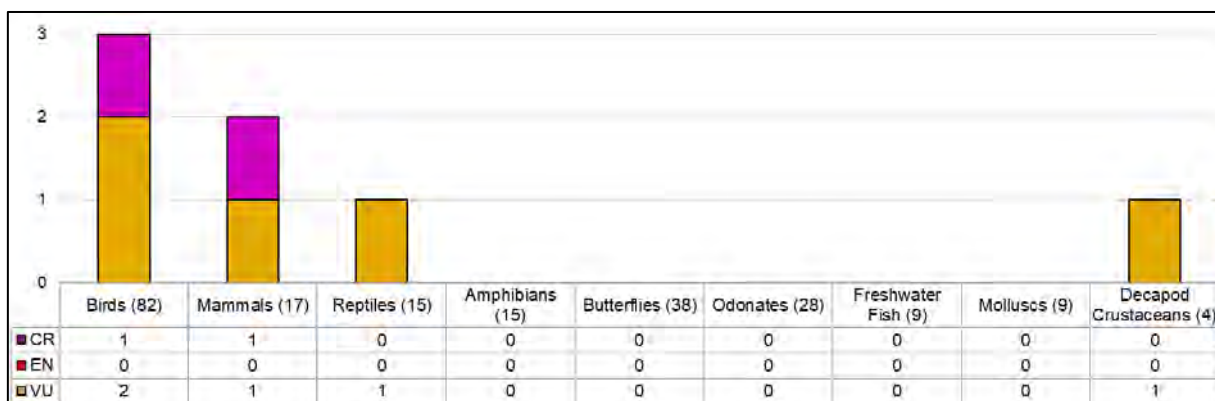
### Summary of Species of Conservation Significance

In summary, 17 fauna species of local conservation value (defined as Vulnerable / VU and above, see Figure 4.29) were found in this current survey, comprising 6 birds, 3 mammals, 2 reptiles, 1 amphibian, and 2 odonates. Together with an additional 4 that were found from the DHI EIS 2016 (1 mammal, 1 reptile, 1 butterfly, and 1 decapod crustacean), a total of 21 fauna species of local conservation significance are known from the project site.



**Figure 4.29.** Local conservation status overview of species recorded within project study area

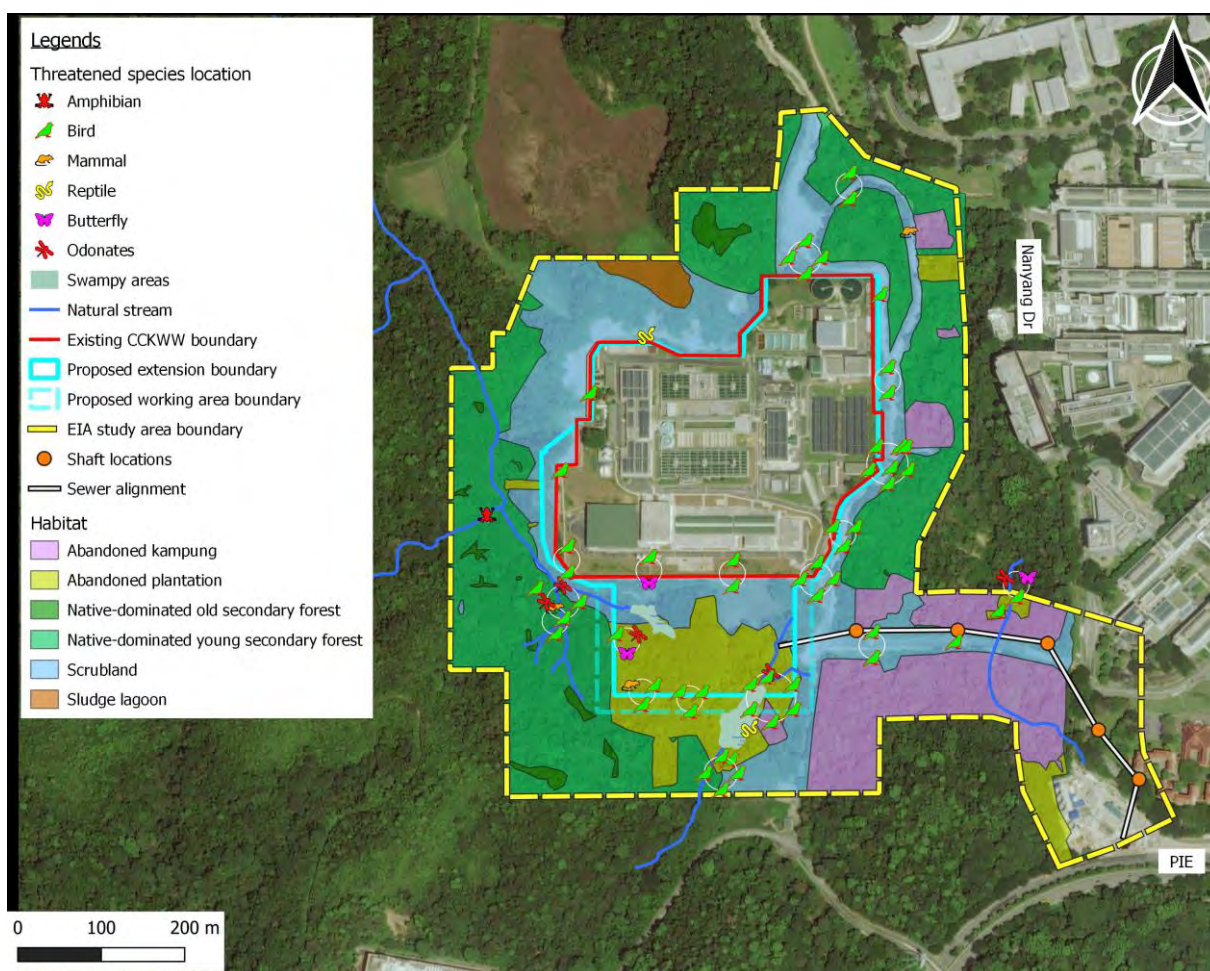
The surveys also recorded 6 species of global conservation significance (defined as Vulnerable / VU and above, see Figure 4.30) based on the IUCN Red List (2021), comprising 3 birds, 2 mammals, 1 reptile. With data from the DHI EIS 2016 (1 decapod crustacean), this accounts to a total of 7 species of global conservation significance.



**Figure 4.30.** Global conservation status overview of species recorded within project study area

Fauna not identified to species level have been excluded from this analysis.

Overall locations of species of conservation values, both local and global, with regard to habitat types found in the project study area are presented in Figure 4.31.



**Figure 4.31.** Location of species of local and global conservation value vis-à-vis habitat types

A large number of the non-avian species of conservation value detected are forest dependent species. Based on our surveys, many of these species were mostly concentrated around the freshwater streams within the forested parts of the project site. This included several species of

amphibians and odonates.

The Johnson's freshwater crab (*Irmengardia johnsoni*) was previously found in small numbers in the vicinity of Stream 3 of current study area. In addition to the crab species being listed as nationally Endangered and globally Vulnerable, it is endemic (not found anywhere else in the world) to Singapore. As such, the safeguarding of population of this species in Singapore is of tantamount importance.

However, the Johnson's freshwater crab was not detected during the current round of aquatic surveys. This could be due to several reasons. The stream profile where the crab was found in DHI EIS 2016 has been altered since then as part of the previous development. This could have led to the reduced or complete loss of the crab population within the stream, hence leading to reduced detectability. However, biodiversity surveys rarely yield all species present in a given site, and it is possible that the Johnson's freshwater crab is still found in the streams of the project site but were undetected. Also, since the previous development, the crab might have moved to other parts of the stream north of the study area. Nevertheless, the presence of the Johnson's freshwater crab in recent years still illustrates the importance of stream habitats within this site for the existence of rare and threatened aquatic species.

Large-scale deforestation has resulted in a loss of much of Singapore's forested areas (Corlett, 1992), and the forested freshwater stream habitat is a particularly threatened habitat type due to its current rarity in Singapore. The presence of a relatively high proportion of threatened species within the forested parts of the project site makes it significant for the continued protection of these species in Singapore.

#### **4.4.3 Summary of Flora and Fauna Baseline**

At least 307 flora species and 191 fauna species (birds, mammals, reptiles, amphibians, butterflies, odonates, freshwater fish, molluscs, and decapod crustaceans) were observed during current study within the project study area.

Floristically, the fern *Helminthostachys zeylanica* is the most notable finding as the individual recorded in the present study is the first record of its kind in Singapore. This species is considered Critically Endangered and Endangered in several other countries in Asia. Other Critically Endangered species include the climber *Gynochthodes ridleyi* (previously recorded only at Jurong, Kranji, and Western Catchment) and *Melicope lunu-ankenda*, all of which are located in the proposed CCKWW extension area. Given that the land used to be a plantation, species cultivated for domestic and commercial purposes can be found here such as Borneo mahogany (*Calophyllum inophyllum*) that is prized in the woodworking industry, para rubber (*Hevea brasiliensis*), cocoa (*Theobroma cacao*), and oil palm (*Elaeis guineensis*).

Native-dominated young secondary forest is the most common habitat type at the southwestern part of the CCKWW within project study area. Small patches of native-dominated old secondary forest are also found distributed within the young secondary forests. Scrubland is another fairly common habitat type, occurring around the perimeter of the CCKWW as well as along all the roads in the area. The abandoned kampung habitat type occurs mainly to the southeast of CCKWW, while abandoned plantation is found mainly in the south of study area. The importance of the vegetation within the site can be explained in a recent local study. This study reported that remnant, spontaneous native vegetation, more than managed forms of green spaces, are

key to support biodiversity, which corroborates evidence from elsewhere in the world that conservation of native vegetation is a key action for urban biodiversity conservation (Chong et al., 2014; Hahs et al., 2009). The large number of recorded conservation significant flora species at study area provides yet another example of such native vegetation providing refugia for plant species that remained since the clearance of old Western Catchment vegetation.

In terms of forest structure and succession stage, the native-dominated old secondary forests on this site, which amounts to approximately 0.44 ha, or 1.4% of the EIA study area, are highly complex and diverse. The structure of these forests are comparable to other late-successional secondary forests near nature reserves, while hosting a unique composition of plant species. Such mature secondary forests are pivotal for survival of forest-dependent native fauna species. To look at individual flora, fauna, or habitat does not fully encompass the ecological importance of the whole site; the site must be assessed holistically so as to not miss the forest for the trees.

In terms of fauna, this area consists of several threatened species across all taxonomic groups, including straw-headed bulbul (*Pycnonotus zeylanicus*) and Sunda pangolin (*Manis javanica*).

Many of these threatened species and depend on specific habitat requirements for their continued survival. Of particular significance is a number of stream specialists found within the project site. Running through these habitats are three forested freshwater streams that provide vital habitats for aquatic fauna, such as the locally Critically Endangered elf dragonfly (*Tetrathemis hyalina*). This species is very rare in Singapore, and only a few individuals have ever been recorded in Singapore. Other stream/swamp specialists found include the locally Vulnerable four-ridged toad and puff-faced water snake (*Homalopsis buccata*), and the locally Near-threatened Malesian frog (*Limnonectes malesianus*) and masked rough-sided frog (*Pulchrana laterimaculata*). The locally Endangered smooth-coated otter (*Lutrogale perspicillata*) was also recorded in Stream 3 located at the west of CCKWW. While the Johnson's freshwater crab (*Irmengardia johnsoni*) was previously recorded on the site, it was not detected during current surveys.

There was a presence of feral dogs (*Canis lupus*) and an abundance of wild pigs (*Sus scrofa*) found in the site. These animals may pose a human-wildlife conflict issue during the construction and operation phase of the project. The wild pigs, in particular, may require further management as they are known to be fed by humans in the project site.

## 4.5 Impact Assessment

The proposed development works includes three distinct work locations as explained in Chapter 2 (Section 2.4). As the specific work activities and corresponding impacts for each of these work locations are different, they will be studied separately in this section.

Area 1 (Existing CCKWW Plant) includes construction of new facilities and treatment units within the existing CCKWW plant. Currently, the area is already built up with turf as the main vegetation found within the area. Also, demolition of existing facilities are planned within this area.

Area 2 (CCKWW Extension Area) includes the green-field forested site south of the existing CCKWW plant. The area within the project footprint will be hoarded, completely cleared of vegetation, and levelled for the construction of proposed facilities and treatment units.

Area 3 (Sewerage Alignment) includes the construction of new sewerage pipeline leading from the existing CCKWW plant to Nanyang Drive. The pipeline will be installed using the pipejacking method, with three pipejacking shafts being located within existing forested area. Besides pipejacking shaft locations, as well as access routes, no other vegetation clearance or associated works will take place above ground level.

#### **4.5.1 Identification of Sources of Potential Impacts**

##### **Area 1 (Existing CCKWW Plant)**

The main activity that may impact sensitive biodiversity receptors is the generation of light and noise during the construction phase, as well as potential impacts due to surface runoff into nearby drains which empty into the forested area.

##### **Area 2 (CCKWW Extension Area)**

The pre-construction, construction, and operational phases of the CCKWW extension project may have impacts on the natural environment. The main activity that may impact the sensitive biodiversity receptors is vegetation clearance, which may result in species mortality (for both flora and fauna) and habitat loss (mainly for fauna). Earthworks to level the existing terrain will also result in species mortality and habitat loss. Sources of impacts during the different phases of the project are identified as the following:

###### Pre-Construction and Construction Phase

- Clearance of vegetation for site access, storage of equipment, setting up of temporary hoarding, and other forms of working space (temporary)
- Clearance of vegetation for soil investigation works (temporary)
- Clearance of vegetation and damage to habitats due to construction of plant (permanent)
- Earthworks (backfilling/excavation) for plant and associated infrastructure (permanent)
- Heavy construction activities such as piling (temporary)

###### Operational Phase

- Fragmentation of habitats due to position of plant (permanent)
- Increased light and noise due to operation of the plant (permanent)

##### **Area 3 (Sewerage Alignment)**

Impacts to the natural environment from the sewer installation in Area 3 will be mostly limited to the pre-construction and construction phases. As the method for installation of the pipeline will be pipejacking, as opposed to open-cut installation, impacts will be mostly limited to the areas where working shafts are placed. The main activity that may impact the sensitive biodiversity receptors is vegetation clearance, which may result in species mortality and disturbance. Noise generated from the pipejacking may also cause some levels of species disturbance. Sources of impacts during the different phases of the project are identified as the following:

###### Pre-Construction Phase and Construction Phase

- Vegetation clearance for placement of working shafts at manhole locations (permanent)
- Vegetation clearance for site access, and other forms of working space (temporary)
- Vibration and noise from pipe jacking (temporary)

#### **4.5.2 Identification of Potential Impacts**

This section summarises the potential impacts affecting biodiversity receptors that may take place during the pre-construction, construction, and post-construction phases.

##### ***Pre-construction / Construction Phase***

##### **Area 1 (Existing CCKWW Plant)**

As Area 1 is located within the existing CCKWW plant, and the works envisioned will be contained within plant boundaries, the impacts to the natural environment are likely to be of a small magnitude. The following section describes the likely impacts due to the works in Area 1.

##### **Species Disturbance**

For the purpose of impact assessment, focus is given to species that are either rare or globally/locally threatened according to the local and international databases. Given the location of the proposed works within a built-up area, species mortality is not deemed to be a significant potential impact. However, disturbance of fauna species in the vicinity surrounding the existing CCKWW plant is likely, due to increased vibration, noise, and light generated by the proposed works.

##### **Degradation of Habitats**

The main habitats of concern are the stream habitats directly west and south of the existing CCKWW. Impacts to these stream habitats may be observed should uncontrolled runoff of chemicals and sediment generated from the construction activities is allowed to enter the drainage network that flows into the forest streams.

##### **Area 2 (CCKWW Extension Area)**

Area 2 involves the clear felling of a patch of forested area, which will result in direct impacts to the species and habitats present within the area. The following section details these impacts.

##### **Species Disturbance and Mortality**

For the purpose of impact assessment, focus is given to species that are either rare or globally/locally threatened according to the local and international databases. Section 4.4 discussed the composition of flora and fauna on the project site. Given location of the project site within a core biodiversity area, the project site contains a number of species of conservation value, both flora and fauna.

Fauna species, particularly reptiles and amphibians, may become entrapped when navigating through the worksite, especially by getting stuck in ECM blankets or falling into pits.

The impact of species mortality will be direct, and most prominent within the project footprint, where 3.2ha of vegetated area will be cleared for the placement of the CCKWW extension facility. Approximately 500 surveyed trees are likely to be affected by the extension area, with 18 flora species of conservation value found within the extension area (Table 4.16). More trees are expected to be affected due to ad-hoc site clearance to accommodate various construction needs including but not limited to construction footprints, equipment storage, staging grounds, soil-levelling works, other ad-hoc site issues, etc.

**Table 4.16.** List of threatened flora species in the Area 2

| No | Species                           | Conservation Status   |
|----|-----------------------------------|-----------------------|
| 1  | <i>Helminthostachys zeylanica</i> | Newly Discovered      |
| 2  | <i>Baccaurea motleyana</i>        | Critically Endangered |
| 3  | <i>Cissus repens</i>              | Critically Endangered |
| 4  | <i>Dioscorea cf polyclados</i>    | Critically Endangered |
| 5  | <i>Eurycoma longifolia</i>        | Critically Endangered |
| 6  | <i>Melicope lunu-ankenda</i>      | Critically Endangered |
| 7  | <i>Aphanamixis polystachya</i>    | Endangered            |
| 8  | <i>Callicarpa longifolia</i>      | Endangered            |
| 9  | <i>Jasminum elongatum</i>         | Endangered            |
| 10 | <i>Pellacalyx axillaris</i>       | Endangered            |
| 11 | <i>Cyrtococcum cf patens</i>      | Vulnerable            |
| 12 | <i>Dracaena cantleyi</i>          | Vulnerable            |
| 13 | <i>Ficus aurata</i>               | Vulnerable            |
| 14 | <i>Goniophlebium percussum</i>    | Vulnerable            |
| 15 | <i>Lygodium circinnatum</i>       | Vulnerable            |
| 16 | <i>Porterandia anisophylla</i>    | Vulnerable            |
| 17 | <i>Pternandra coerulescens</i>    | Vulnerable            |
| 18 | <i>Symplocos fasciculata</i>      | Vulnerable            |

Despite the project footprint occurring largely within mixed regenerated forest and scrubland, and that the native-dominated forest west of the extension area will remain relatively untouched, mortality of plants of conservation value is still of concern. While several species of Conservation Significance are likely of cultivated stock and thus have low conservation value, such as *Baccaurea motleyana* and *Nephelium lappaceum*, there are still multiple species of interest, such as *Helminostachys zeylanica*, which is a newly discovered fern in the country, as well as a Critically Endangered fern in other countries in Asia.

The project site contains several fauna species of conservation value (see Table 4.17). Significance should be placed on non-volant animals of conservation value.

**Table 4.17.** List of threatened fauna species in the Area 2

| No. | Species name                 | Common Name                 | Local Status    | Global Status |
|-----|------------------------------|-----------------------------|-----------------|---------------|
| 1   | <i>Psittacula longicauda</i> | Long-tailed parakeet        |                 | VU            |
| 2   | <i>Loriculus galgulus</i>    | Blue-crowned hanging parrot | EN              | LC            |
| 3   | <i>Pycnonotus zeylanicus</i> | Straw-headed bulbul         | EN              | CR            |
| 4   | <i>Gallus gallus</i>         | Red junglefowl              | EN              | LC            |
| 5   | <i>Nisaetus cirrhatus</i>    | Changeable hawk-eagle       | EN              | LC            |
| 6   | <i>Copsychus saularis</i>    | Oriental magpie-robin       | EN              | LC            |
| 7   | <i>Homalopsis buccata</i>    | Puff-faced water snake      | VU              | LC            |
| 8   | <i>Manis javanica</i>        | Sunda pangolin              | CR              | CR            |
| 9   | <i>Copera vittata</i>        | Variable featherlegs        | VU              | LC            |
| 10  | <i>Catopyrops ancya</i>      | Ancyra blue                 | Moderately Rare |               |

| No. | Species name              | Common Name       | Local Status    | Global Status |
|-----|---------------------------|-------------------|-----------------|---------------|
| 11  | <i>Tagiades calligana</i> | Malayan snow flat | Moderately Rare |               |

This project site is home to the critically endangered Sunda pangolin. Sunda pangolins are known to use hollows in trees to sleep and are thus not often detected during the day. Injury and death of such species may take place in the event that individuals are trapped within working boundaries when vegetation clearance commences. Several other mammal species including the leopard cat, Malayan porcupine, and the greater mousedeer, while not observed during this baseline survey, are also known from the general Western Catchment area, and might use the study area.

The presence of several rare or threatened bird species, such as the endangered straw-headed bulbul, barred eagle owl, and changeable hawk eagle, can be found on this site. These species are particularly vulnerable while they are nesting and will be impacted if schedules of vegetation clearance do not consider the possibility of nesting birds.

Several fauna species, including the elf dragonfly, masked rough-sided frog, and puff-faced water snake, were found on the study area. Most of these species of conservation value were found within the section of Stream 3 which will not be directly impacted by the works in Area 2. (see Figure 4.25, Figure 4.22, and Figure 4.19 respectively). Indirect impacts to these species may occur should their habitats be compromised. Stream 2 will be filled for the proposed development, resulting in the loss of aquatic species. However, Stream 2 faces disturbance from drainage from the existing CCKWW plant, and most species found in Stream 2 are common species found in many other sites around Singapore. There was one sighting of a puff-faced water snake, sightings of variable featherlegs, and a Malesian frog in the vicinity, indicating that despite the ongoing disturbance on this stream, it is still a potential habitat for rare or forest-dependent species.

As there is a possibility that some night work activities may happen within Area 2 as described in section 2.7, this may create species disturbance from light. Alteration of natural cycles of light and dark by artificial light sources can negatively impact the ecosystem if not managed properly. As some of fauna species, particularly nocturnal species which rely on their hearing for movement, communication, and foraging, are vulnerable to increased night-time noise levels.

#### Degradation/loss of Habitats and Connectivity between Habitats

Vegetation clearance leads to the loss of habitats for fauna species and reduction of foraging habitat, food sources, roosting, breeding and nesting sites, and other resources needed for the continued survival of a species. As the Area 2 consists mostly of abandoned rubber plantation and scrubland habitat types, the loss of such habitats is of relatively low consequence. Such forest habitat is common throughout other forested patches in Singapore. However, this area also contains many species of Conservation Significance, and as such should not be cleared without taking these species into consideration. The lack of records of *Helminthostachys zeylanica* outside of this Area 2 throughout the duration of the survey is also of concern, as the loss of this habitat may prevent the further propagation of this species.

As Area 2 is part of a contiguous patch of forest, and the area to be cleared (3.2 ha out of the whole Western catchment area that covers 8,384 ha) is relatively small, it is expected that most terrestrial fauna species found on this site will be able to disperse to surrounding habitats not

affected by the works. However, at least two streams are found within Area 2. Stream 2, which runs along Dunbar Walk and is part of the project footprint, will be filled for the proposed development. This will lead to the loss of stream habitat within the project area, and a potential deterioration or loss of the same stream south (downstream) of the Area 2.

Stream 3 has been identified as a habitat of high priority for conservation, given the presence of several rare and threatened stream specialists in the area. Potential impacts to the stream habitat include loss of habitat due to backfilling, siltation and changes of stream levels due to earthworks and vegetation clearance, and increased surface runoff into the stream from the development of sealed surfaces, which could lead to pollution or changes in water quality.

Loss of connectivity between forested areas east and west of the proposed development may lead to reduced movement between populations on either side of the plant. Such fragmentation of populations can in turn lead to reduced gene flow and local extinctions of species. However, as the area south of the proposed development will still be forested, impacts from loss of connectivity are likely to be low.

#### Edge Effects

Vegetation clearance may also have indirect impacts to ecology, including the creation of edge effects in the surrounding habitats, where forest edges are exposed to abiotic and biotic changes due to vegetation clearance. These changes include increased light intensity and temperature, increased soil nutrient content, and changes in air and soil moisture levels. The edge effect may lead to changes in microclimates, forest structure, and ecological interactions. Gradual deterioration of habitats and changes in flora and fauna communities in the areas adjacent to the proposed development due to edge effects may also occur.

This may be cause for concern as Area 2 has native-dominated young secondary forest habitats to the west of the area, a habitat type that has been steadily decreasing throughout Singapore, as well as multiple patches of native-dominated old secondary forest interspersed throughout this area. The aforementioned edge effects could result in the introduction of invasive species as well as a reduced survivability of the native trees along the new edges of the forest.

#### Changes in Soil and Topography

Vegetation plays an important role in soil stability. Where vegetation clearance has taken place, soil will be left vulnerable to erosion, particularly during rainy periods. Erosion may lead to siltation of waterbodies, and also the runoff of nutrients in topsoil, leading to lowered nutrient levels of the remaining soil on the site.

Where there are planned earthworks within the project footprint, impacts to habitats, including streams and other localized habitats, and species are likely to be felt. Impacts on the biodiversity may go beyond the actual footprint of working boundaries due to indirect impacts caused by associated works such as backfilling, compaction, and soil excavation.

Concretization of the Area 2 could also result in increased surface runoff and therefore reduced groundwater recharge, as well as increased soil erosion. The surface runoff may also contribute to increased nutrient, chemical levels or change in pH levels in the surrounding soils due to other substances on the concrete the surface runoff may collect.

### Invasive Species

Invasive species have many impacts on biodiversity, including the displacement of native species (Peh, 2010), hybridization of distinct populations (Vuillaume et al., 2015; Peh, 2010), and degradation of ecosystem services (Çinar, 2014).

In the case of construction activities, invasive plants are sometimes introduced through seeds embedded in construction equipment and the boots of construction personnel. Non-native and/or invasive species may also be introduced outside of the future extension area if these species are planted in the future, such as rain trees (*Samanea saman*), trumpet tree (*Tabebuia rosea*), common lantana (*Lantana camara*), etc.

There are several non-native species currently found on project site, including the Mukhlesur's narrow-mouthed frog and Gunther's frog. The clearance of forest may allow such generalist non-native species to proliferate in the newly cleared environment.

### Human–Wildlife Conflict

There is a presence of numerous wild pigs and feral dogs within the project site. Given that most of the forested area will remain undeveloped and inaccessible to the general public, human wildlife issues are most probable during the construction phase. Other animals in the area include the long-tailed macaque, smooth-coated otter, king cobra, and equatorial spitting cobra, all of which may result in human-wildlife conflict. If measures are not taken to ensure that fauna species are not able to enter the working area, injury to construction personnel or these fauna species is possible during the construction phase.

### Potential Roadkill

While Area 2 is not near any major road network, it is adjacent to Dunbar Walk, a small road with low traffic, utilised by employees and visitors of the existing CCKWW plant, as well as shuttlebuses from NTU. During the construction phase, traffic volume is also likely to increase from construction vehicles accessing the site. If mitigation measures are not taken, animals are likely to run out of the forest, particularly during periods of high disturbance such as tree felling and heavy construction (e.g. piling). This may lead to an increased incidence of roadkill, which poses negative impacts on both the wildlife present on the site as well as drivers along the road.

### Area 3 (Sewerage Alignment)

The proposed sewer installation for Area 3 will be done through pipejacking method. As such, the extent of the works will be mostly limited to vegetation clearance for the placement of three shafts which will be within forested area. The remaining two shafts will not be within forested area. The general impacts in Area 3 are likely to be similar to Area 2, although the scale of the works in Area 3 is much smaller, and as such, the magnitude of these impacts is likely to be lower.

### Species Disturbance and Mortality

Given location of the project site within a core biodiversity area, the project site contains a number of species of conservation value, both flora and fauna. The impact of species mortality and disturbance will be direct and occur where vegetation needs to be cleared within working boundaries. However, the pipejacking operation does not require vegetation clearance along the proposed sewerage alignment, at areas between proposed shaft working areas (refer to Figure 2.8). Depending on the depth of the sewerage line, the operation is also unlikely to affect tree roots along the proposed pipeline at areas between manholes as local tree structural roots

in general do not extend too deep underground and are relatively close to the soil surface. Most of the construction impact are expected to come from the excavation at shaft locations and associated land clearance & trenching works for machinery access and other construction needs.

The presence of several rare or threatened bird species can be found on this site. Two barred eagle-owls were observed within Area 3. These species are particularly vulnerable while they are nesting and will be impacted if schedules of vegetation clearance do not consider the possibility of nesting birds.

Fauna species, particularly reptiles and amphibians, may become entrapped when navigating through the worksite, especially by getting stuck in ECM blankets or falling into pits.

#### *Degradation/loss of Habitats and Connectivity between Habitats*

The sewerage alignment is located mostly within scrubland, with parts of it overlapping with abandoned kampong forest vegetation. There is also a stream running across the sewer alignment. Direct impacts to these habitats due to vegetation clearance is likely to be minimal. However, any uncontrolled runoff of chemicals and sediment generated from the construction activities may impact the stream habitat. However, given the small scope of works, impacts to habitats are expected to be temporary and minor. Due to the nature of pipejacking works, the installation of the sewerage pipeline is also unlikely to have long term impacts to the ecological connectivity of the site.

#### *Edge Effects*

Given that vegetation clearance for the sewer works is likely to be minimal and confined to the working area of the shaft locations, and vegetation will be reinstated with native trees in working areas, impacts related to edge effects are unlikely to be major.

#### *Changes in Soil and Topography*

Vegetation plays an important role in soil stability. Where vegetation clearance has taken place, soil will be left vulnerable to erosion, particularly during rainy periods. Erosion may lead to siltation of waterbodies, and also the runoff of nutrients in topsoil, leading to lowered nutrient levels of the remaining soil on the site. For Area 3, these impacts might be observed where the working shafts are placed. Erosion or siltation, if uncontrolled, may also indirectly impact areas beyond the project footprint.

Soil compaction during the construction of the manholes and boreholes may result in reduced soil health, such as reduced nutrient availability, porosity and hydraulic conductivity, which may result in negative impacts to plant growth and mortality around the manholes and boreholes. Concretization around these areas would increase surface runoff and reduce groundwater recharge, which may result in increased soil erosion while facilitating the transport of any chemicals and nutrients from the concretized surface to the surrounding soil.

#### *Invasive Species*

Invasive species have many impacts on biodiversity, including the displacement of native species (Peh, 2010), hybridization of distinct populations (Vuillaume et al., 2015; Peh, 2010), and degradation of ecosystem services (Çinar, 2014).

In the case of construction activities, invasive plants are sometimes introduced through seeds embedded in construction equipment and the boots of construction personnel.

### Human–Wildlife Conflict

Area 3 had a large number of wild pigs. Other animals in the area include the long-tailed macaque, king cobra, and equatorial spitting cobra, all of which may result in human-wildlife conflict. If measures are not taken to ensure that fauna species are not able to enter the working area, injury to construction personnel or these fauna species is possible during the construction phase. During the operation period, given that the area will still largely be forested, human-wildlife conflict may be possible when maintenance personnel enter the area for their maintenance regimes.

### Potential Roadkill

The planned sewerage alignment is located along Peng Kang Avenue connecting Dunbar Walk to Nanyang Drive. Traffic along this road is infrequent and comprises mostly of employees and visitors of the existing CCKWW plant, as well as shuttlebuses from NTU. Traffic volume is likely to increase during the construction phase due to an increase in the number of construction vehicles accessing the site.

While sewerage works are of a relatively small scale, there is a possibility of animals running onto the road, particularly during periods of high disturbance such as site clearance. This may lead to an increased incidence of roadkill, which poses negative impacts on both the wildlife present on the site as well as drivers along the road.

### **Operation Phase**

Impacts from the operation phase of the CCKWW plant include those associated with edge effect, especially the increase of light and noise impacts into the forest, as well as the high volume of stormwater run-off from new facilities at Area 2. No impact are anticipated for the sewerage line operation.

### **Overall Impact**

Based on the assessment above, some of the potential impacts on site's biodiversity are deemed to be permanent in nature while the others are short-term and reversible. Appropriate measures are to be proposed to mitigate these impacts to acceptable level.

Additionally, critical impacts of other physical parameters are discussed in the relevant chapters. Impacts of hydrology and surface water quality on biodiversity are addressed in Chapter 5, impacts of noise are discussed in Chapter 5, impacts of air quality on biodiversity are discussed in Chapter 7, impacts of light on biodiversity are discussed in Chapter 9, and impacts of waste management are discussed in Chapter 10.

### **Cumulative Impacts**

There is a possibility that proposed LTA JRL development (refer to Section 2.10) located adjacent to Area 3, happen concurrently with this project. While the scale of works for the current project in Area 3 is small, its working area is in close vicinity to this JRL development.

Impacts from these JRL development work include vegetation and connectivity loss, loss and mortality of fauna, soil disturbance, changes to hydro-ecology, changes to ecological processes, and general habitat degradation due to improper waste disposal (T. Y. Lin, 2018). Impacts from the sewerage alignment work, while of a much smaller scale to the JRL works, also include limited vegetation loss and soil disturbance. As the JRL development work is of a much larger

scale compared to this project's sewer alignment works, should the project timelines coincide, cumulative impacts, mainly from vegetation clearance of both sites, are expected.

In order to address the cumulative impacts of both developments, all impacts from Area 3 were given a score of 3 in the 'Cumulativity' column of RIAM table to calculate the ES and assess overall impact.

#### **4.5.3 Evaluation of Potential Impacts**

Table 4.18 and Table 4.19 summarises the impacts for all Areas with their corresponding Environmental Scores before and after the implementation of mitigation measures which are elaborated in the following section.

### **4.6 Mitigation Measures**

Mitigation measures are to be implemented wherever significant negative impacts are predicted, in order to reduce the impacts of the works on the environment. Most of biodiversity mitigation measures are covered in this chapter. Additionally, mitigation measures related to other environmental aspects are covered in relevant chapters as well.

#### **4.6.1 Pre-construction / Construction Phase**

##### **Area 1 (Existing CCKWW Plant)**

Although works in Area 1 are expected to be small scale, some impacts to the natural environment, including disturbance to fauna species due to construction activities, and potential impacts to the surrounding habitats due to runoff, have been envisioned.

The main impacts to species include increased noise, light, and vibration disturbance. Noise barriers / sound proofing should surround all working areas to decrease the impact of noise to the surrounding fauna. Demolition activities should be avoided during the night (i.e., 6 pm to 8 am). Heavy construction activities (e.g., piling) should also be minimised in the night (i.e., 6 pm to 8 am), except for work exigencies, and planned activities, with the implementation of the lighting management plan (LMP) approved by NParks i.e. with construction lights (if any) face inwards and away from forested areas. All demobilisation has to be completed by 6 pm. To reduce vibration impacts to fauna located in proximity to construction areas, machineries causing vibration may be placed on isolators. The details of these impacts are further elaborated upon in their respective chapters.

Indirect impact to the important habitats should also be avoided. Care must be taken to prevent soil erosion or sediment run-off into adjacent habitats, particularly waterbodies. As such, proper Earth Control Measures (ECM) Plan approved by PUB should be in place prior to start of (pre-) construction activities and effectively implemented throughout the duration of activities. The ECM tanks utilised for the project will have additional 25% surplus capacities to avoid overflow and silty discharge. As far as possible, the project should minimise the chemical storage within work areas. All hazardous / flammable chemicals to be used during construction phase have to be properly labelled, stored within bund containment and under shelter.

##### **Area 2 (CCKWW Extension Area)**

Although the habitat of Area 2 is mostly classified as Abandoned Plantation, the area consists

of multiple flora species of Conservation Significance, especially at western part of the area.

Given the nature of the project, the clearance of threatened plant species within this Area 2 is unavoidable, and several Conservation Significant flora species will be affected. The whole extension area (3.2 ha) will be cleared, with potentially additional areas in range of 20 m may be cleared temporarily to cater for construction phase working area needs.

The layout of the proposed CCKWW extension area (Section 2.5.2) doesn't allow for the retention of trees within the planned project footprint, though retention of large trees in additional construction phase working area should be possible.

#### Plant Salvaging and Planting of Native Species

To remedy the adverse impact to biodiversity, it is recommended salvaging saplings and small individuals of plants of conservation value. We recommend the transplanting of 15 Conservation Significant species found within CCKWW extension area: *Callicarpa longifolia* (climber), *Dioscorea polyclados* (climber), *Dracaena cantleyi* (shrub), *Eurycoma longifolia* (small tree), *Ficus aurata* (shrub), *Goniophlebium percussum* (fern), *Helminthostachys zeylanica* (fern), *Jasminum elongatum* (climber), *Lygodium circinnatum* (fern), *Melicope lunu-ankenda* (small trees), *Pellacalyx axillaris* (saplings), *Porterandia anisophylla* (seedlings to saplings), and *Symplocos fasciculata* (sapling). Additionally, some more common native plants may also be targeted for translocation, including *Syzygium polyanthum* (tree), *Syzygium lineatum* (tree), *Terminalia catappa* (tree), *Aporosa frutescens* (tree), *Cinnamomum iners* (tree), *Elaeocarpus mastersii* (tree), *Ficus fistulosa* (tree), *Litsea elliptica* (tree), *Macaranga bancana* (tree), *Macaranga conifera* (tree), *Morella esculenta* (tree). For this project, there should be discussions with NParks on their interest in salvaging and collecting fertile fruits and seeds of some of the abovementioned species prior to the commencement of construction.

Salvaged native species can be incorporated as far as reasonably practical as a nature-based component for replanting requirements such as for reinstatement, slope stabilizing, or fauna connectivity purposes. Compared to conventional method of using cowgrass (*Axonopus compressus*) to turf back the site after temporary site-clearance, using salvaged native-plants for reinstatement works brings three layers of ecological and biodiversity benefits. Firstly, parts of the project area include abandoned kampong habitat with low native plant diversity, therefore introducing the native trees salvaged from extension area into the abandoned kampong will help enhance the forest vegetation structure by facilitating forest succession processes from abandoned kampong to native-dominated forests, potentially leading to net positive environmental impact for the abandoned kampong. Secondly, the salvaged plants will be already of certain height, which will help rapidly close the forest canopy gap created by sewer line works or ad-hoc site clearance and prevent invasion of non-native plants into more sensitive habitat of native-dominated secondary forests. Thirdly, using salvaged native trees to strengthen forest edges and reinstate site after completion of construction works is an effort to preserve and disperse native flora germplasm rather than introducing foreign species which might deter conservation of native flora germplasm. Overall, these measures will help to mitigate biodiversity loss, conserve certain degree of biodiversity, and even enhance forest habitat by utilising salvaged native trees for to strengthen forest gaps created due to establishment of temporary working area boundary.

#### Flora Management Plan

The detailed plant salvaging and tree protection guidelines are provided in Section 13.4 as part

of overall EMMP framework, including the requirements of personnel qualification, specifications of nursery, transplanting, tree assessment, tree felling and so on. A summary of flora management plan is provided below.

Prior to any site clearance and plant salvaging processes, the Contractor is to liaise with NParks on plants they plan to salvage. Additionally, it is recommended to engage a nursery to maintain an adequate number of saplings. Once the nursery is confirmed, the native species to be salvaged are to be tagged and recorded. After the tagging and recording of native flora on site have been completed, the whole process of individual saplings being salvaged from site, transported, and planted at destination site should be performed within a day.

A tree assessment report recording tree information such as site and tree photos, species, height, girth, crown spread, tree health, form, structure, site observations and assessment for the trees affected by proposed development footprint is to be prepared and it should also include mitigation measures to reduce construction impact on trees. This tree assessment report will then serve as record of pre-development tree condition, and to be utilised performing monthly monitoring for trees.

Trees identified for retention at the boundaries of the construction footprint of the working area should be demarcated by Tree Protection Zones (TPZs) determined by Arborist (see Figure 13.9). TPZ should be installed prior to construction zone, and monthly monitoring should be conducted by the Arborist to maintain records of tree health and TPZ integrity. If the tree needs to be pruned due to machinery access, the pruning will need to be carried out by landscape contractors with relevant qualifications under supervision of the Arborist.

#### Restricted Work Timings

Due to the sensitive nature of forest animals, particularly nocturnal animals, construction works within Area 2 are to be limited to daylight hours (8 am – 6 pm), including demobilisation, except in cases of work exigency and planned night activities, as explained in Section 2.7.2, with the implementation of the lighting management plan (LMP) approved by NParks. Following specific measures are recommended to manage these work exigency scenarios.

- Complete all pile drilling before 6pm and only casting of piles to be carried out past 6pm.
- Complete all rebar fixing works before 6pm and only necessary concreting work for slab/ wall to be carried out past 6pm.
- Installation of 6m high noise barrier along project boundary.
- Additional localized noise barriers surrounding the works where possible.
- Only maintain minimum number/ intensity of lights needed for safety and security.

Also, should there be any change in type of construction activities to continue at night, authorities' approval should be obtained.

Additionally, a detailed LMP shall be developed as part of the Construction EMMP which is further elaborated in Section 9.3.1 of the chapter on light impacts.

#### Pre-felling Fauna Inspections

Mortality of fauna species of conservation value should be avoided. In general, most of the fauna of conservation value recorded include birds and mammals that are highly motile, hence the works should not greatly affect their survivability. However, several measures can be put in place to further decrease species mortality. Prior to any tree felling, the project site should be

inspected for active bird nests or holes, and chicks should be allowed to fledge prior to tree felling. Tree hollows should also be inspected for species of conservation value such as the Sunda Pangolin, and such species should be translocated prior to tree felling.

#### *Directional Clearance and Wildlife Translocation*

Trees should be felled in stages, in one direction. Such directional clearance will give time for less motile fauna to move into other forested areas. Tree felling works are also to avoid the peak nesting period of many birds (between the months of February and July).

The northern section of Stream 2, which is running along Dunbar Walk, will be filled as part of the development footprint. The eastern branch of Stream 3 will also be backfilled. Where Stream 2 is backfilled, an alternative route for the flow of water should be established. Where feasible, this diversion should consist of naturalised banks so as to act as a replacement habitat for the backfilled stream. Additionally, due to the presence of native and nationally threatened species within the stream (including the puff-faced water snake), a dedicated period for targeted fauna translocation is recommended for areas to be backfilled. Targeted efforts should include the placement of overnight traps to trap native species, including but not limited to fish, reptiles, and decapod crustaceans. Species caught should be translocated to the other unaffected streams in the vicinity. Targeted relocation should take place before any works have commenced in the vicinity.

#### *Site Hoarding*

Mortality from roadkills of highly motile fauna species such as Sunda pangolins and monitor lizards should be avoided. Hoarding along the perimeter of Dunbar Walk, as well as phased clearance of trees in the direction away from Dunbar Walk, could help to prevent animals from running onto the road. The full clearance area should be hoarded, with hoardings installed at a depth of 300 mm to prevent wild boars from digging under the hoarding. The hoarding line should be located a distance away from the forest edge to prevent the inadvertent damage of major roots of trees during installation, which may lead to the trees falling onto the site. Hoarding should be monitored regularly to ensure that no animals are entering the site either through gaps or by digging under the hoarding. Protocol in the case of animal encounter is elaborated in Section 13.3.6.

#### *Minimising Working Area Footprint*

Direct impacts on important habitats, in this case Stream 3 west of the development site and native-dominated old secondary forest to the southwest, should be avoided. Direct impacts to other habitats should be minimised where possible by adjusting working area footprint. The design of access routes, storage areas, and other working areas should also aim to minimise the amount of land area used. All working areas should be hoarded up so as to prevent any inadvertent damage to these habitats.

Due to the importance of Stream 3, as determined by the EIA, the project working area footprint has been altered to minimise any direct impacts on the stream (see Section 5.8.1 and Figure 5.8). Stream 3 should be regularly monitored during the construction phase, and for three months after the construction phase, to ascertain that the stream is not adversely impacted by the works. Details of regular monitoring recommendation can be found in Chapter 13.

Impacts to the important habitats, including the stream west of the project site, and the native-dominated old secondary forest southwest of the project site, should also be avoided. Care must

be taken to prevent soil erosion or sediment run-off into adjacent habitats, particularly waterbodies. As such, proper Earth Control Measures (ECM) Plan approved by PUB should be in place prior to start of (pre-)construction activities and effectively implemented throughout the duration of activities. The ECM tanks utilised for the project will have additional 25% surplus capacities to avoid overflow and silty discharge. As far as possible, all hazardous / flammable chemicals to be used during construction shall be properly labelled, stored within bund containment, and under shelter.

#### Management of Vehicular Traffic Speed

Although Dunbar Walk is not near any major road networks, the volume of traffic is likely to increase during the construction phase. Potential roadkills can be minimised through the implementation of speed bumps, and the erection of road signs to inform drivers of potential animal crossings. Alternatively, speed limits may be implemented and enforced to ensure that vehicles slow down while driving near forested areas.

#### Management of Human–Wildlife Encounters

The presence of many wild pigs and feral dogs may pose a threat to human health and safety. Dependence on humans for food could lead to aggressive behaviour in wild animals. Wild pigs and feral dogs encountered on site should be removed. To prevent wild animals from scavenging on construction site, areas for food consumption, storage, and waste disposal should be demarcated and managed. The personnel should also be briefed on what to do should they encounter specific wild animals. Wildlife Response and Rescue Protocol is elaborated in Section 13.3.6.

#### Management of Invasive Species

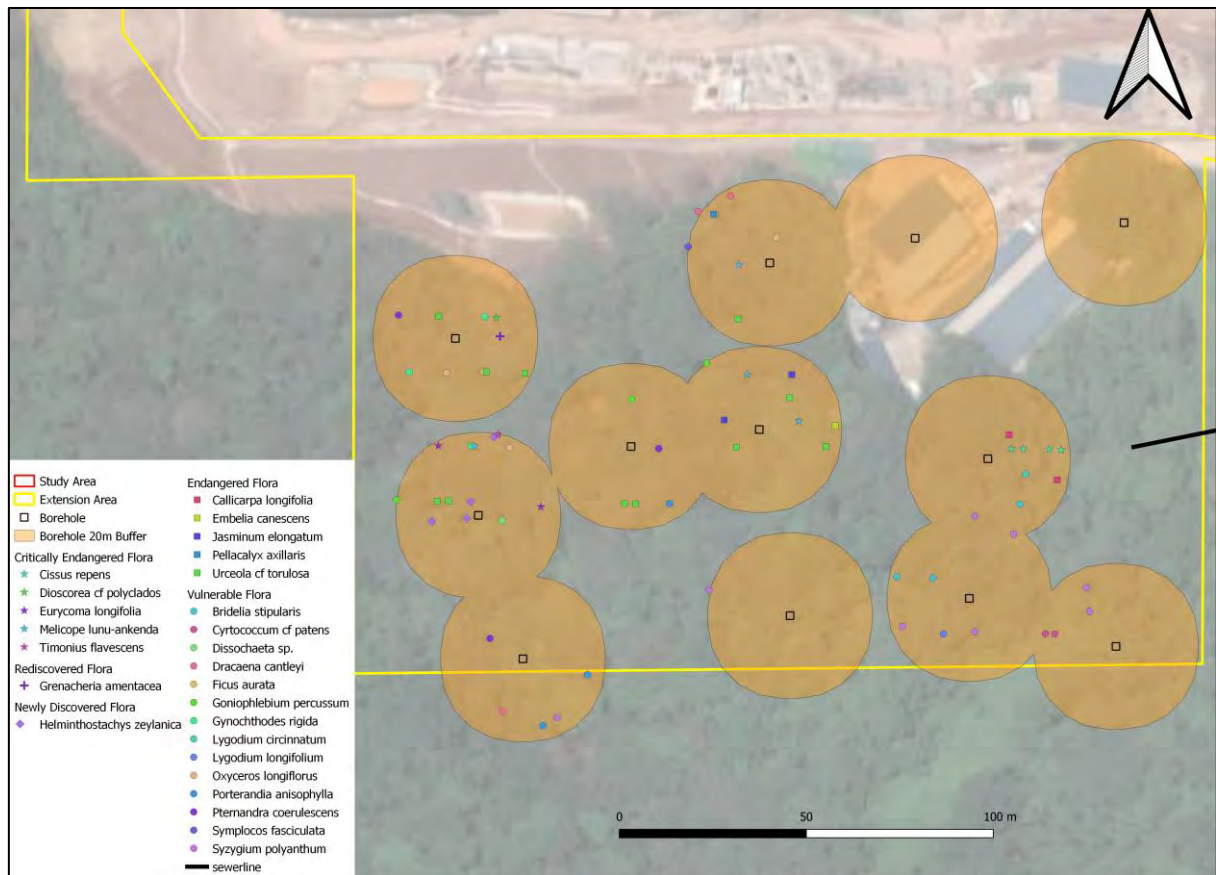
In order to minimise the chance of invasive plant species establishing within the project site and dispersing to the nearby forested areas, periodic checks by the flora specialist should be made within the hoarding boundaries of the construction site. Any introduced plant species found to have spontaneously established within the hoarding area should be removed and disposed of properly.

#### Minimising Fauna Entrapment

To prevent fauna mortality due to entrapment within the worksite, such as in ECM blankets or pits, daily site checks should be conducted to identify and rescue entrapped fauna. Moreover, only non-plastic, biodegradable ECM mats are to be installed throughout the site. These mats should minimise entrapment of wild animals like snakes and lizards.

#### Mitigation for Soil Investigation Works

Soil investigation works are planned for Area 2 (Figure 4.32). There is the possibility that these works will take place substantially earlier than the clearance of the remaining forest within Area 3. If this is so, mitigation measures should be implemented to mitigate the initial impacts of the project stemming from the soil investigation works.



**Figure 4.32.** Conservation significant flora affected by planned borehole works

Borehole locations should be shifted to avoid waterbodies or tree roots. All working areas – borehole locations, access routes, and other working areas – should be pegged out prior to soil investigation works. This will allow for species of conservation value within the pegged areas to be salvaged prior to the works.

Additionally, no equipment should be stored within vegetated areas, and workers should be briefed on wildlife encounter protocol. There should be no eating of food within forested areas.

In summary, the mitigation measures that should be implemented to mitigate the impacts on sensitive biodiversity receptors for Area 2 are as follow:

- Avoid nightworks (i.e., limit work activities to 8 am – 6 pm) except in for work exigency and planned night activities, with the implementation of the lighting management plan (LMP) approved by NParks
- Ensure project layout avoids targeted sensitive receptors where possible
- Identify plants of high conservation significance to be transplanted
- Establish Tree Protection Zones (TPZ) for trees to be retained
- Hoard up entire work site.
- Visually inspect trees and holes for nesting birds and species of conservation value prior to felling
- Replant new planting verges with native plant species
- Reinstate habitats where possible upon completion of works
- Remove introduced plant species that have spontaneously established within hoarding boundaries.

- Conduct targeted relocation exercises for native aquatic fauna species.
- Conduct phased /directional clearance of vegetation in the direction away from Dunbar Walk.
- Conduct daily checks of ECM blankets and pits to rescue any entrapped fauna.
- Implement proper ECMs to prevent impact to adjacent habitats
- Utilise inert construction material
- Ensure equipment, vehicles, and footwear are clean prior to commencing works
- Establish designated areas for food and waste disposal
- Demarcate and manage areas for food consumption, storage, and waste disposal and install monkey-proof bins.
- Erect speed bumps or signages informing drivers of potential animal crossings along Dunbar Walk
- Implement and enforce speed limits
- During the construction phase, conduct information sessions for workers on what to do upon encountering wildlife
- Conduct regular monitoring of trees and vegetation health during construction phase within project site

### **Area 3 (Sewerage Alignment)**

#### **Avoiding Impacts on Sensitive Habitats**

Direct impacts on important habitats, should be avoided. Direct impacts to other habitats should be minimised where possible. The proposed sewer alignment crosses one stream, which is of high priority for conservation. Placement of working shafts, trenchwork, and all other working spaces should avoid the stream. The other habitats the alignment crosses include abandoned kampong and scrubland vegetation. While these habitats are of lower priority for conservation, they serve as important refuge area for fauna species. Hence, the design of access routes, storage areas, and other working areas should be minimised where possible. Should any threatened native, mature tree species be affected by the placement of working areas or working shafts, a slight shift of placement could help to prevent impacts on these species. The full working area should be hoarded, with hoardings installed at a depth of 300 mm to prevent wild boars from digging under the hoarding. The hoarding line should be located a distance away from the forest edge to prevent the inadvertent damage of major roots of trees during installation, which may lead to the trees falling onto the site. Hoarding should be monitored regularly to ensure that no animals are entering the site either through gaps or by digging under the hoarding. Protocol in the case of animal encounter is elaborated in Section 13.3.6.

Indirect impacts to the important habitats, including the stream west of the project site, and the native-dominated old secondary forest southwest of the project site, should also be avoided. Care must be taken to prevent soil erosion or sediment run-off into adjacent habitats, particularly waterbodies. As such, proper Earth Control Measures (ECM) Plan approved by PUB should be in place prior to start of (pre-)construction activities and effectively implemented throughout the duration of activities. The ECM tanks utilised for the project must have surplus capacities to avoid overflow and silty discharge. As far as possible, all hazardous / flammable chemicals to be used during construction shall be properly labelled, stored within bund containment, and under shelter.

#### **Restricted Work Timings**

Due to the sensitive nature of forest animals, particularly nocturnal animals, construction works

for the boreholes and three working shafts located within the forested area should be limited to daylight hours (8 am – 6 pm), including demobilisation for the day, except in cases of work exigency (e.g., structural integrity). Should there be a need for light construction activities to continue at night, authorities' approval should be obtained.

#### Flora Management Plan

Any slope levelling work that may be required should also take into account any flora immediately upslope of the planned work area, especially trees, as their root zones would likely be affected by any excavation works that are carried out. Any waste material should also be disposed of properly, as the leaching of contaminants into the soil or wastewater could result in negative impacts south of the pipeline, which is downslope and has a higher biodiversity compared to the area north of the road.

Trees identified for retention at the boundaries of the construction footprint of the working area should be demarcated by Tree Protection Zones (TPZs) determined by Arborist (see Flora Management Plan for diagram example). TPZ should be installed prior to construction zone, and monthly monitoring should be conducted by the Arborist to maintain records of tree health and TPZ integrity. If the tree needs to be pruned due to machinery access, the pruning will need to be carried out by landscape contractors with relevant qualifications under supervision of the Arborist.

Replanting should utilise native plant species to increase the overall diversity of the area, as well as prevent the possibility of invasive species from invading the adjacent forested areas. Replanting can be done using salvaged saplings and plants from Area 2.

#### Pre-felling Fauna Inspections

Mortality of fauna species of conservation value should be avoided. In general, most of the fauna of conservation value recorded include birds and mammals that are highly motile, hence the works should not greatly affect their survivability. However, several measures can be put in place to further decrease species mortality. Prior to any tree felling, the project site should be inspected for active bird nests or holes, and chicks should be allowed to fledge prior to tree felling. Tree hollows should also be inspected for species of conservation value such as the Sunda Pangolin, and such species should be translocated prior to tree felling. Trees should be felled in stages, in one direction. Such directional clearance will give time for less motile fauna to move into other forested areas. Tree felling works are also to avoid the peak nesting period of many birds (between the months of February and July).

#### Management of Human Wildlife Encounters

Mortality from roadkills of highly motile fauna species such as Sunda pangolins and monitor lizards should be avoided. Hoarding along the road adjacent to the works should be erected to prevent animals from running onto the road.

To prevent other wild animals from scavenging on construction site, areas for food consumption, storage, and waste disposal should be demarcated and managed. The personnel should also be briefed on what to do should they encounter specific wild animals.

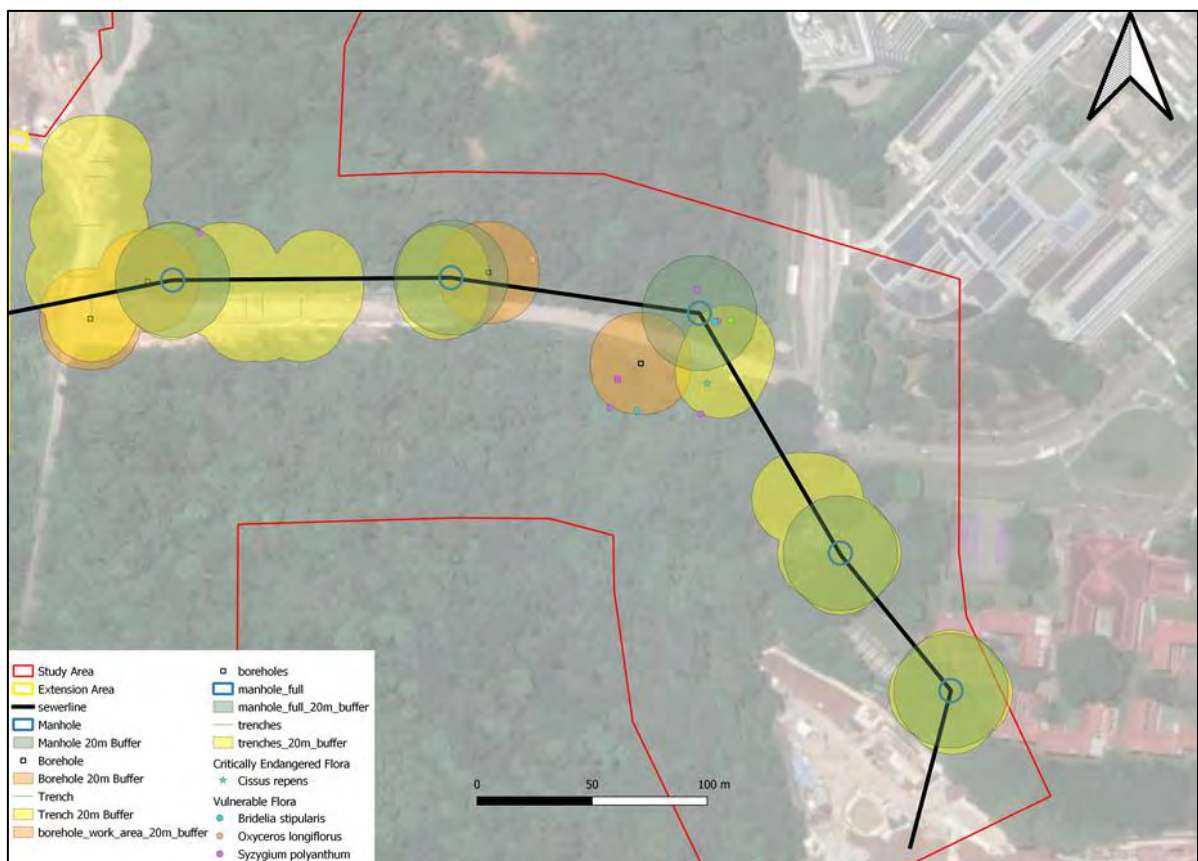
#### Minimising Fauna Entrapment

To prevent fauna mortality due to entrapment within the worksite, such as in ECM blankets or pits, daily site checks should be conducted to identify and rescue entrapped fauna. Moreover,

only non-plastic, biodegradable ECM mats are to be installed throughout the site. These mats should minimise entrapment of wild animals like snakes and lizards.

#### Mitigation for Soil Investigation Works

Works in Area 3 also include associated soil investigation borehole, manhole and trench works that are to be conducted. It is likely that soil investigation and trial trench works will take place prior to the actual sewerage alignment construction work. As can be seen in Figure 4.33, taking into account a 20-metre buffer for all types of work, it is recommended that *Cissus repens*, a Critically Endangered climber, and *Syzygium polyanthum*, a Vulnerable tree species, are salvaged where possible based on the criteria and methods described previously. *Syzygium polyanthum* saplings in particular may be used for site reinstatement works after construction is completed. The Contractors should also take measures to ensure that there is no other conservation significant flora within any vegetated area that is to be cleared prior to work commencement, as well as taking the appropriate action should any salvageable Conservation Significant species are found as detailed in Flora Management Plan.



**Figure 4.33.** Conservation significant flora affected by planned borehole, manhole and trench works

In summary, the mitigation measures that should be implemented to mitigate the impacts on sensitive biodiversity receptors for Area 3 are as follow:

- Avoid nightworks (i.e., limit work activities to 8 am – 6 pm), except in case of work exigency.
- Where nightworks are unavoidable, implement additional mitigation measures, and development a Lighting Management Plan (LMP).
- Ensure project layout avoids targeted sensitive receptors where possible

- Identify plants of high conservation significance to be transplanted
- Establish Tree Protection Zones (TPZ) for trees to be retained
- Hoard up entire work site.
- Avoid felling of trees during the peak bird nesting period (February to July)
- Visually inspect trees and holes for nesting birds and species of conservation value prior to felling
- Replant new planting verges with native trees species whenever possible
- Reinstate habitats with salvaged native species where possible upon work completion
- Conduct daily checks of ECM blankets and pits to rescue any entrapped fauna.
- Implement proper ECMs to prevent impact to adjacent habitats
- Utilise inert construction material
- Ensure equipment, vehicles, and footwear are clean prior to commencing works
- Establish designated areas for food and waste disposal
- Demarcate and manage areas for food consumption, storage, and waste disposal
- Provide monkey-proof waste bins in the project area
- During the construction phase, conduct information sessions for workers on what to do upon encountering wildlife
- Conduct regular monitoring of trees and vegetation health during construction phase within project site

#### **4.6.2 Operation Phase**

For the operation of plant extension area, edge effects are likely to degrade remaining habitats on the edge of the development. In order to mitigate the edge effect, restoration of working areas along the boundary of the project area should be done. Reinstatement of this working area should include the planting of native plant species of diverse structures, with the aim of simulating the structure of natural forests.

Mitigation measures for the impact of noise and light during the operation phase are addressed in their respective chapters.

During the operation phase of the sewer pipeline, no impacts are expected, hence no mitigation measures are recommended.

#### **4.7 Residual Impacts**

The residual impacts were evaluated using the RIAM method (Table 4.18 and Table 4.19) with due consideration that the recommended mitigation measures are implemented by Contractor. The residual impacts are likely to be in the band of slight negative to moderate negative.

Area 1 works are limited to the existing, developed section of the CCKWW plant. With the implementation of mitigation measures, there might be slight negative impacts on species due to the increased levels of noise and light into the surrounding areas.

Area 2 involves the permanent clear-felling of the project area. With the implementation of mitigation measures, some moderate negative impacts on flora and fauna species, and habitats, will remain, especially during the construction period. Slight residual negative impacts are expected on biodiversity during the pre-construction period, and from other impacts during the

construction period. Lastly, some residual slight negative human-wildlife conflict impacts may be present during the construction period.

Area 3 works involve the installation of the sewerage pipeline through pipejacking method. With the implementation of mitigation measures, some minor negative impacts on biodiversity may be present during the construction period. Lastly, some residual slight negative human-wildlife conflict impacts may be present during the construction period.

**Table 4.18.** Environmental Scores of the identified impacts on site's biodiversity with corresponding mitigation measures for Area 2 activities

| Phase            | Impact component                                                                                     | Predicted impact before mitigation |    |   |   |   |      |                   | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Predicted impact after mitigation |    |   |   |   |      |                   |
|------------------|------------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|------|-------------------|
|                  |                                                                                                      | I                                  | M  | P | R | C | ES   | Impact            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | I                                 | M  | P | R | C | ES   | Impact            |
| Ecology          |                                                                                                      |                                    |    |   |   |   |      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                   |    |   |   |   |      |                   |
| Pre-Construction | Flora and fauna species damage and mortality due to vegetation clearance for temporary working space | 4                                  | -3 | 2 | 2 | 3 | -84  | Moderate negative | <ul style="list-style-type: none"><li>Erect temporary hoarding</li><li>Identify plants to be transplanted</li><li>Avoid tree roots</li><li>Establish Tree Protection Zones (TPZ) for trees to be retained</li><li>Visually inspect trees and holes for nesting birds and animals prior to felling</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4                                 | -2 | 2 | 2 | 2 | -48  | Minor negative    |
|                  | Loss of habitats due to vegetation clearance for temporary working space                             | 3                                  | -3 | 2 | 2 | 3 | -63  | Minor negative    | <ul style="list-style-type: none"><li>Erect temporary hoarding to limit vegetation clearance</li><li>Reinstate habitats where possible upon completion of works</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                                 | -2 | 2 | 2 | 2 | -36  | Slight negative   |
| Construction     | Flora and fauna species mortality due to vegetation clearance for development of CCKWW extension     | 4                                  | -4 | 3 | 3 | 3 | -144 | Major negative    | <ul style="list-style-type: none"><li>Alter project footprint to avoid direct impacts on Stream 3</li><li>Erect temporary hoarding</li><li>Identify trees to be transplanted</li><li>Conduct targeted translocation of aquatic fauna from Stream 2 and Stream 3</li><li>Conduct phased, directional vegetation clearance along with wildlife shepherding</li><li>Conduct daily checks of ECM blankets and pits to rescue any entrapped fauna</li><li>Utilise only non-plastic, biodegradable ECM mats throughout the site.</li><li>Reinstate working areas with selected native species</li><li>Avoid felling trees during the peak bird nesting period (February to July)</li><li>Establish Tree Protection Zones (TPZ) for trees to be retained</li><li>Visually inspect trees for nesting birds and animals prior to felling</li></ul> | 4                                 | -3 | 3 | 3 | 3 | -108 | Moderate negative |

| Phase | Impact component                                                                                    | Predicted impact before mitigation |    |   |   |   |      |                   | Mitigation measures                                                                                                                                                                                                                                                                                                             | Predicted impact after mitigation |    |   |   |   |     |                   |
|-------|-----------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-------------------|
|       |                                                                                                     | I                                  | M  | P | R | C | ES   | Impact            |                                                                                                                                                                                                                                                                                                                                 | I                                 | M  | P | R | C | ES  | Impact            |
|       | Loss of habitats due to vegetation clearance for development of CCKWW extension                     | 3                                  | -4 | 3 | 3 | 3 | -108 | Moderate negative | <ul style="list-style-type: none"> <li>Alter project footprint to minimise direct impacts on Stream 3</li> <li>Erect temporary hoarding to limit vegetation clearance</li> <li>Reinstate habitats with native flora species where possible upon completion of works</li> </ul>                                                  | 3                                 | -3 | 3 | 3 | 3 | -81 | Moderate negative |
|       | Impact on habitats including waterbodies due to soil erosion and runoff                             | 4                                  | -3 | 3 | 2 | 3 | -96  | Moderate negative | <ul style="list-style-type: none"> <li>Implement Earth Control Measures (ECMs) that are approved by PUB</li> <li>Alter project footprint to minimise direct impacts on Stream 3</li> </ul>                                                                                                                                      | 3                                 | -2 | 2 | 2 | 2 | -36 | Slight negative   |
|       | Indirect impact on habitats due to edge effect                                                      | 2                                  | -2 | 3 | 3 | 3 | -36  | Slight negative   | <ul style="list-style-type: none"> <li>Avoid night works (i.e., limit construction activities to 8 am – 6 pm), except in the case of work exigencies and planned nightwork, with the implementation of the LMP approved by NParks</li> <li>Construction lights (if any) should face inwards and away from the forest</li> </ul> | 2                                 | -2 | 3 | 3 | 3 | -36 | Slight negative   |
|       | Impact on native species due to the introduction of invasive species                                | 2                                  | -3 | 3 | 3 | 3 | -54  | Minor negative    | <ul style="list-style-type: none"> <li>Ensure equipment, vehicles, and footwear used are clean prior to commencing works</li> <li>Use native plants for replanting of roadside verges</li> <li>Periodic checks within hoarding boundary to ensure that no introduced plant species have spontaneously established</li> </ul>    | 2                                 | -2 | 3 | 3 | 3 | -36 | Slight negative   |
|       | Impacts to flora and fauna due to illegal encroachment of vegetated areas by construction personnel | 2                                  | -3 | 2 | 2 | 3 | -42  | Minor negative    | <ul style="list-style-type: none"> <li>Incorporate signs prohibiting entrance into vegetated areas</li> <li>Erect fences to prevent illegal entry</li> </ul>                                                                                                                                                                    | 2                                 | -2 | 2 | 2 | 3 | -28 | Slight negative   |

| Phase                           | Impact component                                                                                                            | Predicted impact before mitigation |    |   |   |   |     |                   | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Predicted impact after mitigation |    |   |   |   |     |                 |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                                 |                                                                                                                             | I                                  | M  | P | R | C | ES  | Impact            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | I                                 | M  | P | R | C | ES  | Impact          |
|                                 | Roadkill or road injury to fauna escaping from forested site                                                                | 4                                  | -3 | 3 | 3 | 2 | -96 | Moderate negative | <ul style="list-style-type: none"> <li>Erect physical barriers to direct fauna movement into unaffected forested areas and prevent fauna from crossing the road</li> <li>Erect speed bumps or signages to alert drivers about potential animal crossings</li> <li>Enforce speed limit on the roads</li> </ul>                                                                                                                                                              | 4                                 | -2 | 3 | 3 | 2 | -64 | Minor negative  |
|                                 | Impact to ecological processes due to disturbance from night works                                                          | 3                                  | -3 | 2 | 2 | 3 | -63 | Minor negative    | <ul style="list-style-type: none"> <li>Complete all pile drilling before 6pm and only casting of piles to be carried out past 6pm.</li> <li>Complete all rebar fixing works before 6pm and only necessary concreting work for slab/ wall to be carried out past 6pm.</li> <li>Installation of 6m high noise barrier along project boundary.</li> <li>Additional localized noise barriers surrounding the works where possible</li> <li>Lighting management plan</li> </ul> | 3                                 | -2 | 2 | 2 | 2 | -36 | Slight negative |
| Operation                       | Indirect impact on habitats due to edge effect                                                                              | 3                                  | -2 | 2 | 2 | 2 | -36 | Slight negative   | <ul style="list-style-type: none"> <li>Use native plant for successful reinstatement of working areas</li> <li>Restore vegetation area with native trees of diverse structures and species.</li> </ul>                                                                                                                                                                                                                                                                     | 3                                 | -1 | 2 | 2 | 2 | -18 | Slight negative |
| <b>Human Health and Safety</b>  |                                                                                                                             |                                    |    |   |   |   |     |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Attacks of construction personnel by animals including wild pigs, feral dogs, long-tailed macaques, and smooth-coated otter | 3                                  | -3 | 2 | 3 | 2 | -63 | Minor negative    | <ul style="list-style-type: none"> <li>Establish designated areas for food and waste disposal</li> <li>Install monkey-proof bins</li> <li>Conduct information sessions on what to do upon encountering wildlife</li> </ul>                                                                                                                                                                                                                                                 | 3                                 | -2 | 2 | 3 | 1 | -36 | Slight negative |
|                                 | Impact to health of personnel due to tick/ mite bites from feral dogs                                                       | 3                                  | -3 | 2 | 3 | 2 | -63 | Minor negative    | <ul style="list-style-type: none"> <li>Implement proper use of Personal Protective Equipment (PPE)</li> </ul>                                                                                                                                                                                                                                                                                                                                                              | 3                                 | -2 | 2 | 3 | 1 | -36 | Slight negative |
|                                 | Injury caused by tree falls due to damaged roots                                                                            | 3                                  | -3 | 2 | 3 | 2 | -63 | Minor negative    | <ul style="list-style-type: none"> <li>Establish Tree Protection Zones (TPZ) for trees to be retained</li> </ul>                                                                                                                                                                                                                                                                                                                                                           | 3                                 | -2 | 2 | 3 | 1 | -36 | Slight negative |

| Phase | Impact component | Predicted impact before mitigation |   |   |   |   |    |        | Mitigation measures                                                                                 | Predicted impact after mitigation |   |   |   |   |    |        |
|-------|------------------|------------------------------------|---|---|---|---|----|--------|-----------------------------------------------------------------------------------------------------|-----------------------------------|---|---|---|---|----|--------|
|       |                  | I                                  | M | P | R | C | ES | Impact |                                                                                                     | I                                 | M | P | R | C | ES | Impact |
|       |                  |                                    |   |   |   |   |    |        | <ul style="list-style-type: none"> <li>Regularly monitor health of trees on project site</li> </ul> |                                   |   |   |   |   |    |        |

**Table 4.19.** Environmental Scores of the identified impacts on site's biodiversity with corresponding mitigation measures for Area 3 activities

| Phase            | Impact component                                                                             | Predicted impact before mitigation |    |   |   |   |     |                   | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Predicted impact after mitigation |    |   |   |   |     |                 |
|------------------|----------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                  |                                                                                              | I                                  | M  | P | R | C | ES  | Impact            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | I                                 | M  | P | R | C | ES  | Impact          |
| Ecology          |                                                                                              |                                    |    |   |   |   |     |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                   |    |   |   |   |     |                 |
| Pre-Construction | Flora and fauna damage and mortality due to vegetation clearance for temporary working space | 3                                  | -2 | 2 | 2 | 3 | -42 | Minor negative    | <ul style="list-style-type: none"><li>Erect temporary hoarding</li><li>Avoid tree roots in placement of working areas</li><li>Establish Tree Protection Zones (TPZ) for trees to be retained</li><li>Visually inspect trees and holes for nesting birds and animals prior to felling</li></ul>                                                                                                                                                                                  | 3                                 | -1 | 2 | 2 | 3 | -21 | Slight negative |
|                  | Loss of habitats due to vegetation clearance for temporary working space                     | 3                                  | -2 | 2 | 2 | 3 | -42 | Minor negative    | <ul style="list-style-type: none"><li>Erect temporary hoarding to limit vegetation clearance</li><li>Avoid placement of working areas in waterbodies</li><li>Reinstate habitats where possible upon completion of works</li></ul>                                                                                                                                                                                                                                               | 3                                 | -1 | 2 | 2 | 2 | -18 | Slight negative |
| Construction     | Flora and fauna mortality due to vegetation clearance for working shaft locations            | 3                                  | -3 | 3 | 3 | 3 | -81 | Moderate negative | <ul style="list-style-type: none"><li>Avoid placement of shafts near native, mature trees, where possible</li><li>Identify trees and plants to be transplanted</li><li>Conduct daily checks of ECM blankets and pits to rescue any entrapped fauna.</li><li>Utilise only non-plastic, biodegradable ECM mats throughout the site.</li><li>Reinstate working areas with selected native species</li><li>Establish Tree Protection Zones (TPZ) for trees to be retained</li></ul> | 3                                 | -2 | 3 | 3 | 3 | -54 | Minor negative  |

| Phase                           | Impact component                                                                                       | Predicted impact before mitigation |    |   |   |   |      |                   | Mitigation measures                                                                                                                                                                                                                              | Predicted impact after mitigation |    |   |   |   |     |                 |
|---------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                                 |                                                                                                        | I                                  | M  | P | R | C | ES   | Impact            |                                                                                                                                                                                                                                                  | I                                 | M  | P | R | C | ES  | Impact          |
|                                 |                                                                                                        |                                    |    |   |   |   |      |                   | <ul style="list-style-type: none"> <li>Visually inspect trees for nesting birds and animals prior to felling</li> </ul>                                                                                                                          |                                   |    |   |   |   |     |                 |
|                                 | Loss of habitats due to vegetation clearance for working shaft locations                               | 3                                  | -2 | 3 | 3 | 3 | -54  | Minor negative    | <ul style="list-style-type: none"> <li>Erect temporary hoarding to limit vegetation clearance</li> <li>Reinstate habitats where possible upon completion of works</li> </ul>                                                                     | 3                                 | -2 | 3 | 2 | 3 | -42 | Minor negative  |
|                                 | Impact on habitats including waterbodies due to discharge of silty water                               | 2                                  | -3 | 2 | 2 | 3 | -42  | Minor negative    | <ul style="list-style-type: none"> <li>Implement proper Earth Control Measures (ECM) plan prepared by QECP and approved by PUB</li> </ul>                                                                                                        | 2                                 | -2 | 2 | 2 | 3 | -28 | Slight negative |
|                                 | Impact on native species due to the introduction of invasive species                                   | 2                                  | -3 | 3 | 3 | 3 | -54  | Minor negative    | <ul style="list-style-type: none"> <li>Ensure equipment, vehicles, and footwear used are clean prior to commencing works</li> <li>Use native plants for replanting of roadside verges</li> </ul>                                                 | 2                                 | -2 | 3 | 3 | 3 | -36 | Slight negative |
|                                 | Impacts to flora and fauna due to illegal encroachment of vegetated areas by construction personnel    | 2                                  | -3 | 2 | 2 | 3 | -42  | Minor negative    | <ul style="list-style-type: none"> <li>Incorporate signs prohibiting entrance into vegetated areas.</li> <li>Erect fences to prevent illegal entry</li> </ul>                                                                                    | 2                                 | -2 | 2 | 2 | 3 | -28 | Slight negative |
|                                 | Roadkill or road injury to fauna escaping from forested site                                           | 4                                  | -3 | 3 | 3 | 3 | -108 | Moderate negative | <ul style="list-style-type: none"> <li>Erect physical barriers along road to prevent fauna from crossing the road</li> <li>Erect signages to alert drivers about potential animal crossings</li> <li>Enforce speed limit on the roads</li> </ul> | 4                                 | -2 | 3 | 3 | 3 | -72 | Minor negative  |
| <b>Human Health and Safety</b>  |                                                                                                        |                                    |    |   |   |   |      |                   |                                                                                                                                                                                                                                                  |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Attacks of construction personnel by animals including wild pigs, feral dogs, and long-tailed macaques | 3                                  | -3 | 2 | 3 | 2 | -63  | Minor negative    | <ul style="list-style-type: none"> <li>Establish designated areas for food and waste disposal</li> <li>Install monkey-proof bins</li> <li>Conduct information sessions on what to do upon encountering wildlife</li> </ul>                       | 3                                 | -2 | 2 | 3 | 1 | -36 | Slight negative |
|                                 | Impact to health of personnel due to tick/mite bites from feral dogs                                   | 3                                  | -3 | 2 | 3 | 2 | -63  | Minor negative    | <ul style="list-style-type: none"> <li>Implement proper use of Personal Protective Equipment (PPE)</li> </ul>                                                                                                                                    | 3                                 | -2 | 2 | 3 | 1 | -36 | Slight negative |
|                                 | Injury caused by tree falls due to damaged roots                                                       | 3                                  | -3 | 2 | 3 | 2 | -63  | Minor negative    | <ul style="list-style-type: none"> <li>Establish Tree Protection Zones (TPZ) for trees to be retained</li> </ul>                                                                                                                                 | 3                                 | -2 | 2 | 3 | 1 | -36 | Slight negative |

| Phase | Impact component | Predicted impact before mitigation |   |   |   |   |    |        | Mitigation measures                                                                                 | Predicted impact after mitigation |   |   |   |   |    |        |
|-------|------------------|------------------------------------|---|---|---|---|----|--------|-----------------------------------------------------------------------------------------------------|-----------------------------------|---|---|---|---|----|--------|
|       |                  | I                                  | M | P | R | C | ES | Impact |                                                                                                     | I                                 | M | P | R | C | ES | Impact |
|       |                  |                                    |   |   |   |   |    |        | <ul style="list-style-type: none"> <li>Regularly monitor health of trees on project site</li> </ul> |                                   |   |   |   |   |    |        |

## **5 HYDROLOGY AND SURFACE WATER QUALITY**

### **5.1 Introduction**

This Section describes the relevant laws and standard for water quality that are applicable to the project, the methodology and results for the baseline hydrological conditions and water quality at the study area. It also describes potential impacts on hydrology and water quality due to the construction works of the project, and recommendation for mitigation measures. A quantitative impact assessment approach is being used for assessment of impacts.

### **5.2 Relevant Environmental Legislation, Guidelines and Standards**

The Sewerage and Drainage Act 2001 authorizes PUB to construct, maintain and improve sewerage and drainage systems, to regulate the discharge into these systems, and to issue codes of practice or specifications.

Sewerage and Drainage (Surface Water Drainage) Regulation 2007 specifies a maximum discharge limit for total suspended solids (TSS) as 50 milligrams per liter (mg/L) of the discharge. This regulation also requires every contractor to comply with the Code of Practice (COP) on Surface Water Drainage.

Sewerage and Drainage (Trade Effluent) Regulation 2007 requires contractors to obtain approval from PUB for discharge of trade effluent into a public sewer. The nature or type of effluent discharge must be approved by PUB. Effluent quality must comply with those specified in the regulations, and if required by PUB, must be treated prior to discharge.

The discharge of wastewater into open drains, canals and rivers is regulated by the Environmental Protection and Management (EPM) Act 2002 and the EPM (Trade Effluent) Regulation 2008. The Act and its regulations prescribe allowable limits for trade effluent discharge to controlled watercourse and are administered by NEA.

COPs and guidelines relevant to public utilities and watercourse are listed below.

- COP for Environmental Control Officers;
- COP on Surface Water Drainage 2018;
- COP on Pollution Control (SS 593: 2013);
- Guidebook for Qualified Erosion Control Professional (QECP) 2006;
- Guidebook on Erosion and Sediment Control at Construction Sites 2018.

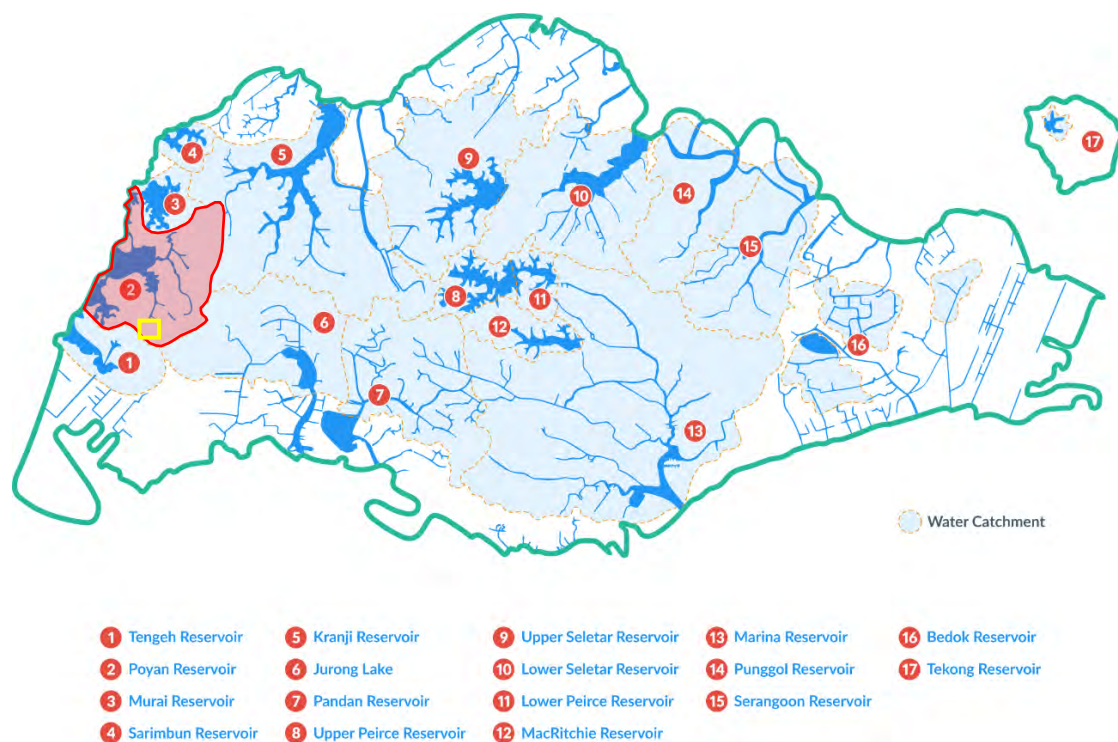
### **5.3 Hydrology Profile of Project Study Area**

#### **5.3.1 Catchment Profile**

The project study area is located within the catchment area of Poyan Reservoir and also near Tengeh Reservoir and Jurong Lake (Figure 5.1), which was created by damming Sungei Poyan and its delta in 1977. The reservoir's main body covers an area of approximately 356 hectares and is roughly 2.3 km away from the project study area. Together with three other reservoirs (Murai, Sarimbun, Tengeh), it forms the Western

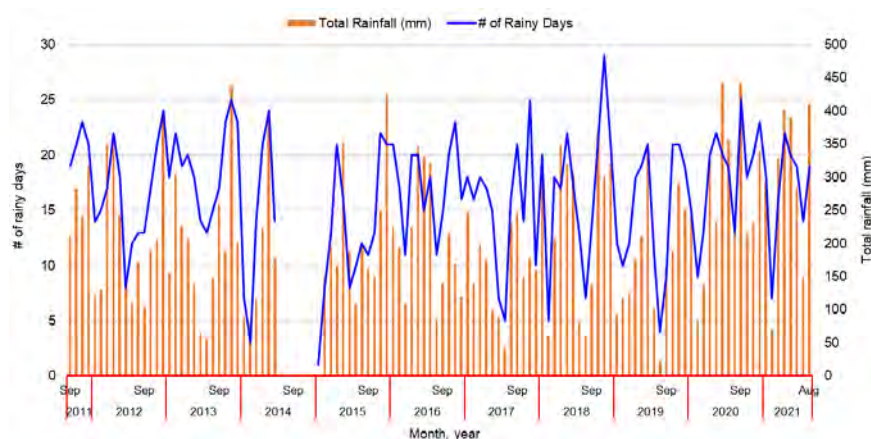
Water Catchment Area, a planning region designated to catch and reserve rainwater in western Singapore for national usage. As such, Poyan Reservoir catchment area contains both natural and man-modified drainages to benefit from its topography for maximum water retention. These include the natural streams in the project study area, the run-offs of which enter the drinking water reservoirs.

Annual monthly rainfall over the last decade in the project study area is presented in Figure 5.2 using records at Jurong West weather station as representative of the site.



**Figure 5.1.** Catchment areas of Singapore (Public Utilities Board, 2019)

Approximate location of the project site is demarcated in the yellow box within the Poyan Reservoir catchment area (shaded red).

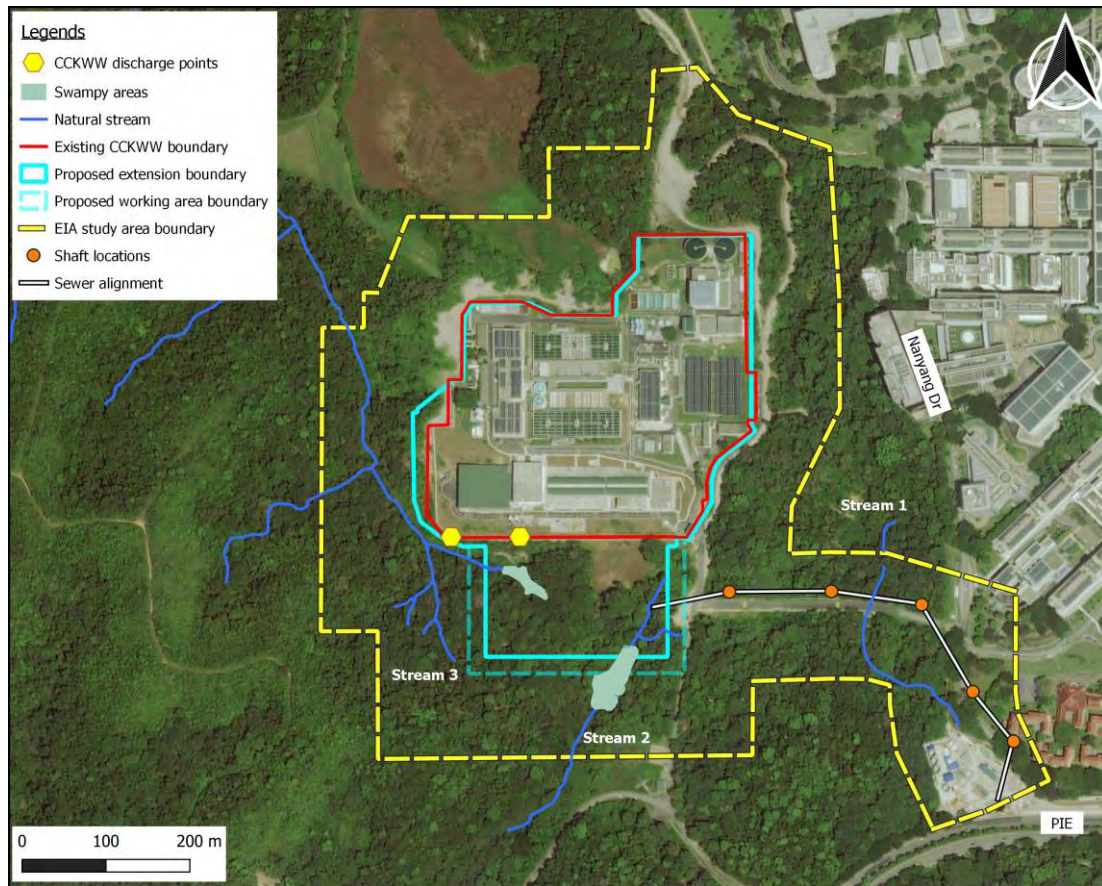


**Figure 5.2.** Annual monthly rainfall at Jurong West weather station during 2011–2021

N. B. there was no rainfall record from July to December 2014

### 5.3.2 Drainage Profile

Based on the study by NParks (2016) as well as data collected from current topography survey, three streams were identified within the study area (Figure 5.3).



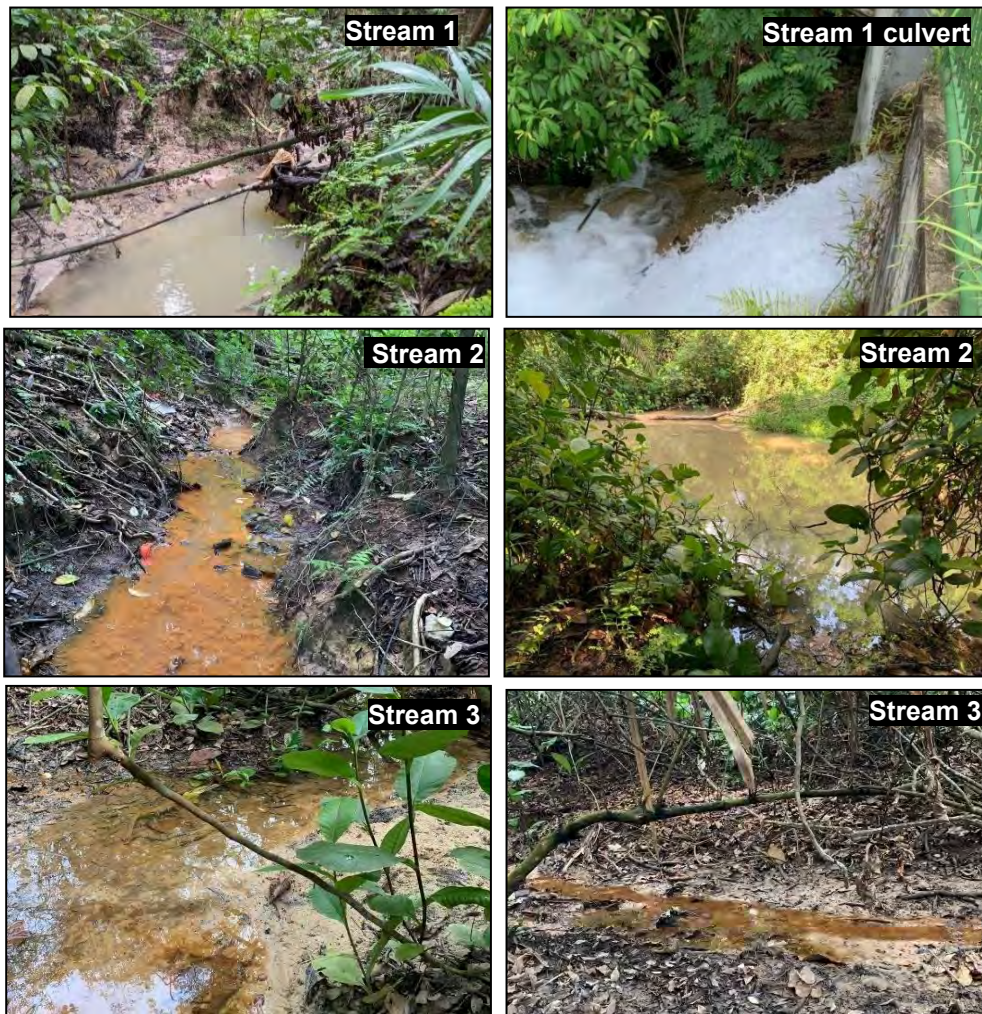
**Figure 5.3.** Location of natural streams in the project study area

Stream 1 likely originates from its northern edge that is of higher elevation and flows southward (Figure 5.4), eventually connecting to a drainage system that empties into Jurong Lake catchment area. The northern stretch carries a lower volume of discharge compared to its southern stretch. This increased discharge in the lower portion of Stream 1 is due to it running through a culvert under a road connecting Nanyang Drive with Peng Kang Avenue, where Stream 1 receives large volumes of input from the roadside run-off and CCKWW plant raw water discharge.

The flow of Stream 2 appears to be fed by run-off from rainwater as well as discharge from CCKWW plant (Figure 5.4). This stream feeds into a swampy area in its mid part before continuing as a stream southward. Stream 2 flows to the Tengeh Reservoir at the south-west of the project study area.

Lastly, Stream 3 originates from a point at its south and flows northward, emptying into Poyan Reservoir (Figure 5.4). The present study found two discharge points along the CCKWW plant perimeter that feed into Stream 3. In addition to surface run-off from rainfall, the varying elevation of study area seems to cause differences in water table,

enabling groundwater flow to recharge the stream and keep them running constantly.



**Figure 5.4.** Photographs showing various streams within project study area

NParks (2016) observed that these streams were commonly lined by tree ferns along their banks with simpoh air (*Dillenia suffruticosa*) and para rubber (*Hevea brasiliensis*) dominating the secondary forest housing these streams. The present study also found oil palm (*Elaeis guineensis*) to be the dominant riparian vegetation along the southern stretch of Stream 2. Elevation in project study area varies greatly with approximately 30 meters difference between its highest and lowest points. Rainfall is either intercepted by vegetation canopy or falls directly onto ground surface.

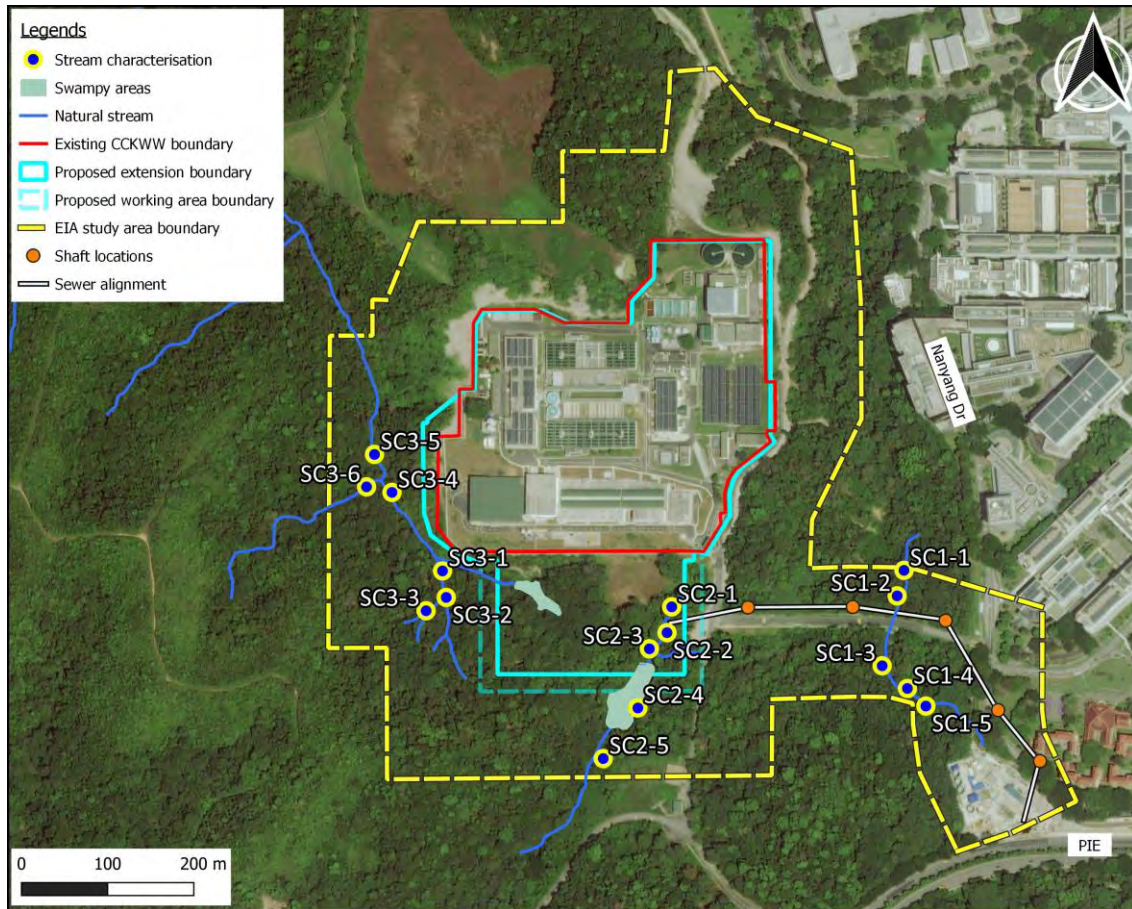
#### **5.4 Natural Stream Characterisation**

There are three natural streams within EIA study area as noted earlier. This section provides approach as well as findings of stream characterisation survey.

##### **5.4.1 Stream Characterization Approach**

The stream characterisation survey covered details such as substrate type, riparian vegetation composition, wetted width and flow velocity of streams as explained below.

The survey locations were selected based on coverage of the stream profile variations, site conditions and safe access.



**Figure 5.5.** Location of stream characterisation survey points

### Wetted Depth and Width

The depths and widths of the stream channels were measured using a scale ruler as well as a metric transect tape depending upon site conditions.

### Canopy Cover

To measure the canopy cover at each measuring point, a photo of the canopy was taken parallel to the ground at eye-level. The percentage of canopy cover derived is an estimate of the total area covered by vegetation over the entire frame of in the taken canopy photo.

### Substrate Type

The soil substrate on the banks of the stream and in the waterbody were determined using visual assessment as well as the “texture by feel” method (Thein, 1979). The “texture by feel” method involves picking up roughly 25 g of moistened soil and shaping it into a ball, which is then squeezed between the thumb and index finger upwards beyond the fingers and allowed to break from its own weight. The length of the resultant ribbon is measured to determine the soil substrate. Following that, a pinch of the soil was wet and rubbed to determine the soil texture and composition.

Table 5.1 shows the soil substrate types that can be determined based on the soil textures and ribbon length.

**Table 5.1.** Classification of substrate types (Thein, 1979)

| Ribbon Length  | Soil Texture              | Soil Substrate  |
|----------------|---------------------------|-----------------|
| No ribbon      | NA                        | Loamy sand      |
| 2.5 cm or less | Gritty                    | Sandy loam      |
|                | Smooth                    | Silt loam       |
|                | Neither gritty nor smooth | Loam            |
| 2.5 to 5 cm    | Gritty                    | Sandy clay loam |
|                | Smooth                    | Silty clay loam |
|                | Neither gritty nor smooth | Clay loam       |
| More than 5 cm | Gritty                    | Sandy clay      |
|                | Smooth                    | Silty clay      |
|                | Neither gritty nor smooth | Clay            |

Subsequently, sandy substrates were further categorized visually according to their estimated particle size in Table 5.2 using the “texture by feel” method as well as by placing a sample of the soil and spreading it across a scale card.

**Table 5.2.** Classification of sand particle sizes (Wentworth, 1922)

| Sand Particle Type | Particle Size              |
|--------------------|----------------------------|
| Fine sand          | $\leq 0.5$ mm              |
| Coarse sand        | $> 0.5$ mm and $\leq 2$ mm |
| Fine gravel        | $\leq 4$ mm                |
| Coarse gravel      | $> 4$ mm and $\leq 32$ mm  |
| Cobble             | $\leq 256$ mm              |

### Riparian Vegetation

The species of plants that are immediately next to, within or overhead each measuring point are identified by qualified arborist.

### Flow Velocity

To measure the flow velocity of streams, the float and measure method was employed with the use of a ping pong ball as a float, a metric transect tape to set the travel distance, and a stopwatch to measure the duration of travel. To begin the measurement, the transect tape is laid down parallel to the stream flow, and the start and end points determined. When the float enters the start point of the stretch, the stopwatch would start timing, and when the float passes the end point of the stretch, the stopwatch would stop timing and the duration is recorded. Each measurement was taken three times and then averaged out in order to obtain the more accurate approximate time taken for the float to cover the predetermined stretch.

To calculate the flow velocity using the recorded data, the formula below is used:

$$\text{Flow Velocity} = \frac{\text{Distance}}{\text{Time}} \times \text{Friction Correction}$$

Friction correction refers to the numerical factors used in order to factor in the difference in flow due to the channel surfaces, where 0.9 is used for smooth surfaces (e.g., drains, canals, and sandy bottoms) and 0.85 is used for rough surface rock bottoms. Since the top of a stream typically flows faster than the bottom due to friction against the stream bed, applying friction correction events out the flow velocity (Harrelson, Rawlins, & Potyondy, 1994). The unit used for flow velocity is meter per second (m/s).

## 5.4.2 Stream Characterization Results

### Characterization of Stream 1

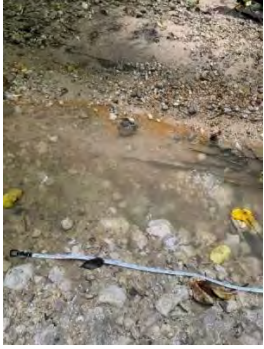
Stream 1 is identified as a major earth stream within project study area with the highest discharge. This stream is connected to an outlet that is linked to the CCKWW in the area and has been observed to release large amounts of treated water discharge occasionally. With a higher discharge volume and higher velocity, this stream is also one of the few streams that have significant bank heights flanking the stream and carry much larger and heavier sediment loads such as cobble and anthropogenic debris, such as bricks and ceramic. It should be noted that a major portion of the stream has banks and beds that consist of clay, resulting in a stream that has portions of large alluvial sediments next to clear beds of clay bedrock.

Table 5.3 provides the results of Stream 1 characterization survey.

**Table 5.3.** Results of stream 1 characterization survey

| Survey Point ID | Stream Cross-Section & Canopy Photograph                                                                             | Wetted Width & Depth (m)      | Soil Substrate                                                 | Flow Velocity (m/s) | Riparian Vegetation                                                                                                                                                                                                                                    | Associated Fauna    |
|-----------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| SC1-1           | Cross-section:<br><br>Canopy: 70% | Width: 1m<br><br>Depth: 0.09m | Stream:<br>Loamy coarse sand<br><br>Bank: Mud, loamy fine sand | 0 m/s;<br>Stagnant  | <i>Dillenia suffruticosa</i> ,<br><i>Cyathea sp.</i> ,<br><i>Davallia denticulata</i> ,<br><i>Gynotroches axillaris</i> ,<br><i>Christella subpubescens</i> ,<br><i>Blechnum orientale</i> ,<br><i>Taenitis blechnoides</i> ,<br><i>Clidemia hirta</i> | Copper-cheeked frog |

| Survey Point ID | Stream Cross-Section & Canopy Photograph                                                                                                                                                                        | Wetted Width & Depth (m)                                        | Soil Substrate                                                                                   | Flow Velocity (m/s) | Riparian Vegetation                                                                                                                                                                                   | Associated Fauna                                     |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
|                 |                                                                                                                                |                                                                 |                                                                                                  |                     |                                                                                                                                                                                                       |                                                      |
| SC1-2           | <p>Cross-section:</p>  <p>Canopy: 75%</p>  | <p>Width: 0.5m</p> <p>Depth: 0.02m</p>                          | <p>Stream: Mud, Loamy coarse sand</p> <p>Bank: Mud, loamy fine sand</p>                          | 0.025 m/s           | <p><i>Davallia denticulata</i>,<br/><i>Macaranga bancana</i>, <i>Hevea brasiliensis</i></p>                                                                                                           | <p>Copper-cheeked frog</p> <p>Malayan giant frog</p> |
| SC1-3           | <p>Cross-Section:</p>  <p>Canopy: 80%</p>                                                                                    | <p>Width: 1.1m</p> <p>Depth: 0.13m</p> <p>Bank Height: 2.1m</p> | <p>Stream: Loamy sand, Coarse sand to cobble, debris, clay beds</p> <p>Bank: Clay walls, mud</p> | 0.0149 m/s          | <p><i>Adiantum latifolium</i>,<br/><i>Asplenium longissimum</i>,<br/><i>Vittaria ensiformis</i>,<br/><i>Caryota mitis</i>,<br/><i>Pellacalix axillaris</i>,<br/><i>Urophyllum</i><br/>117isturbed</p> | Guppy sp.                                            |

| Survey Point ID | Stream Cross-Section & Canopy Photograph                                                                                                                                                                        | Wetted Width & Depth (m)                                        | Soil Substrate                                                                                          | Flow Velocity (m/s) | Riparian Vegetation                                                                                                    | Associated Fauna                                                  |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
|                 |                                                                                                                                |                                                                 |                                                                                                         |                     |                                                                                                                        |                                                                   |
| SC1-4           | <p>Cross-Section:</p>  <p>Canopy: 55%</p>  | <p>Width: 1.2m</p> <p>Depth: 0.16m</p> <p>Bank Height: 1.9m</p> | <p>Stream: Loamy sand, coarse sand to gravel to cobble</p> <p>Bank: Fine sand to gravel, mud</p>        | 0.157 m/s           | <i>Spathodea campanulata</i>                                                                                           | <p>Malaysian trumpet snail</p> <p>Guppy sp.</p> <p>Asian toad</p> |
| SC1-5           | <p>Cross-Section:</p>  <p>Canopy: 90%</p>                                                                                    | <p>Width: 2.7m</p> <p>Depth: 0.2m</p> <p>Bank Height: 1.8m</p>  | <p>Stream: Loamy sand, coarse sand to gravel to cobble</p> <p>Bank: Loamy sand, fine to coarse sand</p> | 0.0254 m/s          | <i>Piper sarmentosum</i> ,<br><i>Dieffenbachia amoena</i> ,<br><i>Cyclosorus sp.</i> ,<br><i>Spathodea campanulata</i> | <p>Malaysian trumpet snail</p> <p>Guppy sp.</p>                   |

| Survey Point ID | Stream Cross-Section & Canopy Photograph                                          | Wetted Width & Depth (m) | Soil Substrate | Flow Velocity (m/s) | Riparian Vegetation | Associated Fauna |
|-----------------|-----------------------------------------------------------------------------------|--------------------------|----------------|---------------------|---------------------|------------------|
|                 |  |                          |                |                     |                     |                  |

### Characterization of Stream 2

Stream 2 varied in widths along the different stream portions, from shorter widths of half a metre to as wide as an estimate of 10 metres. While the upper portion of the stream consists of small discharge and small widths, it opens up to a large channel with high discharge, and eventually filters through a grassy wetland to a swampy environment at the lower portion of the stream.

Table 5.4 provides the results of Stream 2 characterization survey.

**Table 5.4.** Results of stream 2 characterization survey

| Survey Point ID | Stream Cross-Section & Canopy Photograph                                                                                     | Wetted Width & Depth (m)               | Soil Substrate                                            | Flow Velocity (m/s) | Riparian Vegetation                                                                       | Associated Fauna   |
|-----------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------|--------------------|
| SC2-1           | <p>Cross-section:</p>  <p>Canopy: 85%</p> | <p>Width: 0.6m</p> <p>Depth: 0.03m</p> | <p>Stream: Cobble</p> <p>Bank: Loamy fine gravel, mud</p> | 0.00547 m/s         | <i>Cinnamomum iners</i> ,<br><i>Spathodea campanulata</i> ,<br><i>Adiantum latifolium</i> | Malayan giant frog |

| Survey Point ID | Stream Cross-Section & Canopy Photograph                                                                                                                                                                        | Wetted Width & Depth (m)                                      | Soil Substrate                                                                   | Flow Velocity (m/s) | Riparian Vegetation                                                                                                                                                                                                                        | Associated Fauna                                                                                                                                                    |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                 |                                                                                                                                |                                                               |                                                                                  |                     |                                                                                                                                                                                                                                            |                                                                                                                                                                     |
| SC2-2           | <p>Cross-section:</p>  <p>Canopy: 80%</p>  | <p>Width: ~10m</p> <p>Depth: NA</p> <p>Bank Height: 0.75m</p> | <p>Stream: NA</p> <p>Bank: Mud, loamy fine sand</p>                              | NA                  | <p><i>Dillenia suffruticosa</i>, <i>Macaranga</i> sp., <i>Taenitis blechnoides</i>, <i>Nephrolepis</i> sp., <i>Cyclosorus</i> sp., <i>Leea indica</i>, <i>Adiantum latifolium</i>, <i>Cinnamomum iners</i>, <i>Fibraurea tinctoria</i></p> | <p>Three-spot gourami</p> <p><i>Caridina</i> sp.</p> <p>Gunther's frog</p> <p><i>Caridina</i> sp.</p> <p><i>Bithyniidea</i> sp.</p> <p><i>Macrobrachium</i> sp.</p> |
| SC2-3           | <p>Cross-section:</p>                                                                                                        | <p>Width: 1m</p> <p>Depth: 0.06m</p>                          | <p>Stream: Mud, loamy fine sand</p> <p>Bank: Mud, loamy sand and fine gravel</p> | 0.01 m/s            | —                                                                                                                                                                                                                                          | —                                                                                                                                                                   |

| Survey Point ID | Stream Cross-Section & Canopy Photograph                                                                                                                                                                   | Wetted Width & Depth (m)                                                                           | Soil Substrate                                                                                            | Flow Velocity (m/s) | Riparian Vegetation                                                                       | Associated Fauna                                       |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------|
|                 | Canopy: 75%<br>                                                                                                           |                                                                                                    |                                                                                                           |                     |                                                                                           |                                                        |
| SC2-4           | Cross-section:<br><br>Canopy: 55%<br> | Width: ~5m, inland water retention area with temporary pond<br><br>Depth: Fluctuating and temporal | Stream: Mud, organic debris, loamy fine sand<br><br>Bank: mud, organic debris, loamy fine sand and gravel | 0 m/s; negligible   | <i>Elaeis guineensis</i> , <i>Struchium sparganophorum</i> , <i>Ludwigia hyssopifolia</i> | Malaysian trumpet snail<br>Guppy sp.<br>Ramshorn snail |
| SC2-5           | Cross-section:<br>                                                                                                      | Width: 0.7m<br><br>Depth: 0.06m                                                                    | Stream: Mud, loamy fine sand<br><br>Bank: Mud, loamy fine sand                                            | 0.0109 m/s          | <i>Adiantum latifolium</i> , <i>Struchium sparganophorum</i>                              | —                                                      |

| Survey Point ID | Stream Cross-Section & Canopy Photograph                                                         | Wetted Width & Depth (m) | Soil Substrate | Flow Velocity (m/s) | Riparian Vegetation | Associated Fauna |
|-----------------|--------------------------------------------------------------------------------------------------|--------------------------|----------------|---------------------|---------------------|------------------|
|                 | Canopy: 70%<br> |                          |                |                     |                     |                  |

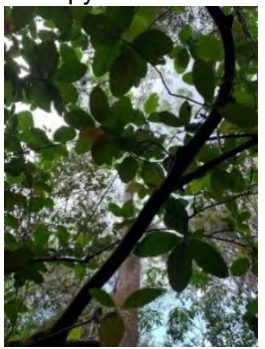
### Characterization of Stream 3

Stream 3 is an earth stream that braid into several branches, with some branches that contain temporal discharge during periods of significant precipitation events. Earth streams in this area are generally in a swampy environment with substrate that consist of fine sand to mud, characterised by shallow wetted areas with low discharge and flow velocity.

Table 5.5 provides the results of Stream 3 characterization survey.

**Table 5.5.** Results of stream 3 characterization survey

| Survey Point ID | Stream Cross-Section & Canopy Photograph                                                                             | Wetted width & Depth (m)        | Soil Substrate                                               | Flow Velocity (m/s) | Riparian Vegetation                                             | Associated Fauna                                                                             |
|-----------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------|---------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| SC3-1           | Cross-section:<br><br>Canopy: 85% | Width: 0.5m<br><br>Depth: 1-2cm | Stream: Loamy coarse sand<br><br>Bank: Clay, loamy fine sand | 0.129 m/s           | <i>Dillenia suffruticosa</i> ,<br><i>Stenochlaena palustris</i> | Copper-cheeked frog<br>Brilliant rasbora<br>Forest betta<br>Whitespot<br>Copper-cheeked frog |

| Survey Point ID | Stream Cross-Section & Canopy Photograph                                                                                                                                                                        | Wetted width & Depth (m)               | Soil Substrate                                                | Flow Velocity (m/s)          | Riparian Vegetation                                                   | Associated Fauna                                                        |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------------|------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------|
|                 |                                                                                                                                |                                        |                                                               |                              |                                                                       |                                                                         |
| SC3-2           | <p>Cross-section:</p>  <p>Canopy: 85%</p>  | <p>Width: 0.6m</p> <p>Depth: 0.03m</p> | <p>Stream: Loamy coarse sand</p> <p>Bank: Loamy fine sand</p> | <p>0 m/s;<br/>Negligible</p> | <p><i>Dillenia suffruticosa</i>,<br/><i>Taenitis blechnoides</i></p>  | <p>Brilliant rasbora</p> <p>Forest betta</p> <p>Copper-cheeked frog</p> |
| SC3-3           | <p>Cross-Section:</p>                                                                                                        | <p>Width: 0.6m</p> <p>Depth: 0.03m</p> | <p>Stream: Loamy sand</p> <p>Bank: Clay, loamy fine sand</p>  | <p>0.00137 m/s</p>           | <p><i>Dillenia suffruticosa</i>,<br/><i>Gynotroches axillaris</i></p> | <p>Copper-cheeked frog</p> <p>Brilliant rasbora</p> <p>Forest betta</p> |

| Survey Point ID | Stream Cross-Section & Canopy Photograph                                                                                                                                                                   | Wetted width & Depth (m)        | Soil Substrate                                                   | Flow Velocity (m/s)  | Riparian Vegetation                                                                                                                                  | Associated Fauna                                                          |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
|                 | Canopy: 85%<br>                                                                                                           |                                 |                                                                  |                      |                                                                                                                                                      |                                                                           |
| SC3-4           | Cross-section:<br><br>Canopy: 60%<br> | Width: 0.5m<br><br>Depth: 0.07m | Stream: Loamy coarse sand<br><br>Bank: Clay, loamy fine sand     | 0.0148 m/s           | <i>Dillenia suffruticosa</i> ,<br><i>Cyclosorus triphyllus</i> ,<br><i>Cyclosorus sp.</i> , <i>Flagellaria indica</i> ,<br><i>Acrostichum aureum</i> | Brilliant rasbora<br><i>Macobranchium sp.</i><br>Saddle barb<br>Whitespot |
| SC3-5           | Cross-Section:<br>                                                                                                      | Width: 0.8m<br><br>Depth: 0.07m | Stream: Clay, loamy fine sand<br><br>Bank: Clay, fine sandy loam | 0 m/s;<br>Negligible | <i>Dillenia suffruticosa</i> ,<br><i>Asplenium nidus</i> ,<br><i>Asplenium nidus</i>                                                                 | Malaysian trumpet snail<br>Saddle barb<br>Whitespot                       |

| Survey Point ID | Stream Cross-Section & Canopy Photograph                                                                                                                                                                   | Wetted width & Depth (m)      | Soil Substrate                                                  | Flow Velocity (m/s) | Riparian Vegetation                                          | Associated Fauna                                                                                           |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------------------------------|---------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
|                 | Canopy: 75%<br>                                                                                                           |                               |                                                                 |                     |                                                              |                                                                                                            |
| SC3-6           | Cross-Section:<br><br>Canopy: 90%<br> | Width: 0.4m<br><br>Depth: 3cm | Stream:<br>Loamy coarse sand<br><br>Bank: Clay, loamy fine sand | 0.108 m/s           | <i>Dillenia suffruticosa</i> ,<br><i>Pandanus atrocarpus</i> | Forest betta<br>Brilliant rasbora<br><i>Macrobrachium</i> sp.<br>Copper-cheeked frog<br>Malayan giant frog |

## 5.5 Surface Water Quality Baseline Methodology

Per the scoping exercise, the following parameters were tested from the surface water quality samples:

**Table 5.6.** Surface water quality parameters to be surveyed

| Parameter        | Unit    | Limit              | Test Method                          |
|------------------|---------|--------------------|--------------------------------------|
| <b>In-situ</b>   |         |                    |                                      |
| Temperature      | °C      | 45 <sup>[1]</sup>  | MLS-SOP-ES-004 using multi-parameter |
| Conductivity     | µS/cm   | –                  |                                      |
| pH               | pH unit | 6–9 <sup>[1]</sup> |                                      |
| Turbidity        | NTU     | –                  |                                      |
| Dissolved Oxygen | mg/L    | ≥ 4 <sup>[2]</sup> |                                      |

| Parameter                    | Unit      | Limit                                                    | Test Method                   |
|------------------------------|-----------|----------------------------------------------------------|-------------------------------|
| <b>Ex-situ</b>               |           |                                                          |                               |
| Total Phosphorus             | mg/L      | –                                                        | APHA 4500-P (J)               |
| Total Nitrogen               | mg/L      | –                                                        | APHA 4500-P (J)               |
| Nitrate as NO <sub>3</sub>   | mg/L      | 20 <sup>[1]</sup>                                        | APHA 4500-NO <sub>3</sub> (I) |
| Phosphate as PO <sub>4</sub> | mg/L      | 2 <sup>[1]</sup>                                         | APHA 4500-P (G)               |
| Total Suspended Solids       | mg/L      | 30 <sup>[1]</sup>                                        | APHA 2540D                    |
| Total Dissolved Solids       | mg/L      | 1,000 <sup>[1]</sup>                                     | APHA 2540C                    |
| Total Organic Carbon         | mg/L      | –                                                        | APHA 5310B                    |
| Ammoniacal Nitrogen          | mg/L      | 1 <sup>[2]</sup>                                         | APHA 4500-NH <sub>3</sub> (H) |
| Enterococcus                 | cfu/100mL | 40–40,000 <sup>[3]</sup><br>4,000–200,000 <sup>[4]</sup> | APHA 9230C                    |
| Arsenic as As                | mg/L      | 0.01 <sup>[1]</sup>                                      | APHA 3120B                    |
| Cadmium as Cd                | mg/L      | 0.003 <sup>[1]</sup>                                     | APHA 3120B                    |
| Chromium as Cr               | mg/L      | 0.05 <sup>[1]</sup>                                      | APHA 3120B                    |
| Iron as Fe                   | mg/L      | 1 <sup>[1]</sup>                                         | APHA 3120B                    |
| Nickel as Ni                 | mg/L      | 0.1 <sup>[1]</sup>                                       | APHA 3120B                    |
| Copper as Cu                 | mg/L      | 0.1 <sup>[1]</sup>                                       | APHA 3120B                    |
| Lead as Pb                   | mg/L      | 0.1 <sup>[1]</sup>                                       | APHA 3120B                    |
| Zinc as Zn                   | mg/L      | 0.5 <sup>[1]</sup>                                       | APHA 3120B                    |
| Mercury as Hg                | mg/L      | 0.001 <sup>[1]</sup>                                     | USEPA 245.1 (FIMS) (1994)     |

<sup>[1]</sup> NEA's Allowable Limits for Trade Effluent Discharge to Controlled Watercourse guideline

<sup>[2]</sup> Vietnam's National Technical Regulation on Surface Water Quality for Protection of Aquatic Lives (QCVN 38: 2011/TTBTNMT) (2011)

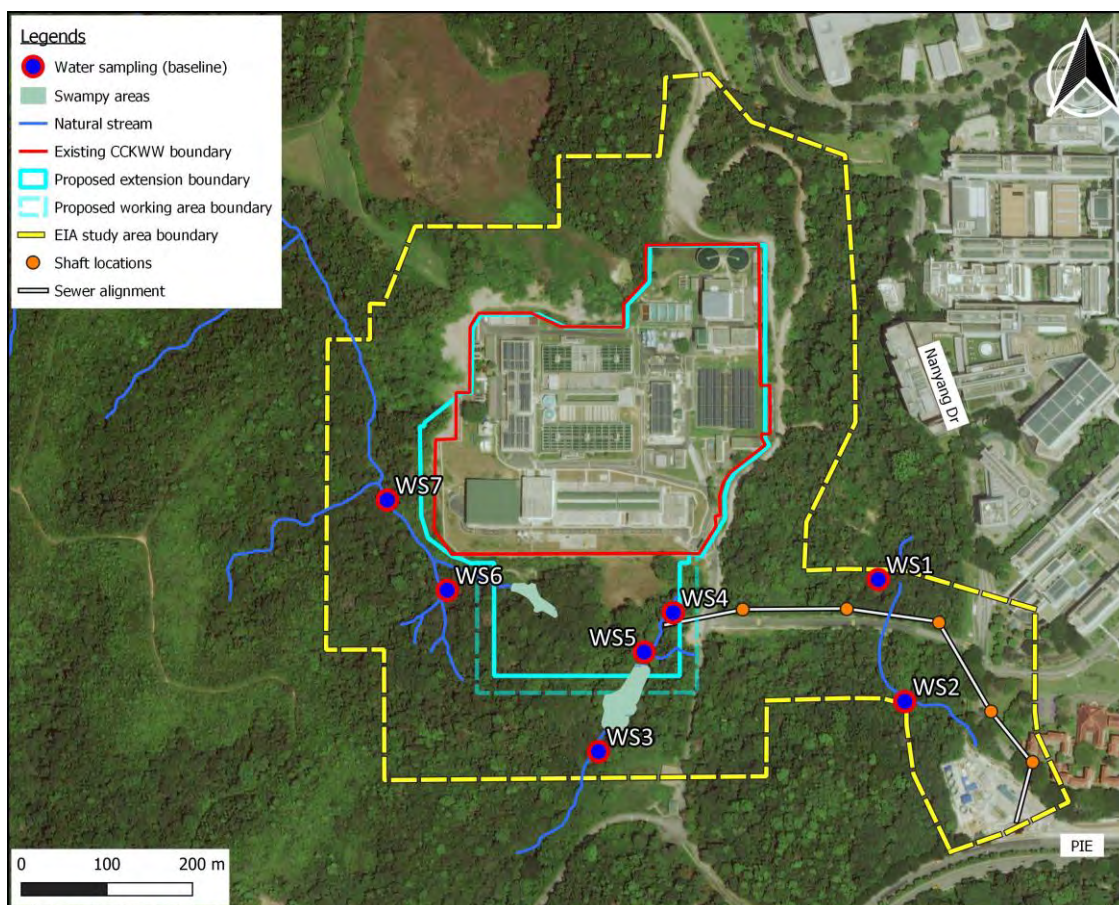
<sup>[3]</sup> Values for dry weather sample in Australian Guideline for Urban Stormwater Management (2000)

<sup>[4]</sup> Values for wet weather sample in Australian Guideline for Urban Stormwater Management (2000)

Sampling was conducted at seven (7) accessible points covering all three identified natural streams (Table 5.7, Figure 5.6). A total of three rounds of water quality monitoring were carried out. The dry weather samples were collected on 17–18 August 2021 and 7 September 2021, while the wet weather samples were collected on 21 September 2021. The SINGLAS-accredited Marchwood Laboratory Services Pte Ltd was appointed to conduct the surface water quality sampling and analysis of the collected water samples. Figure 5.7 depicts the surface water sampling activities.

**Table 5.7.** Coordinate of surface water quality sampling locations

| ID  | Coordinates |            | Date and Time of Sampling |                  |                 |
|-----|-------------|------------|---------------------------|------------------|-----------------|
|     |             |            | Dry Weather #1            | Dry Weather #2   | Wet Weather     |
| WS1 | 1.342220    | 103.678285 | 18/8/2021, 10:31          | 18/8/2021, 10:31 | 7/9/2021, 11:17 |
| WS2 | 1.341083    | 103.678298 | 18/8/2021, 11:00          | 18/8/2021, 11:00 | 7/9/2021, 11:30 |
| WS3 | 1.340453    | 103.675160 | 17/8/2021, 09:50          | 17/8/2021, 09:50 | 7/9/2021, 10:57 |
| WS4 | 1.341960    | 103.675856 | 18/8/2021, 09:37          | 18/8/2021, 09:37 | 7/9/2021, 10:38 |
| WS5 | 1.341404    | 103.675331 | 17/8/2021, 10:58          | 17/8/2021, 10:58 | 7/9/2021, 10:25 |
| WS6 | 1.342067    | 103.673358 | 17/8/2021, 10:48          | 17/8/2021, 10:48 | 7/9/2021, 10:06 |
| WS7 | 1.343307    | 103.672880 | 17/8/2021, 10:38          | 17/8/2021, 10:38 | 7/9/2021, 09:49 |



**Figure 5.6.** Location of surface water quality sampling points in project study area



**Figure 5.7.** Surface water sampling activities

## 5.6 Surface Water Quality Baseline Results and Discussion

The main standard used for surface water quality comparison was the Allowable Limits for Trade Effluent Discharge under EPM (Trade Effluent) Regulation (2020). The results were compared with Controlled Watercourse<sup>2</sup> limits. Moreover, the values of parameters

<sup>2</sup> Defined as “a watercourse from which water supplied by PUB under the Public Utilities Act is obtained but does not include a watercourse from which water is pumped into a main of the PUB” (National Environment Agency, 2020).

not regulated by Singapore standards were compared with relevant standards from other countries in the region. Hence, Dissolved Oxygen values were compared with the limit in Vietnam's National Technical Regulation QCVN 38:2011/ TTBTNMT on Surface Water Quality for Protection of Aquatic Lives (2011), while *Enterococcus* values were compared with the Australian Guideline for Urban Stormwater Management (2000).

The results of surface water quality analysis in comparison with the relevant standards are tabulated in Table 5.8, Table 5.9, and Table 5.10 respectively for the first round of dry weather sampling, second round of dry weather sampling, and wet weather sampling. The laboratory analysis reports are provided in **Appendix G**.

**Table 5.8.** Result of first dry weather period surface water quality samples

| Parameter                    | Unit           | Threshold Limit          | WS1         | WS2        | WS3         | WS4     | WS5         | WS6        | WS7         |
|------------------------------|----------------|--------------------------|-------------|------------|-------------|---------|-------------|------------|-------------|
| <b><i>In-situ</i></b>        |                |                          |             |            |             |         |             |            |             |
| Temperature                  | °C             | 45 <sup>[1]</sup>        | 25.9        | 26.4       | 25.9        | 26.0    | 25.9        | 25.9       | 25.7        |
| Conductivity                 | µS/cm          | –                        | 69.4        | 263        | 315         | 421     | 281         | 42.7       | 115         |
| pH                           | pH unit        | 6–9 <sup>[1]</sup>       | <b>3.8</b>  | <b>5.7</b> | <b>5.6</b>  | 6.9     | <b>5.6</b>  | <b>4.9</b> | 6.4         |
| Turbidity                    | NTU            | –                        | 12.2        | 31.7       | 44.1        | 14.1    | 50.8        | 21.1       | 19.1        |
| Dissolved Oxygen             | mg/L           | ≥ 4 <sup>[2]</sup>       | 7.14        | 7.66       | <b>3.35</b> | 5.70    | 5.37        | 7.02       | 6.87        |
| <b><i>Ex-situ</i></b>        |                |                          |             |            |             |         |             |            |             |
| Total Phosphorus             | mg/L           | –                        | 0.046       | 0.10       | 0.33        | 0.31    | 0.30        | 0.11       | 0.077       |
| Total Nitrogen               | mg/L           | –                        | 1.14        | 0.71       | 1.34        | 0.60    | 0.94        | 0.65       | 0.74        |
| Nitrate as NO <sub>3</sub>   | mg/L           | 20 <sup>[1]</sup>        | 4.46        | 1.54       | <0.015      | 0.68    | 0.37        | 2.26       | 1.76        |
| Phosphate as PO <sub>4</sub> | mg/L           | 2 <sup>[1]</sup>         | 0.13        | 0.17       | 0.33        | 0.58    | 0.32        | 0.14       | 0.16        |
| Total Suspended Solids       | mg/L           | 30 <sup>[1]</sup>        | <10         | 10         | <b>40.5</b> | <10     | 27.3        | 20.3       | <10         |
| Total Dissolved Solids       | mg/L           | 1,000 <sup>[1]</sup>     | 35          | 134        | 176         | 217     | 158         | 22         | 68          |
| Total Organic Carbon         | mg/L           | –                        | 2.05        | 4.41       | 10.1        | 9.23    | 10.3        | 1.16       | 4.02        |
| Ammoniacal Nitrogen          | mg/L           | 1 <sup>[2]</sup>         | 0.024       | 0.029      | <0.01       | <0.01   | <0.01       | 0.034      | 0.15        |
| Enterococcus                 | cfu/<br>100 mL | 40–40,000 <sup>[3]</sup> | <1          | <1         | <1          | 300     | <1          | 10         | 4           |
| Arsenic as As                | mg/L           | 0.01 <sup>[1]</sup>      | <0.023      | <0.023     | 0.016       | <0.023  | 0.011       | 0.0088     | <0.023      |
| Cadmium as Cd                | mg/L           | 0.003 <sup>[1]</sup>     | <0.0012     | <0.0012    | <0.0012     | <0.0012 | <0.0012     | <0.0012    | <0.0012     |
| Chromium as Cr               | mg/L           | 0.05 <sup>[1]</sup>      | <0.003      | <0.003     | <0.003      | <0.003  | <0.003      | <0.003     | <0.003      |
| Iron as Fe                   | mg/L           | 1 <sup>[1]</sup>         | <b>1.65</b> | 0.86       | <b>5.97</b> | 0.51    | <b>2.19</b> | 0.81       | <b>1.29</b> |
| Nickel as Ni                 | mg/L           | 0.1 <sup>[1]</sup>       | 0.0049      | <0.0045    | <0.0045     | <0.0045 | <0.0045     | <0.0045    | <0.0045     |
| Copper as Cu                 | mg/L           | 0.1 <sup>[1]</sup>       | <0.0015     | 0.0078     | 0.0019      | 0.0048  | 0.0022      | <0.0015    | <0.0015     |
| Lead as Pb                   | mg/L           | 0.1 <sup>[1]</sup>       | <0.009      | <0.009     | <0.009      | <0.009  | <0.009      | <0.009     | <0.009      |
| Zinc as Zn                   | mg/L           | 0.5 <sup>[1]</sup>       | 0.06        | 0.035      | 0.019       | 0.044   | 0.028       | 0.038      | 0.033       |
| Mercury as Hg                | mg/L           | 0.001 <sup>[1]</sup>     | <0.0001     | <0.0001    | <0.0001     | <0.0001 | <0.0001     | <0.0001    | <0.0001     |

Note: Values in Bold denote non-conformity to the respective water quality standard.

<sup>[1]</sup> NEA's Allowable Limits for Trade Effluent Discharge to Controlled Watercourse guideline (2020)

<sup>[2]</sup> Vietnam's National Technical Regulation on Surface Water Quality for Protection of Aquatic Lives (QCVN 38:2011/TTBTNMT) (2011)

<sup>[3]</sup> Values for dry weather sample in Australian Guideline for Urban Stormwater Management (2000)

**Table 5.9.** Result of second dry weather period surface water quality samples

| Parameter             | Unit | Threshold Limit   | WS1  | WS2  | WS3  | WS4  | WS5  | WS6  | WS7  |
|-----------------------|------|-------------------|------|------|------|------|------|------|------|
| <b><i>In-situ</i></b> |      |                   |      |      |      |      |      |      |      |
| Temperature           | °C   | 45 <sup>[1]</sup> | 26.9 | 27.1 | 26.4 | 27.2 | 25.9 | 26.1 | 25.8 |

| Parameter                    | Unit           | Threshold Limit          | WS1        | WS2        | WS3         | WS4     | WS5         | WS6         | WS7     |
|------------------------------|----------------|--------------------------|------------|------------|-------------|---------|-------------|-------------|---------|
| Conductivity                 | µS/cm          | –                        | 64         | 256        | 222         | 370     | 234         | 65.8        | 88.1    |
| pH                           | pH unit        | 6–9 <sup>[1]</sup>       | <b>4.7</b> | <b>5.7</b> | 6.4         | 6.1     | <b>5.6</b>  | <b>3.7</b>  | 6.5     |
| Turbidity                    | NTU            | –                        | 14.4       | 58.4       | 69.1        | 14.0    | 78.6        | 13.2        | 24.5    |
| Dissolved Oxygen             | mg/L           | ≥ 4 <sup>[2]</sup>       | 6.55       | 7.58       | <b>2.31</b> | 5.96    | 4.89        | 6.63        | 6.47    |
| <b><u>Ex-situ</u></b>        |                |                          |            |            |             |         |             |             |         |
| Total Phosphorus             | mg/L           | –                        | 0.057      | 0.073      | 0.26        | 0.23    | 0.2         | 0.055       | 0.044   |
| Total Nitrogen               | mg/L           | –                        | 0.87       | 0.67       | 0.77        | 0.39    | 0.68        | 1.2         | 0.76    |
| Nitrate as NO <sub>3</sub>   | mg/L           | 20 <sup>[1]</sup>        | 3.01       | 1.52       | 0.088       | 0.17    | 0.43        | 4.78        | 2.17    |
| Phosphate as PO <sub>4</sub> | mg/L           | 2 <sup>[1]</sup>         | 0.096      | 0.11       | 0.24        | 0.27    | 0.22        | 0.083       | 0.096   |
| Total Suspended Solids       | mg/L           | 30 <sup>[1]</sup>        | <10        | 29.3       | <b>34.7</b> | <10     | <b>33</b>   | <10         | <10     |
| Total Dissolved Solids       | mg/L           | 1,000 <sup>[1]</sup>     | 32         | 132        | 119         | 218     | 118         | 33          | 44      |
| Total Organic Carbon         | mg/L           | –                        | 2.07       | 3.61       | 10.9        | 7.57    | 7.81        | 1.69        | 3.62    |
| Ammoniacal Nitrogen          | mg/L           | 1 <sup>[2]</sup>         | 0.033      | 0.092      | 0.18        | 0.043   | 0.14        | 0.013       | 0.1     |
| Enterococcus                 | cfu/<br>100 mL | 40–40,000 <sup>[3]</sup> | 23         | 6          | 2,400       | 260     | 2,100       | 7           | 40      |
| Arsenic as As                | mg/L           | 0.01 <sup>[1]</sup>      | <0.023     | <0.023     | <0.023      | <0.023  | <0.023      | <0.023      | <0.023  |
| Cadmium as Cd                | mg/L           | 0.003 <sup>[1]</sup>     | <0.0012    | <0.0012    | <0.0012     | <0.0012 | <0.0012     | <0.0012     | <0.0012 |
| Chromium as Cr               | mg/L           | 0.05 <sup>[1]</sup>      | <0.003     | <0.003     | <0.003      | <0.003  | <0.003      | <0.003      | <0.003  |
| Iron as Fe                   | mg/L           | 1 <sup>[1]</sup>         | 0.43       | 0.78       | <b>1.6</b>  | 0.82    | <b>1.66</b> | <b>1.03</b> | 0.85    |
| Nickel as Ni                 | mg/L           | 0.1 <sup>[1]</sup>       | <0.0045    | <0.0045    | <0.0045     | <0.0045 | <0.0045     | 0.0047      | <0.0045 |
| Copper as Cu                 | mg/L           | 0.1 <sup>[1]</sup>       | 0.0018     | 0.006      | 0.0054      | 0.0026  | 0.002       | 0.0026      | <0.0015 |
| Lead as Pb                   | mg/L           | 0.1 <sup>[1]</sup>       | <0.015     | <0.015     | <0.015      | <0.015  | <0.015      | <0.015      | <0.015  |
| Zinc as Zn                   | mg/L           | 0.5 <sup>[1]</sup>       | 0.053      | 0.0017     | 0.031       | 0.041   | 0.019       | 0.043       | 0.025   |
| Mercury as Hg                | mg/L           | 0.001 <sup>[1]</sup>     | <0.0001    | <0.0001    | <0.0001     | <0.0001 | <0.0001     | <0.0001     | <0.0001 |

Note: Values in Bold denote non-conformity of the respective water quality standard.

<sup>[1]</sup> NEA's Allowable Limits for Trade Effluent Discharge to Controlled Watercourse guideline (2020)

<sup>[2]</sup> Vietnam's National Technical Regulation on Surface Water Quality for Protection of Aquatic Lives (QCVN 38:2011/TTBTNMT) (2011)

<sup>[3]</sup> Values for dry weather sample in Australian Guideline for Urban Stormwater Management (2000)

**Table 5.10.** Result of wet weather period surface water quality samples

| Parameter                    | Unit           | Threshold Limit          | WS1        | WS2       | WS3         | WS4   | WS5   | WS6        | WS7        |
|------------------------------|----------------|--------------------------|------------|-----------|-------------|-------|-------|------------|------------|
| <b><u>In-situ</u></b>        |                |                          |            |           |             |       |       |            |            |
| Temperature                  | °C             | 45 <sup>[1]</sup>        | 27.9       | 28.6      | 26.9        | 27.3  | 27.4  | 27.1       | 26.9       |
| Conductivity                 | µS/cm          | –                        | 50         | 248       | 315         | 377   | 304   | 45.4       | 154        |
| pH                           | pH unit        | 6–9 <sup>[1]</sup>       | <b>5.6</b> | 6.8       | 6.6         | 6.4   | 6.7   | <b>5.1</b> | <b>5.6</b> |
| Turbidity                    | NTU            | –                        | 17.5       | 74.5      | 33.1        | 22    | 53.5  | 57.3       | 75.8       |
| Dissolved Oxygen             | mg/L           | ≥ 4 <sup>[2]</sup>       | 6.13       | 7.37      | <b>2.59</b> | 5.2   | 4.81  | 6.87       | 6.78       |
| <b><u>Ex-situ</u></b>        |                |                          |            |           |             |       |       |            |            |
| Total Phosphorus             | mg/L           | –                        | 0.055      | 0.057     | 0.3         | 0.14  | 0.13  | 0.061      | 0.054      |
| Total Nitrogen               | mg/L           | –                        | 0.63       | 0.65      | 1.16        | 0.46  | 0.64  | 0.66       | 0.66       |
| Nitrate as NO <sub>3</sub>   | mg/L           | 20 <sup>[1]</sup>        | 1.91       | 1.25      | <0.07       | 0.37  | <0.07 | 1.46       | 0.77       |
| Phosphate as PO <sub>4</sub> | mg/L           | 2 <sup>[1]</sup>         | 0.14       | 0.14      | 0.32        | 0.26  | 0.2   | 0.1        | 0.13       |
| Total Suspended Solids       | mg/L           | 30 <sup>[1]</sup>        | <10        | <b>32</b> | 19.3        | <10   | 23.7  | 26.3       | 23         |
| Total Dissolved Solids       | mg/L           | 1,000 <sup>[1]</sup>     | 27         | 121       | 154         | 187   | 146   | 23         | 78         |
| Total Organic Carbon         | mg/L           | –                        | 2.76       | 3.2       | 11.3        | 7.32  | 10.5  | 4.5        | 6.45       |
| Ammoniacal Nitrogen          | mg/L           | 1 <sup>[2]</sup>         | 0.064      | 0.07      | 0.51        | 0.14  | 0.24  | 0.13       | 0.26       |
| Enterococcus                 | cfu/<br>100 mL | 40–40,000 <sup>[3]</sup> | 54         | 1,300     | 4,900       | 1,200 | 1,100 | 470        | 600        |

| Parameter      | Unit | Threshold Limit      | WS1     | WS2     | WS3         | WS4         | WS5         | WS6     | WS7        |
|----------------|------|----------------------|---------|---------|-------------|-------------|-------------|---------|------------|
| Arsenic as As  | mg/L | 0.01 <sup>[1]</sup>  | <0.023  | <0.023  | <0.023      | <0.023      | <0.023      | <0.023  | <0.023     |
| Cadmium as Cd  | mg/L | 0.003 <sup>[1]</sup> | <0.0012 | <0.0012 | <0.0012     | <0.0012     | <0.0012     | <0.0012 | <0.0012    |
| Chromium as Cr | mg/L | 0.05 <sup>[1]</sup>  | <0.003  | <0.003  | <0.003      | <0.003      | <0.003      | <0.003  | <0.003     |
| Iron as Fe     | mg/L | 1 <sup>[1]</sup>     | 0.6     | 0.93    | <b>3.18</b> | <b>1.06</b> | <b>1.65</b> | 0.91    | <b>1.6</b> |
| Nickel as Ni   | mg/L | 0.1 <sup>[1]</sup>   | <0.0045 | <0.0045 | <0.0045     | <0.0045     | <0.0045     | <0.0045 | <0.0045    |
| Copper as Cu   | mg/L | 0.1 <sup>[1]</sup>   | <0.0015 | 0.011   | 0.0019      | 0.0049      | 0.0017      | 0.0026  | 0.0021     |
| Lead as Pb     | mg/L | 0.1 <sup>[1]</sup>   | <0.015  | <0.015  | <0.015      | <0.015      | <0.015      | <0.015  | <0.015     |
| Zinc as Zn     | mg/L | 0.5 <sup>[1]</sup>   | 0.04    | 0.032   | 0.012       | 0.042       | 0.014       | 0.037   | 0.026      |
| Mercury as Hg  | mg/L | 0.001 <sup>[1]</sup> | <0.0001 | <0.0001 | <0.0001     | <0.0001     | <0.0001     | <0.0001 | <0.0001    |

Note: Values in Bold denote non-conformity to the respective water quality standard.

<sup>[1]</sup> NEA's Allowable Limits for Trade Effluent Discharge to Controlled Watercourse guideline (2020)

<sup>[2]</sup> Vietnam's National Technical Regulation on Surface Water Quality for Protection of Aquatic Lives (QCVN 38:2011/TTBTNMT) (2011)

<sup>[3]</sup> Values for dry weather sample in Australian Guideline for Urban Stormwater Management (2000)

In general, the surface water quality of streams within the project study area complied with the applicable standards except for the pH readings at most of the points (WS1–3 and WS5–7), Dissolved Oxygen measured at WS3, and Iron as Fe at WS3, 5, and 7.

While some of the points within the project site had low pH readings (i.e., WS2 and WS5 were slightly more acidic in dry weather while only a sample each of WS3 and WS7 was more acidic than the recommended range), forested streams in Singapore are known be mildly acidic. Also, all pH readings at WS1 and WS6 fell below NEA's (2020) standard range. These consistent results indicate that the surface water quality at these points had been more acidic than the recommended range in the first place. Several factors may contribute to this condition, such as low buffering related to acidic granite and anthropogenic acid rain (Nguyen, Wasson, & Ziegler, 2018).

On the other hand, all Dissolved Oxygen values measured at WS3 were under 4 mg/L, which is below the threshold considered ideal for the survival of aquatic life (Ministry of Natural Resources and Environment of Vietnam, 2011). This implies that the southern stretch of Stream 2 may not be favourable for aquatic life.

Surface water quality sampling was conducted by previous EIS (2016) at the Stream 3 locations which also found that the stream water quality was acidic.

These results provide the short-term trend based on current surveys. A longer-term water quality monitoring data can help to define the water quality of these streams more accurately based on long-term trend.

## **5.7 Impact Assessment**

### **5.7.1 Identification of Potential Impacts**

#### ***Pre-construction / Construction Phase***

##### **Area 1 (Existing CCKWW Plant)**

Project activities planned in this Area include the demolition of existing infrastructures and construction of new facilities. Based on these activities, the following potential impacts on hydrology and surface water quality are identified:

##### **Soil Erosion and Surface Run-off**

Construction run-offs may cause physical, biological, and/or chemical impacts to the water streams within study area during the construction of new facilities within CCKWW plant. The sources of construction run-offs include:

- Land clearance and site preparation of construction work areas
- Run-off from exposed soil surface and earth working areas.
- Run-off from dust suppression sprays

The physical effects include potential blockage of drainage channels and increased concentrations of suspended solids in the drainage channels. Adverse impacts to drainage pattern may occur from soil erosion and sediment transport during storm events. Impacts would be greatest in areas with exposed, disturbed soil. This impact could worsen if there are soil stockpiles left unprotected in the construction area and no mitigation measures in place. Run-offs containing particulates will increase concentrations of suspended solids, sediments, and other contaminants and thus create additional burdens on the existing drainage system in the project site.

##### **Trade Effluent**

The construction activities such as piling works and soil investigation are expected to generate trade effluent from construction works at Area 1. These includes process wastewater such as bentonite slurry, concrete washout, and excess grouting materials from construction activities. Such wastewater is alkaline, consists of fine particles and may contain heavy metals. Discharge of the trade effluent into the watercourse either directly or indirectly could adversely impact the water quality of the receiving watercourse.

This trade effluent shall not be discharged to any receiving watercourse without prior treatment. It must be treated separately from surface runoff before being discharged. The treated trade effluent from construction activities must have written approval from NEA if it will be discharged into public storm water or PUB if it will be discharged into public sewers. The trade effluent must also be treated to levels prescribed by NEA if it will be discharged into the public drain or PUB if it will be discharged into the public sewer.

##### **Fuel & Chemical Spillage and Waste Disposal**

The construction activities may require on-site storage and handling of potential polluting material such as bentonite, fuel, lubricant, cement, as well as packaging materials. The following sources may also affect the water quality within the project area:

- Fuel and lubricants from maintenance of construction vehicles and equipment
- Illegal dumping of debris and rubbish such as packaging, construction materials, and refuse; and
- Accidental spillage of liquids stored on-site; e.g., oil/grease, solvents.

No hazardous chemicals as specified under EPM (Hazardous Substances) regulations are expected to be used in the construction activities in this Area. It is expected that the Contractor will deploy adequate portable chemical toilets for construction personnel. These toilets shall be cleaned and maintained by approved sanitary waste collector. Hence, no impact is expected from sewage discharge on surface water quality.

The impacts and effects on water quality from the construction activities is expected to be controlled with the implementation of a well-designed temporary drainage system and proposed mitigation measure provided in following Section.

### **Area 2 (CCKWW Extension Area)**

Area 2 is expected to be the most impacted site with its secondary forest and two natural streams. Project activities planned during pre-construction and construction phase in this Area include soil investigation works consisting of twelve boreholes, site clearance, soil filling and earthworks for the specified platform level, and construction of post-treatment units and supporting infrastructures.

Based on these activities, the following potential impacts on hydrology and surface water quality are identified:

#### **Disturbance to Local Hydrological Pattern**

Two natural streams are located within or nearby the project footprint of this Area 2, namely Stream 2 and 3 (see Figure 5.3) which will be disturbed. Northern part of stream 2 including swampy areas within Area 2 will be backfilled. For Stream 3, the section of eastern branch and swampy area will be backfilled to cater for the construction of new facilities within Area 2. The streams are currently partially fed by drainage from the existing CCKWW plant. As these streams provide natural channels for water circulation in the project study area, their loss is likely to disturb its hydrological pattern. Additionally, these natural streams are important habitats for aquatic fauna and construction activities may cause degradation and loss of these habitats.

#### **Impact on Catchment Area/Reservoirs**

The catchment area is undulating, with the project area being in general of higher elevation than the reservoirs within the Western Catchment Area. Stream 3 flows in the general direction of Poyan Reservoir. Change in drainage patterns may alter water flow to reservoir.

Lastly, impacts on water quality due to sediment runoff and chemical spillage described in the preceding paragraphs may have adverse impacts on reservoir water quality.

#### **Soil Erosion and Surface Run-off**

Construction run-offs may cause physical, biological, and/or chemical impacts to the

water streams within study area during the construction of new facilities within CCKWW plant. The sources of construction run-offs include:

- Land clearance and site preparation of construction work areas
- Run-off from exposed soil surface and earth working areas.
- Run-off from dust suppression sprays
- Soil investigation works

The physical effects include potential blockage of drainage channels and increased concentrations of suspended solids in the drainage channels. Adverse impacts to drainage pattern may occur from soil erosion and sediment transport during storm events. Impacts would be greatest in areas with exposed, disturbed soil. This impact could worsen if there are soil stockpiles left unprotected in the construction area and no mitigation measures in place. Run-offs containing particulates will increase concentrations of suspended solids, sediments, and other contaminants and thus create additional burdens on the existing drainage system in the project site.

Significant amounts of silt discharge into water streams may affect the local ecology if proper mitigation measures are not implemented.

#### Trade Effluent

The construction activities such as piling works and soil investigation are expected to generate trade effluent from construction works at Area 1, which include processed wastewater such as bentonite slurry, concrete washout, and excess grouting materials from construction activities. The trade effluent containing significant amount of bentonite, concrete, and cement-derived materials may cause primary chemical effects such as increasing turbidity, discoloration, elevation in pH, and accretion of solids. A number of secondary effects may also result in intoxication of water biota due to elevated pH value, and reduced decay rate of microorganisms and photosynthetic rate due to the decreased light penetration.

Hence, this trade effluent from construction site shall not be discharged to any receiving watercourse without prior treatment. It must be treated separately from surface runoff before being discharged. The treated trade effluent from construction activities must have written approval from NEA if it will be discharged into public storm water or PUB if it will be discharged into public sewers. The trade effluent must also be treated to levels prescribed by NEA if it will be discharged into the public drain or PUB if it will be discharged into the public sewer.

#### Fuel & Chemical Spillage and Waste Disposal

The construction activities may require on-site storage and handling of potential polluting material such as bentonite, fuel, lubricant, cement, as well as packaging materials. The following sources can affect the water quality within project area.

- Fuel and lubricants from maintenance of construction vehicles and equipment;
- Illegal dumping of debris and rubbish (such as packaging, construction materials, and refuse); and
- Accidental spillage of liquids stored on-site; e.g., oil/grease, solvents.

No hazardous chemicals as specified under EPM (Hazardous Substances) regulations are expected to be used in the construction activities in this Area.

### Sewage Discharge

The onsite sanitary facilities within construction work areas will generate sewage. As the Area 2 is located within ecological sensitive area, the discharge of sewage into the surrounding water streams either directly or indirectly could adversely impact the water quality of the receiving water streams. It is expected that the Contractor will deploy adequate portable chemical toilets for construction personnel. These toilets shall be cleaned and maintained by approved sanitary waste collector. Hence, no impact is expected from sewage discharge on surface water quality.

The impacts and effects on water quality from the construction activities is expected to be controlled with the implementation of a well-designed temporary drainage system within work area and proposed mitigation measure provided in the next Section.

### Area 3 (Sewerage Alignment)

As noted earlier, the new sewerage alignment will be constructed by pipejacking method. Hence, impacts on the hydrology and surface water quality during construction phase in this Area are expected to not be as significant as the other Areas.

Nevertheless, pipejacking works are likely to have some small-scale impacts on water quality. The construction activities may require on-site storage and handling of potential polluting material such as bentonite, fuel, lubricant, cement, as well as packaging materials. The following sources can affect the water quality within project area.

- Effluent from bentonite slurry;
- Discharge of silty groundwater dewatering effluent;
- Illegal dumping of debris and rubbish (e.g., packaging, construction materials, and refuse); and
- Accidental spillage of liquids stored on-site; e.g., oil/grease, solvents.

Overall, potential impacts on hydrology and water quality are expected to be of slight negative to minor negative nature, direct, and local. Some of the potential impacts are deemed to be of long term and permanent in nature while the others are short-term and reversible. Appropriate measures are to be proposed to mitigate these impacts to acceptable level.

### Operation Phase

During the operation phase of CCKWW plant, the process effluent will be discharged to DTSS system and new plant facilities will have adequately designed internal drainage system. There is currently a plan to discharge storm water runoff from the new plant facility into Stream 2 when it is operational. This storm water runoff will not include any process effluent and hence not likely to affect the surface water quality of the balance section of Stream 2.

No impacts on the hydrology and surface water quality of the surrounding environment are expected to be generated during operation of sewerage system.

### **Cumulative Impacts**

There is a possibility that proposed LTA JRL development (see Section 2.10) located adjacent to Area 3, would happen concurrently with this project. Impacts to water quality and hydrology for this JRL development include sedimentation in water courses due to silty runoff, contamination of water courses due to chemical spills, and degradation of water quality due to sewerage leakage (TY Lin International, 2018).

As the JRL development works are of a much larger scale compared to this project's sewerage alignment works, should the projects be implemented concurrently, cumulative impacts on water quality are expected. In order to address the cumulative impacts of both developments, all impacts from Area 3 were given a score of 3 in the 'Cumulativity' column of RIAM table to calculate the ES and assess overall impact.

#### **5.7.2 Evaluation of Potential Impacts**

Table 5.11 to Table 5.13 summarise the impacts with their corresponding Environmental Scores before and after the implementation of mitigation measures, which are elaborated in the following section.

### **5.8 Mitigation Measures**

#### **5.8.1 Pre-construction / Construction Phase**

##### ***Area 1 (Existing CCKWW Plant)***

In order to mitigate impacts to hydrology and water streams from Area 1 activities, the following measures are recommended.

##### **Implementation of Earth Control Measures (ECM)**

An appropriate ECM plan is to be prepared and endorsed by a Qualified Erosion Control Professional (QECP) before the commencement of construction works in all Areas. Moreover, the ECM plan should cover the following items:

- Contractor to engage QECP to prepare the ECM plan and PUB's approval to be obtained for ECM plan.
- ECM plan to be implemented by the Contractor before starting the work.
- ECM tanks utilised for the project must have surplus capacities to minimise overflow and silty discharge.
- Proper sediment control measures designed to capture and retain silt are to be implemented which may include perimeter cut-off drains, perimeter silt fence, silt traps and silt treatment systems.
- QECP to review ECM plan implementation regularly during the construction period to ensure that the measures put in place remain effective.
- Regular monitoring of ECM treatment plant performance is to be carried out by the Environmental Control Officer (ECO).
- Regular maintenance of ECM is to be carried out.
- After rain events, earth control measures in the Area are to be inspected and maintained over the course of construction.
- To ensure that site discharges are treated to comply with respective allowable limits (e.g., 50 mg/L of TSS for discharges processed through ECM and 30 mg/L

of TSS for trade effluent) before discharging into a public drain.

- Exposed earth areas shall be reduced by phasing earth works and effective scheduling of construction activities to minimise bare surfaces, planning of site layout and effective site drainage systems to facilitate flow. Measures may include the use of paving, biodegradable erosion control blankets, and canvas covers.

#### *Fuel & Chemical Spillage and Waste Disposal Management*

- All slurry preparation activities are to be conducted within a designated area that has adequate containment measures.
- To ensure that slurry water, including bentonite slurry, is contained within the working area and does not enter any watercourses or surface water drains.
- Bentonite storage silos and supply lines should be located as far as possible from surface water drains or watercourses.
- Storage silos should have secondary containment.
- Areas where bentonite is mixed are to be surrounded with a small boundary wall or contained within a bund. This action will help to control the slurry produced and prevent it from entering surface water drains or watercourses.
- Bentonite use should be monitored so that excessive use could be identified to prevent bentonite materials from escaping into the ground and potentially polluting groundwater.
- Waste bins to be provided within the construction work area for proper disposal of construction debris and rubbish. Foreign material should not be illegally disposed at the water bodies but removed from site by a licensed waste collector.
- All liquids should be properly labelled and stored in appropriate containers at safe location on site (i.e., under a shelter with kerb surrounding). Personnel handling these liquids should also be suitably trained.
- Vehicle fuelling and major maintenance to be minimised within the project area.
- Appropriate concrete wash out areas should be provided and should not be performed in or near any water body.
- Water used for dust control should not be allowed to cause erosion within the work area or to run offsite.
- Excess loose soil and rock to be contained prior to the commencement of works.

#### *Trade Effluent Management*

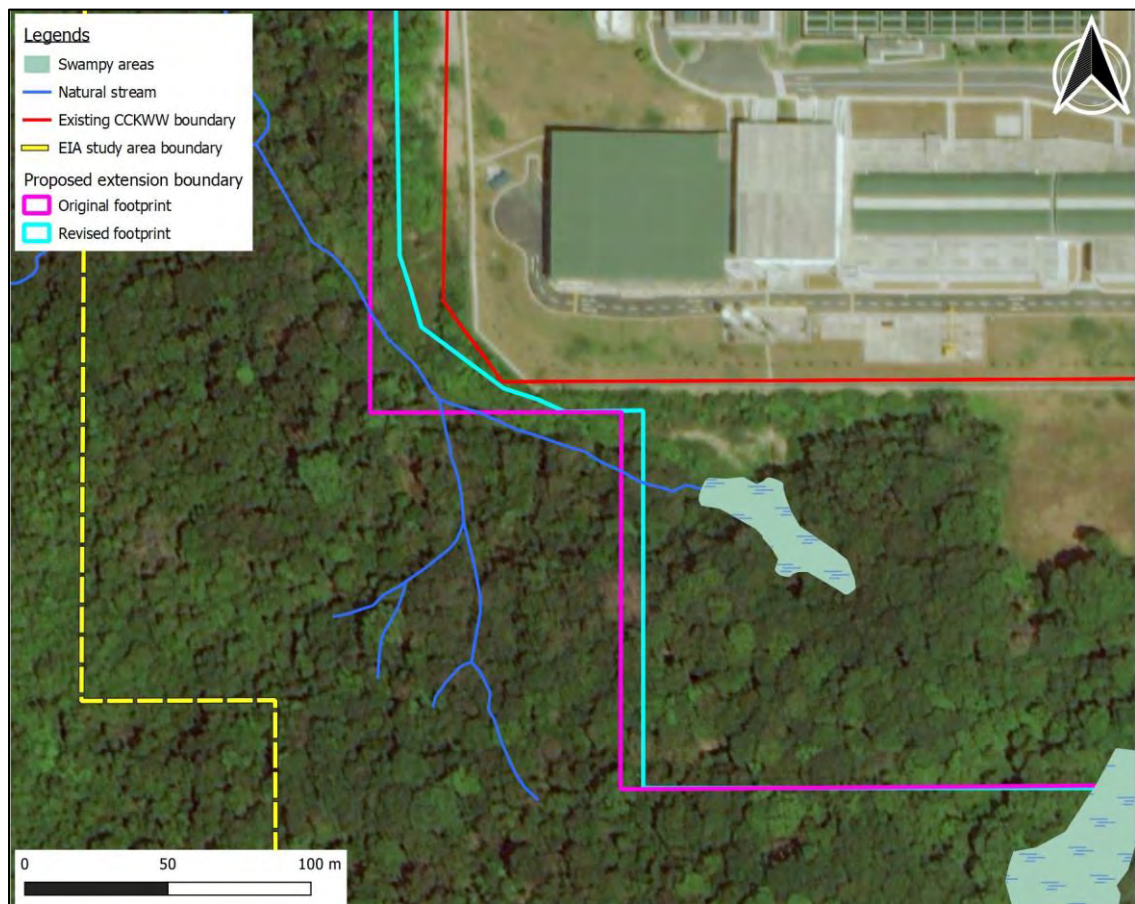
- Contractor to provide storage tanks of adequate size and sufficient numbers to temporarily store trade effluent before treatment and disposal.
- Contractor to provide sampling point and inspection for collection of trade effluent to be tested for parameters stipulated in the Regulations, depending on the discharge point (i.e., sewer, watercourse, controlled watercourse).
- Regular water quality monitoring is to be conducted according to the proposed Environmental Monitoring Plan during construction and post-construction stage.
- Contractor should install and maintain trade effluent treatment facilities (e.g., slurry treatment plant, grease trap and suspended solid settling tank) as per PUB's requirements.

## Area 2 (CCKWW Extension Area)

In order to mitigate impacts to hydrology and water streams from Area 2 activities, the following measures can be adopted.

### Disturbance to Local Hydrological Pattern

Due to the importance of Stream 3 as a habitat for aquatic fauna, the planned project footprint has been amended to include a vegetated buffer of at least 10 m between the working area boundary of the development and Stream 3 (Figure 5.8). The edges of the stream could be further enhanced with native riparian vegetation to increase the resilience of the stream habitat to impacts from changes due to the proposed development.



**Figure 5.8.** Project boundary, including original footprint (magenta) and revised footprint (cyan)  
N.B. sections of Stream 3 within the revised footprint will be backfilled.

To ensure that mitigation measures are effective in protecting the stream habitat, regular monitoring of Stream 3 is to be conducted. Details of the specific monitoring requirement is provided in Section 13.5 as part of EMMP framework.

Other mitigation measures for changes to the local hydrological pattern include:

- The flow of the remaining Stream 2 is to be maintained by stormwater drainage discharge of the new facility.
- Targeted translocation of native aquatic fauna from Stream 2 and Stream 3 to be

conducted (see Section 4.6).

- Ensure that none of the site investigation boreholes are located within 10 m of the natural streams.

#### Earth Control Measures (ECM)

An appropriate ECM plan is to be prepared and endorsed by a Qualified Erosion Control Professional (QECP) before the commencement of construction works in all Areas. The ECM plan should cover the following items:

- Contractor to engage QECP to prepare the ECM plan and PUB's approval to be obtained for ECM plan.
- ECM plan to be implemented by the Contractor before starting the work.
- ECM tanks utilised for the project must have surplus capacities to minimise overflow and silty discharge.
- Proper sediment control measures designed to capture and retain silt are to be implemented which may include perimeter cut-off drains, perimeter silt fence, silt traps and silt treatment systems.
- QECP to review ECM plan implementation regularly during the construction period to ensure that the measures put in place remain effective.
- Regular monitoring of ECM treatment plant performance is to be carried out by the Environmental Control Officer (ECO).
- Regular maintenance of ECM is to be carried out (e.g., cleaning of discharge pipes).
- The Contractor shall access weather forecast from the NEA website to take ECM actions before rain events.
- After rain events, earth control measures in the Area are to be inspected and maintained over the course of construction.
- To ensure that site discharges are treated to comply with respective allowable limits (e.g., 50 mg/L of TSS for discharges processed through ECM and 30 mg/L of TSS for trade effluent) before discharging into a public drain.
- Exposed earth areas shall be reduced by phasing earth works and effective scheduling of construction activities to minimise bare surfaces, planning of site layout and effective site drainage systems to facilitate flow. Measures may include the use of biodegradable erosion control blankets, paving, and canvas covers.
- ECM should only be removed after completion of the works. The QECP should authorize removal of the ECM.

Moreover, a detailed approach for the ECM plan has been prepared to address potential issues of siltation in Area 2 during backfilling works and construction stage in general. Implementation of this approach is expected to prevent silty water from discharging/spilling into natural streams in the area. The proposed approach can be found in **Appendix H** of this report.

#### Trade Effluent Management

- Trade effluent is forbidden from being discharged into Stream 2 and 3.
- Contractor to provide storage tanks of adequate size and sufficient numbers to temporarily store trade effluent before treatment and disposal.

- Area where trade effluent is being transported out to be kerbed to minimise spills
- Contractor to provide sampling point and inspection for collection of trade effluent to be tested for parameters stipulated in the Regulations, depending on the type of discharge point (i.e., sewer, watercourse).
- Regular water quality monitoring is to be conducted according to the proposed Environmental Monitoring Plan during construction and post-construction stage.
- Contractor should install and maintain trade effluent treatment facilities (e.g., slurry treatment plant, grease trap and suspended solid settling tank) as per PUB's requirements.

#### *Oil & Fuel Spillage and Waste Disposal*

- To ensure that bentonite is contained within the working area and does not enter any watercourses or surface water drains.
- Bentonite storage silos and supply lines should be located as far as possible from surface water drains or watercourses.
- Storage silos should have secondary containment.
- Areas where bentonite is mixed are to be surrounded with a small boundary wall or contained within a bund. This action will help to control the slurry produced and prevent it from entering surface water drains or watercourses.
- Bentonite use should be monitored so that excessive use could be identified to prevent bentonite materials from escaping into the ground and potentially polluting groundwater.
- Waste bins to be provided within the construction work area for proper disposal of construction debris and rubbish. Foreign material should not be illegally disposed at the water bodies but removed from site by a licensed waste collector.
- All liquids should be properly labelled and stored in appropriate containers at safe location on site (i.e., under a shelter with kerb surrounding). Personnel handling these liquids should also be suitably trained.
- Vehicle fuelling and major maintenance to be minimised within the project area.
- Appropriate concrete wash out areas should be provided and should not be performed in or near any water body.
- Water used for dust control should not be allowed to cause erosion within the work area or to run offsite.
- Excess loose soil and rock to be contained prior to the commencement of the works.
- Emergency spill kits should be provided by contractor on site in the event of any chemical spillages. Emergency response team should also be trained in the use of these spill kits.

#### *Area 3 (Sewerage Alignment)*

In order to mitigate impacts to hydrology and water streams from Area 3 activities, the following measures can be adopted.

#### *Earth Control Measures (ECM)*

An appropriate ECM plan is to be prepared and endorsed by a Qualified Erosion Control Professional (QECP) before the commencement of construction works in all Areas. ECM

tanks utilised for the project must have surplus capacities to minimise overflow and silty discharge.

#### *Chemical & Fuel Spillage and Waste Disposal*

- Regular water quality monitoring is to be conducted according to the proposed Environmental Monitoring Plan during construction and post-construction stage.
- All slurry preparation activities are to be conducted within a designated area that has adequate containment measures.
- To ensure that slurry water, including bentonite slurry, is contained within the working area and does not enter any watercourses or surface water drains.
- Bentonite storage silos and supply lines should be located as far as possible from surface water drains or watercourses.
- Storage silos should have secondary containment.
- Foreign material should not be illegally disposed at the water bodies but removed from site by a licensed waste collector.
- All liquids should be properly labelled and stored in appropriate containers at safe location on site. Personnel handling these liquids should also be suitably trained.
- No vehicle fuelling and maintenance to be allowed within the project area.
- Excess loose soil and rock to be contained prior to the commencement of works.

#### **5.8.2 Operation Phase**

There is currently a plan to discharge storm water runoff from the new plant facility into Stream 2 when it is operational. The design of this discharge outlet is yet to be finalised. It is expected that it will be designed in consultation with relevant authorities (e.g., NParks, PUB).

As no impact are expected on the hydrology and surface water quality of surrounding environment during operation of sewerage alignment, no mitigation measures are proposed.

#### **5.9 Residual Impacts**

The residual impacts were evaluated using the RIAM method (Table 5.11 to Table 5.13) with due consideration that the recommended mitigation measures are implemented by the Contractor. The residual impacts are likely to be in the slight negative band to minor negative and considered acceptable.

**Table 5.11.** Environmental Scores of the identified impacts on site's hydrology & water quality with corresponding mitigation measures (Area 1)

| Phase                           | Impact component                                                                                 | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Predicted impact after mitigation |    |   |   |   |     |                 |
|---------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                                 |                                                                                                  | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | I                                 | M  | P | R | C | ES  | Impact          |
| Ecology                         |                                                                                                  |                                    |    |   |   |   |     |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Impact to water quality due to sediment runoff and siltation and trade effluent discharge        | 2                                  | -3 | 2 | 2 | 3 | -42 | Minor negative  | <ul style="list-style-type: none"><li>• Implementation proper Earth Control Measures (ECMs).</li><li>• Contractor to provide storage tanks of adequate size and sufficient numbers to temporarily store trade effluent before treatment and disposal.</li></ul> Contractor to provide sampling point and inspection for collection of trade effluent to be tested for parameters stipulated in the Regulations, depending on the discharge point. <ul style="list-style-type: none"><li>• Regular water quality monitoring is to be conducted according to the proposed Environmental Monitoring Plan during construction and post-construction stage.</li><li>• Contractor should install and maintain trade effluent treatment facilities as per NEA's requirements</li><li>• Appropriate concrete wash out areas should be provided and should not be performed in or near any water body.</li></ul> | 2                                 | -2 | 2 | 2 | 3 | -28 | Slight negative |
|                                 | Impact to water quality due to accidental spillage of construction materials, fuel, and solvents | 2                                  | -2 | 2 | 2 | 3 | -28 | Slight negative | <ul style="list-style-type: none"><li>• To ensure that bentonite is contained within the working area and does not enter any watercourses or surface water drains.</li><li>• Storage silos should have secondary containment.</li><li>• Bentonite use should be monitored so that excessive use could be identified to prevent bentonite materials from escaping into the ground and potentially polluting groundwater.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2                                 | -1 | 2 | 2 | 3 | -14 | Slight negative |

| Phase                           | Impact component                                                                                 | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Predicted impact after mitigation |    |   |   |   |     |                 |
|---------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                                 |                                                                                                  | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | I                                 | M  | P | R | C | ES  | Impact          |
|                                 |                                                                                                  |                                    |    |   |   |   |     |                 | <ul style="list-style-type: none"> <li>Waste bins to be provided within the construction work area for proper disposal of construction debris and rubbish.</li> <li>Vehicle fuelling and major maintenance to be minimised within the project area.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                             |                                   |    |   |   |   |     |                 |
| <b>Human Health and Safety</b>  |                                                                                                  |                                    |    |   |   |   |     |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Impact to water quality due to accidental spillage of construction materials, fuel, and solvents | 2                                  | -2 | 2 | 2 | 2 | -24 | Slight negative | <ul style="list-style-type: none"> <li>Contractor to provide storage tanks of adequate size and sufficient numbers to temporarily store trade effluent before treatment and disposal.</li> <li>Contractor to provide sampling point and inspection for collection of trade effluent to be tested for parameters stipulated in the Regulations, depending on the discharge point (i.e., sewer, watercourse, controlled watercourse).</li> <li>Contractor should install and maintain trade effluent treatment facilities (e.g., slurry treatment plant, grease trap and suspended solid settling tank) as per NEA's requirements</li> </ul> | 2                                 | -1 | 2 | 2 | 2 | -12 | Slight negative |

**Table 5.12.** Environmental Scores of the identified impacts on site's hydrology & water quality with corresponding mitigation measures (Area 2)

| Phase              | Impact component                        | Predicted impact before mitigation |    |   |   |   |     |                   | Mitigation measures                                                                                                                                                              | Predicted impact after mitigation |    |   |   |   |     |                |
|--------------------|-----------------------------------------|------------------------------------|----|---|---|---|-----|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|----------------|
|                    |                                         | I                                  | M  | P | R | C | ES  | Impact            |                                                                                                                                                                                  | I                                 | M  | P | R | C | ES  | Impact         |
| Ecology            |                                         |                                    |    |   |   |   |     |                   |                                                                                                                                                                                  |                                   |    |   |   |   |     |                |
| Pre-construction / | Degradation and loss of stream habitats | 3                                  | -4 | 3 | 3 | 2 | -96 | Moderate negative | <ul style="list-style-type: none"><li>Alter project footprint to minimise direct impacts on Stream 3</li><li>Enhance edges of Stream 3 with native riparian vegetation</li></ul> | 3                                 | -2 | 3 | 3 | 2 | -48 | Minor negative |

| Phase | Impact component                                             | Predicted impact before mitigation |    |   |   |   |     |                   | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Predicted impact after mitigation |    |   |   |   |     |                 |
|-------|--------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|       |                                                              | I                                  | M  | P | R | C | ES  | Impact            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | I                                 | M  | P | R | C | ES  | Impact          |
|       |                                                              |                                    |    |   |   |   |     |                   | <ul style="list-style-type: none"> <li>Ensure that none of the site investigation boreholes are located within 10 m of the natural streams.</li> <li>Regular water quality monitoring is to be conducted according to the proposed Environmental Monitoring Plan during construction and post-construction stage.</li> <li>Design appropriately to ensure that stormwater flow can be controlled during heavy rain period.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                   |    |   |   |   |     |                 |
|       | Impact to water quality due to sediment runoff and siltation | 2                                  | -3 | 2 | 2 | 3 | -42 | Minor negative    | <ul style="list-style-type: none"> <li>Implementation proper Earth Control Measures (ECMs)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2                                 | -2 | 2 | 2 | 3 | -28 | Slight negative |
|       | Impact to water quality due to trade effluent                | 3                                  | -4 | 2 | 2 | 3 | -84 | Moderate negative | <ul style="list-style-type: none"> <li>Contractor to provide storage tanks of adequate size and sufficient numbers to temporarily store trade effluent before treatment and disposal.</li> <li>Area where trade effluent is being transported out to be kerbed to minimise spills</li> <li>Contractor to provide sampling point and inspection for collection of trade effluent to be tested for parameters stipulated in the Regulations, depending on the discharge point (i.e., sewer, watercourse).</li> <li>Regular water quality monitoring is to be conducted according to the proposed Environmental Monitoring Plan during construction and post-construction stage.</li> <li>Contractor should install and maintain trade effluent treatment facilities (e.g., slurry treatment plant, grease trap and suspended solid settling tank) as per NEA's requirements.</li> </ul> | 3                                 | -3 | 2 | 2 | 3 | -63 | Minor negative  |

| Phase                          | Impact component                                                                                 | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Predicted impact after mitigation |    |   |   |   |     |                 |
|--------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                                |                                                                                                  | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | I                                 | M  | P | R | C | ES  | Impact          |
|                                | Impact to water quality due to accidental spillage of construction materials, fuel, and solvents | 2                                  | -2 | 2 | 2 | 3 | -28 | Slight negative | <ul style="list-style-type: none"> <li>To ensure that bentonite is contained within the working area and does not enter any watercourses or surface water drains.</li> <li>Storage silos should have secondary containment.</li> <li>Areas where bentonite is mixed are to be surrounded with a small boundary wall or contained within a bund. This action will help to control the slurry produced and prevent it from entering surface water drains or watercourses.</li> <li>Bentonite use should be monitored so that excessive use could be identified to prevent bentonite materials from escaping into the ground and potentially polluting groundwater.</li> <li>Foreign material should not be illegally disposed at the water bodies but removed from site by a licensed waste collector.</li> <li>Vehicle fuelling and major maintenance to be minimised within the project area.</li> <li>Appropriate concrete wash out areas should be provided and should not be performed in or near any water body.</li> <li>Emergency spill kits should be provided by contractor on site in the event of any chemical spillages. Emergency response team should also be trained in the use of these spill kits.</li> </ul> | 2                                 | -1 | 2 | 2 | 3 | -14 | Slight negative |
| Operation                      | Impact to drainage pattern and water quality in Stream 2                                         | 2                                  | -3 | 2 | 2 | 3 | -42 | Minor negative  | <ul style="list-style-type: none"> <li>No discharge of process effluent or release of pollutants through this outlet to the stream.</li> <li>Flow rate of discharge is to be regulated to prevent erosion within receiving stream.</li> <li>Water quality checks if there is suspicion of discharge of pollutant to the stream.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2                                 | -2 | 2 | 2 | 3 | -28 | Slight negative |
| <b>Human Health and Safety</b> |                                                                                                  |                                    |    |   |   |   |     |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                   |    |   |   |   |     |                 |

| Phase                           | Impact component                                                                                 | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Predicted impact after mitigation |    |   |   |   |     |                 |
|---------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                                 |                                                                                                  | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | I                                 | M  | P | R | C | ES  | Impact          |
| Pre-construction / Construction | Impact to water quality due to accidental spillage of construction materials, fuel, and solvents | 2                                  | -2 | 2 | 2 | 2 | -24 | Slight negative | <ul style="list-style-type: none"> <li>Contractor to provide storage tanks of adequate size and sufficient numbers to temporarily store trade effluent before treatment and disposal.</li> <li>Contractor to provide sampling point and inspection for collection of trade effluent to be tested for parameters stipulated in the Regulations, depending on the discharge point (i.e., sewer, watercourse, controlled watercourse).</li> <li>Contractor should install and maintain trade effluent treatment facilities (e.g., slurry treatment plant, grease trap and suspended solid settling tank) as per PUB's requirements</li> </ul> | 2                                 | -1 | 2 | 2 | 2 | -12 | Slight negative |

**Table 5.13.** Environmental Scores of the identified impacts on site's hydrology & water quality with corresponding mitigation measures (Area 3)

| Phase                           | Impact component                                                                                 | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                 | Predicted impact after mitigation |    |   |   |   |     |                 |
|---------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                                 |                                                                                                  | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                     | I                                 | M  | P | R | C | ES  | Impact          |
| Ecology                         |                                                                                                  |                                    |    |   |   |   |     |                 |                                                                                                                                                                                                                                                                     |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Degradation of stream habitats                                                                   | 3                                  | -2 | 2 | 2 | 3 | -42 | Minor negative  | <ul style="list-style-type: none"><li>Implement proper Earth Control Measures (ECMs)</li><li>Regular water quality monitoring is to be conducted according to the proposed Environmental Monitoring Plan during construction and post-construction stage.</li></ul> | 3                                 | -1 | 2 | 2 | 3 | -21 | Slight negative |
|                                 | Impact to water quality due to accidental spillage of construction materials, fuel, and solvents | 2                                  | -2 | 2 | 2 | 3 | -28 | Slight negative | <ul style="list-style-type: none"><li>To ensure that bentonite is contained within the working area and does not enter any watercourses or surface water drains.</li><li>Storage silos should have secondary containment.</li></ul>                                 | 2                                 | -1 | 2 | 2 | 3 | -14 | Slight negative |

| Phase                           | Impact component                                                                                 | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Predicted impact after mitigation |    |   |   |   |     |                 |
|---------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                                 |                                                                                                  | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | I                                 | M  | P | R | C | ES  | Impact          |
|                                 |                                                                                                  |                                    |    |   |   |   |     |                 | <ul style="list-style-type: none"> <li>Foreign material should not be illegally disposed at the water bodies but removed from site by a licensed waste collector.</li> <li>No vehicle fuelling and maintenance to be allowed within the project area.</li> <li>Excess loose soil and rock to be contained prior to the commencement of the works.</li> </ul>                                                                                                                                                                                                                            |                                   |    |   |   |   |     |                 |
| <b>Human Health and Safety</b>  |                                                                                                  |                                    |    |   |   |   |     |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Impact to water quality due to accidental spillage of construction materials, fuel, and solvents | 2                                  | -2 | 2 | 2 | 3 | -28 | Slight negative | <ul style="list-style-type: none"> <li>Contractor to provide storage tanks of adequate size and sufficient numbers to temporarily store trade effluent before treatment and disposal.</li> <li>Contractor to provide sampling point and inspection for collection of trade effluent to be tested for parameters stipulated in the Regulations, depending on the discharge point.</li> <li>Contractor should install and maintain trade effluent treatment facilities (e.g., slurry treatment plant, grease trap and suspended solid settling tank) as per NEA's requirements</li> </ul> | 2                                 | -1 | 2 | 2 | 3 | -14 | Slight negative |

## 6 NOISE

### 6.1 Introduction

The project study area is largely vegetated with an internal network of roads for CCKWW plant usage. Main source of noise comes from the traffic in Nanyang Drive at the east of the site, with occasional source being the activities conducted in the area.

This section describes the relevant standard for the ambient noise that are applicable to the Project activities, the methodology and results for the baseline noise study. It also described potential noise impacts from construction works, and recommendation for mitigation measures. A quantitative impact assessment approach is being used for assessment of impacts.

### 6.2 Relevant Environmental Legislation, Guidelines and Standards

The Environmental Protection and Management (Control of Noise at Construction Sites) Regulation 2008 prescribes the maximum noise levels permissible at construction sites for different days of the week, different periods of the day, and different types of premises affected. The permissible levels are listed in Table 6.1.

**Table 6.1.** Maximum permissible noise levels from construction sites (source: SSO)

| Types of affected buildings                                                        | Leq        | 7 am – 7 pm | 7–10 pm  | 10 pm – 7 am |
|------------------------------------------------------------------------------------|------------|-------------|----------|--------------|
| <b><i>Mondays–Saturdays</i></b>                                                    |            |             |          |              |
| (a) Hospitals, schools, institutions of higher learning, homes for aged sick, etc. | Leq 12 hrs | 60 dB(A)    | 50 dB(A) |              |
|                                                                                    | Leq 5 mins | 75 dB(A)    | 55 dB(A) |              |
| (b) Residential buildings located less than 150 m from the construction site       | Leq 12 hrs | 75 dB(A)    | -        |              |
|                                                                                    | Leq 1 hr   | -           | 65 dB(A) | 55 dB(A)     |
|                                                                                    | Leq 5 mins | 90 dB(A)    | 70 dB(A) | 55 dB(A)     |
| (c) Buildings other than those in (a) and (b) above                                | Leq 12 hrs | 75 dB(A)    | 65 dB(A) |              |
|                                                                                    | Leq 5 mins | 90 dB(A)    | 70 dB(A) |              |
| <b><i>Sundays and Public Holidays</i></b>                                          |            |             |          |              |
| (a) Hospitals, schools, institutions of higher learning, homes for aged sick, etc. | Leq 12 hrs | 60 dB(A)    | 50 dB(A) |              |
|                                                                                    | Leq 5 mins | 75 dB(A)    | 55 dB(A) |              |
| (b) Residential buildings located less than 150m from the construction site        | Leq 12 hrs | 75 dB(A)    | -        |              |
|                                                                                    | Leq 5 mins | 75 dB(A)    | 55 dB(A) |              |
| (c) Buildings other than those in (a) and (b) above                                | Leq 12 hrs | 75 dB(A)    | 65 dB(A) |              |
|                                                                                    | Leq 5 mins | 90 dB(A)    | 70 dB(A) |              |

#### No Work Rule on Sundays and Public Holidays

In addition to the setting of the permissible noise limits, NEA has also implemented a rule prohibiting work on Sundays and Public Holidays (PH) for construction sites located within 150 m away from residential premises and noise sensitive premises as follows:

- a) Construction Work Commenced on or after 1<sup>st</sup> September 2010
  - No work allowed from 10 pm on Saturday/eve of Public Holiday to 10 am on Sunday/Public Holiday
- b) Construction Work Commenced on or after 1<sup>st</sup> September 2011
  - No work allowed from 10 pm on Saturday/eve of Public Holiday to 7 am on

the following Monday/day after the Public Holiday

### **Correction Factor for Baseline Background Noise Exceeding the Regulations**

In certain areas, the baseline background noise levels may already exceed permissible noise levels outlined in Table 6.1. As such, the maximum permissible noise levels during the project activities will be adjusted according to a correction factor. The value of this factor is subject to the difference between the background noise level and permissible noise level, as shown in Table 6.2 below. This correction factor is added to the higher of the two noise levels to produce a new permissible value during the project activities.

**Table 6.2.** Value of correction factor accounting for baseline noise level exceedance

| <b>Difference between Permissible and Baseline Background Noise Levels (dB(A))</b> | <b>Correction Factor to be Added to the Higher of the Two Noise Levels (dB(A))</b> |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| < 2                                                                                | 3                                                                                  |
| $2 \leq \text{difference} < 4$                                                     | 2                                                                                  |
| $4 \leq \text{difference} < 10$                                                    | 1                                                                                  |
| $\geq 10$                                                                          | 0                                                                                  |

The EPM (Control of Noise at Construction Sites) Regulation 2008 is aimed to manage noise impacts on humans, rather than animals including birds, which have different hearing ranges.

As noted, the project site is inhabited by many fauna species (see Section 4.4.2) which may be sensitive to noise. Some of these species, particularly nocturnal species which rely on their hearing for movement, communication, and foraging, are likely to be impacted by increased noise levels. Chronic and frequent noise such as traffic noise interferes with animals' abilities to detect important sounds, whereas intermittent and unpredictable noise such as piling, honking from vehicles and machinery and shouting is often perceived as a threat (Francis & Barber, 2013) This can alter species' behaviour and impair their ability to forage and avoid predation, leading to decreased survivability.

Hence, following EPM (Control of Noise at Construction Sites) Regulation 2008 noise levels during construction phase may not protect the fauna species from noise impacts. The effects of noise on fauna are poorly understood since both stimuli and responses can vary. Further discussion on the suggested threshold noise level to be considered during construction phase for the protection of fauna is provided in Section 6.5.2.

## **6.3 Baseline Methodology**

### **6.3.1 Sensitive Receptors Identification**

Noise can lead to annoyance due to interference with communication or disturbance to receptors involved in leisure activities, as well as sleep disturbance. The effects of noise may vary with the individual receptor and is dependent on many factors such as the activity that the receptor is engaged in, as well as the duration of noise exposure.

The criteria applied for identification of Noise Sensitive Receptor (NSR) are based on EPM (Control of Noise at Construction Sites) Regulation 2008 which are listed below:

- (a) Hospitals, schools, institutions of higher learning, homes for aged sick, etc.
- (b) Residential buildings located less than 150 m from the construction site
- (c) Buildings other than those in (a) and (b) above

The sensitive receptor potentially affected by the proposed project have been identified (Table 6.3) through a combination of desktop study and visual surveys at the study areas.

**Table 6.3.** List of identified noise-sensitive receptors

| ID   | Type                           | Description                                       | Approximate Distance* |
|------|--------------------------------|---------------------------------------------------|-----------------------|
| NSR1 | Institution of higher learning | Experimental Medicine Building                    | 300m from Area 1      |
| NSR2 | Institution of higher learning | Research Techno Plaza                             | 120m from Area 3      |
| NRS3 | Institution of higher learning | Wee Kim Wee School of Communication & Information | 70m from Area 3       |
| NSR4 | Residential building           | Block 36 of the former NTU Hall of Residence 7    | 10m from Area 3       |
| NSR5 | Residential building           | Block 37 of the former NTU Hall of Residence 7    | 20m from Area 3       |

\*Approximate distance from the nearest project work area

### 6.3.2 Baseline Field Survey

The baseline ambient noise level is monitored at five (5) locations for a week (24 hours x 7 days) duration. Monitoring locations were chosen based on the factors such as sensitive receptor, relevance to project activity, site access and equipment security. Type 1 noise level meter–Svantek Model SVAN 971–was installed at each location to record the sound pressure level (Leq) with 5-minute sampling intervals for seven (7) consecutive days, yielding the data of Leq 12 hrs, Leq 1 hr, and Leq 5 mins.

Figure 6.1 shows the photographs of on-site monitoring equipment setup and Figure 6.2 the location map of the noise monitoring stations. Table 6.4 summarises the coordinates of monitoring stations, applicable noise limit that corresponds to the receptors the station represents, and short description of the location of each station. The permissible noise limits follow Environmental Protection & Management (Control of Noise at Construction Sites) Regulation 2008 which prescribes the maximum noise levels permissible at construction sites for different periods and types of premises affected.

The complete baseline noise monitoring report is available as **Appendix I** of this report.

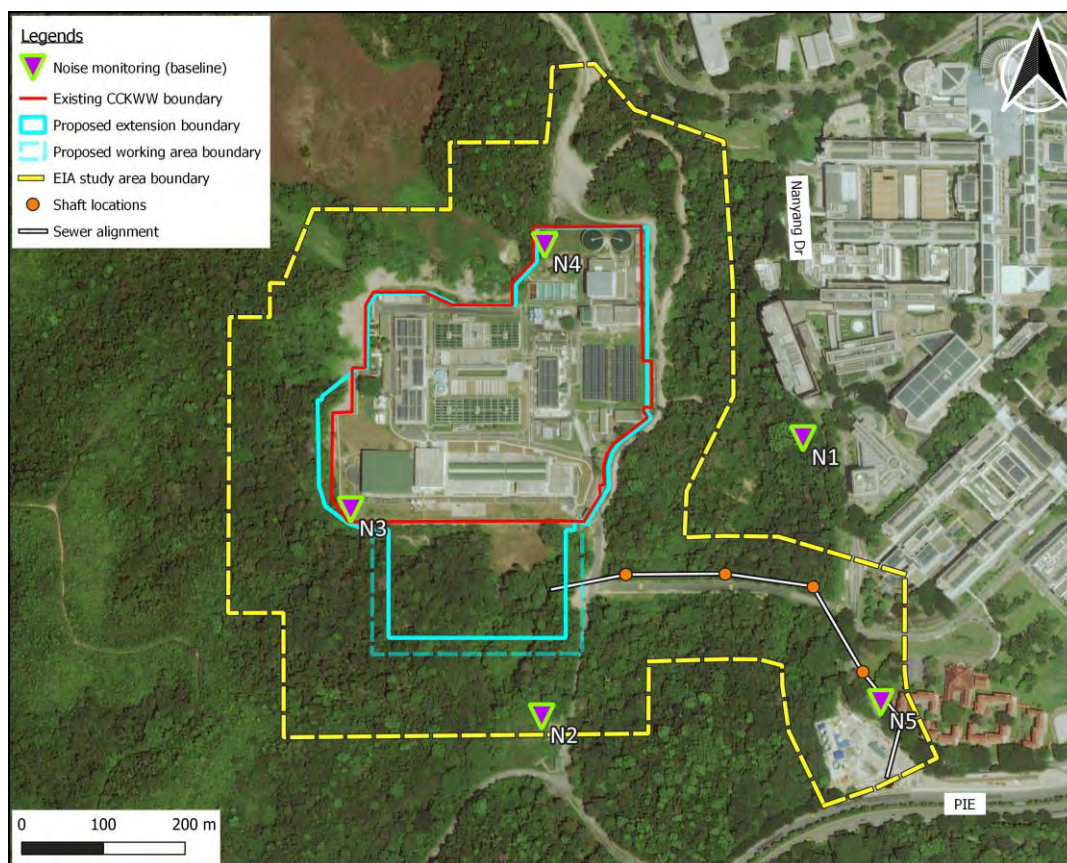
**Table 6.4.** Location of baseline noise monitoring stations

| Station ID | Monitoring Period   | Latitude | Longitude  | Noise Limit Type | Description                                                                                                                    |
|------------|---------------------|----------|------------|------------------|--------------------------------------------------------------------------------------------------------------------------------|
| N1         | 26 Aug – 2 Sep 2021 | 1.343694 | 103.678806 | (a)              | Open field at the top of the slope just next to the unloading/loading bay of Experimental Medicine Building and Nanyang Avenue |

| Station ID | Monitoring Period   | Latitude | Longitude  | Noise Limit Type | Description                                                                                                             |
|------------|---------------------|----------|------------|------------------|-------------------------------------------------------------------------------------------------------------------------|
| N2         | 18–25 Aug 2021      | 1.340417 | 103.675972 | (c)              | South of CCKWW, next to Dunbar Walk                                                                                     |
| N3         | 3–10 Sep 2021       | 1.342722 | 103.673528 | (c)              | At the south-western perimeter fence of CCKWW, near Ceramic Membrane Filtration plant                                   |
| N4         | 3–10 Sep 2021       | 1.345694 | 103.675722 | (c)              | At the open field west to the sedimentation tanks at the northern part of CCKWW                                         |
| N5         | 26 Aug – 2 Sep 2021 | 1.340528 | 103.679417 | (b)              | Beside the pedestrian walkway of Nanyang Avenue and nearby a bicycle parking shed next to Blk 36 of Hall of Residence 7 |



**Figure 6.1.** Setup of the baseline noise monitoring equipment



**Figure 6.2.** Location of baseline noise monitoring stations within project study area

#### 6.4 Baseline Results and Discussion

Table 6.5 and Figure 6.3 show the baseline noise level monitored at Station N1. While the noise level in the location during the day complied with the noise limit for type (a), its night-time noise level exceeded the noise limit throughout the week. Such high readings may be attributed to the traffic in Nanyang Avenue, a road just next to Station 1 that links the northern and southern part of NTU and is commonly used by the university's internal shuttle bus services. Another potential cause is the station's proximity to the unloading/loading bay of NTU's Experimental Medicine Building.

**Table 6.5.** Summary of the baseline Leq 5 mins noise monitoring results at Station N1

| Date           | Monitoring Period           | Noise Level (dB(A)) |                   |                 | Type (a) Noise Limit (dB(A)) |
|----------------|-----------------------------|---------------------|-------------------|-----------------|------------------------------|
|                |                             | Min. Leq 5 mins     | Median Leq 5 mins | Max. Leq 5 mins | Leq 5 mins                   |
| Thu, 26/8/2021 | 7 am – 7 pm                 | 55.7                | 59.5              | 64.7            | 75                           |
|                | 7 pm – 7 am <sup>(+1)</sup> | 54.6                | <b>55.4</b>       | <b>60.6</b>     | 55                           |
| Fri, 27/8/2021 | 7 am – 7 pm                 | 56.2                | 60.2              | 68.7            | 75                           |
|                | 7 pm – 7 am <sup>(+1)</sup> | 55                  | <b>55.6</b>       | <b>61.2</b>     | 55                           |
| Sat, 28/8/2021 | 7 am – 7 pm                 | 55.4                | 58.1              | 83.7            | 75                           |
|                | 7 pm – 7 am <sup>(+1)</sup> | 54.9                | <b>56.3</b>       | <b>59</b>       | 55                           |
| Sun, 29/8/2021 | 7 am – 7 pm                 | 55.2                | 57.1              | 67              | 75                           |
|                | 7 pm – 7 am <sup>(+1)</sup> | 54.9                | <b>56.6</b>       | <b>71.4</b>     | 55                           |

| Date           | Monitoring Period           | Noise Level (dB(A)) |                   |                 | Type (a) Noise Limit (dB(A)) |
|----------------|-----------------------------|---------------------|-------------------|-----------------|------------------------------|
|                |                             | Min. Leq 5 mins     | Median Leq 5 mins | Max. Leq 5 mins | Leq 5 mins                   |
| Mon, 30/8/2021 | 7 am – 7 pm                 | 57                  | 61.1              | 70.4            | 75                           |
|                | 7 pm – 7 am <sup>(+1)</sup> | <b>55.5</b>         | <b>61.2</b>       | <b>67</b>       | 55                           |
| Tue, 31/8/2021 | 7 am – 7 pm                 | 56.3                | 60.4              | 67.5            | 75                           |
|                | 7 pm – 7 am <sup>(+1)</sup> | 54.9                | <b>59.6</b>       | <b>65.6</b>     | 55                           |
| Mon, 1/9/2021  | 7 am – 7 pm                 | 55.7                | 59.5              | 65.5            | 75                           |
|                | 7 pm – 7 am <sup>(+1)</sup> | 54.7                | <b>60</b>         | <b>67.2</b>     | 55                           |

Note: Bold results shows exceedance.

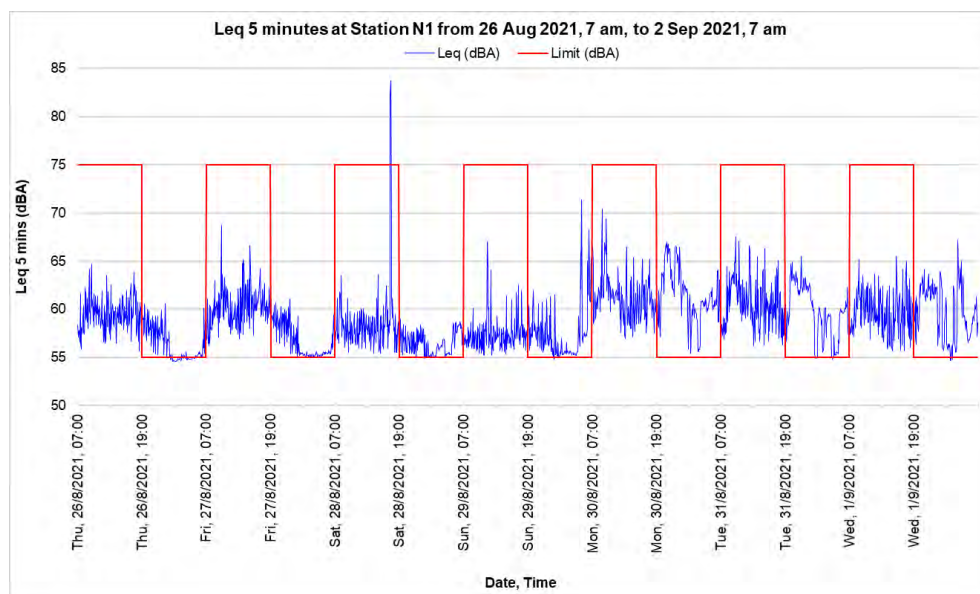


Figure 6.3. Baseline Leq 5 mins monitoring results at Station N1 over one-week period

Table 6.6 and Figure 6.4 show the baseline noise level monitored by Station N2, all of which fell within the noise limit of type (c).

Table 6.6. Summary of the baseline Leq 5 mins noise monitoring results at Station N2

| Date           | Monitoring Period           | Monitored Noise Level (dB(A)) |                   |                 | Type (c) Noise Limit (dB(A)) |
|----------------|-----------------------------|-------------------------------|-------------------|-----------------|------------------------------|
|                |                             | Min. Leq 5 mins               | Median Leq 5 mins | Max. Leq 5 mins | Leq 5 mins                   |
| Wed, 18/8/2021 | 7 am – 7 pm                 | 44.1                          | 51                | 67.5            | 90                           |
|                | 7 pm – 7 am <sup>(+1)</sup> | 44.9                          | 52                | 67.4            | 70                           |
| Thu, 19/8/2021 | 7 am – 7 pm                 | 46                            | 51.2              | 65.6            | 90                           |
|                | 7 pm – 7 am <sup>(+1)</sup> | 45.7                          | 54.7              | 66.8            | 70                           |
| Fri, 20/8/2021 | 7 am – 7 pm                 | 46.1                          | 50.6              | 71.8            | 90                           |
|                | 7 pm – 7 am <sup>(+1)</sup> | 46.1                          | 55.8              | 64.73           | 70                           |
| Sat, 21/8/2021 | 7 am – 7 pm                 | 47.3                          | 51.6              | 68.1            | 90                           |
|                | 7 pm – 7 am <sup>(+1)</sup> | 47.3                          | 57                | 66              | 70                           |
| Sun, 22/8/2021 | 7 am – 7 pm                 | 42.8                          | 50                | 59.8            | 90                           |
|                | 7 pm – 7 am <sup>(+1)</sup> | 46.6                          | 51.8              | 60.2            | 70                           |

| Date           | Monitoring Period           | Monitored Noise Level (dB(A)) |                   |                 | Type (c) Noise Limit (dB(A)) |
|----------------|-----------------------------|-------------------------------|-------------------|-----------------|------------------------------|
|                |                             | Min. Leq 5 mins               | Median Leq 5 mins | Max. Leq 5 mins | Leq 5 mins                   |
| Mon, 23/8/2021 | 7 am – 7 pm                 | 45.3                          | 51.1              | 64.9            | 90                           |
|                | 7 pm – 7 am <sup>(+1)</sup> | 48.7                          | 55                | 69.8            | 70                           |
| Tue, 24/8/2021 | 7 am – 7 pm                 | 47.2                          | 54.9              | 79.9            | 90                           |
|                | 7 pm – 7 am <sup>(+1)</sup> | 46.9                          | 53.5              | 62.6            | 70                           |

Note: Bold results shows exceedance.

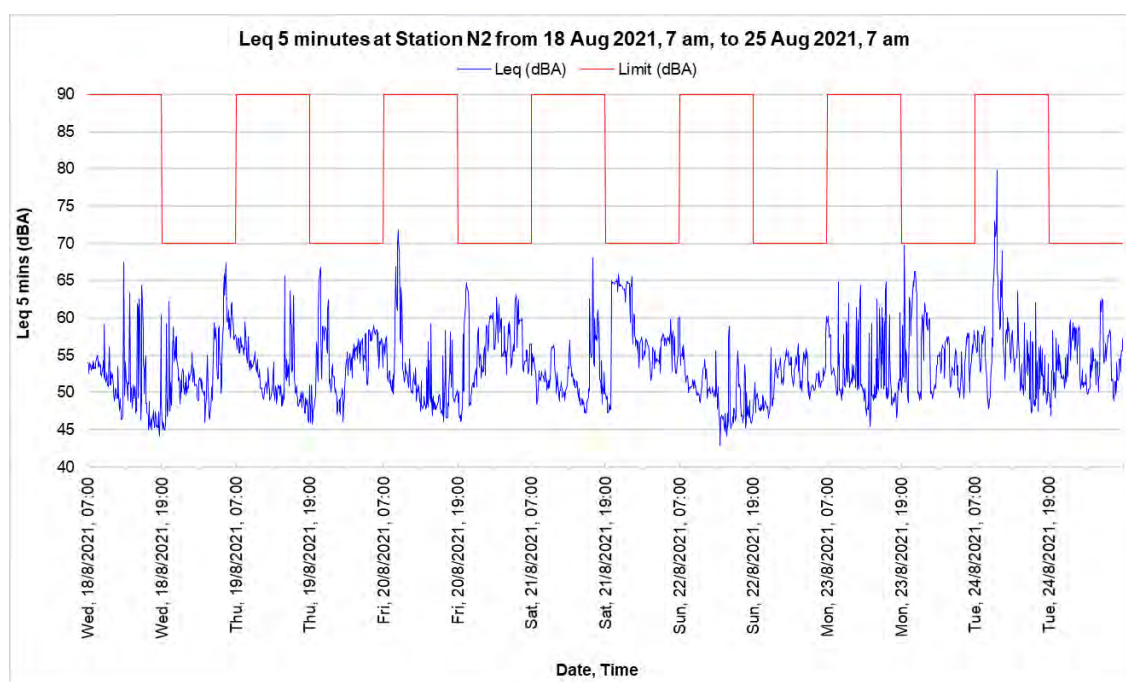


Figure 6.4. Baseline Leq 5 mins monitoring results at Station N2 over one-week period

Table 6.7 and Figure 6.5 show the baseline noise level monitored by Station N3. Except for three outlier readings of 76 dB(A) on 4 September 2021, at around 19:30, and 70.05 and 70.03 dB(A) on 10 September 2021 between 5:00–5:05, the noise level fell within the limit of type (c).

Table 6.7. Summary of the baseline Leq 5 mins noise monitoring results at Station N3

| Date          | Monitoring Period           | Monitored Noise Level (dB(A)) |                   |                 | Type (c) Noise Limit (dB(A)) |
|---------------|-----------------------------|-------------------------------|-------------------|-----------------|------------------------------|
|               |                             | Min. Leq 5 mins               | Median Leq 5 mins | Max. Leq 5 mins | Leq 5 mins                   |
| Fri, 3/9/2021 | 7 am – 7 pm                 | 45.5                          | 48.5              | 64.4            | 90                           |
|               | 7 pm – 7 am <sup>(+1)</sup> | 46.2                          | 50.5              | 55.3            | 70                           |
| Sat, 4/9/2021 | 7 am – 7 pm                 | 46.2                          | 49.1              | 58.2            | 90                           |
|               | 7 pm – 7 am <sup>(+1)</sup> | 49.9                          | 52.6              | <b>76</b>       | 70                           |
| Sun, 5/9/2021 | 7 am – 7 pm                 | 46.7                          | 48.2              | 55.3            | 90                           |
|               | 7 pm – 7 am <sup>(+1)</sup> | 47.5                          | 51.6              | 56              | 70                           |
| Mon, 6/9/2021 | 7 am – 7 pm                 | 46.2                          | 50.4              | 81.6            | 90                           |
|               | 7 pm – 7 am <sup>(+1)</sup> | 49.5                          | 52.5              | 55.6            | 70                           |

| Date          | Monitoring Period           | Monitored Noise Level (dB(A)) |                   |                 | Type (c) Noise Limit (dB(A)) |
|---------------|-----------------------------|-------------------------------|-------------------|-----------------|------------------------------|
|               |                             | Min. Leq 5 mins               | Median Leq 5 mins | Max. Leq 5 mins | Leq 5 mins                   |
| Tue, 7/9/2021 | 7 am – 7 pm                 | 45.4                          | 49.2              | 65              | 90                           |
|               | 7 pm – 7 am <sup>(+1)</sup> | 46.3                          | 52.7              | 68.1            | 70                           |
| Wed, 8/9/2021 | 7 am – 7 pm                 | 47                            | 50                | 65.7            | 90                           |
|               | 7 pm – 7 am <sup>(+1)</sup> | 47.4                          | 53.3              | 58.2            | 70                           |
| Thu, 9/9/2021 | 7 am – 7 pm                 | 46.2                          | 48.2              | 54.7            | 90                           |
|               | 7 pm – 7 am <sup>(+1)</sup> | 47.2                          | 52.4              | 70              | 70                           |

Note: Bold results shows exceedance.

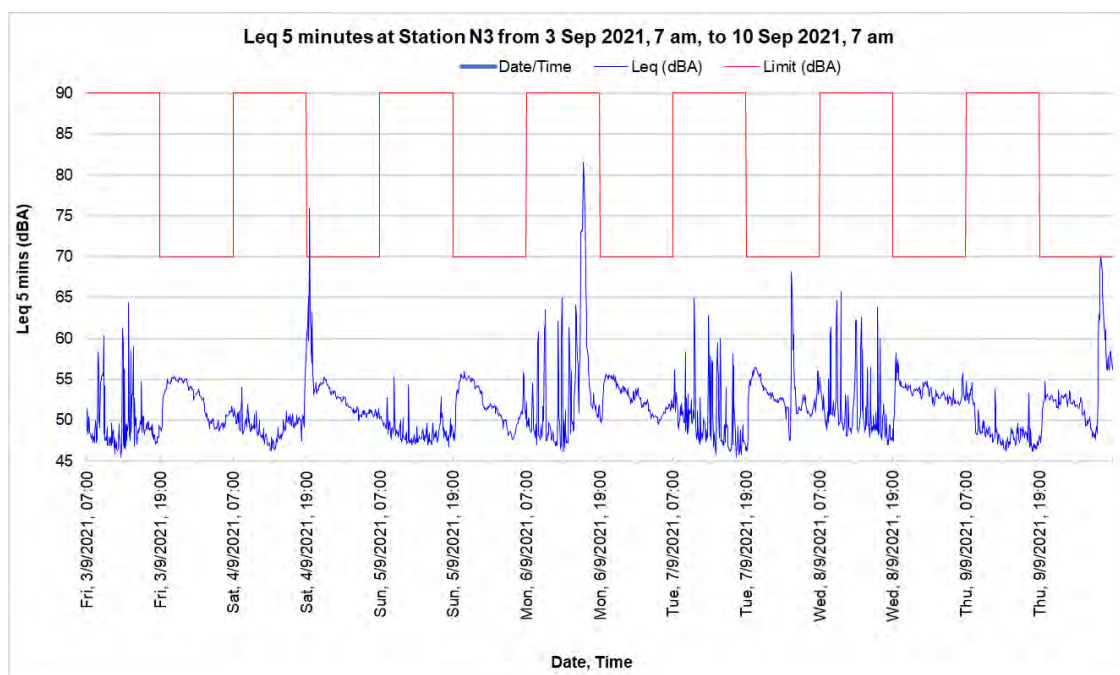


Figure 6.5. Baseline Leq 5 mins monitoring results at Station N3 over one-week period

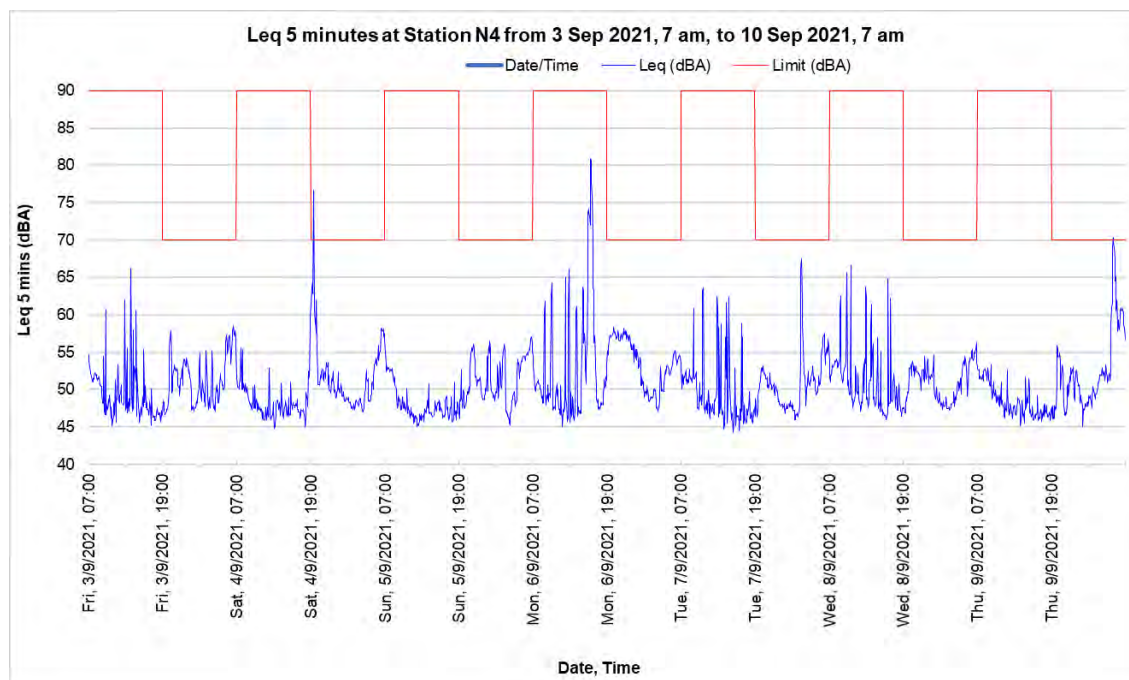
Table 6.8 and Figure 6.6 show the baseline noise level monitored by Station N4. Except for three outlier readings of 76.3 and 76.7 dB(A) on 4 September 2021 between 19:25–19:30, and 70.4 dB(A) on 10 September 2021 at 5:00, the noise level fell within the limit of type (c). Given that Station N4 was located within the vicinity of CCKWW, it is likely that the outliers were caused by plant's activities as well.

Table 6.8. Summary of the baseline Leq 5 mins noise monitoring results at Station N4

| Date          | Monitoring Period           | Monitored Noise Level (dB(A)) |                   |                 | Type (c) Noise Limit (dB(A)) |
|---------------|-----------------------------|-------------------------------|-------------------|-----------------|------------------------------|
|               |                             | Min. Leq 5 mins               | Median Leq 5 mins | Max. Leq 5 mins | Leq 5 mins                   |
| Fri, 3/9/2021 | 7 am – 7 pm                 | 45.1                          | 48.2              | 66.2            | 90                           |
|               | 7 pm – 7 am <sup>(+1)</sup> | 44.8                          | 49.8              | 58.6            | 70                           |
| Sat, 4/9/2021 | 7 am – 7 pm                 | 44.8                          | 47.7              | 59.8            | 90                           |
|               | 7 pm – 7 am <sup>(+1)</sup> | 47                            | 50.6              | <b>76.7</b>     | 70                           |
| Sun,          | 7 am – 7 pm                 | 45.1                          | 47.4              | 56.6            | 90                           |

| Date             | Monitoring Period           | Monitored Noise Level (dB(A)) |                      |                    | Type (c) Noise Limit (dB(A)) |
|------------------|-----------------------------|-------------------------------|----------------------|--------------------|------------------------------|
|                  |                             | Min.<br>Leq 5 mins            | Median<br>Leq 5 mins | Max.<br>Leq 5 mins | Leq 5 mins                   |
| 5/9/2021         | 7 pm – 7 am <sup>(+1)</sup> | 45.3                          | 50.9                 | 57.1               | 70                           |
| Mon,<br>6/9/2021 | 7 am – 7 pm                 | 45.1                          | 49.6                 | 80.9               | 90                           |
|                  | 7 pm – 7 am <sup>(+1)</sup> | 47.1                          | 53.9                 | 58.4               | 70                           |
| Tue,<br>7/9/2021 | 7 am – 7 pm                 | 44.3                          | 48.1                 | 63.6               | 90                           |
|                  | 7 pm – 7 am <sup>(+1)</sup> | 46                            | 50.3                 | 67.6               | 70                           |
| Wed,<br>8/9/2021 | 7 am – 7 pm                 | 46                            | 50.1                 | 66.7               | 90                           |
|                  | 7 pm – 7 am <sup>(+1)</sup> | 46.9                          | 51.4                 | 56.3               | 70                           |
| Thu,<br>9/9/2021 | 7 am – 7 pm                 | 45.2                          | 47.8                 | 53.4               | 90                           |
|                  | 7 pm – 7 am <sup>(+1)</sup> | 45                            | 51.2                 | <b>70.4</b>        | 70                           |

Note: Bold results shows exceedance.



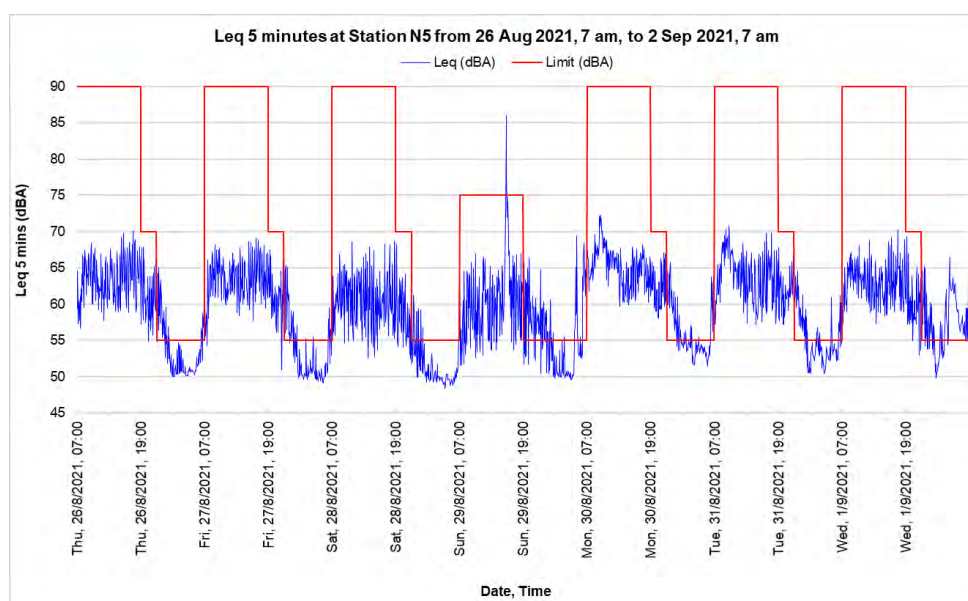
**Figure 6.6.** Baseline Leq 5 mins monitoring results at Station N4 over one-week period

Table 6.9 and Figure 6.7 show the baseline noise level monitored by Station N5. While daytime noise level at this location generally complied with the limit for type (b), the case is the contrary for the noise levels at late night-time (10 pm to 7 am on the next day) and on Sunday. Potential causes of these readings include the nearby construction of Deep Tunnel Sewerage System (DTSS) Phase 2 that may take place on the weekend and the daily traffic in Nanyang Avenue.

**Table 6.9.** Summary of the baseline Leq 5 mins noise monitoring results at Station N5

| Date           | Monitoring Period            | Noise Level (dB(A)) |                   |                 | Type (b) Noise Limit (dB(A)) |
|----------------|------------------------------|---------------------|-------------------|-----------------|------------------------------|
|                |                              | Min. Leq 5 mins     | Median Leq 5 mins | Max. Leq 5 mins | Leq 5 mins                   |
| Thu, 26/8/2021 | 10 am – 7 pm                 | 56.7                | 62.9              | 70.1            | 90                           |
|                | 7 pm – 10 pm                 | 54.8                | 60.7              | 67.9            | 70                           |
|                | 10 pm – 7 am <sup>(+1)</sup> | 50                  | 51.8              | <b>65.1</b>     | 55                           |
| Fri, 27/8/2021 | 7 am – 7 pm                  | 54.3                | 63.3              | 69.1            | 90                           |
|                | 7 pm – 10 pm                 | 51                  | 61.1              | 67.5            | 70                           |
|                | 10 pm – 7 am <sup>(+1)</sup> | 49.1                | 51.3              | <b>65.4</b>     | 55                           |
| Sat, 28/8/2021 | 7 am – 7 pm                  | 52.6                | 60.5              | 68.7            | 90                           |
|                | 7 pm – 10 pm                 | 51.8                | 57.9              | 68.3            | 70                           |
|                | 10 pm – 7 am <sup>(+1)</sup> | 48.4                | 50.2              | <b>61.2</b>     | 55                           |
| Sun, 29/8/2021 | 7 am – 7 pm                  | 51.6                | 60.5              | <b>86</b>       | 75                           |
|                | 7 pm – 7 am <sup>(+1)</sup>  | 49.6                | 54.7              | <b>69.5</b>     | 55                           |
| Mon, 30/8/2021 | 7 am – 7 pm                  | 59.7                | 65.4              | 72.2            | 90                           |
|                | 7 pm – 10 pm                 | 58                  | 61.8              | 67.1            | 70                           |
|                | 10 pm – 7 am <sup>(+1)</sup> | 51.5                | 54.4              | <b>65.8</b>     | 55                           |
| Tue, 31/8/2021 | 7 am – 7 pm                  | 57.2                | 64.3              | 70.9            | 90                           |
|                | 7 pm – 10 pm                 | 57.2                | 61.4              | 67.8            | 70                           |
|                | 10 pm – 7 am <sup>(+1)</sup> | 50.4                | 53.9              | <b>64.6</b>     | 55                           |
| Mon, 1/9/2021  | 7 am – 7 pm                  | 57.3                | 63                | 70.2            | 90                           |
|                | 7 pm – 10 pm                 | 52.9                | 59.8              | 69.4            | 70                           |
|                | 10 pm – 7 am <sup>(+1)</sup> | 49.8                | <b>57.3</b>       | <b>66.5</b>     | 55                           |

Note: Bold results shows exceedance.



**Figure 6.7.** Baseline Leq 5 mins monitoring results at Station N5 over one-week period

## **6.5 Impact Assessment**

### **6.5.1 Potential Sources of Impacts**

The following activities are identified as potential sources of noise during the pre-construction and construction phases of CCKWW facilities:

- Site clearing, earthworks, and general construction activities;
- Stockpiling and materials handling;
- Vehicle movements in and out of construction work areas; and
- Use of high-impact stationary and mobile equipment such as pile drivers, mobile cranes, excavators, dozers, etc. leading to variable and sporadic noise levels, typically repeating over time.

Specifically, the potential main sources of noise emission during construction of various CCKWW facilities at Area 1 and 2 are:

- Vibro hammer for installing and removing sheet piles (temporary retaining walls);
- Bored Piling rigs to be used to pre-bore the ground for installation of the piles;
- Excavators for earth handling; and
- Demolition activities

Main sources of ground vibration emissions during construction of sewerage alignment Area 3 are:

- Pipe jacking machine, and Micro Tunnel Boring Machine (MTBM) during pipe jacking method

### **6.5.2 Methodology and Assessment Criteria**

#### **Methodology**

The prediction of the potential noise levels generated by the construction activities at all CCKWW work areas were based on the available project information and simulated using noise modelling software SoundPLAN version 2021 developed by SoundPLAN GmbH. Predicted noise levels at sensitive receptor locations were used to determine the impact magnitude.

At the time of this EIA report preparation, exact and final information on the construction methodology, type of construction equipment and their approximate numbers that will be used during construction activities were not yet available as these details are expected to be available only in construction phase when the construction Contractor is onboard. As such, the noise modelling was conducted by utilising available information and relevant assumptions about the potential equipment that would be used for work areas.

Prediction of the noise levels yielded two sets of results. There are firstly the non-mitigated construction noise impacts at sensitive receptor locations and surrounding areas. The second set of results predicted the mitigated construction noise impacts at the sensitive receptor locations and surrounding areas after the implementation of practicable direct mitigation measures (i.e., noise barrier).

The following assumptions were made in undertaking this assessment:

- Noise levels were predicted for construction activities which employ the (sets of) highest noise-generating equipment (e.g., piling rig, excavator).
- Elevation data was incorporated into the noise modelling software for prediction of noise level.
- The noise modelling assessment did not take monitored background noise into consideration. The assessment only incorporated the baseline noise level to correct the noise limits according to the regulations.
- The construction equipment operation hours from 9 am to 7 pm with 3 hours break (effectively operating for 7 hours) for day-time and from 7 pm to 10 pm with no break (effectively operating for 3 hours) for night-time are considered for worst case scenario.
- Noise levels at NSR were estimated with vertical grids with a 1 x 1 m spacing.
- A reflection loss of 11 dB and continuous height of 6 m were considered for noise barriers.

### Assessment Criteria

In light of the baseline background noise level recorded in the study area, the respective permissible noise limit during the project activities for human receptor has been adjusted using the values in Table 6.2. The corrected limits are summarised in Table 6.10.

**Table 6.10.** Adjusted permissible noise limit at respective receptors on Monday–Saturday

| ID   | Sensitive Receptor                                | Noise Limit Type | Leq        | 7 am – 7 pm | 7–10 pm  | 10 pm – 7 am |
|------|---------------------------------------------------|------------------|------------|-------------|----------|--------------|
| NSR1 | Experimental Medicine Building                    | (a)              | Leq 12 hrs | 64 dB(A)    | 60 dB(A) |              |
|      |                                                   |                  | Leq 5 mins | 75 dB(A)    | 60 dB(A) |              |
| NSR2 | Research Techno Plaza                             | (a)              | Leq 12 hrs | 64 dB(A)    | 60 dB(A) |              |
|      |                                                   |                  | Leq 5 mins | 75 dB(A)    | 60 dB(A) |              |
| NRS3 | Wee Kim Wee School of Communication & Information | (a)              | Leq 12 hrs | 64 dB(A)    | 60 dB(A) |              |
|      |                                                   |                  | Leq 5 mins | 75 dB(A)    | 60 dB(A) |              |
| NSR4 | Block 36 of the former NTU Hall of Residence 7    | (b)              | Leq 12 hrs | 75 dB(A)    | -        |              |
|      |                                                   |                  | Leq 1 hr   | -           | 65 dB(A) | 55 dB(A)     |
|      |                                                   |                  | Leq 5 mins | 90 dB(A)    | 70 dB(A) | 55 dB(A)     |
| NSR5 | Block 37 of the former NTU Hall of Residence 7    | (b)              | Leq 12 hrs | 75 dB(A)    | -        |              |
|      |                                                   |                  | Leq 1 hr   | -           | 65 dB(A) | 65 dB(A)     |
|      |                                                   |                  | Leq 5 mins | 90 dB(A)    | 70 dB(A) | 55 dB(A)     |

As highlighted in Section 6.2, achieving applicable maximum permissible noise level for construction work based on regulatory requirements may not be adequate to address impact on fauna.

The increase in noise levels during construction phase may impact the fauna species inhabiting the surrounding environment. The frequency and loudness of noise produced by construction works may interfere with the communication calls of certain animal groups, particularly birds. A large number of birds use calls to communicate between members for the purposes of territory marking, courtship, mating, and predator alarms.

Increased environmental noise may mask the birds' song and impact their courting (Swaddle & Page, 2007). Arthropod abundance may also be impacted by the noise pollution due to cascading effects within the ecological network, as well as direct impacts for insects such as crickets that rely on audio signals for mating. This could also result in indirect negative impacts on plants that rely on these organisms for pollination or seed dispersal or have prey-predator relationships with other impacted species (Bunkley et al., 2017).

Particular attention needs to be paid to nocturnal animals. Many nocturnal animals rely more on other senses besides sight to navigate, and increased noise levels may impact their movement and other activities. For example, insectivorous bats rely on sound to locate and catch their prey. This process might be affected by night-time construction noise, which could fall within the spectrum of their auditory signals (Bunkley et al., 2017). It is anticipated that night work will be avoided within near forested area which will help to reduce the impact.

As described below, literature research was carried out to review available scientific knowledge and to suggest a noise limit which can be considered for construction phase activities for animals.

- Studies on laboratory animals exposed to noise indicated that nonhuman primates exposed chronic noise levels above 85 dB for developed cardiovascular conditions. Other effects also included behavioural sensitization of 'fear' in rats when exposed to background noise levels of 80 dB (Turner et al., 2005).
- Results of a study conducted on the impacts of road noise on bat movements highlighted the effects noise has on commuting insectivorous bats. Based on the findings, the study suggested that the noise disturbance threshold of approximately 88 dB (Bennett & Zurcher, 2012).
- A review conducted on various studies and research focused on the effects of noise indicated that terrestrial mammals respond to noise levels ranging from 65 to 130 dB (Kight & Swaddle, 2011).
- Various studies have put forward recommended guidelines stating that continuous construction noise levels should be kept below 60-70 dB(A), as birds may be able to get accustomed to ambient noise below these levels (Dooling & Popper, 2007; Cutts & Allen, 1999).

Considering monitored baseline levels at forested area and practicality of achieving the limit during construction phase, it is recommended that average noise levels to be maintained at construction work boundary adjacent to forest areas should be 65 dB(A) for the protection of fauna.

### **6.5.3 Prediction of Non-mitigated Impacts**

As noted earlier, the potential sources of noise impact during construction phase would mainly be the use of equipment in various construction activities. Airborne noise levels generated by construction equipment vary based on the type, model and condition of equipment, and duration of operation. The airborne noise levels are affected by distance, location (i.e., stationary or mobile), and variations in the power rating of equipment. The airborne noise levels are also affected by noise characteristics such as either continuous

or intermittent noise of the equipment. The noise emissions from construction equipment during construction activities and their typical sound power levels<sup>3</sup> are presented in Table 6.11 below, which were used in the noise model.

**Table 6.11.** Typical sound power levels of construction equipment

| Equipment Name          | Typical A-weighted Sound Power Levels dB(A) |
|-------------------------|---------------------------------------------|
| Rotary Bored Piling rig | 112–124                                     |
| Hydraulic Breaker       | 110–119                                     |
| Crawler Crane (80 hp)   | 100–115                                     |
| Lorry Crane (10 t)      | 118–120                                     |
| Air Compressor (50 hp)  | 93–107                                      |
| Compactor (300 hp)      | 117–124                                     |
| Bulldozer (55hp)        | 101–108                                     |
| Excavator (60 hp)       | 103–116                                     |
| Vibro hammer            | 119–130                                     |
| Dump Truck (20 t)       | 102–107                                     |
| Air compressor (50 hp)  | 93–107                                      |
| Water pump              | 99–105                                      |
| Road Roller (10 t)      | 98–106                                      |
| Generator (110 kVA)     | 108                                         |
| Concrete pump           | 102–107                                     |
| Concrete Mixer          | 89–93                                       |
| Electric Drill          | 94                                          |
| Welding set             | 96                                          |

The following approach was used for the prediction of noise levels.

- Daytime construction activities at Area 1 (Existing CCKWW plant) and Area 2 (CCKWW Extension Area) would be conducted concurrently.
- Night works (from 7 to 12 am) would be conducted at Area 1 (Existing CCKWW plant) and manhole 4 and 5 of Area 3 (Sewerage alignment).
- As activities at Area 3 (Sewerage alignment) would employ pipe jacking method, only the equipment used at ground level (e.g., mobile crane, air compressor) were considered as potential noise sources.
- A list of potential equipment that would be used at each work area, its operating periods, and numbers were assumed based on available information.
- Noise generating equipment (modelled as point sources) were placed close to the project boundary as a conservative approach.
- Noise modelling during daytime was carried out under normal weather condition (i.e., no rain event). It is expected that during the rain event, the construction activity will cease and resume after the rain has stopped.

Results of non-mitigated noise prediction levels during construction phase for all three Areas are summarised in the following sections.

<sup>3</sup> Singapore Standard SS602: 2014 – Code of Practice for Noise Control on Construction and Demolition Sites

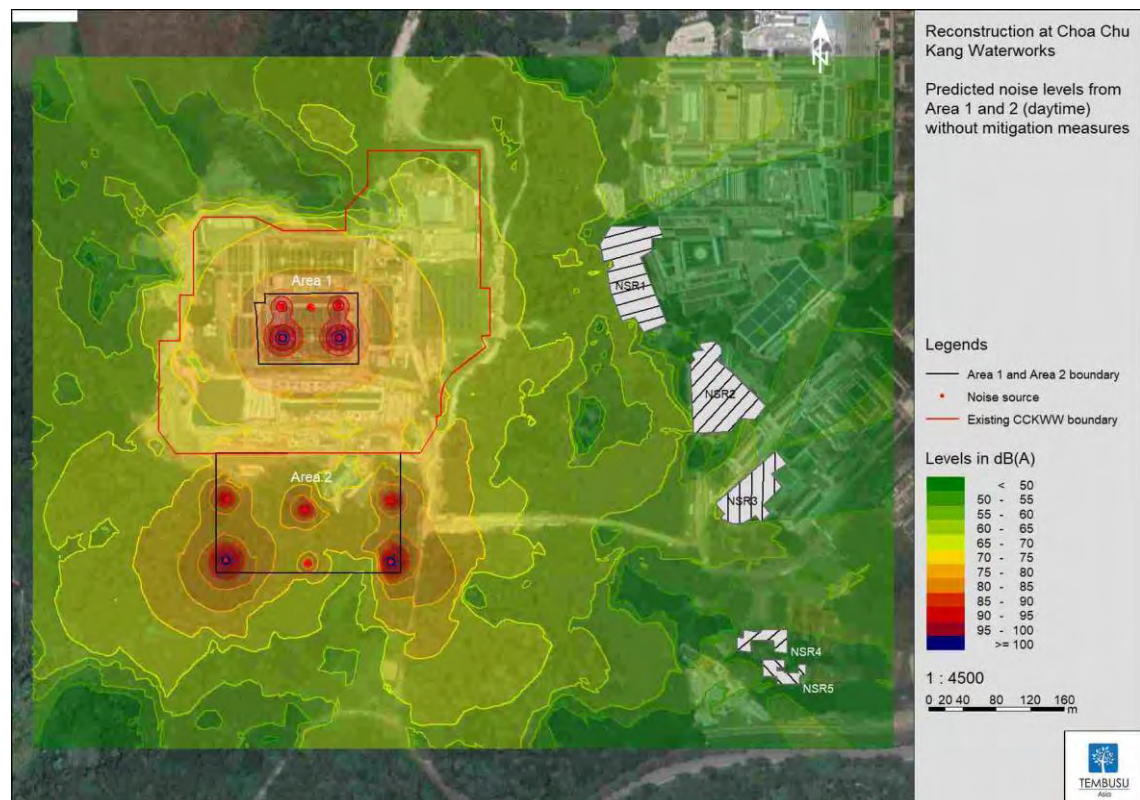
### Area 1 and Area 2

The results of non-mitigated noise prediction levels during construction phase for Area 1 and 2 are summarised in Table 6.12, Figure 6.9 for daytime noise levels, and Figure 6.10 for night-time noise levels.

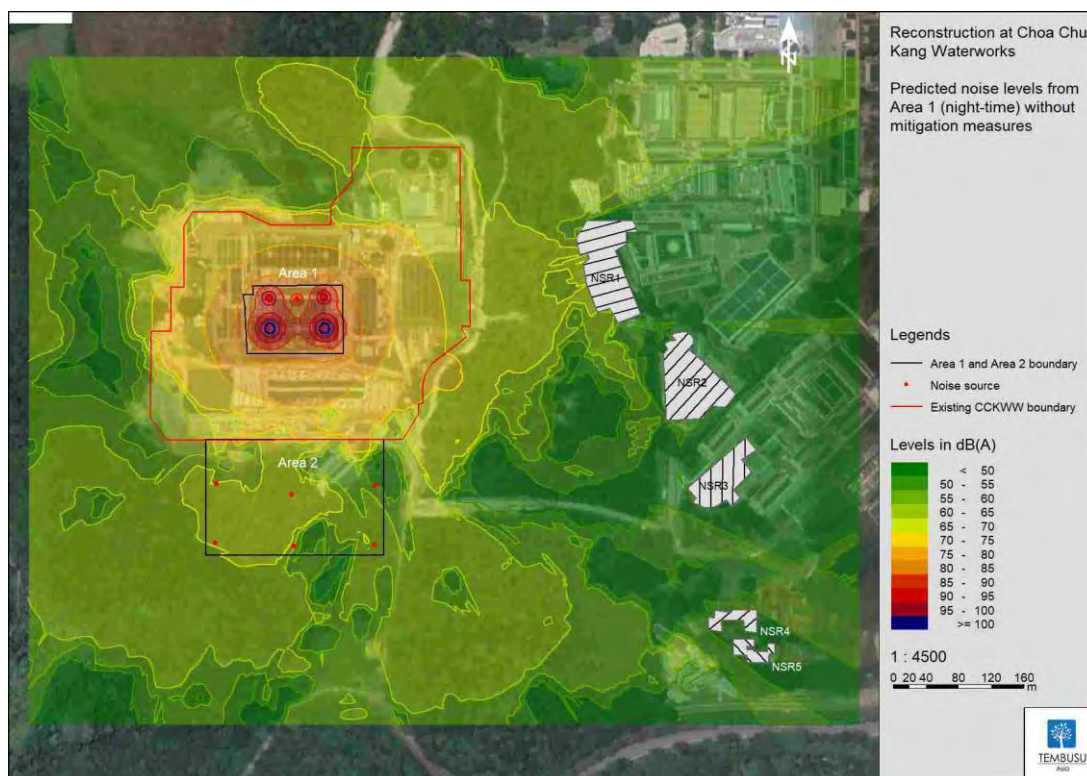
**Table 6.12.** Predicted noise levels at Area 1 and Area 2 without mitigation measures

| ID   | Noise Limit (dB(A)) |               | Noise Level without Mitigation Measures (dB(A)) |               |
|------|---------------------|---------------|-------------------------------------------------|---------------|
|      | 7 am to 7 pm        | 7 pm to 12 am | 7 am to 7 pm                                    | 7 pm to 12 am |
| NSR1 | 64                  | 60            | <b>65</b>                                       | <b>66.5</b>   |
| NSR2 | 64                  | 60            | 60.2                                            | 51.4          |
| NSR3 | 64                  | 60            | 62.2                                            | 57.3          |
| NSR4 | 75                  | 65            | 56.4                                            | 56.5          |
| NSR5 | 75                  | 65            | 53.1                                            | 50.4          |

Note: Bold results shows the exceedance.



**Figure 6.8.** Predicted daytime noise level from Area 1 & 2 activities without mitigation measures



**Figure 6.9.** Predicted night-time noise level from Area 1 activities without mitigation measures

During both day-time and night-time period, the predicted noise levels at most of the receptors are below the maximum adjusted permissible noise level. Only the predicted noise levels at NSR1 (type a limit) exceed the maximum adjusted permissible noise level for both periods. Thus, it is anticipated that there will be no impact on human receptors except at NSR1 due to work activities at Area 1 and Area 2.

On the other hand, the noise levels in the forested area around the construction areas range between 65 to 85 dB(A) during daytime construction activities and between 60 to 70 dB(A) during night-time activities.

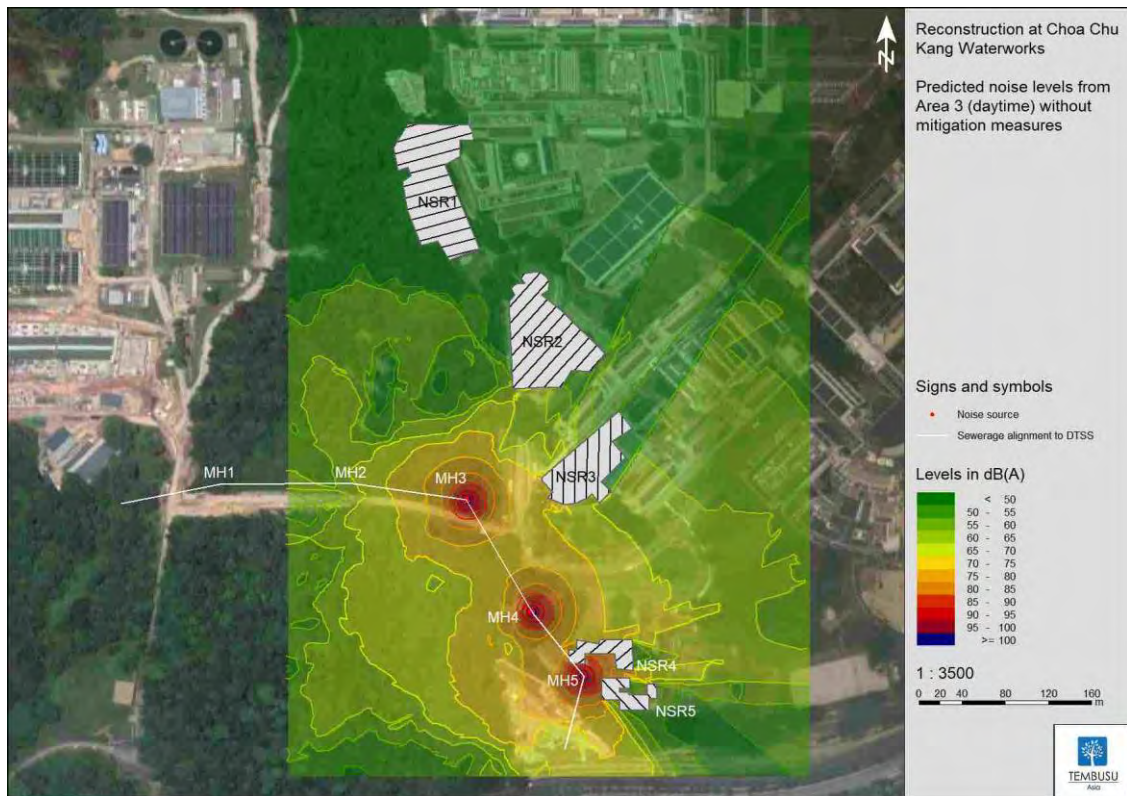
### Area 3

Summary results of non-mitigated noise prediction levels during construction phase for Area 3 are presented in Table 6.13, Figure 6.10 for daytime noise levels, and Figure 6.11 night-time noise levels.

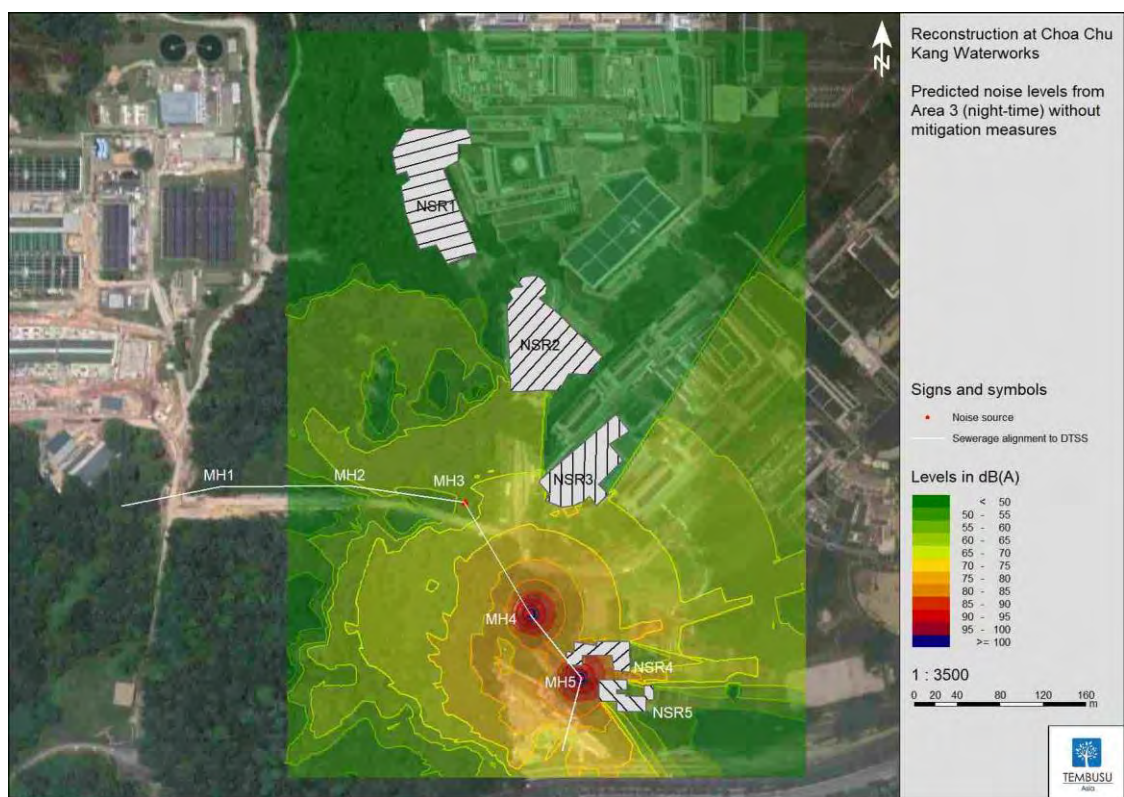
**Table 6.13.** Predicted noise levels at Area 3 without mitigation measures

| ID   | Noise Limit (dB(A)) |               | Noise Level without Mitigation Measures (dB(A)) |               |
|------|---------------------|---------------|-------------------------------------------------|---------------|
|      | 7 am to 7 pm        | 7 pm to 12 am | 7 am to 7 pm                                    | 7 pm to 12 am |
| NSR1 | 64                  | 60            | 54.3                                            | 55.9          |
| NSR2 | 64                  | 60            | <b>68.4</b>                                     | <b>64.7</b>   |
| NSR3 | 64                  | 60            | <b>75.3</b>                                     | <b>70.5</b>   |
| NSR4 | 75                  | 65            | <b>89.1</b>                                     | <b>91.5</b>   |
| NSR5 | 75                  | 65            | <b>86.1</b>                                     | <b>88.5</b>   |

Note: Bold results shows the exceedance.



**Figure 6.10.** Predicted daytime noise level from Area 3 activities without mitigation measures



**Figure 6.11.** Predicted night-time noise level from Area 3 activities without mitigation measures

During both day-time and night-time periods, the predicted noise levels at most of the receptors exceed the maximum adjusted permissible noise level. Only the predicted noise levels at NSR1 (type a limit) are within the maximum adjusted permissible noise level for both periods. This exceedance can be attributed to NSR1 being the receptor farthest to area of development. Thus, it is anticipated that there will be an impact on human receptors except at NSR1 due to work activities at Area 3.

On the other hand, the noise level in the forested area around the construction areas range between 60 to 85 dB(A) during daytime construction activities and between 50 to 75 dB(A) during night-time activities.

Overall, potential noise impacts on NSR without any mitigation measures are expected to be of negative nature, direct, and local. Duration is anticipated to be short term and reversible as the impact will cease upon completion of the noise generating construction activities.

#### 6.5.4 Prediction of Mitigated Impacts

The predicted noise levels during construction phase presented above indicated that, there would be exceedances to the adjusted maximum noise level for human receptors. Also, the predicted noise levels within forested area range higher than the suggested noise threshold levels for fauna.

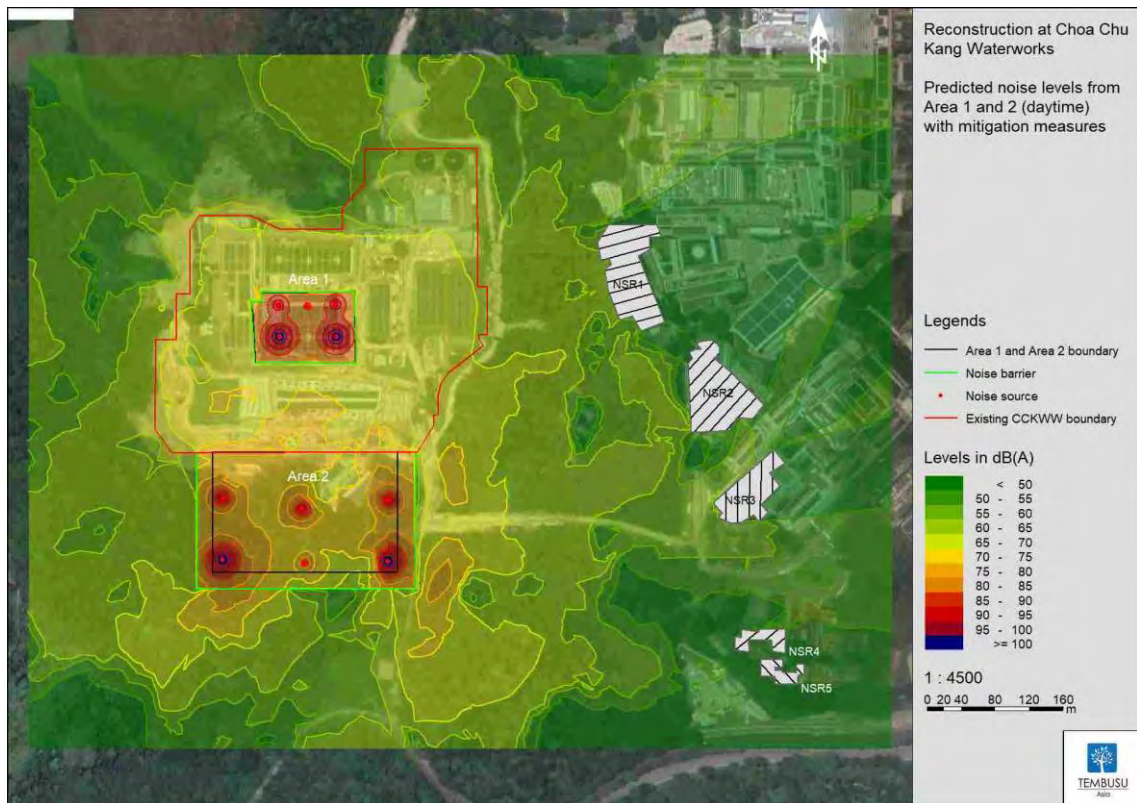
It is recommended to use noise barrier (STC 20) of at least 6m height at the construction work boundary as part of mitigation strategy to reduce noise impacts. Noise barrier is considered to be implemented around Area 1 work boundary within CCKWW plant, along forested areas at Area 2 work boundary, and at respective manhole locations along Nanyang Drive and Nanyang Crescent.

Results of the predicted noise levels with the implementation of noise barriers at the affected NSRs are summarised respectively in Table 6.14, Figure 6.12 (daytime noise levels), and Figure 6.13 (night-time noise levels) for Area 1 and 2 as well as Table 6.15, Figure 6.14 (daytime noise levels), and Figure 6.15 (night-time noise levels) for Area 3.

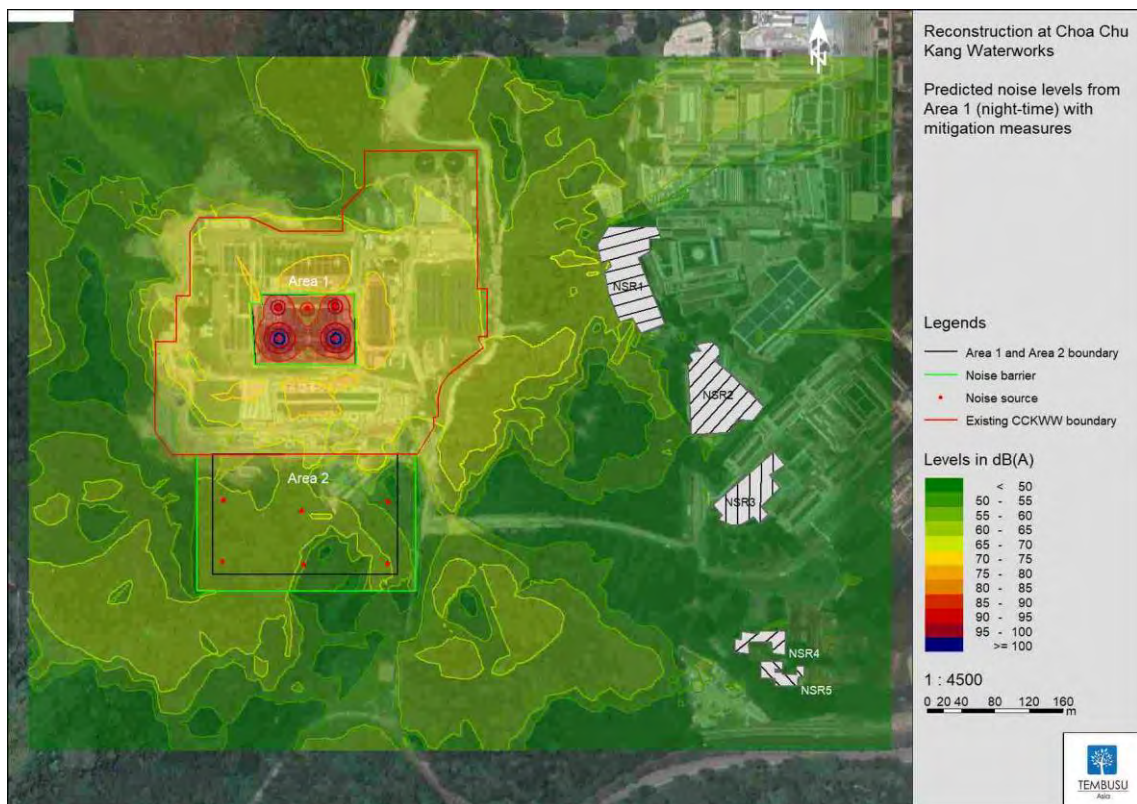
**Table 6.14.** Predicted noise levels at Area 1 and Area 2 with mitigation measures

| ID   | Noise Limit (dB(A)) |               | Noise Level with Mitigation Measures (dB(A)) |               |
|------|---------------------|---------------|----------------------------------------------|---------------|
|      | 7 am to 7 pm        | 7 pm to 12 am | 7 am to 7 pm                                 | 7 pm to 12 am |
| NSR1 | 64                  | 60            | <b>64.3</b>                                  | <b>65.6</b>   |
| NSR2 | 64                  | 60            | 60.1                                         | 50.6          |
| NSR3 | 64                  | 60            | 61.8                                         | 56.5          |
| NSR4 | 75                  | 65            | 52.8                                         | 51.8          |
| NSR5 | 75                  | 65            | 50.8                                         | 47.7          |

*Note: Bold results shows the exceedance.*



**Figure 6.12.** Predicted day-time noise level at Area 1 and 2 with mitigation measures

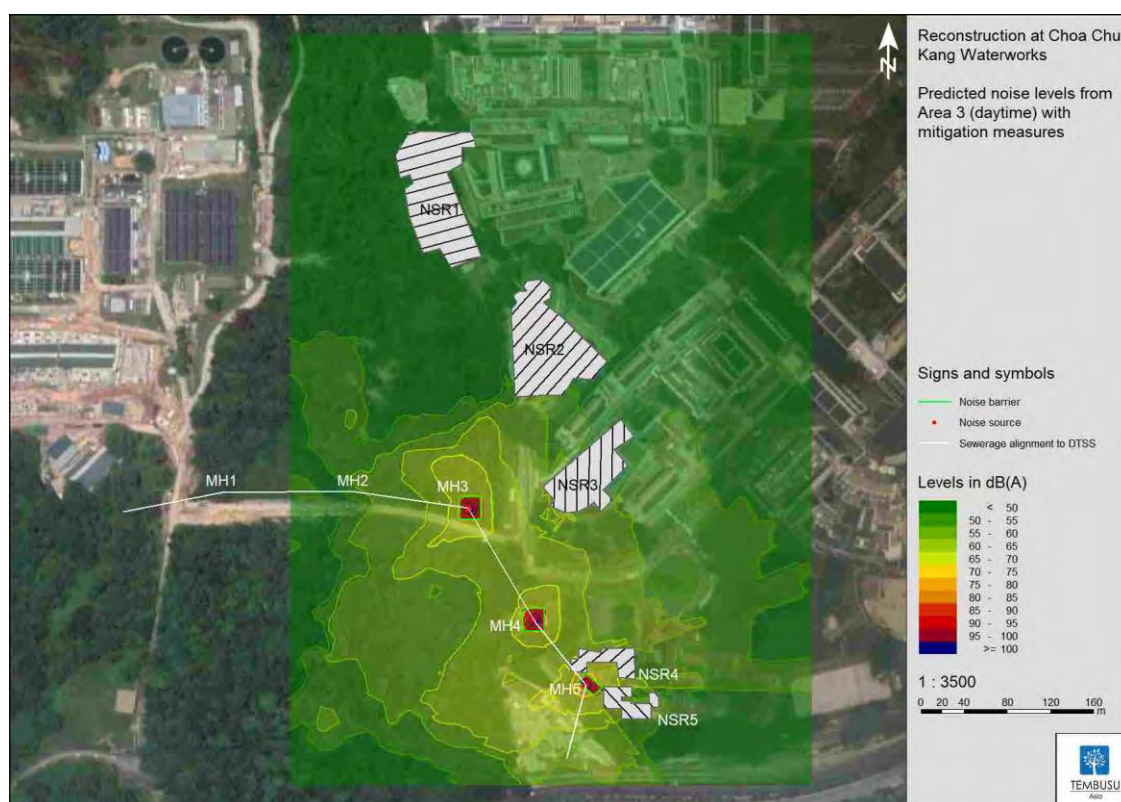


**Figure 6.13.** Predicted night-time noise level at Area 1 with mitigation measures

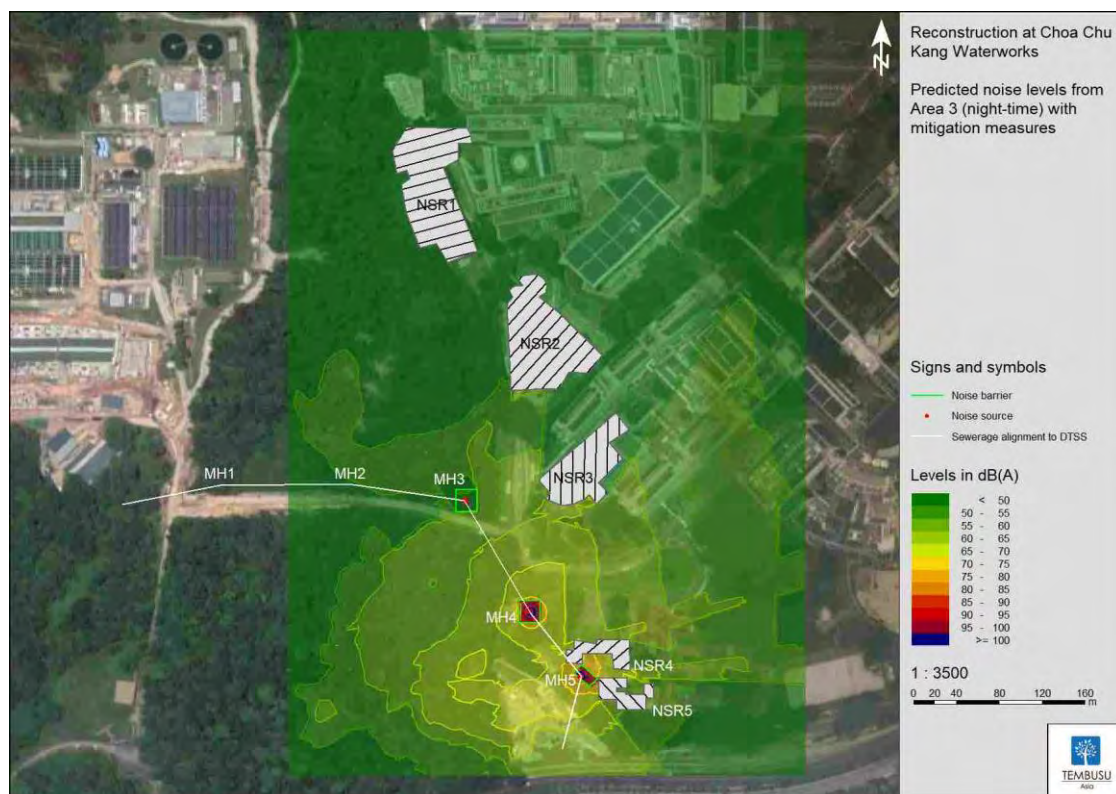
**Table 6.15.** Predicted noise levels at Area 3 with mitigation measures

| ID   | Noise Limit (dB(A)) |               | Noise Level with Mitigation Measures (dB(A)) |               |
|------|---------------------|---------------|----------------------------------------------|---------------|
|      | 7 am to 7 pm        | 7 pm to 12 am | 7 am to 7 pm                                 | 7 pm to 12 am |
| NSR1 | 64                  | 60            | 47.4                                         | 46.8          |
| NSR2 | 64                  | 60            | 58.8                                         | 56.1          |
| NSR3 | 64                  | 60            | 62.9                                         | <b>61.3</b>   |
| NSR4 | 75                  | 65            | <b>76.1</b>                                  | <b>78.4</b>   |
| NSR5 | 75                  | 65            | 74.5                                         | <b>76.8</b>   |

Note: Bold results shows the exceedance.



**Figure 6.14.** Predicted day-time noise level at Area 3 with mitigation measures



**Figure 6.15.** Predicted night-time noise level at Area 3 with mitigation measures

The modelling results show that after the implementation of noise barriers at Area 1 and 2 work boundaries, there would be no exceedances of the maximum adjusted permissible noise level at NSR2 to NSR5 and a marginal exceedance at NSR1. Also, the predicted noise level in the forested area around the Area 1 and 2 work boundaries range between 60 to 70 dB(A) during daytime and between 50 to 65 dB(A) during night-time which are similar to the suggested threshold of 65 dB(A) for fauna. After the implementation of noise barriers at Area 3 work boundaries, there would be exceedances of the maximum adjusted permissible noise level at NSR3, NSR4, and NSR5 during night-time period.

It is to be noted that these predicted noise levels are for worst case scenario and hence conservative in nature. Nevertheless, additional mitigation measures are proposed in next section to address noise impacts. These measures will help to lower the noise levels to acceptable levels.

#### **6.5.5 Additional Mitigation Measures**

This section aims to provide additional mitigation measures / best site practices to further reduce the excessive noise impacts from the project. The following recommendations should be implemented during pre-construction and construction stage.

- For soil investigation works, to use acoustic enclosures on rig engines and install a silencer at the exhaust pipe of the engines throughout the drilling operations and to erect portable hoarding around drilling rigs.
- All the work areas in Area 1 involving demolition activities are to be surrounded by sound barriers.

- Demolition activities in Area 1 should be avoided during the night (i.e., 6 pm to 8 am), with the demobilisation to be completed by 6 pm.
- Construction works within Area 2 are to be limited to daylight hours (8 am – 6 pm), including demobilisation, except in cases of work exigency and planned night activities, as explained in Section 2.7.2, with the implementation of the lighting management plan (LMP) approved by NParks. The specific measures recommended in Section 4.6.1 to manage these work exigency scenarios within Area 2 are to be implemented strictly.
- For piling work, to consider the use of quieter piling methods.
- To consider erection of temporary acoustic barriers around piling equipment. To consider use of a temporary barrier which will allow for relocation as the piling activity moves around site.
- Use of silent piler for sheet piling work in Area 2 (CCKWW Extension Area) where possible.
- Use hydraulic and electric tools in place of pneumatic equipment such as concrete breakers for demolition work at Area 1 (Existing CCKWW plant) where possible.
- Erect noise barriers at work boundaries for all Area 3 shaft locations within forested areas.
- Acoustic enclosures to be considered for where necessary specific plant such as compressors, generators, drilling tools etc.
- Means of improving the sound reduction for any exhaust systems in bulldozers, dump trucks and excavators should be considered.
- All noise and acoustic barriers should be able to achieve at least 10dB(A) noise reduction from source noise levels.
- Construction personnel to be trained in noise-reduction behaviours such as reducing the drop height of materials. Daily toolbox briefings should include reminders on the need to implement noise-reduction behaviours.
- All construction personnel should be educated about sensitive ecological nature of work areas before commencing the work
- To communicate with sensitive receptors about the proposed construction works including the proposed working hours, and the expected level and duration of noise.
- Where alternatives are available, only equipment and vehicles that emit lower noise levels are to be used.
- Care shall be taken when loading or unloading vehicles, dismantling scaffolding or moving materials to reduce impact noise.
- Start-up plant and vehicles sequentially rather than all together;
- Intermittently used vehicles and machinery are to be shut down between work periods or be throttled down to a minimum noise level emission.
- Where possible, vehicles and machinery known to emit high levels of noise in one direction are to be orientated away from sensitive receptors.
- Only well-maintained machineries should be operated on-site and should be serviced regularly during construction.
- Noise level monitoring throughout all phases to ensure that the construction noises remain within acceptable limits.
- Maintain a complaints procedure to log and track response to complaints

received from residents.

- Personnel are to wear appropriate Personal Protective Equipment (PPE) all the time while on the construction site.

Effective implementation of these mitigation measures should be able to reduce the noise impacts from project on sensitive receptors to acceptable levels.

#### **6.5.6 Operation Phase**

As the all the new CCKWW plant equipment and units will be designed as per applicable regulations & standards for noise limits and to be operated accordingly, there are no anticipated impact at operation phase and no mitigation measures are recommended.

#### **Cumulative Impacts**

There is a possibility that a cumulative negative impact to the ambient noise levels could occur if noise emissions from construction works of this project and nearby LTA JRL development happen simultaneously (see Section 2.10).

In order to accommodate the cumulative impacts of both developments, all impacts were given a score of 3 in the “Cumulativity” column of RIAM table to calculate the ES and assess overall impact. It is suggested that any possible coordination between the development agencies (i.e., LTA and PUB) is explored, which can help to avoid the concurrent construction activities or schedule in staggered manner.

#### **6.5.7 Evaluation of Impacts**

Table 6.16 and Table 6.17 summarise the impacts with their corresponding Environmental Scores before and after the implementation of mitigation measures, which are elaborated in the above section.

### **6.6 Residual Impacts**

The residual impacts were evaluated using the RIAM method (Table 6.16 and Table 6.17) with due consideration that the recommended mitigation measures are implemented by the Contractor. The residual impacts are likely to be in the band of slight negative and considered acceptable.

**Table 6.16.** Environmental Scores of the identified noise impacts with corresponding mitigation measures for Area 2

| Phase                           | Impact component                                                                               | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Predicted impact after mitigation |    |   |   |   |     |                 |
|---------------------------------|------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                                 |                                                                                                | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | I                                 | M  | P | R | C | ES  | Impact          |
| Ecology                         |                                                                                                |                                    |    |   |   |   |     |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Disturbance to fauna species in and around the project site                                    | 3                                  | -2 | 2 | 2 | 3 | -42 | Minor negative  | <ul style="list-style-type: none"><li>No night works (i.e., limit construction activities to 8 am – 6 pm), except in the case of work exigencies and planned nightwork, with the implementation of the LMP approved by NParks</li><li>Noise barrier with 6 m height as hoarding</li><li>Quieter equipment and vehicles with low noise levels to be used</li><li>To consider erection of temporary acoustic barriers around piling equipment</li><li>For piling work, to consider the use of quieter piling methods</li><li>Acoustic enclosures to be considered for compressors, generators, drilling tools etc.</li><li>All noise and acoustic barriers to be able to achieve at least 10dB(A) noise reduction from source noise levels.</li><li>Noise level monitoring throughout all phases to ensure that the construction noises remain within acceptable limits</li></ul> | 3                                 | -1 | 2 | 2 | 3 | -21 | Slight negative |
| Human Health and Safety         |                                                                                                |                                    |    |   |   |   |     |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Impact to construction workers due to exposure to high noise levels of construction activities | 2                                  | -2 | 2 | 2 | 3 | -28 | Slight negative | <ul style="list-style-type: none"><li>Personnel are to wear appropriate Personal Protective Equipment (PPE) at all times while on the construction site</li><li>Quieter equipment and vehicles with low noise levels to be used.</li><li>All noise and acoustic barriers to be able to achieve at least 10 dB(A) noise reduction from source noise levels.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2                                 | -1 | 2 | 2 | 3 | -14 | Slight negative |
|                                 | Impact to NSR due to noise pollution from construction activities                              | 2                                  | -2 | 2 | 2 | 3 | -28 | Slight negative | <ul style="list-style-type: none"><li>No night works (i.e., limit construction activities to 8 am – 6 pm), except in the case of work exigencies and planned</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2                                 | -1 | 2 | 2 | 3 | -14 | Slight negative |

| Phase | Impact component | Predicted impact before mitigation |   |   |   |   |    |        | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Predicted impact after mitigation |   |   |   |   |    |        |
|-------|------------------|------------------------------------|---|---|---|---|----|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|---|---|---|---|----|--------|
|       |                  | I                                  | M | P | R | C | ES | Impact |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | I                                 | M | P | R | C | ES | Impact |
|       |                  |                                    |   |   |   |   |    |        | nightwork, with the implementation of the LMP approved by NParks <ul style="list-style-type: none"> <li>Noise barrier with 6 m height as hoarding</li> <li>Quieter equipment and vehicles with low noise levels to be used</li> <li>To consider erection of temporary acoustic barriers around piling equipment</li> <li>For piling work, to consider the use of quieter piling methods</li> <li>Acoustic enclosures to be considered for compressors, generators, drilling tools etc.</li> <li>All noise and acoustic barriers to be able to achieve at least 10dB(A) noise reduction from source noise levels.</li> <li>Noise level monitoring throughout all phases to ensure that the construction noises remain within acceptable limits</li> </ul> |                                   |   |   |   |   |    |        |

**Table 6.17.** Environmental Scores of the identified noise impacts with corresponding mitigation measures for Area 3

| Phase                           | Impact component                                            | Predicted impact before mitigation |    |   |   |   |     |                | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Predicted impact after mitigation |    |   |   |   |     |                 |
|---------------------------------|-------------------------------------------------------------|------------------------------------|----|---|---|---|-----|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                                 |                                                             | I                                  | M  | P | R | C | ES  | Impact         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | I                                 | M  | P | R | C | ES  | Impact          |
| Ecology                         |                                                             |                                    |    |   |   |   |     |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Disturbance to fauna species in and around the project site | 2                                  | -2 | 2 | 2 | 3 | -42 | Minor negative | <ul style="list-style-type: none"><li>No night works (i.e., limit construction activities to 8 am – 6 pm) within the forested area, except in the case of work exigencies and planned nightwork, with the implementation of the LMP approved by NParks</li><li>Noise barrier with 6 m height as hoarding</li><li>Quieter equipment and vehicles with low noise levels to be used</li><li>To consider erection of temporary acoustic barriers around piling equipment</li></ul> | 2                                 | -1 | 2 | 2 | 3 | -21 | Slight negative |
|                                 |                                                             |                                    |    |   |   |   |     |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                   |    |   |   |   |     |                 |

| Phase                           | Impact component                                                                               | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Predicted impact after mitigation |    |   |   |   |     |                 |
|---------------------------------|------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                                 |                                                                                                | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | I                                 | M  | P | R | C | ES  | Impact          |
|                                 |                                                                                                |                                    |    |   |   |   |     |                 | <ul style="list-style-type: none"> <li>For piling work, to consider the use of quieter piling methods</li> <li>Acoustic enclosures to be considered for compressors, generators, drilling tools etc.</li> <li>All noise and acoustic barriers to be able to achieve at least 10 dB(A) noise reduction from source noise levels.</li> <li>Noise level monitoring throughout all phases to ensure that the construction noises remain within acceptable limits</li> </ul>                                                                                                                                                                                                                                                                                                                                    |                                   |    |   |   |   |     |                 |
| <b>Human Health and Safety</b>  |                                                                                                |                                    |    |   |   |   |     |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Impact to construction workers due to exposure to high noise levels of construction activities | 2                                  | -2 | 2 | 2 | 3 | -28 | Slight negative | <ul style="list-style-type: none"> <li>Personnel are to wear appropriate Personal Protective Equipment (PPE) at all times while on the construction site</li> <li>Quieter equipment and vehicles with low noise levels to be used</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2                                 | -1 | 2 | 2 | 3 | -14 | Slight negative |
|                                 | Impact to NSR due to noise pollution from construction activities                              | 1                                  | -3 | 2 | 2 | 3 | -21 | Slight negative | <ul style="list-style-type: none"> <li>No night works (i.e., limit construction activities to 8 am – 6 pm), except in the case of work exigencies</li> <li>Noise barrier with 6m height as hoarding</li> <li>Quieter equipment and vehicles with low noise levels to be used</li> <li>To consider erection of temporary acoustic barriers around piling equipment</li> <li>For piling work, to consider the use of quieter piling methods</li> <li>Acoustic enclosures to be considered for compressors, generators, drilling tools etc.</li> <li>All noise and acoustic barriers to be able to achieve at least 10dB(A) noise reduction from source noise levels.</li> <li>Noise level monitoring throughout all phases to ensure that the construction noises remain within acceptable limits</li> </ul> | 1                                 | -2 | 2 | 2 | 3 | -14 | Slight negative |

## 7 AMBIENT AIR QUALITY

### 7.1 Introduction

The main air pollution in Singapore comes from stationary sources (e.g., power stations, industries, and refineries) and mobile sources (e.g., motor vehicles and visiting marine vessels). The NEA monitors air quality in Singapore and publishes the 24-hour Pollutant Standards Index (PSI) reading that provides an indication of air quality at any point of time.

Currently there is not any stationary source of air pollution in forested area within project study area. The nearest source of air pollution outside project study area includes movement of vehicles within the area and exhaust emissions from vehicular traffic along the Nanyang Drive and Peng Kang Avenue. The project study area's natural setting, and the presence of forested area contribute to the relatively good prevailing air quality of the area.

This Section describes the relevant standard for the ambient air quality that are applicable to the Project activities, the methodology and results for the baseline ambient air quality study. It also described potential air quality impacts from construction works, and recommendation for mitigation measures. A qualitative impact assessment approach is being used for assessment of impacts.

### 7.2 Relevant Environmental Legislation, Guidelines and Standards

Singapore ambient air quality targets are provided in Table 7.1.

**Table 7.1.** Applicable Singapore Ambient Air Quality Targets (NEA, 2021)

| Pollutant                               | Targets by 2020                                                                  | Long Term Targets                                       |
|-----------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------|
| Particulate Matter (PM <sub>10</sub> )  | 24-hour mean: 50 µg/m <sup>3</sup> (WHO Final)                                   |                                                         |
| Particulate Matter (PM <sub>2.5</sub> ) | 24-hour mean: 37.5 µg/m <sup>3</sup> (WHO Interim Target)                        | 24-hour mean: 25 µg/m <sup>3</sup> (WHO Interim Target) |
| Sulphur Dioxide (SO <sub>2</sub> )      | 24-hour mean: 50 µg/m <sup>3</sup> (WHO Interim Target)                          | 24-hour mean: 20 µg/m <sup>3</sup> (WHO Final)          |
| Carbon Monoxide (CO)                    | 1-hour mean: 30 mg/m <sup>3</sup> (WHO Final); 8-hour mean: 10 mg/m <sup>3</sup> |                                                         |
| Nitrogen Dioxide (NO <sub>2</sub> )     | 1-hour mean: 200 µg/m <sup>3</sup> (WHO Final)                                   |                                                         |
| Ozone (O <sub>3</sub> )                 | 8-hour mean: 100 µg/m <sup>3</sup> (WHO Final)                                   |                                                         |

### 7.3 Baseline Methodology

#### 7.3.1 Sensitive Receptors Identification

The Air Sensitive Receptor (ASR) potentially affected by the proposed project have been identified (Table 7.2) through a combination of desktop study and visual surveys at the study areas.

**Table 7.2.** List of identified air-sensitive receptors

| ID   | Type                           | Description                                       | Approximate Distance* |
|------|--------------------------------|---------------------------------------------------|-----------------------|
| ASR1 | Institution of higher learning | Experimental Medicine Building                    | 300m from Area 1      |
| ASR2 | Institution of higher learning | Research Techno Plaza                             | 120m from Area 3      |
| ASR3 | Institution of higher learning | Wee Kim Wee School of Communication & Information | 70m from Area 3       |
| ASR4 | Residential building           | Block 36 of the former NTU Hall of Residence 7    | 10m from Area 3       |
| ASR5 | Residential building           | Block 37 of the former NTU Hall of Residence 7    | 20m from Area 3       |

\*Approximate distance from the nearest project work area

### 7.3.2 Baseline Field Survey

The baseline ambient air quality monitoring was carried out at Two (2) locations for a week (24 hours x 7 days) duration. The monitoring locations were chosen based on the factors such as sensitive receptor, relevance to project activity, site access and equipment security. Six (6) parameters were monitored: Particulate Matters with diameters of less than 10 and 2.5 microns (PM<sub>10</sub> and PM<sub>2.5</sub>), Sulphur Dioxide (SO<sub>2</sub>), Carbon Monoxide (CO), Nitrogen Dioxide (NO<sub>2</sub>), and Ozone (O<sub>3</sub>).

PM<sub>10</sub> and PM<sub>2.5</sub> were monitored using the UKA M-CERTS-certified TSI DustTrak Aerosol Monitor Model 8543-M. This equipment was powered by solar panels, enabling its light-scattering laser photometer to operate continuously and record real-time aerosol mass readings of the Particulate Matters. Data were recorded at 10-minute intervals, with their 24-hour averages calculated and compared with the Ambient Air Quality Targets (NEA, 2021). The remaining four parameters were monitored using electrochemical sensors outfitted in the battery-powered AQMesh Air Quality Monitoring System. This equipment recorded monitoring data at 15-minute intervals, from which hourly and 24-hour averages were calculated and compared with Ambient Air Quality Targets (NEA, 2021).

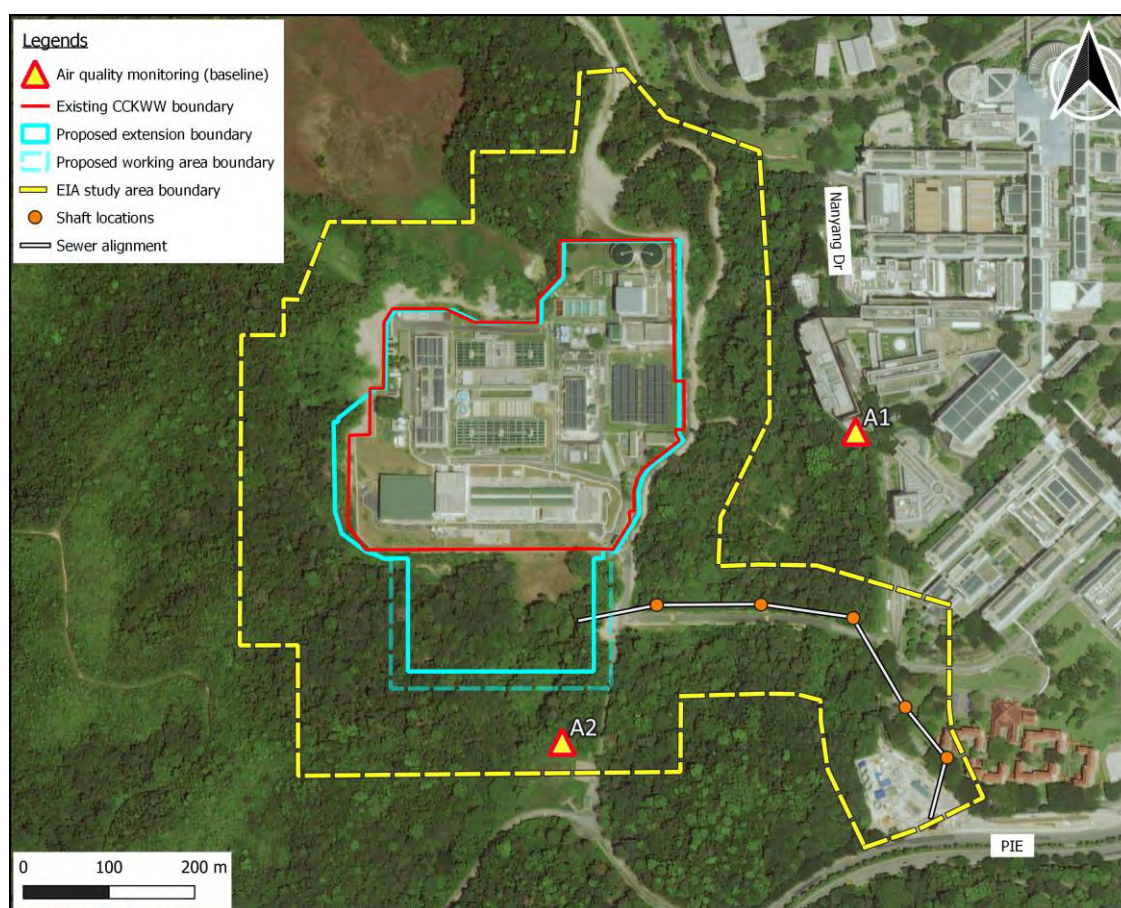
Table 7.3 lists down the coordinates of monitoring stations and short description of the location of each station. Figure 7.1 the photographs of on-site monitoring equipment and Figure 7.2 shows the location map of monitoring stations. The baseline ambient air quality monitoring report is available as **Appendix J**.

**Table 7.3.** Location of ambient air quality monitoring stations

| Station | Monitoring Period | Latitude | Longitude  | Description                                                                                                                        |
|---------|-------------------|----------|------------|------------------------------------------------------------------------------------------------------------------------------------|
| A1      | 24–30 Aug 2021    | 1.343694 | 103.678806 | Open field at the top of the slope just next to the unloading/loading bay of NTU Experimental Medicine Building and Nanyang Avenue |
| A2      | 1–7 Sep 2021      | 1.340417 | 103.675972 | South of CCKWW, next to Dunbar Walk                                                                                                |



**Figure 7.1.** Setup of the baseline ambient air quality monitoring equipment



**Figure 7.2.** Location of baseline ambient air quality monitoring stations within project area

## 7.4 Baseline Results and Discussion

All ambient air quality monitoring results complied with Singapore Ambient Air Quality Targets (NEA, 2021). Representative monitoring data are depicted in the following order:

- 24-hour average of PM<sub>10</sub>, PM<sub>2.5</sub>, and SO<sub>2</sub> in Table 7.4 for Station A1 and Table 7.7 for Station A2;

- 8-hour average and maximum daily hourly average of CO in Table 7.5 for Station A1 and
- Table 7.8 for Station A2; and
- 8-hour average of O3 and maximum daily hourly average of NO2 in
- Table 7.6 for Station A1 and Table 7.9 for Station A2.

#### 7.4.1 Results of Station A1

**Table 7.4.** 24-hour average of PM<sub>10</sub>, PM<sub>2.5</sub>, and SO<sub>2</sub> levels at Station A1

| Monitoring Period                          | 24-hour PM <sub>10</sub><br>average (µg/m <sup>3</sup> ) | 24-hour PM <sub>2.5</sub><br>average (µg/m <sup>3</sup> ) | 24-hour SO <sub>2</sub><br>average (µg/m <sup>3</sup> ) |
|--------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------|
| Tuesday, 24/8/2021                         | 11                                                       | 10                                                        | 11                                                      |
| Wednesday, 25/8/2021                       | 12                                                       | 10                                                        | 14                                                      |
| Thursday, 26/8/2021                        | 16                                                       | 14                                                        | 18                                                      |
| Friday, 27/8/2021                          | 19                                                       | 16                                                        | 15                                                      |
| Saturday, 28/8/2021                        | 10                                                       | 9                                                         | 15                                                      |
| Sunday, 29/8/2021                          | 13                                                       | 12                                                        | 10                                                      |
| Monday, 30/8/2021                          | 12                                                       | 10                                                        | 6                                                       |
| <b>2020 Target (µg/m<sup>3</sup>)</b>      | <b>50</b>                                                | <b>37.5</b>                                               | <b>50</b>                                               |
| <b>Long Term Target (µg/m<sup>3</sup>)</b> | <b>50</b>                                                | <b>25</b>                                                 | <b>20</b>                                               |

**Table 7.5.** 8-hour and maximum daily hourly average of CO level at Station A1

| Monitoring Period                          | Duration    | 8-hour CO average<br>(mg/m <sup>3</sup> ) | Max. hourly CO average<br>(mg/m <sup>3</sup> ) |
|--------------------------------------------|-------------|-------------------------------------------|------------------------------------------------|
| Tuesday, 24/8/2021                         | 00:00–08:00 | 0.4                                       | 1                                              |
|                                            | 08:00–16:00 | 0.6                                       |                                                |
|                                            | 16:00–00:00 | 0.7                                       |                                                |
| Wednesday, 25/8/2021                       | 00:00–08:00 | 0.8                                       | 1                                              |
|                                            | 08:00–16:00 | 0.5                                       |                                                |
|                                            | 16:00–00:00 | 0.5                                       |                                                |
| Thursday, 26/8/2021                        | 00:00–08:00 | 0.6                                       | 1                                              |
|                                            | 08:00–16:00 | 0.6                                       |                                                |
|                                            | 16:00–00:00 | 0.5                                       |                                                |
| Friday, 27/8/2021                          | 00:00–08:00 | 0.8                                       | 1.1                                            |
|                                            | 08:00–16:00 | 0.8                                       |                                                |
|                                            | 16:00–00:00 | 0.4                                       |                                                |
| Saturday, 28/8/2021                        | 00:00–08:00 | 0.4                                       | 0.7                                            |
|                                            | 08:00–16:00 | 0.4                                       |                                                |
|                                            | 16:00–00:00 | 0.3                                       |                                                |
| Sunday, 29/8/2021                          | 00:00–08:00 | 0.4                                       | 0.7                                            |
|                                            | 08:00–16:00 | 0.3                                       |                                                |
|                                            | 16:00–00:00 | 0.4                                       |                                                |
| Monday, 30/8/2021                          | 00:00–08:00 | 0.6                                       | 0.9                                            |
|                                            | 08:00–16:00 | 0.4                                       |                                                |
|                                            | 16:00–00:00 | 0.3                                       |                                                |
| <b>Singapore Target (mg/m<sup>3</sup>)</b> |             | <b>10</b>                                 | <b>30</b>                                      |

**Table 7.6.** 8-hour average of O<sub>3</sub> and maximum daily hourly average of NO<sub>2</sub> at Station A1

| Monitoring Period                     | Duration    | 8-hour O <sub>3</sub> average (µg/m <sup>3</sup> ) | Max. hourly NO <sub>2</sub> average (µg/m <sup>3</sup> ) |
|---------------------------------------|-------------|----------------------------------------------------|----------------------------------------------------------|
| Tuesday, 24/8/2021                    | 00:00–08:00 | 9                                                  | 36                                                       |
|                                       | 08:00–16:00 | 5                                                  |                                                          |
|                                       | 16:00–00:00 | 5                                                  |                                                          |
| Wednesday, 25/8/2021                  | 00:00–08:00 | 6                                                  | 36                                                       |
|                                       | 08:00–16:00 | 26                                                 |                                                          |
|                                       | 16:00–00:00 | 35                                                 |                                                          |
| Thursday, 26/8/2021                   | 00:00–08:00 | 16                                                 | 37                                                       |
|                                       | 08:00–16:00 | 40                                                 |                                                          |
|                                       | 16:00–00:00 | 29                                                 |                                                          |
| Friday, 27/8/2021                     | 00:00–08:00 | 4                                                  | 34                                                       |
|                                       | 08:00–16:00 | 30                                                 |                                                          |
|                                       | 16:00–00:00 | 31                                                 |                                                          |
| Saturday, 28/8/2021                   | 00:00–08:00 | 28                                                 | 32                                                       |
|                                       | 08:00–16:00 | 15                                                 |                                                          |
|                                       | 16:00–00:00 | 50                                                 |                                                          |
| Sunday, 29/8/2021                     | 00:00–08:00 | 19                                                 | 33                                                       |
|                                       | 08:00–16:00 | 32                                                 |                                                          |
|                                       | 16:00–00:00 | 15                                                 |                                                          |
| Monday, 30/8/2021                     | 00:00–08:00 | 4                                                  | 34                                                       |
|                                       | 08:00–16:00 | 9                                                  |                                                          |
|                                       | 16:00–00:00 | 35                                                 |                                                          |
| Singapore Target (µg/m <sup>3</sup> ) |             | 100                                                | 200                                                      |

#### 7.4.2 Result of Station A2

**Table 7.7.** 24-hour average of PM<sub>10</sub> and PM<sub>2.5</sub> levels at Station A2

| Monitoring Period                          | 24-hour PM <sub>10</sub> average (µg/m <sup>3</sup> ) | 24-hour PM <sub>2.5</sub> average (µg/m <sup>3</sup> ) | 24-hour SO <sub>2</sub> average (µg/m <sup>3</sup> ) |
|--------------------------------------------|-------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------|
| Wednesday, 1/9/2021                        | 10                                                    | 9                                                      | 8                                                    |
| Thursday, 2/9/2021                         | 7                                                     | 6                                                      | 16                                                   |
| Friday, 3/9/2021                           | 8                                                     | 7                                                      | 18                                                   |
| Saturday, 4/9/2021                         | 13                                                    | 11                                                     | 18                                                   |
| Sunday, 5/9/2021                           | 11                                                    | 9                                                      | 11                                                   |
| Monday, 6/9/2021                           | 13                                                    | 11                                                     | 8                                                    |
| Tuesday, 7/9/2021                          | 13                                                    | 11                                                     | 9                                                    |
| <b>2020 Target (µg/m<sup>3</sup>)</b>      | <b>50</b>                                             | <b>37.5</b>                                            | <b>50</b>                                            |
| <b>Long Term Target (µg/m<sup>3</sup>)</b> | <b>50</b>                                             | <b>25</b>                                              | <b>20</b>                                            |

**Table 7.8.** 8-hour and maximum daily hourly average of CO level at Station A2

| Monitoring Period   | Duration    | 8-hour CO average (mg/m <sup>3</sup> ) | Max. hourly CO average (mg/m <sup>3</sup> ) |
|---------------------|-------------|----------------------------------------|---------------------------------------------|
| Wednesday, 1/9/2021 | 00:00–08:00 | 0.3                                    | 0.6                                         |
|                     | 08:00–16:00 | 0.3                                    |                                             |
|                     | 16:00–00:00 | 0.2                                    |                                             |

| Monitoring Period                     | Duration    | 8-hour CO average (mg/m <sup>3</sup> ) | Max. hourly CO average (mg/m <sup>3</sup> ) |
|---------------------------------------|-------------|----------------------------------------|---------------------------------------------|
| Thursday, 2/9/2021                    | 00:00–08:00 | 0.4                                    | 0.7                                         |
|                                       | 08:00–16:00 | 0.3                                    |                                             |
|                                       | 16:00–00:00 | 0.4                                    |                                             |
| Friday, 3/9/2021                      | 00:00–08:00 | 0.3                                    | 0.6                                         |
|                                       | 08:00–16:00 | 0.4                                    |                                             |
|                                       | 16:00–00:00 | 0.3                                    |                                             |
| Saturday, 4/9/2021                    | 00:00–08:00 | 0.4                                    | 0.6                                         |
|                                       | 08:00–16:00 | 0.4                                    |                                             |
|                                       | 16:00–00:00 | 0.3                                    |                                             |
| Sunday, 5/9/2021                      | 00:00–08:00 | 0.5                                    | 0.8                                         |
|                                       | 08:00–16:00 | 0.5                                    |                                             |
|                                       | 16:00–00:00 | 0.3                                    |                                             |
| Monday, 6/9/2021                      | 00:00–08:00 | 0.4                                    | 0.7                                         |
|                                       | 08:00–16:00 | 0.5                                    |                                             |
|                                       | 16:00–00:00 | 0.3                                    |                                             |
| Tuesday, 7/9/2021                     | 00:00–08:00 | 0.6                                    | 0.8                                         |
|                                       | 08:00–16:00 | 0.4                                    |                                             |
|                                       | 16:00–00:00 | 0.2                                    |                                             |
| Singapore Target (mg/m <sup>3</sup> ) |             | 10                                     | 30                                          |

**Table 7.9.** 8-hour average of O<sub>3</sub> and maximum daily hourly average of NO<sub>2</sub> at Station A2

| Monitoring Period                     | Duration    | 8-hour O <sub>3</sub> average (µg/m <sup>3</sup> ) | Max. hourly NO <sub>2</sub> average (µg/m <sup>3</sup> ) |
|---------------------------------------|-------------|----------------------------------------------------|----------------------------------------------------------|
| Wednesday, 1/9/2021                   | 00:00–08:00 | 10                                                 | 35                                                       |
|                                       | 08:00–16:00 | 38                                                 |                                                          |
|                                       | 16:00–00:00 | 35                                                 |                                                          |
| Thursday, 2/9/2021                    | 00:00–08:00 | 35                                                 | 32                                                       |
|                                       | 08:00–16:00 | 11                                                 |                                                          |
|                                       | 16:00–00:00 | 31                                                 |                                                          |
| Friday, 3/9/2021                      | 00:00–08:00 | 6                                                  | 38                                                       |
|                                       | 08:00–16:00 | 18                                                 |                                                          |
|                                       | 16:00–00:00 | 54                                                 |                                                          |
| Saturday, 4/9/2021                    | 00:00–08:00 | 20                                                 | 37                                                       |
|                                       | 08:00–16:00 | 45                                                 |                                                          |
|                                       | 16:00–00:00 | 60                                                 |                                                          |
| Sunday, 5/9/2021                      | 00:00–08:00 | 5                                                  | 37                                                       |
|                                       | 08:00–16:00 | 23                                                 |                                                          |
|                                       | 16:00–00:00 | 69                                                 |                                                          |
| Monday, 6/9/2021                      | 00:00–08:00 | 11                                                 | 38                                                       |
|                                       | 08:00–16:00 | 25                                                 |                                                          |
|                                       | 16:00–00:00 | 56                                                 |                                                          |
| Tuesday, 7/9/2021                     | 00:00–08:00 | 11                                                 | 34                                                       |
|                                       | 08:00–16:00 | 24                                                 |                                                          |
|                                       | 16:00–00:00 | 35                                                 |                                                          |
| Singapore Target (µg/m <sup>3</sup> ) |             | 100                                                | 200                                                      |

## 7.5 Impact Assessment

### 7.5.1 Identification of Potential Impacts

#### *Pre-construction / Construction Phase*

##### **Area 1 (Existing CCKWW Plant)**

Project activities planned for this Area include the demolition of existing infrastructures and construction of new facilities as explained in Chapter 2. Based on these activities, the following potential impacts on ambient air quality are identified:

##### **Fugitive Dust Emissions from Construction Works**

The following construction works are expected to create dust emission:

- Demolition of existing buildings within CCKWW plant
- Construction of new buildings within CCKWW plant
- Vehicle movements on access roads

Fugitive dust emissions from construction activities are unavoidable to be generated during the demolition and construction works. These fugitive dust emissions are expected to have a moderate and direct impact on local air quality and affect air sensitive receptor located near dust generating construction activities (i.e., construction workers and staff on-site CCKWW).

The impact duration of fugitive emissions of construction dust is anticipated to be short term and expected to vary significantly from day to day depending upon the duration of dust-generating construction activities within project site. Actual concentrations of fugitive dust emissions from construction site will depend on length of operations, effectiveness of control measures and ambient weather conditions (e.g., rainfall, wind speed, and wind direction). Extent of fugitive emissions is considered to be local, within the project as forested area around project study area will act as barrier for further dispersion. Also, the impacts are reversible when dust-generating construction activities stop.

Hence, fugitive dust emissions impact is considered to be unavoidable, of local importance, having a local geographical extent, short term duration, reversible, and with minor impact magnitude.

##### **Exhaust Emissions from Construction Machinery and Heavy Vehicles**

Project activities in this Area are anticipated to use fuel-burning (i.e., mainly diesel) machinery such as piling rig, road roller machine, generator sets, mobile cranes, and other heavy vehicles such as grader, dozer, excavator, front-end loader, and haul truck that may cause exhaust emission. These exhaust emissions can cause direct and negative effects on local air quality by potentially increasing the concentrations of CO, NO<sub>2</sub>, SO<sub>2</sub>, O<sub>3</sub>, PM<sub>10</sub> and PM<sub>2.5</sub>. This could cause the local air quality to deteriorate.

The impact is expected to be short-term, depending on the duration of operation of machinery and heavy vehicles. The geographical extent of exhaust emissions is considered to be local, within the project site, even its dispersion will be affected by a number of factors such as ambient weather and wind direction. Air quality impacts due

to exhaust emissions of fuel burning machinery and vehicles are reversible when they stop operating.

### **Area 2 (CCKWW Extension Area)**

Project activities in the site include soil investigation works consisting of twelve boreholes, site clearance, soil filling and earthworks for the specified platform level, and construction of post-treatment units and supporting infrastructures.

Based on these activities, the following potential impacts on ambient air quality are identified:

#### **Fugitive Dust Emissions from Construction Works**

The following construction works are expected to create dust emission:

- Site clearance
- Soil investigation work
- Soil filling and earthworks for the platform level
- Temporary stacking of excavated soil within project site
- Erection of temporary (e.g., site office) structures
- Bored piling work
- Earthworks for pile cap
- Backfill, soil mixing and compaction
- Cutting and grinding work for pile head
- Construction of the new infrastructures
- Vehicle movements on access roads

Fugitive dust emissions from construction activities are unavoidable to be generated during soil excavation, backfilling activities, and vehicle movement within the project site on unpaved surfaces. These fugitive dust emissions are expected to have a moderate and direct impact on local air quality and affect air sensitive receptor located near dust generating construction activities (i.e., construction workers).

Additionally, deteriorated ambient air quality also has negative impacts on biodiversity. For example, particulate matter settling on leaves may cause leaf injury through abrasion or chemical interactions. Dust particles entering stomata may also reduce photosynthetic functions. Dust landing directly on soil may affect nutrient cycling processes (Grantz, Garner, & Johnson, 2003).

The existing literature on the effect of construction dust on birds in particular is limited, with most studies examining effects of air pollution on wildlife in general, rather than the effect of construction dust on birds specifically. Since the avian respiratory system, unlike the mammalian respiratory system, is characterized by unidirectional airflow and cross-current gas exchange, this makes them more susceptible to high concentrations of pollutants in the air. Studies have shown that birds exposed to urban air pollution may exhibit a build-up of cellular and mineral debris leading to characteristic conditions of pneumonia (Sanderfoot & Holloway, 2017).

The impact duration of fugitive emissions of construction dust is anticipated to be short term and expected to vary significantly from day to day depending upon the duration of dust-generating construction activities within project site. Actual concentrations of fugitive dust emissions from construction site will depend on length of operations, effectiveness of control measures and ambient weather conditions (e.g., rainfall, wind speed, and wind direction). Extent of fugitive emissions is considered to be local, within the project as forested area around project study area will act as barrier for further dispersion. Also, the impacts are reversible when dust-generating construction activities stop.

#### *Exhaust Emissions from Construction Machinery and Heavy Vehicles*

Project activities in this Area are anticipated to use fuel-burning (i.e., mainly diesel) machinery such as piling rig, road roller machine, generator sets, mobile cranes, and other heavy vehicles such as grader, dozer, excavator, front-end loader, and haul truck that may cause exhaust emission. These exhaust emissions can cause direct and negative effects on local air quality by potentially increasing the concentrations of CO, NO<sub>2</sub>, SO<sub>2</sub>, O<sub>3</sub>, PM<sub>10</sub> and PM<sub>2.5</sub>. This could cause the local air quality to deteriorate.

The impact is expected to be short-term depending on the duration of operation of machinery and heavy vehicles. The geographical extent of exhaust emissions is considered to be local, within the project site, even its dispersion will be affected by a number of factors such as ambient weather and wind direction. Ambient air quality impacts due to exhaust emissions of fuel burning machinery and vehicles are reversible when they stop operating.

#### *Odour Emissions*

During construction works, it is anticipated that odours generated at project site would be minimal. The potential impact is anticipated to be short term, indirect and insignificant as it would be assumed to be managed by suitable waste management mitigation measures discussed in Chapter 10.

#### **Area 3 (Sewerage Alignment)**

In Area 3, new sewerage alignment to DTSS will be constructed by the means of pipe jacking method. Soil investigation work will be carried out at 3 locations along Peng Kang Avenue. Considering the method of construction and extent of working area requirement, impacts on ambient quality are expected to be minor. Nevertheless, the following potential impacts on ambient air quality are identified:

#### *Fugitive Dust Emissions from Construction Works*

The following construction works are expected to create dust emission:

- Soil investigation work
- Temporary stacking of excavated soil within project site
- Bored piling work
- Earth work for pile cap
- Cutting and grinding work for pile head

Fugitive dust emissions from construction activities are unavoidable within the project

site for the pipe jacking method. These fugitive dust emissions are expected to have a moderate and direct impact on local air quality and affect air sensitive receptor located near dust generating construction activities (i.e., construction workers).

The impact duration of fugitive emissions of construction dust is anticipated to be short term and expected to vary significantly from day to day depending upon the duration of dust-generating construction activities within project site. Actual concentration of fugitive dust emissions from construction site will depend on length of operations, effectiveness of control measures and ambient weather conditions (e.g., wind speed, rainfall, and wind direction). Extent of fugitive emissions is considered to be local, within the project as forested area around project study area will act as barrier for further dispersion. Also, the impacts are reversible when dust-generating construction activities stop.

#### **Exhaust Emissions from Construction Machinery and Heavy Vehicles**

It is anticipated that the pipe jacking activities potentially use fuel-burning (i.e., mainly diesel) machinery such as piling rig, mobile cranes, generator sets, and other heavy vehicles such as excavator and haul truck that may cause exhaust emission. The impact is expected to be short-term depending on the duration of operation of machinery and heavy vehicles. Air quality impacts due to exhaust emissions of fuel burning machinery and vehicles are reversible when they stop operating.

Overall, potential air quality impacts on ASR are expected to be of slight negative nature, direct, and local. Duration is anticipated to be short term and reversible as the impact will cease upon completion of the air pollution generating construction activities. As such, mitigation measures are recommended to further reduce the potential impacts.

#### **Operation Phase**

No impacts on ambient air quality are expected to be generated by during operation of CCKWW plant.

#### **Cumulative Impacts**

There is a possibility that a cumulative negative impact to the ambient air quality could occur if air emissions from construction works of this project and nearby LTA JRL development (see Section 2.10) happen simultaneously.

In order to accommodate the cumulative impacts of both developments, all impacts were given a score of 3 in the “Cumulativity” column of RIAM table to calculate the ES and assess overall impact. It is suggested that any possible coordination between the development agencies (i.e., LTA and PUB) is explored, which can help to avoid the concurrent construction activities or schedule in staggered manner.

### **7.5.2 Evaluation of Potential Impacts**

Table 7.10 summarises the impacts covering of all three areas with their corresponding Environmental Scores before and after the implementation of mitigation measures, which are elaborated in the following section.

## **7.6 Mitigation Measures**

### **7.6.1 Pre-construction / Construction Phase**

The combined mitigation measures are proposed covering all three work Areas which are as under:

For site dust control, the contractor shall prepare and implement the dust control plan covering following dust suppression measures:

- Use of hoarding at project boundary within work areas to minimise dust generation by attenuating wind forces.
- To avoid stockpiles of soil and dusty materials at project site within work areas as far as possible.
- If unavoidable, stockpiles of soil should be located away as far as possible from sensitive receptors (i.e., adjacent flora and fauna as well as residents of NTU Hall of Residence 7).
- Provide additional dust screen at project boundary near sensitive receptors (i.e., adjacent flora and fauna as well as residents of NTU Hall of Residence 7).
- Any soil or stockpiles of dusty material should be properly stored, covered entirely with impervious sheeting or dampened with water to maintain entire surface wet by contractor.
- Soil stockpiles shall not be higher than 0.6 times the nearest hoarding height.
- Excavations should be backfilled or reinstated as soon as practicable following completion of the construction work.
- Material transport of inert solids (excavated materials) should be enclosed using impervious sheeting, minimising the visual dust impacts as well.
- Use of regular watering to reduce dust emissions from exposed site surfaces, particularly during dry weather on open areas.
- Open burning of construction and other wastes are not allowed at the worksite as this is an offence under the Environmental Pollution Control regulation.
- Personal protective equipment such as mask shall be worn during the severe air pollution and/or dust exposure periods by construction personnel.
- Avoid soil disturbing works during dry and/or windy conditions.
- Stabilize/cover all stockpiled materials for longer than one month by turfing, erosion blanketing or other method.
- Vehicle on-site speed restrictions should be imposed to prevent dust being stirred up by vehicle movements.

Other mitigation measures that need to be taken by the contractor include:

- Provide vehicle washing facilities before the construction site exit.
- Pave the area between construction site exit and the vehicle washing facilities.
- Maintain road surface in the construction site wet (e.g., using sprinkler).
- Proper maintenance of construction vehicles and fuel burning equipment.
- Intermittently used vehicles and machinery are to be shut down between work periods to minimum exhaust emission.

The recommended measures above, when implemented effectively, shall mitigate the potential impacts on the ambient air quality of the site to acceptable levels.

### **7.6.2 Operation Phase**

As there are no impacts on ambient air quality are expected to be generated by the operational activities of CCKWW plant, no mitigation measures are proposed.

### **7.7 Residual Impacts**

The residual impacts were evaluated using the RIAM method (Table 7.10) with due consideration that the recommended mitigation measures are implemented by the Contractor. The residual impacts are likely to be in band of slight negative to no impact and considered acceptable.

**Table 7.10.** Environmental Scores of the identified impacts on site's ambient air quality with corresponding mitigation measures

| Phase                           | Impact component                                                                                                      | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Predicted impact after mitigation |    |   |   |   |     |                 |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                                 |                                                                                                                       | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | I                                 | M  | P | R | C | ES  | Impact          |
| Ecology                         |                                                                                                                       |                                    |    |   |   |   |     |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Impact to biodiversity due to generation of fugitive dust                                                             | 2                                  | -2 | 2 | 2 | 2 | -24 | Slight negative | <ul style="list-style-type: none"><li>To implement dust suppression plan</li><li>Use of hoarding at project boundary to minimise dust generation by attenuating wind forces.</li><li>To avoid stockpiles of soil and dusty materials at project site within forested area as far as possible.</li><li>Stockpiles of dusty material should be properly stored, covered entirely with impervious sheeting</li><li>Use of regular watering to reduce dust emissions from exposed site surface</li></ul>                                                                  | 2                                 | -1 | 2 | 2 | 2 | -12 | Slight Negative |
| Human Health and Safety         |                                                                                                                       |                                    |    |   |   |   |     |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Impact on construction workers due to generation of fugitive dust and exhaust emissions from construction machineries | 2                                  | -3 | 2 | 2 | 2 | -36 | Slight Negative | <ul style="list-style-type: none"><li>To implement dust suppression plan</li><li>Personal protective equipment i.e., face mask to be worn during dust exposure</li><li>Stockpiles of dusty material to be properly stored, covered entirely with impervious sheeting, or dampened with water</li><li>Use of regular watering to reduce dust emissions from exposed site surfaces</li><li>Proper maintenance of construction vehicles and fuel burning equipment</li><li>Intermittently-used vehicles and machinery are to be shut down between work periods</li></ul> | 2                                 | -2 | 2 | 2 | 2 | -24 | Slight Negative |
|                                 | Impact on ASR due to generation of fugitive dust                                                                      | 2                                  | -2 | 2 | 2 | 2 | -24 | Slight Negative | <ul style="list-style-type: none"><li>To implement dust suppression plan</li><li>Provide dust screen at project boundary near residential community</li><li>Stockpiles of dusty material to be properly stored, covered entirely with impervious sheeting, or dampened with water</li></ul>                                                                                                                                                                                                                                                                           | 2                                 | -1 | 2 | 2 | 2 | -12 | Slight Negative |

| Phase | Impact component                  | Predicted impact before mitigation |    |   |   |   |    |           | Mitigation measures                                                                                                          | Predicted impact after mitigation |    |   |   |   |    |           |
|-------|-----------------------------------|------------------------------------|----|---|---|---|----|-----------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|----|-----------|
|       |                                   | I                                  | M  | P | R | C | ES | Impact    |                                                                                                                              | I                                 | M  | P | R | C | ES | Impact    |
|       |                                   |                                    |    |   |   |   |    |           | <ul style="list-style-type: none"> <li>Use of regular watering to reduce dust emissions from exposed site surface</li> </ul> |                                   |    |   |   |   |    |           |
|       | Impact due to generation of odour | 1                                  | -1 | 2 | 2 | 2 | -6 | No Impact | <ul style="list-style-type: none"> <li>Refer to Section 10.4</li> </ul>                                                      | 1                                 | -1 | 2 | 2 | 2 | -6 | No Impact |

## 8 GROUND-BORNE VIBRATION

### 8.1 Introduction

This Section describes the relevant standard for the ground-borne vibration that are applicable to the Project activities, the methodology and results for the baseline ground vibration study. It also described potential ground vibration impacts from construction works, and recommendation for ground vibration mitigation measures. A qualitative impact assessment approach is being used for assessment of impacts.

### 8.2 Relevant Environmental Legislation, Guidelines and Standards

As there is no specific laws and standard in Singapore presently that describe the assessment criteria for ground vibration, the following international standards are used as reference for ground vibration assessment criteria associated with human annoyance.

1. BS 5228-2 Code of Practice for Noise and Vibration Control on Construction and Open Sites.
2. United States Federal Transit Authority (FTA) Transit Noise and Vibration Impact Assessment Manual.

The baseline ground vibration data obtained are compared with the ground-borne vibration limits stated in the BS 5228-2 Code of Practice for Noise and Vibration Control on Construction and Open Sites (Table 8.1).

**Table 8.1.** BS 5228 Part 2 Guidance on Human Response to Vibration Levels

| Sr No. | Vibration Level | Effect                  | Description                                                                                                                                                                                     |
|--------|-----------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | 0.14 mm/s       | Threshold of Perception | Vibration might be just perceptible in the most sensitive situations for most vibration frequencies associated with construction. At lower frequencies, people are less sensitive to vibration. |
| 2      | 0.3 mm/s        | Just Perceptible        | Vibration might be just perceptible in residential environment.                                                                                                                                 |
| 3      | 1.0 mm/s        | Likely Complaint        | It is likely that vibration of this level in residential environments will cause complaint, but can be tolerated if warning and explanation is given to residents                               |
| 4      | 10 mm/s         | Likely Intolerable      | Vibration is likely to be intolerable for any more than a brief exposure at this level                                                                                                          |

According to BS 5228-2, human beings are known to be very sensitive to vibration, the threshold of perception being typically in the PPV range of 0.14 mm·s<sup>-1</sup> to 0.3 mm·s<sup>-1</sup>. Vibrations above these values can disturb, startle, cause annoyance or interfere with work activities. At higher levels they can be described as unpleasant or even painful. In residential accommodation, vibrations can promote anxiety lest some structural mishap might occur.

## 8.3 Baseline Methodology

### 8.3.1 Sensitive Receptors Identification

The sensitive receptor identification criteria are based on the United States Federal Transit Authority (FTA) Transit Noise and Vibration Impact Assessment Manual. The criteria are listed below:

- Category 1 (High Sensitivity) covers manufacturing as well as vibration-sensitive research, hospitals with vibration-sensitive equipment, and university research operations.
- Category 2 (Residential) covers residential land uses and any buildings where people sleep, such as hotels and hospitals.
- Category 3 (Institutional) covers institutional land uses with primarily daytime use e.g., schools, churches, other institutions, and quiet offices that do not have vibration-sensitive equipment, but still have the potential for activity interference.
- Special Buildings: This category includes concert halls, TV studios, recording studios, auditoriums, and theatres.

The Vibration Sensitive Receptors (VSRs) potentially affected by the proposed project have been identified (Table 8.2) through a combination of desktop study and visual surveys at the study areas.

**Table 8.2.** List of identified vibration sensitive receptor

| ID   | Type                           | Description                                       | Approximate Distance* |
|------|--------------------------------|---------------------------------------------------|-----------------------|
| VSR1 | Institution of higher learning | Experimental Medicine Building                    | 300m from Area 1      |
| VSR2 | Institution of higher learning | Research Techno Plaza                             | 120m from Area 3      |
| VSR3 | Institution of higher learning | Wee Kim Wee School of Communication & Information | 70m from Area 3       |
| VSR4 | Residential building           | Block 36 of the former NTU Hall of Residence 7    | 10m from Area 3       |
| VSR5 | Residential building           | Block 37 of the former NTU Hall of Residence 7    | 20m from Area 3       |

\*Approximate distance from the nearest project work area

The impacts of anthropogenic-derived vibration on fauna are poorly understood. However, some animals, particularly amphibians, are sensitive to vibrations, as increased vibrations may affect their ability to communicate, and may reduce their reproductive success (Caorsi, et al., 2019).

### 8.3.2 Baseline Field Survey

The ground-borne vibration monitoring is conducted for one day (24 hours) for each monitoring station to generate baseline conditions. The monitoring locations were chosen based on the factors such as sensitive receptor, relevance to project activity, site access and equipment security. Ground-borne vibration level was monitored using one unit of Micromate manufactured by Instantel Inc. at the following locations (Table 8.3):

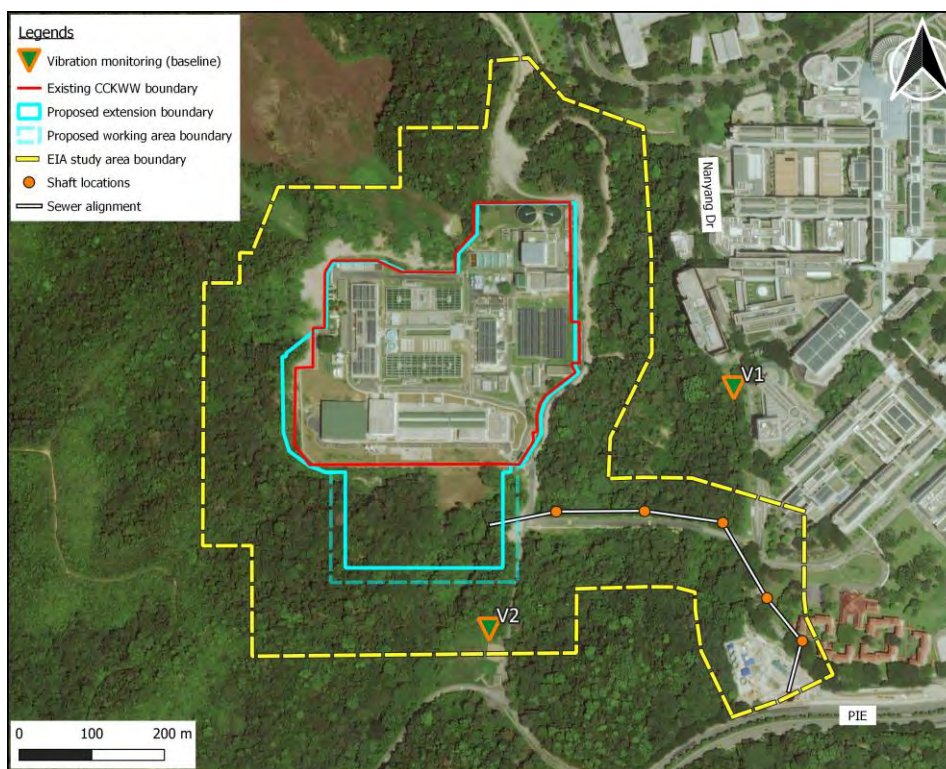
**Table 8.3.** Location of vibration monitoring stations

| Station | Monitoring Period                       | Latitude | Longitude  | Description                                                                                                                        |
|---------|-----------------------------------------|----------|------------|------------------------------------------------------------------------------------------------------------------------------------|
| V1      | 18/8/2021, 10:11 to<br>19/8/2021, 15:20 | 1.343694 | 103.678806 | Open field at the top of the slope just next to the unloading/loading bay of NTU Experimental Medicine Building and Nanyang Avenue |
| V2      | 17/8/2021, 09:20 to<br>18/8/2021, 09:59 | 1.340417 | 103.675972 | South of CCKWW, next to Dunbar Walk                                                                                                |

Figure 8.1 the photographs of on-site monitoring equipment setup and Figure 8.2 shows the location map of monitoring stations. The baseline ground-borne vibration monitoring report is available as **Appendix K**.



**Figure 8.1.** Setup of the baseline ground-borne vibration monitoring equipment



**Figure 8.2.** Location of ambient vibration monitoring stations within project area

## 8.4 Baseline Results and Discussion

The detection level of 1 mm/s was pre-set for ground-borne vibration in consideration of eliminating unknown vibration levels while generating baseline conditions. No ground-borne vibration levels recorded at Station V1 were above detection level of 1 mm/s whereas at Station V2, 14 occurrences were recorded when vibration levels were above 1 mm/s but less than 2 mm/s. Table 8.4 provides the recorded baseline vibration levels at Station V2.

**Table 8.4.** Ground-borne vibration level at Station V2

| Date      | Time     | Peak Vector Sum (mm/s) |
|-----------|----------|------------------------|
| 17/8/2021 | 09:28:52 | 1.25                   |
| 17/8/2021 | 21:48:50 | 1.439                  |
| 17/8/2021 | 21:49:07 | 1.013                  |
| 17/8/2021 | 21:49:33 | 1.242                  |
| 17/8/2021 | 21:49:50 | 1.88                   |
| 17/8/2021 | 21:50:06 | 1.814                  |
| 17/8/2021 | 21:50:18 | 1.163                  |
| 17/8/2021 | 21:51:16 | 1.459                  |
| 17/8/2021 | 21:51:32 | 1.133                  |
| 17/8/2021 | 22:00:46 | 1.037                  |
| 17/8/2021 | 22:01:03 | 1.383                  |
| 17/8/2021 | 22:01:15 | 1.072                  |
| 17/8/2021 | 22:05:01 | 1.076                  |
| 17/8/2021 | 22:05:19 | 1.158                  |

## 8.5 Impact Assessment

### 8.5.1 Identification of Potential Impacts

#### *Pre-construction / Construction Phase*

Construction activities can result in ground-borne vibration, depending on the various type of construction methods and equipment employed. Use of construction equipment could cause ground-borne vibration which spread through the ground and diminish in strength with distance. The ground vibrations generated during construction may create annoyance to people and detrimentally effect structures and sensitive devices. The focus on this vibration impact assessment during construction phase is on human annoyance.

The preliminary construction approach that would be used for this project is given in Chapter 2. The detailed information on the construction methodologies, specific equipment that will be used at each construction stage and their approximate quantity are not available at the time of this report preparation.

Main sources of ground vibration emissions during construction of various CCKWW facilities at Area 1 and 2 are:

- Vibro hammer for installing and removing sheet piles (temporary retaining walls)

- Bored piling rigs to be used to pre-bore the ground for installation of the piles
- Excavators for earth handling
- Demolition activities

Main sources of ground vibration emissions during construction of sewerage alignment Area 3 are:

- Shaft construction and Micro Tunnel Boring Machine (MTBM) during pipe jacking method

Potential ground vibration impacts on VSRs during the construction phase likely arise mainly from vibro hammers, bored piling rigs, excavators, and concrete breakers for Area 1 & Area 2 and pipe jacking machine & MTBM for Area 3.

There are currently no Singapore standards that provide methodology to predict levels of ground vibration from construction activities. As such, to control the impact of vibration during site activities, limits relating to the perceptibility of vibration are typically used (see Section 8.2). BS 5228-2 provides the distances (based on historical field measurements) as which certain activities could give rise to a just perceptible level of ground vibration. These distances are provided in Table 8.5.

**Table 8.5.** Perceptible vibration levels as per BS 5228-2

| <b>Construction Activities</b> | <b>Distance from Activity when Ground Vibration May Just be Perceptible (m)</b> |
|--------------------------------|---------------------------------------------------------------------------------|
| Heavy vehicles                 | 5 - 10                                                                          |
| Rotary bored pilling           | 20 - 30                                                                         |
| Vibro hammer                   | 25 – 35                                                                         |
| Driven sheet piling            | 30 – 35                                                                         |

There is no concern related to construction and demolition activities to be carried out at Area 1 as it is within CCKWW plant and will be contained within plant boundaries. Hence, Area 1 activities are not likely to generate any vibration impacts outside of plant boundaries. The work activities at Area 2 will be within CCKWW extension area and nearest VSR (i.e., NTU Experimental Medicine Building) is approximately 300 m away. Hence, no impact from these activities is anticipated to VSR.

The closest distance between project worksite of Area 3 (Sewerage alignment Shaft 4 & 5) and nearest human related VSRs (i.e., Block 36 and 37 of the former NTU Hall of Residence 7) is between 10 m & 20m respectively. As activities at Area 3 would employ pipe jacking method, only the equipment used at ground level (e.g., mobile crane, excavator) were considered as potential vibration sources. Considering the limited extent of ground activities to be carried out for shaft construction, the potential impacts from these sources are considered short term and reversible as the impact should cease upon termination of the sources or completion of the construction. Other construction activities such as vehicle movements, soil compaction, lifting using mobile cranes are unlikely to generate significant ground-vibration.

Therefore, it is anticipated that human VSRs would not encounter vibration effect above

the perceptible level for construction works and unlikely to be affected. However, it is recognized that Area 3 activities are in close proximity of VSR4 and VSR5, hence Contractor shall implement recommended vibration mitigation measures/best practices for these worksites to minimise the vibration impacts.

There is no extensive research done on how vibration emissions affect fauna. However, it is understood that vibration from anthropogenic sources affects the ability of some fauna, including frogs, to communicate (Caorsi, et al., 2019). Hence, fauna located in proximity to construction areas can be affected when construction vibration sources are located near work boundaries.

It is to be noted that vibration generating activities are already present in the forested project area. As recommended in Chapter 4, the major construction activities are to be restricted to daytime at Area 2 and Area 3 (forested areas). For Area 2, temporary 20m working area will act as buffer as all piling and other main activities will take place within plant footprint. It is anticipated that lower vibration generating construction methods (use of bored piling & silent piler for Area 2 and pipe jacking method for Area 3) will be used. Further, a preliminary analysis of vibration levels likely to be generated from selected activities was carried to ascertain the likely vibration levels from planned construction methods. The vibration analysis adopted a simplified approach taking into consideration the low vibration amplitudes assessed hence further considerations of area affected and habitat sensitivity are not considered in the assessment. The results are presented in **Appendix L**. Considering the findings of the preliminary analysis, vibration impacts are anticipated to be limited to minor negative and is not likely to cause significant impacts on fauna.

Overall, potential vibration impacts on VSR are expected to be of slight negative nature, direct, and local. Duration is anticipated to be short term and reversible as the impact will cease upon completion of the vibration generating construction activities. Further, this impact is to be managed through the mitigation measures/best practices recommended in next section.

### **Operation Phase**

As the all the new CCKWW plant equipment and units will be designed as per applicable regulations & standards for vibration limits and to be operated accordingly, there are no anticipated impacts in operation phase.

### **Cumulative Impact**

There is a possibility that a cumulative impact could occur if ground vibration emissions from construction works of this project and nearby LTA JRL development (see Section 2.10) happen simultaneously.

In order to accommodate the cumulative impacts of both developments, all impacts were given a score of 3 in the “Cumulativity” column of RIAM table to calculate the ES and assess overall impact. It is suggested that any possible coordination between the development agencies (i.e., LTA and PUB) is explored, which can help to avoid the concurrent construction activities or schedule in staggered manner.

### 8.5.2 *Evaluation of Impacts*

Table 8.6 summarises the impacts covering of all three areas with their corresponding Environmental Scores before and after the implementation of mitigation measures, which are elaborated in the following section.

## 8.6 Mitigation Measures

### *Pre-construction / Construction Phase*

The following mitigation measures/ best practices are recommended to manage the impact to fauna surrounding project work areas. The Contractor shall be responsible to implement these measures.

- Construction works involving the use of heavy equipment should be restricted to daytime only (i.e., 8 am – 6 pm) at Area 2 (CCKWW extension area) and Area 3 (near NTU premises) except in case of safety reasons.
- Contractors shall be responsible to comply all applicable regulatory and authority requirements and recommendation in this EIA report on ground-borne vibration.
- Vibrating equipment (e.g., pipe jacking unit) to be placed on isolators such as spring coils.
- Use of equipment or method which generate lower vibration levels, where possible. Examples: use of silent piler instead of vibro hammer, use only low-vibration mode for vibratory compactors.
- To monitor and assess the actual vibration levels near Stream 3 while carrying out sheet piling work within that work area during construction phase.
- The drilling rig engine to be switched off when not in use to reduce vibration.
- Route heavily loaded trucks away from the VSRs as far away as possible.
- To plan and phase excavation, earthmoving, and ground pilling activities in staggered manner where possible to minimize cumulative impact.
- Control speed of vehicle movement at the worksite to 10 km/hour to reduce ground vibration.
- Notify nearby VSRs (e.g., NTU residential building near Area 3) in advance of the construction activities, particularly during pipe jacking works to allay potential concerns.
- To develop the vibration monitoring plan in consultation with Qualified Person (QP) and to monitor the vibration levels throughout construction period.
- Utilize a vibratory compactor that can adjust the impulse energy and frequency, compact soil with static method (no vibrations).
- Keep haul roads within project work site in good condition to provide a smooth traveling surface.

### *Operation Phase*

As there are no anticipated impacts in operation phase, no mitigation measures are recommended.

## **8.7 Residual Impacts**

The residual impacts were evaluated using the RIAM method (Table 8.6) with due consideration that the recommended mitigation measures are implemented by the Contractor. The residual impacts are likely to be in the band of slight negative and considered acceptable.

**Table 8.6.** Environmental Scores of the identified vibration impacts with corresponding mitigation measures

| Phase                           | Impact component                                                             | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Predicted impact after mitigation |    |   |   |   |     |                 |
|---------------------------------|------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                                 |                                                                              | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | I                                 | M  | P | R | C | ES  | Impact          |
| Ecology                         |                                                                              |                                    |    |   |   |   |     |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Disturbance to the fauna due to vibration from construction activities       | 3                                  | -3 | 2 | 2 | 3 | -63 | Minor negative  | <ul style="list-style-type: none"><li>• Use of heavy equipment to be restricted to daytime only at Area 2 (CCKWW extension area).</li><li>• Use of equipment or method which generate lower vibration levels</li><li>• Route heavily loaded trucks away from the VSRs as far away as possible.</li><li>• To plan and phase excavation, earth-moving, and ground pilling activities in staggered manner</li><li>• Control speed of vehicle movement at the worksite</li><li>• Keep haul roads within project work site in good condition</li></ul>                       | 3                                 | -2 | 2 | 2 | 3 | -42 | Minor negative  |
| Human Health and Safety         |                                                                              |                                    |    |   |   |   |     |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Disturbance/ annoyance to the VSR due vibration from construction activities | 2                                  | -2 | 2 | 2 | 3 | -28 | Slight negative | <ul style="list-style-type: none"><li>• Use of heavy equipment to be restricted to daytime only at Area 3 (NTU premises)</li><li>• Use of equipment or method which generate lower vibration levels</li><li>• Route heavily loaded trucks away from the VSRs as far away as possible.</li><li>• Control speed of vehicle movement at the worksite</li><li>• To develop the vibration monitoring plan in consultation QP &amp; to monitor throughout construction period</li><li>• Notify nearby VSRs (NTU premises) in advance of the construction activities</li></ul> | 2                                 | -1 | 2 | 2 | 3 | -14 | Slight negative |

## 9 LIGHT

### 9.1 Introduction

Given its status as a highly urbanised city-state, high levels of light pollution are already being faced throughout Singapore. While unavoidable in cities, such lighting often has adverse impacts on the natural environment and wildlife.

Currently, within the study site, areas with artificial lighting include the existing CCKWW facility and along a portion of Dunbar Walk connecting Nanyang Drive to the entrance of CCKWW for traffic purposes. No artificial lights are present in the rest of the site, especially the forested area.



**Figure 9.1.** Artificial light illuminating from CCKWW facility observed during night surveys

It was observed that along the perimeter fence of CCKWW plant, the artificial lighting used within the plant illuminates the surrounding area including the vegetation directly along the fence line (Figure 9.1). Artificial light originating from CCKWW plant was also observed from within the surrounding forested areas. Light from CCKWW was visible from the surrounding forested area of an estimated distance of 25 meters away from the perimeter fence. Vegetated areas along the portion of Dunbar Walk and Peng Kang Avenue connecting Nanyang Drive to the entrance gate of CCKWW are also illuminated by artificial light used for traffic purposes (Figure 9.2).



**Figure 9.2.** Artificial lighting observed along Dunbar Walk

## **9.2 Impact Assessment**

### **9.2.1 Identification of Potential Impacts**

Impacts to flora and fauna from light pollution should be avoided particularly for sensitive or threatened species highlighted in Chapter 4. Alteration of natural cycles of light and dark by artificial light sources can negatively impact the ecosystem. Increased artificial light during the night disrupts circadian cycles of animals and distorts the day-night cycle of plants. This may lead to increased predation pressure by diurnal carnivores on nocturnal animals, exhaustion from insects attracted to artificial light, disorientation and disruption of foraging by birds, and the alteration of breeding and sleeping cycles of various animals. As a whole, these effects derive from changes in orientation or disorientation, and attraction or repulsion from the altered light environment, which in turn may affect foraging, reproduction, migration, and communication behaviour.

Artificial night lights have also shown to have an impact on migratory birds, a large proportion of which migrate at night. In general, the risk of bird collisions increase with increased light emissions (Ogden, 2002).

Many groups of insects, such as moths, are attracted to lights resulting from their innate navigational behaviour. Visual communication within and between species can also be influenced by artificial lighting. Insects such as fireflies communicate through bioluminescent signals, which can only be achieved in the absence of background light (Longcore & Rich, 2004). Foraging behaviours of animals such as fruit bats are altered by light pollution, fruit bats are less active and forage less in artificially illuminated areas. As fruit bats play the ecological role of seed dispersers and pollinators, their reduced foraging activities may result in the loss of pollinating and seed-dispersal services they provide (Lewanzik & Voigt, 2014). The cumulative effects of such behavioural changes induced by artificial night lighting can have the potential to disrupt key ecosystem functions (Longcore & Rich, 2004).

## ***Pre-construction / Construction Phase***

### **Area 1 (Existing CCKWW Plant)**

Proposed works planned for Area 1 are within the current CCKWW plant which is a built-up and illuminated during the night. Any light impacts to the forested area will take place if there are night construction activities. Nevertheless, as the works are within an already illuminated area, it is expected that additional light impacts from these works will be minimal.

### **Area 2 (CCKWW Extension Area)**

During the pre-construction and construction stage, the source of impacts includes artificial lighting used to illuminate the project site, particularly if night works are conducted. Impacts of light on forested areas will be further exacerbated by edge effects, where vegetation clearance leads to open edges, thus allowing increased amount of light to enter these areas. Area 2 is a forested area used by nocturnal forest species, including the Sunda Pangolin and various species of frogs. These species will be most impacted by nightworks. For this area, it is assumed that all construction activities will be limited to daytime as far as possible except in case of work exigency and planned nightwork, as described in section 2.7, with the implementation of the lighting management plan (LMP) approved by NParks. If left unmitigated, some minor impacts to biodiversity are expected.

### **Area 3 (Sewerage Alignment)**

The proposed works for area 3 are along existing roads, which include part of Peng Kang Avenue and Nanyang Drive areas which already have existing artificial lighting. During the construction phase, particularly if night works are carried out, sources light which include construction lighting. However, as the construction footprint in this area is small, impacts to the surrounding forested areas are expected to be minor.

## ***Operational Phase***

During the operational stage, sources of light impacts will be derived from the operation of the CCKWW facility. These would include lights to illuminate areas within the CCKWW facility as well as lights along the perimeter fence for security reasons.

As the proposed sewerage alignment is to be constructed underground and not above ground, no lighting is required during its operation.

Overall, potential light impacts are expected to be of slight negative nature, direct, and local. Duration is anticipated to be short term & reversible during construction phase and long term & permanent during operation phase. As such, mitigation measures are recommended to further reduce the potential impacts.

### **Cumulative Impacts**

There is a possibility that proposed LTA JRL development (see Section 2.10) located adjacent to Area 3, happen concurrently with this project.

While the scale of works for the current project in Area 3 are small, its working area is in close vicinity to this JRL development. Light pollution for JRL development was

assessed to have impacts on biodiversity, including affecting migratory bird species and nocturnal mammals such as bats. (T. Y. Lin International Pte. Ltd., 2018).

Should the two projects run concurrently, there may be cumulative impacts from construction lights from both sites. In order to address the cumulative impacts of both developments, all impacts were given a score of 3 in the 'Cumulativity' column of RIAM table to calculate the ES and assess overall impact. The cumulativity can be removed by ensuring that there are no nightworks for the current project.

### 9.2.2 Evaluation of Impacts

Table 9.1 summarises the impacts covering of all three areas with their corresponding Environmental Scores before and after the implementation of mitigation measures, which are elaborated in the following section.

## 9.3 Mitigation Measures

Where lighting is necessary for safety and security reasons for the CCKWW facility, mitigation measures can be taken to ensure minimal impact on sensitive wildlife.

### 9.3.1 Pre-construction and Construction Phase

#### Area 1 (Existing CCKWW Plant)

Area 1 is already highly illuminated, but measures must be implemented to ensure that the project activities do not cause additional light impacts on the receptors in surrounding area. In the case for construction lights, such lights must be shielded, facing downwards and away from any forested areas. This will eliminate light spill, which is light that falls outside the area intended to be lit. Examples of ideal shielded lighting can be seen in Figure 9.3 below (Australia Department of the Environment and Energy, 2019).

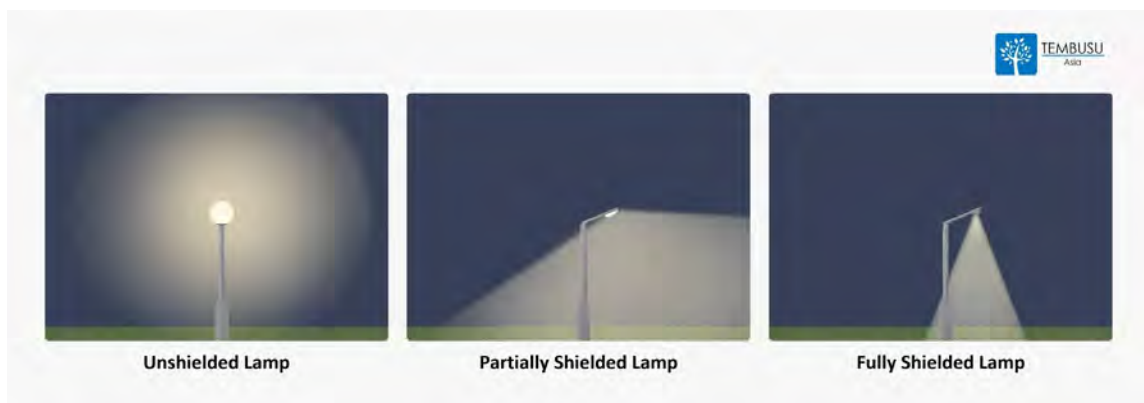


Figure 9.3. Example of light shielding

#### Area 2 (CCKWW Extension Area)

During the construction phase, construction works within Area 2 are to be limited to daylight hours (8 am – 6 pm), including demobilisation, except in cases of work exigency and planned nightwork (Section 2.7.2), with the implementation of the Lighting Management Plan (LMP) approved by NParks to reduce unnecessary stress to biodiversity that are sensitive to light. Specific measures to manage these work exigency

scenarios are described in Section 4.6.1. Also, should there be any change in type of construction activities to continue at night, authorities' approval should be obtained.

Should nightworks be unavoidable in an emergency, construction lights should face downwards and away from forested areas. All artificial lights should be turned off outside working hours except for safety & security reasons.

Additionally, a Lighting Management Plan (LMP) should be developed as part of the Construction EMMP in the case of work exigencies and for planned nightwork, and the LMP is to be approved by NParks. The detailed requirement for LMP is covered in Section 13.5. The LMP should include minimally the following aspects:

- Objectives of Lighting Management Plan
- Purposes and categories of artificial lighting required
- Spatial layout of lighting utilised in project site
- Designs of lighting utilised – including height and shielding design
- Specifications of lighting utilised – lighting colour, spectrum, and brightness level
- Additional measures in place in the event that white or non-shielded lighting is required
- Implementation monitoring plan

### **Area 3 (Sewerage Alignment)**

For the construction phase, it is recommended that works within the forested parts of Area 3 be kept to daytime working hours (i.e., 8 am to 6 pm). To reduce unnecessary stress to biodiversity that are sensitive to light. Where reasonable and practical, all artificial lights should be turned off outside working hours.

Should nightworks be unavoidable in an emergency, construction lights should face downwards and away from forested areas. Additionally, a Lighting Management Plan (LMP) should be developed as part of the Construction EMMP in the case of work exigencies and planned nightwork, and the LMP is to be approved by NParks. The LMP should include minimally the following aspects:

- Objectives of Lighting Management Plan
- Purposes and categories of artificial lighting required
- Spatial layout of lighting utilised in project site
- Designs of lighting utilised – including height and shielding design
- Specifications of lighting utilised – lighting colour, spectrum, and brightness level
- Additional measures in place in the event that white or non-shielded lighting is required
- Implementation monitoring plan

#### **9.3.2 Operational Phase**

During the operation phase, the light impacts from CCKWW plant on the surrounding forested areas can be mitigated through proper lighting system design (Figure 9.3). The lighting layout should use only the minimum number and intensity of lights needed to

provide safety and security. This minimum number of lamps required should be assessed during the detailed design phase. The lighting for operational phase is to utilise non-white lighting as much as possible, as these can reduce potential impact to wildlife.

All lighting should be downward facing and turned away from forested areas. Further, the impact can be reduced through appropriate buffer planting between the CCKWW facility and forested area. The presence of thick and tall hedge plants could minimise the amount of light that enters the forested area, thus reducing its impact on sensitive flora and fauna within the forest (Figure 9.4).



**Figure 9.4.** Example of hedge planting using *Murraya paniculata*

For the sewer alignment, no mitigation measures are recommended as there are no plans for artificial lights to be installed along the alignment.

As such, the summary of mitigation measures that can be used include:

- Construction works to be limited to daylight hours (i.e., 8 am to 6 pm) except in case of work exigency.
- All unnecessary lights should be turned off outside construction working hours.
- Any installed temporary lighting during construction phase for night work shall be directed downwards and inwards, towards the project site (i.e., directly away from forested areas).
- Permanent lighting should be designed to be downward facing and directed away from forested areas.
- To minimum number and intensity of permanent lights.
- Buffer planting should include thick hedge plants which can reduce the amount of light entering the forest.

#### **9.4 Residual Impacts**

The residual impacts were evaluated using the RIAM method (Table 9.1) with due consideration that the recommended mitigation measures are implemented by the Contractor. The residual impacts are likely to be in the band of slight negative to no impact and considered acceptable.

**Table 9.1.** Environmental Scores of the identified light impacts with corresponding mitigation measures

| Phase                           | Impact component                                                             | Predicted impact before mitigation |    |   |   |   |     |                | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Predicted impact after mitigation |    |   |   |   |     |                 |
|---------------------------------|------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                                 |                                                                              | I                                  | M  | P | R | C | ES  | Impact         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | I                                 | M  | P | R | C | ES  | Impact          |
| Ecology                         |                                                                              |                                    |    |   |   |   |     |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Disturbance to the flora and fauna due to light from construction activities | 2                                  | -3 | 2 | 2 | 3 | -42 | Minor negative | <ul style="list-style-type: none"><li>Construction lights in Area 1 shall be directed downwards and inwards, toward project site (i.e., directly away from forested areas)</li><li>Construction in Areas 2 and the forested parts of Area 3 to be kept to daytime working hours (8 am – 6 pm), except in the case of work exigencies and planned nightwork, with the implementation of the LMP approved by NParks</li><li>All unnecessary lights should be turned off outside working hours</li><li>Develop LMP for Areas 2 and 3 for the event of work exigencies and planned nightwork</li></ul> | 2                                 | -2 | 1 | 1 | 1 | -4  | No impact       |
| Operation                       | Disturbance to the flora and fauna due to operation phase lighting           | 2                                  | -3 | 3 | 3 | 3 | -48 | Minor negative | <ul style="list-style-type: none"><li>Lighting installed within Area 2 should be directed downwards and inwards, toward project site (i.e., directly away from forested areas)</li><li>Use the minimum number and intensity of lights required for safety and security</li><li>Area 1 and 2 should utilise non-white lighting where possible</li><li>Buffer planting should include thick hedge plants which can reduce the amount of light entering the forest.</li></ul>                                                                                                                         | 2                                 | -2 | 3 | 3 | 3 | -36 | Slight negative |

## 10 WASTE MANAGEMENT

### 10.1 Introduction

The main impacts related to waste storage, handling, transport, and disposal include:

- Deterioration of the environment (e.g., visual, water quality, biodiversity impacts, increase vectors, contamination etc.); and
- Health and safety risks with regards to hazardous wastes if they are not managed properly.

The main sources of potential impacts during the project activities are identified as the following:

- Waste management requirement for cut vegetation (e.g., from site clearance)
- Waste management requirement for excavated material
- Waste management requirement for general waste
- Waste management requirement for hazardous waste
- Waste management requirement for other construction waste

There are legislations implemented and enforced by the Singapore government for management of general and hazardous wastes which are discussed in next section. Table 10.1 provides anticipated types of waste likely to be generated during construction.

**Table 10.1.** Types of waste likely to be generated during construction phase

| Solid Waste Type      | Source Activity      | Classification |
|-----------------------|----------------------|----------------|
| Cut vegetation        | Site clearance       | Non-hazardous  |
| Excavated material    | Excavation           | Non-hazardous  |
| Plastic/wooden planks | Packaging material   | Mixed          |
| Hazardous waste       | Maintenance activity | Hazardous      |
| Scrap metal           | Form/temporary work  | Non-hazardous  |
| General waste         | Construction workers | Non-hazardous  |

### 10.2 Relevant Environmental Legislation, Guidelines and Standards

The Environmental Public Health (EPHA) Act, 2002 set up the regulatory framework for waste management through following regulations:

- Environmental Public Health (General Waste Collection) Regulations, 2000
- Environmental Public Health (Toxic Industrial Wastes) Regulations, 2000

The Environmental Public Health (General Waste Collection) Regulations set out the requirements for the management of non-hazardous general waste and the duties of generators and companies collecting such wastes. General waste must be managed in an environmentally sound manner and collected by an NEA-licensed general waste collector.

The Environmental Public Health (Toxic Industrial Wastes) Regulations set requirements for the generation, management, and disposal of wastes characterised as hazardous, such as oily sludge, solvents and asbestos containing materials. NEA has set out a list

of specific wastes which are classified as Toxic Industrial Wastes (TIW) under the Schedule of this regulation. Special requirements apply to these TIWs with the key provisions includes appointment of a licensed TIW collection company to collect and treat the waste.

The Code of Practice for ECO stipulates the role of occupiers of construction sites and of the ECO, and their responsibilities pertaining to waste management at construction sites.

### **10.3 Impact Assessment**

#### **10.3.1 Identification of Potential Impacts**

##### ***Pre-Construction and Construction Phase***

###### **Cut Vegetation**

As discussed in Chapter 4 of Biodiversity, potentially large number of trees will be affected during the implementation of this project. Similarly, other vegetation, such as bushes and grassland will need to be cleared within footprint of the project boundary.

Where cutting is required, the waste vegetation needs to be properly managed to prevent fire risks. The preferred disposal route for this type of material is for the extraction of timber for use in the wood industry. Large stumps will require off-site transport for disposal at local disposal sites. The contractor shall not be permitted to dispose of cut vegetation by burning, as specified in the EPM (Prohibition on Use of Open Fires) Order.

###### **Excavated Material**

Due to the nature of construction work involved (i.e., cut and fill), no significant volumes of spoil are expected to be generated from this project. Based on preliminary information, the quantity of excavated material will be approximately 28,880 cubic meters. Most of excavated material is expected to be utilized to backfill within project area as the fill quantity required for Area 2 is approximately 32,820 cubic meters. Any unutilised excavated material should be disposed through a licensed waste collector for further treatment/ disposal.

###### **Other Construction Waste**

Other construction wastes will likely include:

- Broken rock or concrete
- Ferrous and non-ferrous scrap metal items
- Wooden planks, boards, pallets, formwork
- Packaging or wrapping materials such as plastic sheeting, corrugated
- Cardboard, paper
- Sacks and bags
- Metal or plastic containers and cans.

If not managed properly, the storage, handling, transport and disposal of construction wastes has the potential to result in visual, water, dust, noise and general environmental deterioration. In the event of inappropriate management methods, the disposal of construction waste is unlikely to raise any long-term concerns due to the inert nature of these types of materials. However, it is good practice to segregate different categories

of construction waste at source to facilitate recycling/disposal. It is expected that general refuse will be handled, stored, managed and collected for appropriate disposal/treatment in accordance with the EPHA and the EPH (General Waste Collection) Regulations and there shall be no significant adverse impact to the environment.

#### General Waste

General waste will include debris, litter and wastes generated by the construction workers. The storage and handling of general refuse has the potential to give rise to a variety of adverse impacts. These include odour problems if the waste is not collected regularly (i.e., daily), windblown litter, water quality impacts if waste enters watercourses, visual impacts as well as the attraction of pests, disease vectors and scavenging animals (insects, rodents etc.) to the site if waste materials are incorrectly stored onsite. The number of workers on site and subsequently the estimated volume general refuse is unknown at this time. Quantities of general refuse produced by workers are typically in the range of 0.5 – 1 kg per person per day. It is expected that general refuse will be handled, stored, managed and collected for appropriate disposal/ treatment in accordance with the EPHA and EPH (General Waste Collection) Regulations and there shall be no adverse impact to the environment.

#### Hazardous Waste

Hazardous wastes include those listed in the EPHA and EPH (Toxic Industrial Wastes) Regulations. Hazardous wastes are likely to arise principally as a result of maintenance activities for construction machinery and heavy vehicles at the construction site. Estimates of the quantities of hazardous waste are not available during preparation of this report. Types of hazardous construction waste generated include:

- Waste oils and used oil filters
- Paints and solvent residues
- Off-specification chemicals
- Oily water
- Drums, containers, packaging or wrapping materials, or soil contaminated with the TIW.

Hazardous construction wastes can pose serious environmental problems such as air, water, and land pollution unless they are handled, stored, transported and disposed of in an appropriate manner. Potential hazards may include:

- Toxic / adverse health effects on the workforce
- Adverse effects on water quality/ surface water resources, soil and groundwater in the event of spills and leaks
- Fire hazards

The generation of the hazardous construction wastes mentioned above is unavoidable. However, their environmental impact is not expected to raise long-term or irreversible negative effects as hazardous wastes are not routinely generated in big quantities by construction activities. In addition, hazardous wastes will not be generated after the construction phase of the Project has been completed.

Overall, potential waste management impacts are expected to be of slight negative nature, indirect, and local. Duration is anticipated to be short term and reversible as the impact will cease upon completion of the construction activities. As such, mitigation measures are recommended to further reduce the potential impacts.

### **Operation Phase**

During the operation of the project, all construction activities associated with the development will have been completed (other than any maintenance of road as required). Considering this, no waste or hazardous materials management impacts have been identified.

## **10.3.2 Evaluation of Potential Impacts**

Table 10.2 summarises the impacts covering of all three areas with their corresponding Environmental Scores before and after the implementation of mitigation measures, which are elaborated in the following section.

## **10.4 Mitigation Measures**

### **Pre-Construction and Construction Phase**

The various options within waste management can be categorised in terms of preference from an environmental perspective. The options considered to be more preferable have the least impacts and are more sustainable in a long-term context. Hence, the hierarchy is as follows:

- Avoidance and minimization, i.e., not generating waste through changing or improving processes.
- Reuse or recycling of materials, thus avoiding disposal.
- Disposed in a safe and appropriate manner through licensed waste collection and disposal contractors.

It is recommended that the following guidelines and procedures are adopted:

- The contractor should be contractually obligated to develop a solid waste management plan to manage the collection, recycling and ultimate disposal of all generated wastes in an environmentally responsible manner.
- Timber/wood from cut vegetation can be recovered for use in the wood industry as far as possible.
- Surplus excavated material and inert wastes (soil, broken rock etc.) shall be reused within project site as backfill, landscaping, erosion control and restoration features wherever practicable.
- Scrap metals (e.g., welding rods, end caps, off-cuts etc.) can be recovered and sent for recycling as scrap.
- Other inert general waste will be collected and disposed through licensed waste collector.
- General refuse generated on-site must be stored in enclosed bins separate from construction and hazardous wastes. A licensed general waste collector shall be employed by the Contractor to remove general refuse, on a daily or every second day basis to minimise odour, pest and litter impacts.

- Chemical toilet facilities/ septic tank system with collection of accumulated waste for off-site disposal by a licensed general waste collector.
- All non-hazardous wastes that are generated must be handled and disposed of in accordance with the requirements of the EPHA and the EPH (General Waste Collection) Regulations.
- Any hazardous wastes that are generated must be handled and disposed of in accordance with the requirements of the EPHA and the EPH (Toxic Industrial Wastes) Regulations.
- Disposal of hazardous waste must be through a licensed waste collector for hazardous waste.

The recommended measures above, when implemented effectively, shall mitigate the potential impacts of waste management to acceptable levels.

### **Operation Phase**

The operation phase impact is assessed to be insignificant, therefore no mitigation measure is proposed.

## **10.5 Residual Impacts**

The residual impacts were evaluated using the RIAM method (Table 10.2) with due consideration that the recommended mitigation measures are implemented by the Contractor. The residual impacts are likely to be in the band of slight negative to no impact and considered acceptable.

**Table 10.2.** Environmental Scores of the identified waste management impacts with corresponding mitigation measures

| Phase                           | Impact component                   | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                       | Predicted impact after mitigation |    |   |   |   |     |                 |
|---------------------------------|------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                                 |                                    | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                           | I                                 | M  | P | R | C | ES  | Impact          |
| Ecology                         |                                    |                                    |    |   |   |   |     |                 |                                                                                                                                                                                                                                                                                                                                           |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Disposal of cut vegetation         | 2                                  | -2 | 2 | 2 | 2 | -24 | Slight Negative | <ul style="list-style-type: none"><li>• To develop solid waste management plan</li><li>• Timber/wood to be recovered for use in the wood industry as far as possible</li></ul>                                                                                                                                                            | 2                                 | -1 | 2 | 2 | 2 | -12 | Slight Negative |
|                                 | Disposal of excavated material     | 2                                  | -3 | 2 | 2 | 2 | -36 | Slight Negative | <ul style="list-style-type: none"><li>• To develop solid waste management plan</li><li>• Surplus excavated material to be reused within project site as fill, landscaping, erosion control and restoration wherever practicable</li></ul>                                                                                                 | 2                                 | -2 | 2 | 2 | 2 | -24 | Slight Negative |
|                                 | Disposal of general waste material | 1                                  | -2 | 2 | 2 | 2 | -12 | Slight Negative | <ul style="list-style-type: none"><li>• Scrap metals to be recovered and sent for recycling as scrap</li><li>• Inert general waste to be collected and disposed through licensed waste collector</li><li>• All non-hazardous wastes to be handled and disposed of in accordance with EPH (General Waste Collection) Regulations</li></ul> | 1                                 | -1 | 2 | 2 | 2 | -6  | No Impact       |
| Human Health and Safety         |                                    |                                    |    |   |   |   |     |                 |                                                                                                                                                                                                                                                                                                                                           |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Disposal of hazardous waste        | 1                                  | -2 | 2 | 2 | 2 | -12 | Slight Negative | <ul style="list-style-type: none"><li>• All hazardous wastes to be handled and disposed of in accordance with EPH (Toxic Industrial Wastes) Regulations</li><li>• Disposal of hazardous waste to be conducted by a licensed waste collector for hazardous waste</li></ul>                                                                 | 1                                 | -1 | 2 | 2 | 2 | -6  | No impact       |

## 11 VECTOR CONTROL

### 11.1 Introduction

As Singapore has warm tropical climate, the possibility of vector borne diseases is always present. Vectors are organisms that transmit disease. According to NEA, the five main vectors in Singapore and the diseases that they could transmit are summarized in Table 11.1. As the majority of vector-borne diseases in Singapore are transmitted by mosquitoes, vector control management shall mainly focus on mosquito control during construction activities.

**Table 11.1.** Common vectors in Singapore

| Vector     | Disease                                                                                                                                                                                           |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mosquitoes | <ul style="list-style-type: none"><li>- Dengue and Dengue Hemorrhagic Fever</li><li>- Chikungunya</li><li>- Zika</li><li>- Malaria</li><li>- Japanese Encephalitis</li><li>- Filariasis</li></ul> |
| Rat Flea   | <ul style="list-style-type: none"><li>- Plague</li></ul>                                                                                                                                          |
| Rodent     | <ul style="list-style-type: none"><li>- Rat Bite Fever</li><li>- Leptospirosis</li><li>- Murine Typhus</li></ul>                                                                                  |
| Cockroach  | <ul style="list-style-type: none"><li>- Cholera</li><li>- Food-Borne Diseases</li></ul>                                                                                                           |
| Fly        | <ul style="list-style-type: none"><li>- Cholera</li><li>- Typhoid and Para Typhoid</li><li>- Salmonellosis</li><li>- Dysentery</li></ul>                                                          |

Source: <https://www.nea.gov.sg/our-services/pest-control/overview>

### 11.2 Relevant Environmental Legislation, Guidelines and Standards

The Control of Vectors and Pesticides Act, 2002 is the main legislation for control of vectors with the prime objective to prevent related diseases such as dengue fever in Singapore. It aims to prevent the propagation of vectors and stipulates the prohibition of creation of any condition favourable to the propagation and harbouring of vectors. Under Part V of the Act, only companies that are registered with NEA as vector control operators are allowed to be engaged to conduct any vector control treatment or activity. Employees of vector control operators are required to be licensed as vector control technicians or vector control workers to conduct any vector control activity. All public health pesticide/repellent products intended for use against the five vectors (namely rodents, mosquitoes, flies, cockroaches, and rat fleas) must be registered prior to local sales in Singapore as per this Act.

The Environmental Public Health Act, 2002 stipulates requirements pertaining to vector control such as public cleansing, public nuisances, and insanitary premises.

Environmental Public Health Act (Specified Construction Sites) Order 2021, and Code

of Practice for Environmental Control Officers which require construction site occupiers to employ either a part-time or full-time Environmental Control Officer, depending on the contract sum of the construction works. The Code of Practice for Environmental Control Officers sets out requirements on environmental health management of construction sites in the areas of vector control.

### **11.3 Impact Assessment**

#### **11.3.1 Identification of Potential Impacts**

##### ***Pre-construction and Construction Phase***

The main impact is the potential to cause an increase in the vector population within and in surrounding project site areas including CCKWW plant and NTU premises during construction stage. There is a likelihood of vector-borne diseases affecting humans.

An increase in vectors may be caused by:

- Pooled water on the construction site (e.g., potholes, access roads with puddles, equipment)
- Storage of water on the construction site (e.g., water for washing)
- Storage and improper management of garbage and food waste
- Creation of stagnant conditions during drainage diversion
- Entrainment of debris, refuse and silts in stormwater run-off resulting choked drains and stagnant drainage
- Poor cleanliness and unsanitary conditions

The assessment of the potential impacts is described below.

##### **Increase in Number of Mosquitos**

Project construction activities have the potential to create suitable breeding conditions for mosquito populations. Specifically, without control measures construction work can create pools of water due to site conditions (e.g., vehicle potholes), construction work (e.g., trenches) and storage of equipment and wastes favouring the breeding of mosquito populations. The likelihood of mosquitoes breeding without control measures in place is high.

##### **Increase in Number of Other Vectors**

During the construction, construction personnel will generate domestic waste and food waste that if not managed, can attract other vectors (e.g., rats, flies and cockroaches). A lack of correct and secure disposal of these wastes would create suitable breeding conditions for these vectors. The likelihood of rats, flies and cockroaches breeding without control measures in place is high.

Fleas require a host (e.g., rodents or other mammal) in order to breed and therefore fleas are considered to be directly relevant to the construction of the Project. With control of other vectors, fleas and related diseases are not likely to occur.

##### **Increase in the Incidence of Dengue-Fever and Other Vector-related Diseases**

If the number of mosquitoes increase in the construction works areas, the likelihood of

a dengue outbreak occurring in the vicinity of the project can increase. Therefore, it is important to prevent mosquitoes breeding at the construction works areas. The CCKWW plant and NTU premises is in immediate surrounding of the project area. Hence occupiers of these premises can be exposed to vectors if proper control measures are not implemented.

If the number of rats, flies and cockroaches increases in the construction works areas, the likelihood of nuisance issues and vector-related diseases could increase. These impacts associated with these vectors can quickly change from nuisance to vector-disease levels. Rats can cause diseases and also damage equipment.

Overall, potential impacts are expected to be of slight negative nature, indirect, and local. Duration is anticipated to be short term and reversible as the impact will cease upon completion of the construction activities. As such, mitigation measures are recommended to further reduce the potential impacts.

### **Operation Phase**

During the operation of the project, all construction activities associated with the development will have been completed (other than plant maintenance work). Considering this, no vector related impacts have been identified.

## **11.3.2 Evaluation of Potential Impacts**

Table 11.2 summarises the impacts covering of all three areas with their corresponding Environmental Scores before and after the implementation of mitigation measures, which are elaborated in the following section.

## **11.4 Mitigation Measures**

### **Pre-construction / Construction Phase**

#### **Increase in Number of Mosquitoes**

It is essential that the construction contractor proactively implements vector control plan with appropriate requirements. Source reduction and effective drainage are proposed to be the main forms of mosquito control, with the following mitigation measures to be implemented:

- Before construction commences, the construction contractor shall engage a vector control operator registered with NEA to prepare and implement vector management plan.
- Before construction starts, a Vector Management Plan shall be submitted to NEA.
- The construction contractor shall form an in-house vector control team to check construction sites for breeding of mosquitoes.
- During the daily checks, particular attention shall be made to discarded receptacles and building wastes; building materials, canvas sheets, equipment and machinery; puddles on the ground levels; water storage drums, tanks and containers; bulk storage containers; trenches; lift wells; drains and channels temporarily constructed to drain off water; air handling units and air conditioners; and, flat roofs of temporary buildings.

- Empty receptacles, pails, basins and other containers shall be kept indoors.
- Construction worksite shall be kept free of litter; construction wastes shall be disposed promptly into bulk waste containers and the containers shall be emptied daily.
- Building materials shall be stored under shelter as far as practicable; materials shall be stored at least 60 cm above the ground to allow water collected below to be treated by the vector control operator.
- Air-handling units shall be stored under shelter or, the overflow pipe shall be uncapped to allow rainwater to drain out.
- Stagnant water shall be pumped from the works areas and ground depressions shall be covered with earth.
- Anti-mosquito oil and insecticides including BTI shall be applied into stagnant water at least once a week. The application should be repeated after rain as the oil and insecticides would be washed away by the rain.
- Thermal fogging shall not be carried out due to location of project site within forested area.
- Any significant increase in numbers of vectors shall be reported to the NEA and investigated.

#### *Increase in Number of Other Vectors*

Rats, flies and cockroaches shall also be targeted for active preventative measures to reduce their breeding habitats (waste disposal areas and wet areas). General cleanliness and waste disposal protocols will control these vector populations and worker hygiene can help reduce insect attraction. The following mitigation shall be implemented:

- Worksite shall be kept litter-free and refuse bins shall always be covered tightly.
- Construction workers' food provisions shall be stored in rodent-proof rooms or cabinets.
- All food items shall be adequately covered and stored at least 60 cm above the ground.
- The in-house vector control team and the vector control operator should check for rodent burrows every week. Active burrows should be treated with rodenticides for three consecutive days or until the rats are all dead (i.e., no more dead rats found), and then sealed with compacted earth.
- Sanitary waste/domestic waste should be removed from the site in accordance with Singapore's legislation promptly.
- Any significant increase in numbers of vectors shall be reported to the NEA and investigated.

#### *Increase in the Incidence of Dengue-Fever and Other Vector-related diseases*

The mitigation to control mosquito numbers and other vectors will mitigate this secondary impact on increase of dengue. Additionally, the following mitigation is proposed:

- During construction, vector control shall be undertaken as per the NEA Guidebook for "Scope of Works for Mosquito Control".
- If required, assistance shall be provided to the authorities to investigate outbreaks of vector-borne diseases at the construction site.
- Weekly monitoring of the NEA's dengue cluster map shall be undertaken to determine if the workers at construction areas are at risk.

With the implementation of the mitigation measures proposed, it is expected that the impacts can be reduced along with the potential for vector borne diseases.

### ***Operation Phase***

The operation phase impact is assessed to be insignificant, therefore no mitigation measure is proposed.

## **11.5 Residual Impacts**

The residual impacts were evaluated using the RIAM method (Table 11.2) with due consideration that the recommended mitigation measures are implemented by the Contractor. The residual impacts are likely to be in the band of slight negative to no impact and considered acceptable.

**Table 11.2.** Environmental Scores of the identified vector control impacts with corresponding mitigation measures

| Phase                           | Impact component                                                                     | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                        | Predicted impact after mitigation |    |   |   |   |     |                 |
|---------------------------------|--------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|                                 |                                                                                      | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                            | I                                 | M  | P | R | C | ES  | Impact          |
| Human Health and Safety         |                                                                                      |                                    |    |   |   |   |     |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Increase in the number of mosquitoes                                                 | 2                                  | -2 | 2 | 2 | 2 | -24 | Slight Negative | <ul style="list-style-type: none"><li>Contractor to engage an NEA-registered vector control operator to prepare and implement vector management plan</li><li>An in-house vector control team to check construction sites for breeding of mosquitoes</li><li>Construction worksite to be kept free of litter; construction wastes shall be disposed promptly into bulk waste containers and the containers shall be emptied daily</li></ul> | 2                                 | -1 | 2 | 2 | 2 | -12 | Slight Negative |
|                                 | Increase in the number of other vectors (e.g., flies and rodents)                    | 1                                  | -2 | 2 | 2 | 2 | -12 | Slight Negative | <ul style="list-style-type: none"><li>Worksite shall be kept litter-free and refuse bins shall always be covered tightly.</li><li>Construction workers' food provisions shall be stored in rodent-proof rooms or cabinets</li><li>In-house vector control team and vector control operator to check for rodent burrows every week</li></ul>                                                                                                | 1                                 | -1 | 2 | 2 | 2 | -6  | No Impact       |
|                                 | Increase in incidence of dengue fever and vector-related diseases (secondary impact) | 1                                  | -2 | 2 | 2 | 2 | -12 | Slight Negative | <ul style="list-style-type: none"><li>To implement mitigation measures to control mosquito numbers and other vectors</li><li>To aid the authorities to investigate outbreaks of vector-borne diseases if required</li></ul>                                                                                                                                                                                                                | 1                                 | -1 | 2 | 2 | 2 | -6  | No Impact       |

## 12 SUMMARY OF IMPACT ASSESSMENT & MITIGATION MEASURES

**Table 12.1.** Summary of Environmental Scores of the identified impacts with corresponding mitigation measures

| Phase                           | Aspects                  | Impact component                                                                                     | Predicted impact before mitigation |    |   |   |   |      |                   | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Predicted impact after mitigation |    |   |   |   |      |                   |
|---------------------------------|--------------------------|------------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|------|-------------------|
|                                 |                          |                                                                                                      | I                                  | M  | P | R | C | ES   | Impact            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | I                                 | M  | P | R | C | ES   | Impact            |
| Ecology                         |                          |                                                                                                      |                                    |    |   |   |   |      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                   |    |   |   |   |      |                   |
| Pre-construction / Construction | Biodiversity<br>(Area 2) | Flora and fauna species damage and mortality due to vegetation clearance for temporary working space | 4                                  | -3 | 2 | 2 | 3 | -84  | Moderate negative | <ul style="list-style-type: none"><li>Erect temporary hoarding</li><li>Identify plants to be transplanted</li><li>Avoid tree roots</li><li>Establish Tree Protection Zones (TPZ) for trees to be retained</li><li>Visually inspect trees and holes for nesting birds and animals prior to felling</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4                                 | -2 | 2 | 2 | 2 | -48  | Minor negative    |
|                                 |                          | Loss of habitats due to vegetation clearance for temporary working space                             | 3                                  | -3 | 2 | 2 | 3 | -63  | Minor negative    | <ul style="list-style-type: none"><li>Erect temporary hoarding to limit vegetation clearance</li><li>Reinstate habitats where possible upon completion of works</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3                                 | -2 | 2 | 2 | 2 | -36  | Slight negative   |
|                                 |                          | Flora and fauna species mortality due to vegetation clearance for development of CCKWW extension     | 4                                  | -4 | 3 | 3 | 3 | -144 | Major negative    | <ul style="list-style-type: none"><li>Alter project footprint to avoid direct impacts on Stream 3</li><li>Erect temporary hoarding</li><li>Identify trees to be transplanted</li><li>Conduct targeted translocation of aquatic fauna from Stream 2 and Stream 3</li><li>Conduct phased, directional vegetation clearance along with wildlife shepherding</li><li>Conduct daily checks of ECM blankets and pits to rescue any entrapped fauna</li><li>Utilise only non-plastic, biodegradable ECM mats throughout the site.</li><li>Reinstate working areas with selected native species</li><li>Avoid felling trees during the peak bird nesting period (February to July)</li><li>Establish Tree Protection Zones (TPZ) for trees to be retained</li></ul> | 4                                 | -3 | 3 | 3 | 3 | -108 | Moderate negative |

| Phase | Aspects | Impact component                                                                                    | Predicted impact before mitigation |    |   |   |   |      |                   | Mitigation measures                                                                                                                                                                                                                                                                                                             | Predicted impact after mitigation |    |   |   |   |     |                   |
|-------|---------|-----------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-------------------|
|       |         |                                                                                                     | I                                  | M  | P | R | C | ES   | Impact            |                                                                                                                                                                                                                                                                                                                                 | I                                 | M  | P | R | C | ES  | Impact            |
|       |         |                                                                                                     |                                    |    |   |   |   |      |                   | <ul style="list-style-type: none"> <li>Visually inspect trees for nesting birds and animals prior to felling</li> </ul>                                                                                                                                                                                                         |                                   |    |   |   |   |     |                   |
|       |         | Loss of habitats due to vegetation clearance for development of CCKWW extension                     | 3                                  | -4 | 3 | 3 | 3 | -108 | Moderate negative | <ul style="list-style-type: none"> <li>Alter project footprint to avoid direct impacts on Stream 3</li> <li>Erect temporary hoarding to limit vegetation clearance</li> <li>Reinstate habitats where possible upon completion of works</li> </ul>                                                                               | 3                                 | -3 | 3 | 3 | 3 | -81 | Moderate negative |
|       |         | Impact on habitats including waterbodies due to soil erosion and runoff                             | 4                                  | -3 | 3 | 2 | 3 | -96  | Moderate negative | <ul style="list-style-type: none"> <li>Implement Earth Control Measures (ECMs) that are approved by PUB</li> <li>Alter project footprint to avoid direct impacts on Stream 3</li> </ul>                                                                                                                                         | 3                                 | -2 | 2 | 2 | 2 | -36 | Slight negative   |
|       |         | Indirect impact on habitats due to edge effect                                                      | 2                                  | -2 | 3 | 3 | 3 | -36  | Slight negative   | <ul style="list-style-type: none"> <li>Avoid night works (i.e., limit construction activities to 8 am – 6 pm), except in the case of work exigencies and planned nightwork, with the implementation of the LMP approved by NParks</li> <li>Construction lights (if any) should face inwards and away from the forest</li> </ul> | 2                                 | -2 | 3 | 3 | 3 | -36 | Slight negative   |
|       |         | Impact on native species due to the introduction of invasive species                                | 2                                  | -3 | 3 | 3 | 3 | -54  | Minor negative    | <ul style="list-style-type: none"> <li>Ensure equipment, vehicles, and footwear used are clean prior to commencing works</li> <li>Use native trees for replanting of roadside verges</li> <li>Periodic checks within hoarding boundary to ensure that no introduced plant species have spontaneously established</li> </ul>     | 2                                 | -2 | 3 | 3 | 3 | -36 | Slight negative   |
|       |         | Impacts to flora and fauna due to illegal encroachment of vegetated areas by construction personnel | 2                                  | -3 | 2 | 2 | 3 | -42  | Minor negative    | <ul style="list-style-type: none"> <li>Incorporate signs prohibiting entrance into vegetated areas</li> <li>Erect fences to prevent illegal entry</li> </ul>                                                                                                                                                                    | 2                                 | -2 | 2 | 2 | 3 | -28 | Slight negative   |
|       |         | Roadkill or road injury to fauna escaping from forested site                                        | 4                                  | -3 | 3 | 3 | 3 | -108 | Moderate negative | <ul style="list-style-type: none"> <li>Erect physical barriers to prevent fauna from crossing the road</li> <li>Erect speed bumps or signages to alert drivers about potential animal crossings</li> </ul>                                                                                                                      | 4                                 | -2 | 3 | 3 | 3 | -72 | Minor negative    |

| Phase | Aspects               | Impact component                                                                             | Predicted impact before mitigation |    |   |   |   |     |                   | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Predicted impact after mitigation |    |   |   |   |     |                 |
|-------|-----------------------|----------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|       |                       |                                                                                              | I                                  | M  | P | R | C | ES  | Impact            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | I                                 | M  | P | R | C | ES  | Impact          |
|       |                       |                                                                                              |                                    |    |   |   |   |     |                   | <ul style="list-style-type: none"> <li>Enforce speed limit on the roads</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                         |                                   |    |   |   |   |     |                 |
|       |                       | Impact to ecological processes due to disturbance from night works                           | 3                                  | -3 | 2 | 2 | 3 | -63 | Minor negative    | <ul style="list-style-type: none"> <li>Complete all pile drilling before 6pm and only casting of piles to be carried out past 6pm.</li> <li>Complete all rebar fixing works before 6pm and only necessary concreting work for slab/ wall to be carried out past 6pm.</li> <li>Installation of 6m high noise barrier along project boundary.</li> <li>Additional localized noise barriers surrounding the works where possible</li> <li>Lighting management plan</li> </ul> | 3                                 | -2 | 2 | 2 | 2 | -36 | Slight negative |
|       | Biodiversity (Area 3) | Flora and fauna damage and mortality due to vegetation clearance for temporary working space | 3                                  | -2 | 2 | 2 | 3 | -42 | Minor negative    | <ul style="list-style-type: none"> <li>Erect temporary hoarding</li> <li>Avoid tree roots in placement of working areas</li> <li>Establish Tree Protection Zones (TPZ) for trees to be retained</li> <li>Visually inspect trees and holes for nesting birds and animals prior to felling</li> </ul>                                                                                                                                                                        | 3                                 | -1 | 2 | 2 | 3 | -21 | Slight negative |
|       |                       | Loss of habitats due to vegetation clearance for temporary working space                     | 3                                  | -2 | 2 | 2 | 3 | -42 | Minor negative    | <ul style="list-style-type: none"> <li>Erect temporary hoarding to limit vegetation clearance</li> <li>Avoid placement of working areas in waterbodies</li> <li>Reinstate habitats where possible upon completion of works</li> </ul>                                                                                                                                                                                                                                      | 3                                 | -1 | 2 | 2 | 2 | -18 | Slight negative |
|       |                       | Flora and fauna mortality due to vegetation clearance for working shaft locations            | 3                                  | -3 | 3 | 3 | 3 | -81 | Moderate negative | <ul style="list-style-type: none"> <li>Avoid placement of shafts near native, mature trees, where possible</li> <li>Identify trees and plants to be transplanted</li> <li>Conduct daily checks of ECM blankets and pits to rescue any entrapped fauna</li> <li>Utilise only non-plastic, biodegradable ECM mats throughout the site.</li> <li>Reinstate working areas with selected native species</li> </ul>                                                              | 3                                 | -2 | 3 | 3 | 3 | -54 | Minor negative  |
|       |                       |                                                                                              |                                    |    |   |   |   |     |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                   |    |   |   |   |     |                 |

| Phase | Aspects                            | Impact component                                                                                    | Predicted impact before mitigation |    |   |   |   |      |                   | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Predicted impact after mitigation |    |   |   |   |     |                 |
|-------|------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|       |                                    |                                                                                                     | I                                  | M  | P | R | C | ES   | Impact            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | I                                 | M  | P | R | C | ES  | Impact          |
|       |                                    |                                                                                                     |                                    |    |   |   |   |      |                   | <ul style="list-style-type: none"> <li>Establish Tree Protection Zones (TPZ) for trees to be retained</li> <li>Visually inspect trees for nesting birds and animals prior to felling</li> </ul>                                                                                                                                                                                                                                                                                                                                         |                                   |    |   |   |   |     |                 |
|       |                                    | Loss of habitats due to vegetation clearance for working shaft locations                            | 3                                  | -2 | 3 | 3 | 3 | -54  | Minor negative    | <ul style="list-style-type: none"> <li>Erect temporary hoarding to limit vegetation clearance</li> <li>Reinstate habitats where possible upon completion of works</li> </ul>                                                                                                                                                                                                                                                                                                                                                            | 3                                 | -2 | 3 | 2 | 3 | -42 | Minor negative  |
|       |                                    | Impact on habitats including waterbodies due to soil erosion                                        | 2                                  | -3 | 2 | 2 | 3 | -42  | Minor negative    | <ul style="list-style-type: none"> <li>Implement proper Earth Control Measures (ECM) plan prepared by QECP and approved by PUB</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                               | 2                                 | -2 | 2 | 2 | 3 | -28 | Slight negative |
|       |                                    | Impact on native species due to the introduction of invasive species                                | 2                                  | -3 | 3 | 3 | 3 | -54  | Minor negative    | <ul style="list-style-type: none"> <li>Ensure equipment, vehicles, and footwear used are clean prior to commencing works</li> <li>Use native plants for replanting of roadside verges</li> </ul>                                                                                                                                                                                                                                                                                                                                        | 2                                 | -2 | 3 | 3 | 3 | -36 | Slight negative |
|       |                                    | Impacts to flora and fauna due to illegal encroachment of vegetated areas by construction personnel | 2                                  | -3 | 2 | 2 | 3 | -42  | Minor negative    | <ul style="list-style-type: none"> <li>Incorporate signs prohibiting entrance into vegetated areas.</li> <li>Erect fences to prevent illegal entry</li> </ul>                                                                                                                                                                                                                                                                                                                                                                           | 2                                 | -2 | 2 | 2 | 3 | -28 | Slight negative |
|       |                                    | Roadkill or road injury to fauna escaping from forested site                                        | 4                                  | -3 | 3 | 3 | 3 | -108 | Moderate negative | <ul style="list-style-type: none"> <li>Erect physical barriers along road to prevent fauna from crossing the road</li> <li>Erect signages to alert drivers about potential animal crossings</li> <li>Enforce speed limit on the roads</li> </ul>                                                                                                                                                                                                                                                                                        | 4                                 | -2 | 3 | 3 | 3 | -72 | Minor negative  |
|       | Hydrology & Water Quality (Area 1) | Impact to water quality due to sediment runoff and siltation and trade effluent discharge           | 2                                  | -3 | 2 | 2 | 3 | -42  | Minor negative    | <ul style="list-style-type: none"> <li>Implementation proper Earth Control Measures (ECMs).</li> <li>Contractor to provide storage tanks of adequate size and sufficient numbers to temporarily store trade effluent before treatment and disposal.</li> <li>Contractor to provide sampling point and inspection for collection of trade effluent to be tested for parameters stipulated in the Regulations, depending on the discharge point.</li> <li>Regular water quality monitoring is to be conducted according to the</li> </ul> | 2                                 | -2 | 2 | 2 | 3 | -28 | Slight negative |

| Phase | Aspects                            | Impact component                                                                                 | Predicted impact before mitigation |    |   |   |   |     |                   | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Predicted impact after mitigation |    |   |   |   |     |                 |
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|       |                                    |                                                                                                  | I                                  | M  | P | R | C | ES  | Impact            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | I                                 | M  | P | R | C | ES  | Impact          |
|       |                                    |                                                                                                  |                                    |    |   |   |   |     |                   | proposed Environmental Monitoring Plan during construction and post-construction stage. <ul style="list-style-type: none"><li>Contractor should install and maintain trade effluent treatment facilities as per NEA's requirements</li><li>Appropriate concrete wash out areas should be provided and should not be performed in or near any water body.</li></ul>                                                                                                                                                                                                                                                                              |                                   |    |   |   |   |     |                 |
|       |                                    | Impact to water quality due to accidental spillage of construction materials, fuel, and solvents | 2                                  | -2 | 2 | 2 | 3 | -28 | Slight negative   | <ul style="list-style-type: none"><li>To ensure that bentonite is contained within the working area and does not enter any watercourses or surface water drains.</li><li>Storage silos should have secondary containment.</li><li>Bentonite use should be monitored so that excessive use could be identified to prevent bentonite materials from escaping into the ground and potentially polluting groundwater.</li><li>Waste bins to be provided within the construction work area for proper disposal of construction debris and rubbish.</li><li>Vehicle fuelling and major maintenance to be minimised within the project area.</li></ul> | 2                                 | -1 | 2 | 2 | 3 | -14 | Slight negative |
|       | Hydrology & Water Quality (Area 2) | Degradation and loss of stream habitats                                                          | 3                                  | -4 | 3 | 3 | 2 | -96 | Moderate negative | <ul style="list-style-type: none"><li>Alter project footprint to avoid direct impacts on Stream 3</li><li>Enhance edges of Stream 3 with native riparian vegetation</li><li>Ensure that none of the site investigation boreholes are located within 10 m of the natural streams.</li><li>Design appropriately to ensure that stormwater flow can be controlled during heavy rain period.</li></ul>                                                                                                                                                                                                                                              | 3                                 | -2 | 3 | 3 | 2 | -48 | Minor negative  |

| Phase | Aspects | Impact component                                                                                 | Predicted impact before mitigation |    |   |   |   |     |                   | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Predicted impact after mitigation |    |   |   |   |     |                 |
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|       |         |                                                                                                  | I                                  | M  | P | R | C | ES  | Impact            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | I                                 | M  | P | R | C | ES  | Impact          |
|       |         | Impact to water quality due to sediment runoff and siltation                                     | 2                                  | -3 | 2 | 2 | 3 | -42 | Minor negative    | <ul style="list-style-type: none"> <li>Implementation proper Earth Control Measures (ECMs)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2                                 | -2 | 2 | 2 | 3 | -28 | Slight negative |
|       |         | Impact to water quality due to trade effluent                                                    | 3                                  | -4 | 2 | 2 | 3 | -84 | Moderate negative | <ul style="list-style-type: none"> <li>Contractor to provide storage tanks of adequate size and sufficient numbers to temporarily store trade effluent before treatment and disposal.</li> <li>Contractor to provide sampling point and inspection for collection of trade effluent to be tested for parameters stipulated in the Regulations, depending on the discharge point (i.e., sewer, watercourse).</li> <li>Regular water quality monitoring is to be conducted according to the proposed Environmental Monitoring Plan during construction and post-construction stage.</li> <li>Contractor should install and maintain trade effluent treatment facilities (e.g., slurry treatment plant, grease trap and suspended solid settling tank) as per NEA's regulations for trade effluent under EPA.</li> </ul> | 3                                 | -3 | 2 | 2 | 3 | -63 | Minor negative  |
|       |         | Impact to water quality due to accidental spillage of construction materials, fuel, and solvents | 2                                  | -2 | 2 | 2 | 3 | -28 | Slight negative   | <ul style="list-style-type: none"> <li>To ensure that bentonite is contained within the working area and does not enter any watercourses or surface water drains.</li> <li>Storage silos should have secondary containment.</li> <li>Areas where bentonite is mixed are to be surrounded with a small boundary wall or contained within a bund. This action will help to control the slurry produced and prevent it from entering surface water drains or watercourses.</li> <li>Bentonite use should be monitored so that excessive use could be identified to</li> </ul>                                                                                                                                                                                                                                            | 2                                 | -1 | 2 | 2 | 3 | -14 | Slight negative |

| Phase | Aspects                            | Impact component                                                                                 | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Predicted impact after mitigation |    |   |   |   |     |                 |
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|       |                                    |                                                                                                  | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | I                                 | M  | P | R | C | ES  | Impact          |
|       |                                    |                                                                                                  |                                    |    |   |   |   |     |                 | prevent bentonite materials from escaping into the ground and potentially polluting groundwater. <ul style="list-style-type: none"> <li>Foreign material should not be illegally disposed at the water bodies but removed from site by a licensed waste collector.</li> <li>Vehicle fuelling and major maintenance to be minimised within the project area.</li> <li>Appropriate concrete wash out areas should be provided and should not be performed in or near any water body.</li> <li>Emergency spill kits should be provided by contractor on site in the event of any chemical spillages. Emergency response team should also be trained in the use of these spill kits.</li> </ul> |                                   |    |   |   |   |     |                 |
|       | Hydrology & Water Quality (Area 3) | Degradation of stream habitats                                                                   | 3                                  | -2 | 2 | 2 | 3 | -42 | Minor negative  | <ul style="list-style-type: none"> <li>Implement proper Earth Control Measures (ECMs)</li> <li>Regular water quality monitoring is to be conducted according to the proposed Environmental Monitoring Plan during construction and post-construction stage.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                      | 3                                 | -1 | 2 | 2 | 3 | -21 | Slight negative |
|       |                                    | Impact to water quality due to accidental spillage of construction materials, fuel, and solvents | 2                                  | -2 | 2 | 2 | 3 | -28 | Slight negative | <ul style="list-style-type: none"> <li>To ensure that bentonite is contained within the working area and does not enter any watercourses or surface water drains.</li> <li>Storage silos should have secondary containment.</li> <li>Foreign material should not be illegally disposed at the water bodies but removed from site by a licensed waste collector.</li> <li>No vehicle fuelling and maintenance to be allowed within the project area.</li> </ul>                                                                                                                                                                                                                              | 2                                 | -1 | 2 | 2 | 3 | -14 | Slight negative |

| Phase | Aspects        | Impact component                                            | Predicted impact before mitigation |    |   |   |   |     |                | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Predicted impact after mitigation |    |   |   |   |     |                 |
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|       |                |                                                             | I                                  | M  | P | R | C | ES  | Impact         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | I                                 | M  | P | R | C | ES  | Impact          |
|       |                |                                                             |                                    |    |   |   |   |     |                | <ul style="list-style-type: none"> <li>Excess loose soil and rock to be contained prior to the commencement of the works.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                   |    |   |   |   |     |                 |
|       | Noise (Area 2) | Disturbance to fauna species in and around the project site | 3                                  | -2 | 2 | 2 | 3 | -42 | Minor negative | <ul style="list-style-type: none"> <li>No night works (i.e., limit construction activities to 8 am – 6 pm), except in the case of work exigencies and planned nightwork, with the implementation of the LMP approved by NParks</li> <li>Noise barrier with 6 m height as hoarding</li> <li>Quieter equipment and vehicles with low noise levels to be used</li> <li>To consider erection of temporary acoustic barriers around piling equipment</li> <li>For piling work, to consider the use of quieter piling methods</li> <li>Acoustic enclosures to be considered for compressors, generators, drilling tools etc.</li> <li>All noise and acoustic barriers to be able to achieve at least 10dB(A) noise reduction from source noise levels.</li> <li>Noise level monitoring throughout all phases to ensure that the construction noises remain within acceptable limits</li> </ul> | 3                                 | -1 | 2 | 2 | 3 | -21 | Slight negative |
|       | Noise (Area 3) | Disturbance to fauna species in and around the project site | 2                                  | -2 | 2 | 2 | 3 | -42 | Minor negative | <ul style="list-style-type: none"> <li>No night works (i.e., limit construction activities to 8 am – 6 pm) within forested areas, except in the case of work exigencies</li> <li>Noise barrier with 6 m height as hoarding</li> <li>Quieter equipment and vehicles with low noise levels to be used</li> <li>To consider erection of temporary acoustic barriers around piling equipment</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2                                 | -1 | 2 | 2 | 3 | -21 | Slight negative |

| Phase | Aspects             | Impact component                                                       | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Predicted impact after mitigation |    |   |   |   |     |                 |
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|       |                     |                                                                        | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | I                                 | M  | P | R | C | ES  | Impact          |
|       |                     |                                                                        |                                    |    |   |   |   |     |                 | <ul style="list-style-type: none"> <li>For piling work, to consider the use of quieter piling methods</li> <li>Acoustic enclosures to be considered for compressors, generators, drilling tools etc.</li> <li>All noise and acoustic barriers to be able to achieve at least 10 dB(A) noise reduction from source noise levels.</li> <li>Noise level monitoring throughout all phases to ensure that the construction noises remain within acceptable limits</li> </ul>                                                                      |                                   |    |   |   |   |     |                 |
|       | Ambient Air Quality | Impact to biodiversity due to generation of fugitive dust              | 2                                  | -2 | 2 | 2 | 2 | -24 | Slight negative | <ul style="list-style-type: none"> <li>To implement dust suppression plan</li> <li>Use of hoarding at project boundary to minimise dust generation by attenuating wind forces.</li> <li>To avoid stockpiles of soil and dusty materials at project site within forested area as far as possible.</li> <li>Stockpiles of dusty material should be properly stored, covered entirely with impervious sheeting</li> <li>Use of regular watering to reduce dust emissions from exposed site surface</li> </ul>                                   | 2                                 | -1 | 2 | 2 | 2 | -12 | Slight Negative |
|       | Vibration           | Disturbance to the fauna due to vibration from construction activities | 3                                  | -3 | 2 | 2 | 3 | -63 | Minor negative  | <ul style="list-style-type: none"> <li>Use of heavy equipment to be restricted to daytime only at Area 2 (CCKWW extension area).</li> <li>Use of equipment or method which generate lower vibration levels</li> <li>Route heavily loaded trucks away from the VSRs as far away as possible.</li> <li>To plan and phase excavation, earth-moving, and ground pilling activities in staggered manner</li> <li>Control speed of vehicle movement at the worksite</li> <li>Keep haul roads within project work site in good condition</li> </ul> | 3                                 | -2 | 2 | 2 | 3 | -42 | Minor negative  |

| Phase     | Aspects               | Impact component                                                             | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Predicted impact after mitigation |    |   |   |   |     |                 |
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|           |                       |                                                                              | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | I                                 | M  | P | R | C | ES  | Impact          |
|           | Light                 | Disturbance to the flora and fauna due to light from construction activities | 2                                  | -3 | 2 | 2 | 3 | -42 | Minor negative  | <ul style="list-style-type: none"> <li>Construction lights in Area 1 shall be directed downwards and inwards, toward project site (i.e., directly away from forested areas)</li> <li>Construction in Areas 2 and the forested parts of Area 3 to be kept to daytime working hours (8 am – 6 pm), except in the case of work exigencies and planned nightwork, with the implementation of the LMP approved by NParks</li> <li>All unnecessary lights should be turned off outside working hours</li> <li>Develop LMP for Areas 2 and 3 for the event of work exigencies and planned nightwork</li> </ul> | 2                                 | -2 | 1 | 1 | 1 | -4  | No impact       |
|           | Waste Management      | Disposal of cut vegetation                                                   | 2                                  | -2 | 2 | 2 | 2 | -24 | Slight Negative | <ul style="list-style-type: none"> <li>To develop solid waste management plan</li> <li>Timber/wood to be recovered for use in the wood industry as far as possible</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                           | 2                                 | -1 | 2 | 2 | 2 | -12 | Slight Negative |
|           |                       | Disposal of excavated material                                               | 2                                  | -3 | 2 | 2 | 2 | -36 | Slight Negative | <ul style="list-style-type: none"> <li>To develop solid waste management plan</li> <li>Surplus excavated material to be reused within project site as fill, landscaping, erosion control and restoration wherever practicable</li> </ul>                                                                                                                                                                                                                                                                                                                                                                | 2                                 | -2 | 2 | 2 | 2 | -24 | Slight Negative |
|           |                       | Disposal of general waste material                                           | 1                                  | -2 | 2 | 2 | 2 | -12 | Slight Negative | <ul style="list-style-type: none"> <li>Scrap metals to be recovered and sent for recycling as scrap</li> <li>Inert general waste to be collected and disposed through licensed waste collector</li> <li>All non-hazardous wastes to be handled and disposed of in accordance with EPH (General Waste Collection) Regulations</li> </ul>                                                                                                                                                                                                                                                                 | 1                                 | -1 | 2 | 2 | 2 | -6  | No Impact       |
| Operation | Biodiversity (Area 2) | Indirect impact on habitats due to edge effect                               | 3                                  | -2 | 2 | 2 | 2 | -36 | Slight negative | <ul style="list-style-type: none"> <li>Use native plant for successful reinstatement of working areas.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3                                 | -1 | 2 | 2 | 2 | -18 | Slight negative |

| Phase                           | Aspects                            | Impact component                                                                                                            | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Predicted impact after mitigation |    |   |   |   |     |                 |
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|                                 |                                    |                                                                                                                             | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | I                                 | M  | P | R | C | ES  | Impact          |
|                                 |                                    |                                                                                                                             |                                    |    |   |   |   |     |                 | <ul style="list-style-type: none"> <li>Restore vegetation area with native trees of diverse structures and species.</li> </ul>                                                                                                                                                                                                                                                                                                                                                  |                                   |    |   |   |   |     |                 |
|                                 | Biodiversity (Area 3)              | Flora and fauna mortality due to vegetation clearance for maintenance regimes                                               | 2                                  | -2 | 2 | 2 | 2 | -24 | Slight negative | <ul style="list-style-type: none"> <li>Avoid clearance of vegetation during maintenance regimes</li> </ul>                                                                                                                                                                                                                                                                                                                                                                      | 2                                 | -1 | 1 | 1 | 1 | -6  | No Impact       |
|                                 | Hydrology & Water Quality (Area 2) | Impact to drainage pattern and water quality in Stream 2                                                                    | 2                                  | -3 | 2 | 2 | 3 | -42 | Minor negative  | <ul style="list-style-type: none"> <li>No discharge of process effluent or release of pollutants through this outlet to the stream.</li> <li>Flow rate of discharge is to be regulated to prevent erosion within receiving stream.</li> <li>Water quality checks if there is suspicion of discharge of pollutant to the stream.</li> </ul>                                                                                                                                      | 2                                 | -2 | 2 | 2 | 3 | -28 | Slight negative |
|                                 | Light                              | Disturbance to the flora and fauna due to operation phase lighting                                                          | 2                                  | -3 | 3 | 3 | 3 | -48 | Minor negative  | <ul style="list-style-type: none"> <li>Lighting installed within Area 2 should be directed downwards and inwards, toward project site (i.e., directly away from forested areas)</li> <li>Use the minimum number and intensity of lights required for safety and security</li> <li>Area 1 and 2 should utilise non-white lighting where possible</li> <li>Buffer planting should include thick hedge plants which can reduce the amount of light entering the forest.</li> </ul> | 2                                 | -2 | 3 | 3 | 3 | -36 | Slight negative |
| <b>Human Health and Safety</b>  |                                    |                                                                                                                             |                                    |    |   |   |   |     |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                   |    |   |   |   |     |                 |
| Pre-construction / Construction | Biodiversity (Area 2)              | Attacks of construction personnel by animals including wild pigs, feral dogs, long-tailed macaques, and smooth-coated otter | 3                                  | -3 | 2 | 3 | 2 | -63 | Minor negative  | <ul style="list-style-type: none"> <li>Establish designated areas for food and waste disposal</li> <li>Install monkey-proof bins</li> <li>Conduct information sessions on what to do upon encountering wildlife</li> </ul>                                                                                                                                                                                                                                                      | 3                                 | -2 | 2 | 3 | 1 | -36 | Slight negative |
|                                 |                                    | Impact to health of personnel due to tick/ mite bites from feral dogs                                                       | 3                                  | -3 | 2 | 3 | 2 | -63 | Minor negative  | <ul style="list-style-type: none"> <li>Implement proper use of Personal Protective Equipment (PPE)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                   | 3                                 | -2 | 2 | 3 | 1 | -36 | Slight negative |
|                                 |                                    | Injury caused by tree falls due to damaged roots                                                                            | 3                                  | -3 | 2 | 3 | 2 | -63 | Minor negative  | <ul style="list-style-type: none"> <li>Establish Tree Protection Zones (TPZ) for trees to be retained</li> </ul>                                                                                                                                                                                                                                                                                                                                                                | 3                                 | -2 | 2 | 3 | 1 | -36 | Slight negative |

| Phase | Aspects                            | Impact component                                                                                       | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Predicted impact after mitigation |    |   |   |   |     |                 |
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|       |                                    |                                                                                                        | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | I                                 | M  | P | R | C | ES  | Impact          |
|       |                                    |                                                                                                        |                                    |    |   |   |   |     |                 | <ul style="list-style-type: none"> <li>Regularly monitor health of trees on project site</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                   |    |   |   |   |     |                 |
|       | Biodiversity (Area 3)              | Attacks of construction personnel by animals including wild pigs, feral dogs, and long-tailed macaques | 3                                  | -3 | 2 | 3 | 2 | -63 | Minor negative  | <ul style="list-style-type: none"> <li>Establish designated areas for food and waste disposal</li> <li>Install monkey-proof bins</li> <li>Conduct information sessions on what to do upon encountering wildlife</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                 | 3                                 | -2 | 2 | 3 | 1 | -36 | Slight negative |
|       |                                    | Impact to health of personnel due to tick/ mite bites from feral dogs                                  | 3                                  | -3 | 2 | 3 | 2 | -63 | Minor negative  | <ul style="list-style-type: none"> <li>Implement proper use of Personal Protective Equipment (PPE)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3                                 | -2 | 2 | 3 | 1 | -36 | Slight negative |
|       |                                    | Injury caused by tree falls due to damaged roots                                                       | 3                                  | -3 | 2 | 3 | 2 | -63 | Minor negative  | <ul style="list-style-type: none"> <li>Establish Tree Protection Zones (TPZ) for trees to be retained</li> <li>Regularly monitor health of trees on project site</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3                                 | -2 | 2 | 3 | 1 | -36 | Slight negative |
|       | Hydrology & Water Quality (Area 1) | Impact to water quality due to accidental spillage of construction materials, fuel, and solvents       | 2                                  | -2 | 2 | 2 | 2 | -24 | Slight negative | <ul style="list-style-type: none"> <li>Contractor to provide storage tanks of adequate size and sufficient numbers to temporarily store trade effluent before treatment and disposal.</li> <li>Contractor to provide sampling point and inspection for collection of trade effluent to be tested for parameters stipulated in the Regulations, depending on the discharge point (i.e., sewer, watercourse, controlled watercourse).</li> <li>Contractor should install and maintain trade effluent treatment facilities (e.g., slurry treatment plant, grease trap and suspended solid settling tank) as per NEA's requirements</li> </ul> | 2                                 | -1 | 2 | 2 | 2 | -12 | Slight negative |
|       | Hydrology & Water Quality (Area 2) | Impact to water quality due to accidental spillage of construction materials, fuel, and solvents       | 2                                  | -2 | 2 | 2 | 2 | -24 | Slight negative | <ul style="list-style-type: none"> <li>Contractor to provide storage tanks of adequate size and sufficient numbers to temporarily store trade effluent before treatment and disposal.</li> <li>Contractor to provide sampling point and inspection for collection of trade effluent to be tested for parameters</li> </ul>                                                                                                                                                                                                                                                                                                                 | 2                                 | -1 | 2 | 2 | 2 | -12 | Slight negative |

| Phase | Aspects                            | Impact component                                                                                 | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Predicted impact after mitigation |    |   |   |   |     |                 |
|-------|------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|       |                                    |                                                                                                  | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | I                                 | M  | P | R | C | ES  | Impact          |
|       |                                    |                                                                                                  |                                    |    |   |   |   |     |                 | stipulated in the Regulations, depending on the discharge point (i.e., sewer, watercourse, controlled watercourse).<br>• Contractor should install and maintain trade effluent treatment facilities (e.g., slurry treatment plant, grease trap and suspended solid settling tank) as per PUB's requirements                                                                                                                                                                                                                     |                                   |    |   |   |   |     |                 |
|       | Hydrology & Water Quality (Area 3) | Impact to water quality due to accidental spillage of construction materials, fuel, and solvents | 2                                  | -2 | 2 | 2 | 3 | -28 | Slight negative | • Contractor to provide storage tanks of adequate size and sufficient numbers to temporarily store trade effluent before treatment and disposal.<br>• Contractor to provide sampling point and inspection for collection of trade effluent to be tested for parameters stipulated in the Regulations, depending on the discharge point.<br>• Contractor should install and maintain trade effluent treatment facilities (e.g., slurry treatment plant, grease trap and suspended solid settling tank) as per NEA's requirements | 2                                 | -1 | 2 | 2 | 3 | -14 | Slight negative |
|       | Noise (Area 2)                     | Impact to construction workers due to exposure to high noise levels of construction activities   | 2                                  | -2 | 2 | 2 | 3 | -28 | Slight negative | • Personnel are to wear appropriate Personal Protective Equipment (PPE) at all times while on the construction site<br>• Quieter equipment and vehicles with low noise levels to be used.<br>• All noise and acoustic barriers to be able to achieve at least 10dB(A) noise reduction from source noise levels.                                                                                                                                                                                                                 | 2                                 | -1 | 2 | 2 | 3 | -14 | Slight negative |
|       |                                    | Impact to NSR due to noise pollution from construction activities                                | 2                                  | -2 | 2 | 2 | 3 | -28 | Slight negative | • No night works (i.e., limit construction activities to 8 am – 6 pm), except in the case of work exigencies and planned nightwork, with the implementation of the LMP approved by NParks                                                                                                                                                                                                                                                                                                                                       | 2                                 | -1 | 2 | 2 | 3 | -14 | Slight negative |

| Phase | Aspects           | Impact component                                                                               | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Predicted impact after mitigation |    |   |   |   |     |                 |
|-------|-------------------|------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|       |                   |                                                                                                | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | I                                 | M  | P | R | C | ES  | Impact          |
|       |                   |                                                                                                |                                    |    |   |   |   |     |                 | <ul style="list-style-type: none"> <li>Noise barrier with 6 m height as hoarding</li> <li>Quieter equipment and vehicles with low noise levels to be used</li> <li>To consider erection of temporary acoustic barriers around piling equipment</li> <li>For piling work, to consider the use of quieter piling methods</li> <li>Acoustic enclosures to be considered for compressors, generators, drilling tools etc.</li> <li>All noise and acoustic barriers to be able to achieve at least 10dB(A) noise reduction from source noise levels.</li> <li>Noise level monitoring throughout all phases to ensure that the construction noises remain within acceptable limits</li> </ul> |                                   |    |   |   |   |     |                 |
|       | Noise<br>(Area 3) | Impact to construction workers due to exposure to high noise levels of construction activities | 2                                  | -2 | 2 | 2 | 3 | -28 | Slight negative | <ul style="list-style-type: none"> <li>Personnel are to wear appropriate Personal Protective Equipment (PPE) at all times while on the construction site</li> <li>Quieter equipment and vehicles with low noise levels to be used</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2                                 | -1 | 2 | 2 | 3 | -14 | Slight negative |
|       |                   | Impact to NSR due to noise pollution from construction activities                              | 1                                  | -3 | 2 | 2 | 3 | -21 | Slight negative | <ul style="list-style-type: none"> <li>No night works (i.e., limit construction activities to 8 am – 6 pm), except in the case of work exigencies</li> <li>Noise barrier with 6m height as hoarding</li> <li>Quieter equipment and vehicles with low noise levels to be used</li> <li>To consider erection of temporary acoustic barriers around piling equipment</li> <li>For piling work, to consider the use of quieter piling methods</li> </ul>                                                                                                                                                                                                                                    | 1                                 | -2 | 2 | 2 | 3 | -14 | Slight negative |

| Phase | Aspects             | Impact component                                                                                                      | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Predicted impact after mitigation |    |   |   |   |     |                 |
|-------|---------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|       |                     |                                                                                                                       | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | I                                 | M  | P | R | C | ES  | Impact          |
|       |                     |                                                                                                                       |                                    |    |   |   |   |     |                 | <ul style="list-style-type: none"> <li>Acoustic enclosures to be considered for compressors, generators, drilling tools etc.</li> <li>All noise and acoustic barriers to be able to achieve at least 10dB(A) noise reduction from source noise levels.</li> <li>Noise level monitoring throughout all phases to ensure that the construction noises remain within acceptable limits</li> </ul>                                                                                                                                                                               |                                   |    |   |   |   |     |                 |
|       | Ambient Air Quality | Impact on construction workers due to generation of fugitive dust and exhaust emissions from construction machineries | 2                                  | -3 | 2 | 2 | 2 | -36 | Slight Negative | <ul style="list-style-type: none"> <li>To implement dust suppression plan</li> <li>Personal protective equipment i.e., face mask to be worn during dust exposure</li> <li>Stockpiles of dusty material to be properly stored, covered entirely with impervious sheeting, or dampened with water</li> <li>Use of regular watering to reduce dust emissions from exposed site surfaces</li> <li>Proper maintenance of construction vehicles and fuel burning equipment</li> <li>Intermittently-used vehicles and machinery are to be shut down between work periods</li> </ul> | 2                                 | -2 | 2 | 2 | 2 | -24 | Slight Negative |
|       |                     | Impact on ASR due to generation of fugitive dust                                                                      | 2                                  | -2 | 2 | 2 | 2 | -24 | Slight Negative | <ul style="list-style-type: none"> <li>To implement dust suppression plan</li> <li>Provide dust screen at project boundary near residential community</li> <li>Stockpiles of dusty material to be properly stored, covered entirely with impervious sheeting, or dampened with water</li> <li>Use of regular watering to reduce dust emissions from exposed site surface</li> </ul>                                                                                                                                                                                          | 2                                 | -1 | 2 | 2 | 2 | -12 | Slight Negative |
|       |                     | Impact due to generation of odour                                                                                     | 1                                  | -1 | 2 | 2 | 2 | -6  | No Impact       | Refer to Section 10.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1                                 | -1 | 2 | 2 | 2 | -6  | No Impact       |
|       | Vibration           | Disturbance/ annoyance to the VSR due vibration from construction activities                                          | 2                                  | -2 | 2 | 2 | 3 | -28 | Slight negative | Use of heavy equipment to be restricted to daytime only at Area 3 (NTU premises)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2                                 | -1 | 2 | 2 | 3 | -14 | Slight negative |

| Phase | Aspects          | Impact component                                                  | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Predicted impact after mitigation |    |   |   |   |     |                 |
|-------|------------------|-------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|       |                  |                                                                   | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | I                                 | M  | P | R | C | ES  | Impact          |
|       |                  |                                                                   |                                    |    |   |   |   |     |                 | <ul style="list-style-type: none"> <li>Use of equipment or method which generate lower vibration levels</li> <li>Route heavily loaded trucks away from the VSRs as far away as possible.</li> <li>Control speed of vehicle movement at the worksite</li> <li>To develop the vibration monitoring plan in consultation QP &amp; to monitor throughout construction period</li> <li>Notify nearby VSRs (NTU premises) in advance of the construction activities</li> </ul> |                                   |    |   |   |   |     |                 |
|       | Waste Management | Disposal of hazardous waste                                       | 1                                  | -2 | 2 | 2 | 2 | -12 | Slight Negative | <ul style="list-style-type: none"> <li>All hazardous wastes to be handled and disposed of in accordance with EPH (Toxic Industrial Wastes) Regulations</li> <li>Disposal of hazardous waste to be conducted by a licensed waste collector for hazardous waste</li> </ul>                                                                                                                                                                                                 | 1                                 | -1 | 2 | 2 | 2 | -6  | No Impact       |
|       | Vector Control   | Increase in the number of mosquitoes                              | 2                                  | -2 | 2 | 2 | 2 | -24 | Slight Negative | <ul style="list-style-type: none"> <li>Contractor to engage an NEA-registered vector control operator to prepare and implement vector management plan</li> <li>An in-house vector control team to check construction sites for breeding of mosquitoes</li> <li>Construction worksite to be kept free of litter; construction wastes shall be disposed promptly into bulk waste containers and the containers shall be emptied daily</li> </ul>                           | 2                                 | -1 | 2 | 2 | 2 | -12 | Slight Negative |
|       |                  | Increase in the number of other vectors (e.g., flies and rodents) | 1                                  | -2 | 2 | 2 | 2 | -12 | Slight Negative | <ul style="list-style-type: none"> <li>Worksite shall be kept litter-free and refuse bins shall always be covered tightly.</li> <li>Construction workers' food provisions shall be stored in rodent-proof rooms or cabinets</li> <li>In-house vector control team and vector control operator to check for rodent burrows every week</li> </ul>                                                                                                                          | 1                                 | -1 | 2 | 2 | 2 | -6  | No Impact       |

| Phase     | Aspects               | Impact component                                                                                      | Predicted impact before mitigation |    |   |   |   |     |                 | Mitigation measures                                                                                                                                                                                                            | Predicted impact after mitigation |    |   |   |   |     |                 |
|-----------|-----------------------|-------------------------------------------------------------------------------------------------------|------------------------------------|----|---|---|---|-----|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----|---|---|---|-----|-----------------|
|           |                       |                                                                                                       | I                                  | M  | P | R | C | ES  | Impact          |                                                                                                                                                                                                                                | I                                 | M  | P | R | C | ES  | Impact          |
|           |                       | Increase in incidence of dengue fever and vector-related diseases ( <i>secondary impact</i> )         | 1                                  | -2 | 2 | 2 | 2 | -12 | Slight Negative | <ul style="list-style-type: none"> <li>To implement mitigation measures to control mosquito numbers and other vectors</li> <li>To aid the authorities to investigate outbreaks of vector-borne diseases if required</li> </ul> | 1                                 | -1 | 2 | 2 | 2 | -6  | No Impact       |
| Operation | Biodiversity (Area 3) | Attacks of maintenance personnel by animals including wild pigs, feral dogs, and long-tailed macaques | 3                                  | -3 | 2 | 3 | 2 | -63 | Minor negative  | <ul style="list-style-type: none"> <li>Avoid consuming food within forested areas</li> <li>Conduct information sessions on what to do upon encountering wildlife</li> </ul>                                                    | 3                                 | -2 | 2 | 3 | 1 | -36 | Slight negative |

## **13 ENVIRONMENTAL MANAGEMENT & MONITORING PLAN FRAMEWORK**

### **13.1 Overview**

The Environmental Management and Monitoring Plan (EMMP) is a systematic approach to mitigate environmental impacts and monitor the implementation of these mitigation measures to ensure that project implementation will not cause any significant adverse impact to the site and the surrounding environment. It is also a useful tool to assess whether the mitigation measures taken are effective to reduce or mitigate the potential impacts caused by this project to minimal and acceptable levels during the construction.

The EMMP framework presented in this chapter is an outcome from the EIA process. This EMMP framework consolidates the mitigation and monitoring strategy required for this project and appointed Contractor shall adhere to this strategy, develop it further and implement throughout during the construction phase.

### **13.2 Construction EMMP**

Before the construction works start, the appointed Contractor must establish a detailed Construction EMMP (CEMMP) based on this EMMP framework, which is to be implemented and monitored during the construction phase. The CEMMP shall be submitted by the Contractors within one month of award for the construction work. The Contractor shall be responsible to submit and obtain approval for CEMMP from PUB, NParks, and other relevant Technical Agencies before commencement of works. The CEMMP shall also cover all site investigation works (e.g., soil investigation, trial trenching, etc), and constructions works including temporary works.

The CEMMP to include the identified mitigation measures of this EIA and shall address the methodologies of the construction works prior to their commencement. Additionally, Fauna and Flora Management Plans recommended in this chapter need to be incorporated in CEMMP with involvement of biodiversity specialists from Contractor EMMP team to finetune and implement ecological mitigation measures recommended in this report. The CEMMP should also include waste management practices including restricting use & spillage of chemicals during construction phase into surrounding forested area.

The CEMMP shall include environmental monitoring comprises of compliance inspections for prescribed mitigation measures and ambient environmental data collection, generally requiring sample collection and analysis. The environmental monitoring activities should also ensure that the project does not cause any significant long-term environmental impacts, in particular cumulative impacts, and that the existing environmental conditions and biodiversity is maintained.

Monthly environmental monitoring reports with all monitoring results (compliance and ambient monitoring), identified problems and additional actions taken to mitigate these problems should be prepared and submitted to the relevant authorities during the

construction phase. Each subsequent monthly monitoring reports should report on successful or failed follow-up actions until a problem has been effectively mitigated.

Table 13.9 provides the overview of recommended EMMP measures for this project which is to be incorporated into the CEMMP by the Contractor. The environmental monitoring locations are to be finalised by EMMP team during CEMMP formulation in consultation with relevant stakeholders.

### 13.2.1 EMMP Team

The contractors shall be responsible for implementing all the environmental requirements specified in this EIA report including CEMMP conditions as well as requirements mandated by the applicable regulations and relevant authorities. It is recommended that to implement the CEMMP, an EMMP team having necessary qualifications and experience shall be available throughout the construction period to support the Contractor. The EMMP team should include but not limited to the following:

**Table 13.1.** Roles and responsibilities of EMMP team members

| Role Name                                                   | Qualification                                                                                                                                          | Responsibilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Environmental Manager/<br>Environment Control Officer (ECO) | Valid registration with the National Environment Agency (NEA) and demonstrated previous work experience in developments of similar size or complexity. | To act as CEMMP in-charge and lead the implementation and reporting requirements of CEMMP during construction phase. Responsible for managing all environmental issues arising from the construction work which includes the monitoring and ensuring the implementation and management of change of the CEMMP, the environmental performance of the project, investigation of incidents, inspections of site and implementing corrective/ preventive measures. To coordinate with other EMMP team members for advice on specific issues related to CEMMP implementation. To prepare monthly environmental performance monitoring reports. |
| Qualified Erosion control Professional (QECP)               | Valid registration with the Institution of Engineers Singapore (IES)                                                                                   | Responsible to prepare, submit and obtain approval for Earth Control Measures (ECM) Plan from PUB prior to contractors start work. To ensure that the contractors implement the ECM in compliance with the ECM Plan.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Arborist / Flora Specialist /<br>Nursery Manager            | Certified Arborist (CA) with an International Society of Arboriculture (ISA) certification with 5 years' experience.                                   | To prepare project site specific flora management plan as part of CEMMP, utilizing the EMMP framework recommended in the EIA and assist contractor in obtaining authority approval. To review the flora that will be affected, and to provide monitoring and advice to the contractor on matter related to transplantation and reinstatement. To assist in selecting a nursery, carrying out salvaging of native plants, and maintaining the salvaged saplings. To visit the site regularly to oversee the implementation of flora management plan and to provide biodiversity awareness training to site                                 |

| Role Name                             | Qualification                                                            | Responsibilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       |                                                                          | personnel. To advise and implement specific measures in the case of tree pruning prior clearance, tree injury, and construction activities affecting tree roots. To provide checks on Tree Protection Zone (TPZ). To assist in reporting requirements of CEMMP during construction phase. To liaise with NParks on addressing any comments/ requirements related to flora implementation measures.                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Ecologist / Wildlife Specialist       | At least 10 years' experience in wildlife in Singapore and/or the region | To prepare project site specific fauna management plan as part of CEMMP, utilizing the EMMP framework recommended in the EIA and assist contractor in obtaining authority approval. To conduct a pre-clearance wildlife inspection and prepare Wildlife management protocols as necessary during site clearance stage. To provide advice and inspection related to wildlife throughout the construction duration and to identify, rescue & manage any trapped and/or injured wildlife at project site. To visit the site regularly to oversee the implementation of fauna management plan and to provide biodiversity awareness training to site personnel. To assist in reporting requirements of CEMMP during construction phase. To liaise with NParks on addressing any comments/ requirements related to wildlife implementation measures. |
| Earth Control Measures Officer (ECMO) | Valid IES registration                                                   | The ECMO is responsible to implement all ECM requirements in compliance with the ECM Plan approved by PUB.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

### 13.2.2 Environmental Objectives

Environmental Objectives for the project are recommended below. Contractors are required to adhere to the Environmental Objectives during entire construction stage.

- Minimise removal of conservation significant flora species at or adjacent to worksite.
- Maximise harvesting of sapling, and transplant of conservation significant flora species that will be affected.
- Prevent risk of fauna injury related to construction such as from fauna re-entry to worksite.
- Prevent human – wildlife conflicts at or near to worksite.
- Ensure retained, translocated, and planted flora's health.
- Ensure noise impacts comply with adjusted Maximum Permissible Noise Limits.
- Ensure water quality impacts comply with PUB and NEA allowable limits for trade effluent quality.
- Ensure no transmission of vector borne disease.
- Ensure no indiscriminate / illegal disposal of waste.
- Ensure no spillage / leakage of hazardous material.

### 13.2.3 Training Requirements

The site personnel involved in implementation of CEMMP shall be adequately trained. Training needs assessment shall be conducted regularly and should include the concerned sub-contractors also. The contractors shall ensure that trainings are conducted before the starting the construction work and at a regular interval during construction phase for the site personnel. The recommended training program is provided in Table 13.2.

**Table 13.2.** Training Programme for Site Personnel

| Training Schedule                           | Training Topics                                                                   | Conducted By            | Target Audience                                                                                                                        | Frequency                                  |
|---------------------------------------------|-----------------------------------------------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Prior to commencement of activities on site | CEMMP Requirements<br>Biodiversity & Environmental Awareness                      | EMMP Consultant         | Environmental Manager/ ECO/<br>Project Manager /<br>Construction Manager/<br>Construction Engineers/ Site Supervisors/ Sub-contractors | Once                                       |
| Refresher training                          | Biodiversity & Environmental Awareness Briefing                                   | CEMMP In-charge/<br>ECO | Site Personnel including Sub-contractors                                                                                               | Every six months during construction phase |
| Toolbox meetings                            | Briefing to include reminders on wildlife encounters and environmental protection | CEMMP In-charge/<br>ECO | Construction Workers including Sub-contractors                                                                                         | Daily                                      |

### 13.2.4 Environmental Emergency Response Procedure

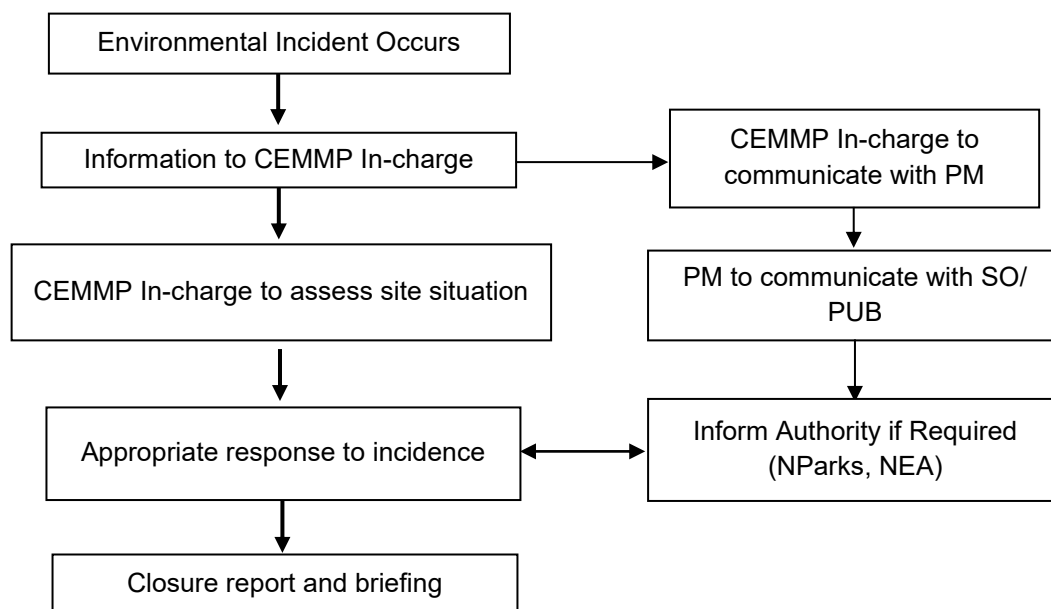
The regular and continual environmental monitoring may result in observations of failed or inadequate mitigation measures. Also, a public complaint/ observation may be received. Examples of environmental emergency are described below.

- Incident related to human – wildlife conflict
- Discovery of wildlife within construction site
- Incident of injury to wildlife due to work activity
- Damage to retained/ translocated/ planted flora
- Earth control measures (i.e., silt fence, cut-off drain, treatment) are ineffective
- Discharge of ECM does not meet regulatory limits
- Illegal water discharge from construction site
- Noise levels from construction activities exceed maximum permissible limits.
- Dust event due to dry weather conditions and high winds.
- Release of hazardous materials to land and watercourse
- Illegal disposal of waste into forested area.
- Community complaint relating to pollution

In the event that a failure is discovered that failure must be reported to the CEMMP In-charge/ Project Manager within shortest possible time. The CEMMP In-charge/ Project Manager will be responsible for ensuring adequate follow-up activities. This may include:

- Consultation with the EMMP specialists/ QECP/ VCO/ PRO.
- Arranging an immediate appropriate response on guidance of EMMP Specialists/ QECP/ VCO as necessary.
- Reporting and consultation with the relevant authorities (i.e., NEA, NParks, PUB) as required.

The emergency response flowchart in handling environmental emergency is presented in Figure 13.1. In the event of violation of relevant standards/ regulations, it is recommended that site environmental management practices are reviewed immediately, and the appropriate mitigation action taken immediately to reduce impacts to acceptable levels.



**Figure 13.1.** Environmental incidence reporting flowchart

### 13.2.5 EMMP Reporting and Documentation

#### *Site Environmental Control Report by Environmental Control Officer*

As per Code of Practice for Environmental Control Officers for Construction Sites, ECO shall prepare the Site Environmental Control Programme before work commences at the worksite. Further, ECO shall submit the Site Environmental Control Report (SECR) to Project Manager covering assessment of the environmental efforts carried out and review the effectiveness of these measures.

The ECO shall inspect the construction activities regularly and routinely to ensure that the appropriate environmental protection and pollution control mitigation measures are properly and timely implemented based on the CEMMP's recommendations. ECO shall record all observations and actions taken to report them in the SECR. This SECR should form part of CEMMP monthly performance report described in next section.

### **Monthly Environmental Performance Report**

A monthly Environmental Performance Report is to be prepared by CEMMP In-charge in assistance with EMMP Team and to be submitted to SO, project management team, PUB, NParks, and other relevant authorities. The Performance Report is to include the description of the project activities being carried out at site during the month and status of implementation of CEMMP including information on environmental incidence if any. Table 13.3 provides the monthly environmental performance reporting framework.

**Table 13.3.** Monthly Environmental Performance Report framework

| Sr. No. | Item                             | Description                                                                                                                                                                                                                                                                          |
|---------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.      | Project Status                   | Update on project activities within project area                                                                                                                                                                                                                                     |
| 2.      | CEMMP Implementation Status      | Daily observations and actions taken, ECO report, ECM performance checklist, Biodiversity specialists' observations and recommendations, Physical monitoring results (Noise, Air, Surface water quality, ECM discharge) and assessment, Vector control report, Waste disposal record |
| 3.      | Environmental Awareness Training | Record of periodic biodiversity awareness training/ toolbox briefings                                                                                                                                                                                                                |
| 4.      | Environmental Incidence          | Environmental Incident report and corrective actions, public feedback & response                                                                                                                                                                                                     |
| 5.      | Authority Inspection & Findings  | Record of Authority inspection visits (i.e., NParks, NEA, PUB) and corrective actions                                                                                                                                                                                                |

### **Environmental Close-off Report**

A final environmental close-off report should be prepared after construction work is completed to confirm that no residual impacts are observed.

## **13.3 Wildlife Management Plan**

Given the location of the site within a forest of significant conservation value, it is necessary to minimise impacts to fauna species on the site. The development of a proper Wildlife Response and Rescue Plan will help to reduce impacts to fauna, while also reducing the risk of human-wildlife conflict, which may pose a human health and safety issue if not managed.

Wildlife management should commence prior to construction works. This is to ensure that animals within the site are safely moved outside the working boundaries, either by passive herding or active relocation. A Wildlife Rescue and Response Plan should then be put in place for the duration of the construction period for animals that may get trapped or injured within the construction site.

All construction personnel are to be sufficiently trained on biodiversity issues on the site and how to respond to sightings of fauna.

### **13.3.1 Biodiversity Awareness Training**

Prior to any construction activities, all construction personnel are to attend a biodiversity awareness training by the EMMP consultant. This training should impart important information on the ecological importance of the site, and the importance of minimising impacts to the natural environment. Also, they should be trained on the common fauna species they may encounter, and what to do should they have any wildlife encounters.

After the initial training, refresher training and toolbox briefings are to be conducted as specified in Section 13.2.3.

### **13.3.2 Wildlife Management during Clearance**

Wildlife management during tree clearance is required for Area 2. Directional clearance will allow for the passive shepherding of mobile fauna species, while active relocation of animals found within the project site may be required for other species. This is done in order to:

- Minimise the risk of road hazards and kills from the terrestrial fauna that are displaced from the project site onto adjacent roads.
- Minimise the risk of human-wildlife conflict from animals remaining within the project boundaries
- Encourage wildlife movement into designated forested areas located outside of the project site's boundary.

Site clearance to be conducted in the following steps:

- Permanent hoarding along proposed working boundary along Dunbar Walk to prevent entrance of fauna species into construction area and prevent animals from running onto the road;
- Manual clearing of undergrowth in phases and the phased installation, and relocation of permanent & temporary hoarding to prevent return of wildlife;
- Inspection of trees and tree holes for arboreal fauna, and relocation by qualified agencies and/or specialists.
- Felling of remaining trees.

Upon completion of the tree felling in the project area, the permanent hoarding will cover the full perimeter of the CCKWW plant extension.

### **13.3.3 Target Species**

A list of target fauna species has been developed based on the findings of the EIA. Besides species encountered during the baseline surveys, other probable species that exist on the site were also included. This list was developed with the following considerations:

- probable presence of species in the Project area prior to construction;
- risks to species from being in close proximity to construction activities;
- practicality of relocating species from the construction site;
- conservation significance of species; and
- risk of road kills, road hazards, and/or human-wildlife conflict arising from uncontrolled species displacement from the project area.

The target species in the list in Table 13.4 have been categorized into two groups depending on their habits, and thus the approach required for wildlife shepherding. Fauna species in the first category are in general highly mobile species in which a passive approach is recommended, while fauna species in the second category are less mobile and would require a more active approach to shepherding.

**Table 13.4.** List of identified target species

| Category            | Species                                                    | Active Hours          |
|---------------------|------------------------------------------------------------|-----------------------|
| Passive Shepherding | Long-tailed macaque                                        | Diurnal               |
|                     | Smooth-coated otter                                        | Diurnal               |
|                     | Leopard cat                                                | Nocturnal             |
|                     | Greater mousedeer                                          | Nocturnal             |
|                     | Common palm civet                                          | Nocturnal             |
| Capture-and-release | Sunda pangolin                                             | Nocturnal             |
|                     | King cobra                                                 | Diurnal               |
|                     | Black spitting cobra                                       | Diurnal               |
|                     | Reticulated python                                         | Nocturnal             |
|                     | Puff-faced water snake                                     | Nocturnal             |
|                     | Malayan porcupine                                          | Nocturnal             |
|                     | Clouded monitor                                            | Diurnal               |
|                     | Malayan water monitor                                      | Diurnal               |
|                     | Other snake species                                        | Diurnal and nocturnal |
|                     | Native aquatic fauna (molluscs, fish, decapod crustaceans) | Diurnal and nocturnal |

For wild pigs found at or around the site at any time during the project, NParks is to be informed as soon as possible at [nparks\\_wildlife\\_management@nparks.gov.sg](mailto:nparks_wildlife_management@nparks.gov.sg) for advice and subsequent action. An approved wild pig removal contractor must also be engaged to trap and remove the said animals, the process of which may take about 4–8 weeks.

#### **13.3.4 Methodology and Approach**

Directional tree clearance should only during scheduled daylight hours (8am to 6pm). It may include a combination of the following activities:

- installation of hoarding along project boundaries, which would help to guide target terrestrial fauna in the intended direction of movement and as a barrier to prevent wildlife displacement onto adjacent roads;
- installation of temporary hoarding to along width of cleared areas to prevent wildlife from re-entering the cleared areas;
- systematic pattern of tree clearance in the site in order to encourage wildlife to encourage wildlife to move in an intended direction of movement towards adjacent refuge habitats, and;
- careful survey to check for the presence of target fauna species and any active nests or dens.

Prior to any tree felling, the site is to be inspected by an ecologist to ensure that no target fauna and active nests or dens remain. After clearance of undergrowth, cleared areas

should be hoarded up to prevent target terrestrial fauna from returning to the site.

The wildlife inspection prior to tree felling and vegetation clearance will also check for any entrapped animals within the hoarded area to be cleared. This inspection will be valid for 7 days only, during which the trapped animals have to be secured away from the site and clearance must be conducted. If more than 7 days has passed and site clearance is not done yet, the inspection should be carried out again.

In the event that any target fauna listed in Table 13.4 are encountered during this process, the following actions which have been developed with the consideration of reducing stress to fauna while ensuring the effectiveness of the exercise shall be taken:

- i) Passive Shepherding: These are highly mobile species where passive shepherding is likely to be effective. When species in this category are encountered, personnel should allow the animal to move on its own accord. If necessary, personnel may talk loudly or make some noise by clapping their hands together to encourage the animal to move. If any individual fauna does not move on its own after sufficient time (i.e., up to one hour) has passed, the EMMP team is to make a decision on whether to call NParks' Animal Rescue Centre or ACRES for the appropriate removal of the animal.

Should the team encounter a visibly injured animal, NParks' Animal Rescue Centre or ACRES should be called immediately for the next course of action.

No attempt should be made by the EMMP team, workers, or other unqualified personnel at any point to handle animals on site. Handling animals without appropriate certification is illegal under the recent Wildlife Act of June 2020.

- ii) Capture-and-release: Species in this group are less mobile and/or venomous, and a passive shepherding approach is deemed to be ineffective and/or unsafe. A capture-and-release approach will be needed to ensure safe relocation of these fauna from the site prior to construction. In the event that these species are encountered, NParks' Animal Rescue Centre or ACRES should be called immediately for the next course of action. Capture-and release of animals encountered should be conducted by an NParks' licensed animal management company.

For trees that are subjected to removal, it is necessary to check for the presence of fauna species before each individual tree is felled.

The ecologist shall inspect the tree for the presence of fauna, including birds, bats, arboreal mammals, and arboreal herpetofauna. The ecologist should do the following:

- Check the crown of the tree for bird nests
- Check along the trunk from the bottom up for holes in which animals could be nesting
- Scan the trunk and all the branches for animals using the tree
- Scan the ground for potential nests, eggs, or burrows

Photographs of all nests, tree holes, and burrows should be taken for record purposes.

In the event that the presence of birds, bats, arboreal mammals and herpetofauna, are found on the tree, tree felling, or transplanting must be postponed until the animal has left the tree on its own accord.

Tree felling or transplanting should not occur during prime breeding season for birds in Singapore, between the months of mid-March to July. Outside these months, if active nests are detected on the tree, nests shall be left undisturbed until the young birds have fledged. Inactive nests should be removed to minimize the possibility of a new nesting attempt. Tree felling or transplanting shall occur only when no active nests are present on the tree.

Once tree felling is completed, the tree should be inspected again for any animals that were not detected earlier. Should an animal be detected after felling, NParks Animal Response Centre or ACRES should be contacted immediately, especially since the animal might be injured.

Through the undertaking of general approach, a register shall be maintained to record:

- the activities that were carried out,
- the species, numbers, GPS locations, dates, timings, and actions taken (if any) for each target fauna which was identified, and
- the description, GPS location, and actions taken (if any) for each burrow, inhabited tree hole and nest that was identified.

#### **13.3.5 Spatial Visualisation of Directional Clearance**

The figures below illustrate the overall scheduled phasing for directional clearance. These activities should take place during daylight hours only (i.e., 8 am to 6 pm) and a minimum of one rest day (i.e., Sunday) per week shall be provided to reduce disturbance to wildlife. Tree felling may commence once understory vegetation has been cleared.

This plan serves as a guide and should be updated prior to actual clearance works to include a more detailed clearance regime.



**Figure 13.2.** Directional Clearance 1

Hoarding is to be erected along the working boundaries of the project, starting with the boundary along the existing CCKWW plant and Dunbar Walk. Temporary hoarding should be erected along the yellow dotted lines. While Dunbar Walk is not used by many vehicles, speed bumps and signs informing drivers to slow down should be placed along Dunbar Walk to encourage drivers to slow down, particularly during the construction phase. Alternatively, speed limits may be implemented and enforced. Prior to any vegetation clearance, all tree holes, nests, and burrows should also be inspected for wildlife. Directional clearance should take place in the direction of the blue arrows, to encourage fauna to move to the forested areas south and west of the cleared section.



**Figure 13.3.** Directional Clearance 2

The first cleared area should be temporarily hoarded up. Permanent hoarding is to be erected along the south-eastern working boundaries of the project. Prior to any vegetation clearance in the south-eastern segment, all tree holes, nests, and burrows should also be inspected for wildlife. Directional clearance should take place in the direction of the blue arrows, to encourage fauna to move to the forested areas west of the cleared section.



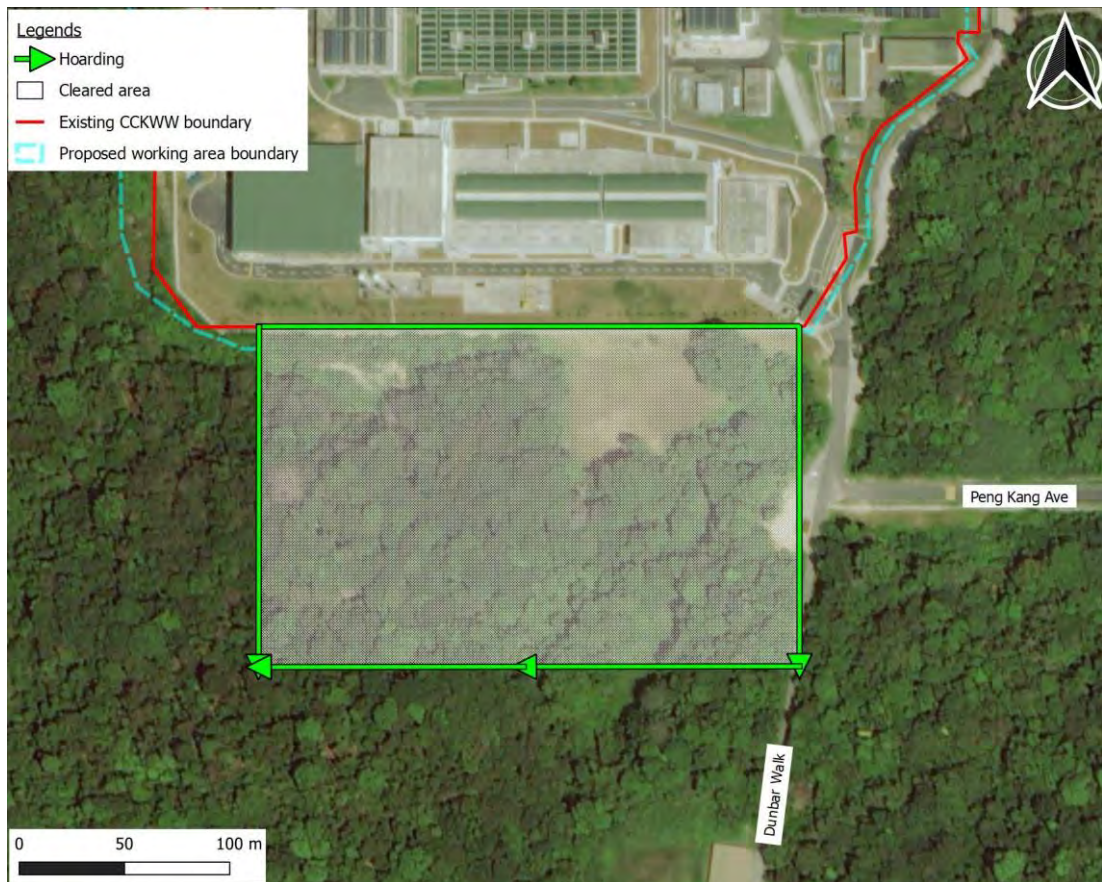
**Figure 13.4.** Directional Clearance 3

The second cleared area should be temporarily hoarded along the dotted yellow lines. Permanent hoarding is to be erected along the western working boundaries of the project. Prior to any vegetation clearance in the north-western segment, all tree holes, nests, and burrows should be inspected for wildlife. Directional clearance should take place in the direction of the blue arrows, to encourage fauna to move to the forested areas south of the cleared section.



**Figure 13.5.** Directional Clearance 4

Upon completion of felling in the north-western segment, the area should be temporarily hoarded along the dotted yellow lines. Prior to any vegetation clearance in the south-western segment, all tree holes, nests, and burrows should be inspected for wildlife. Directional clearance should take place in the direction of the blue arrows, to encourage fauna to move to the forested areas south of the cleared section.



**Figure 13.6.** Directional Clearance 5

Once directional clearance has been completed, the full working area should be hoarded up. There should be no gaps between hoarding so as to prevent fauna species from entering the site.

### 13.3.6 Wildlife Response and Rescue Protocol

Even upon the completion of wildlife shepherding works, it is highly probable that animals might be able to enter the site and get trapped, particularly burrowing or climbing animals. Whenever fauna is encountered within the working areas, all construction activities should be stopped immediately, and the Wildlife Response and Rescue Plan should be followed. Workers are to notify their supervisor, who will in turn contact the designated ecologist. The ecologist will then decide the next appropriate course of action.

All documentations of wildlife are to be captured in photographs, and a Wildlife Incident Form provided in **Appendix M** is to be filled.

**Table 13.5.** Wildlife response and rescue plan

| Particular       | Within the project site                                                                                                                                                                      |                                                                                                                                                                                                           |                                                                                                                                                                                                        |                                                                                                                                                                                                           |                                                                                                                                                                                                        | Outside project site                                              |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Timeframe        | During working hours                                                                                                                                                                         |                                                                                                                                                                                                           |                                                                                                                                                                                                        |                                                                                                                                                                                                           |                                                                                                                                                                                                        | Any time                                                          |
| Animal type      | Highly mobile animals<br>(e.g., wild pig, feral dog, smooth-coated otters, and long-tailed macaques)                                                                                         |                                                                                                                                                                                                           | Non-large animals                                                                                                                                                                                      |                                                                                                                                                                                                           |                                                                                                                                                                                                        | Any                                                               |
|                  |                                                                                                                                                                                              |                                                                                                                                                                                                           | Venomous / poisonous (e.g., king cobra, black spitting cobra)                                                                                                                                          |                                                                                                                                                                                                           | Non-venomous / -poisonous (e.g., Malayan water monitor)                                                                                                                                                |                                                                   |
| Animal condition | Alive / Moving / Resting                                                                                                                                                                     | Dead                                                                                                                                                                                                      | Alive / Moving / Resting                                                                                                                                                                               | Dead                                                                                                                                                                                                      | Any                                                                                                                                                                                                    | –                                                                 |
| Risk To human    | High                                                                                                                                                                                         | Low                                                                                                                                                                                                       | High                                                                                                                                                                                                   | Low                                                                                                                                                                                                       | Low                                                                                                                                                                                                    | –                                                                 |
| Response         | a. Stop work on work site<br>b. Report to PM<br>c. PM to report to CEMMP In-charge<br>d. CEMMP In-charge to inform EMMP Specialist (Fauna), who will contact NParks/ ACRES for next steps if | a. Barricade affected area<br>b. Report to PM<br>c. PM to report to CEMMP In-charge<br>d. CEMMP In-charge to inform EMMP Specialist (Fauna).<br>e. If required, Contractor to assist with transporting of | a. Stop work at affected area; if possible, barricade affected area<br>b. Report to PM<br>c. PM to report to CEMMP In-charge<br>d. CEMMP In-charge to inform EMMP Specialist (Fauna), who will contact | a. Barricade affected area<br>b. Report to PM<br>c. PM to report to CEMMP In-charge<br>d. CEMMP In-charge to inform EMMP Specialist (Fauna).<br>e. If required, Contractor to assist with transporting of | a. Stop work at affected area; if possible, barricade affected area<br>b. Report to PM<br>c. PM to report to CEMMP In-charge<br>d. CEMMP In-charge to inform EMMP Specialist (Fauna), who will contact | a. Notify NParks Animal Rescue Centre/ ACRES hotline if necessary |

| Particular     | Within the project site                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                           | Outside project site                                                                                                                                                                                                                                                                       |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | necessary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | the Wild Animal to Disposal Location                                                            | NParks/ ACRES for next steps if necessary                                                                                                                                                                                                                                                                                                                                | Wild Animal to Disposal Location                                                                 | NParks/ ACRES for next steps if necessary                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                            |
| <b>Remarks</b> | <ul style="list-style-type: none"> <li>• No attempts shall be made by Contractors to handle the animal</li> <li>• Contractor to take photograph of the animal if possible.</li> <li>• Contractors shall allow the animal to leave the site without harassment / handling</li> <li>• If animal is trapped, notify NParks Animal Rescue Centre or ACRES hotline</li> <li>• For wild pigs found at or around the site, NParks is to be informed as soon as possible at for advice and subsequent action. An approved wild pig removal contractor must also be engaged to trap and remove the animal.</li> </ul> | <ul style="list-style-type: none"> <li>• Contractor to take photograph of the animal</li> </ul> | <ul style="list-style-type: none"> <li>• No attempts shall be made by Contractors to handle the animal</li> <li>• Contractor to take photograph of the animal if possible.</li> <li>• Contractors shall allow the animal to leave the site without harassment / handling</li> <li>• If animal is trapped, notify NParks Animal Rescue Centre or ACRES hotline</li> </ul> | <ul style="list-style-type: none"> <li>• Contractor to take photograph of the animal.</li> </ul> | <ul style="list-style-type: none"> <li>• No attempts shall be made by contractors to handle the animal</li> <li>• Contractor to take photographs of the animal if possible.</li> <li>• Contractors shall allow the animal to leave the site without harassment / handling</li> <li>• If animal is trapped, notify NParks Animal Rescue Centre or ACRES hotline</li> </ul> | <ul style="list-style-type: none"> <li>• Contractor is encouraged to report</li> <li>• Reports could be from Public and/or Contractor's Staff</li> <li>• If required, CEMMP In-charge to contact PM for assistance of transferring Wildlife Animal Carcass to Disposal Location</li> </ul> |

### **13.4 Flora Management Plan**

It is recommended to develop a detailed flora management plan as part of CEMMP during construction stage by utilising recommended measures in this Section.

The flora management plan should focus on the CCKWW extension area and the sewerage alignment area. The plan should lay out details on engaging a nursery, how to carry out salvaging of native plants, how to maintain salvaged saplings, how to carry out tree protection works, the required components of tree assessments and monthly monitoring, etc. The responsibility of native plant salvaging process and management from this project should come under the Flora Specialist/Nursery Manager. The tree protection and assessment duties should come under an ISA Certified Arborist.

Within the areas to be cleared, trees to be retained will be confirmed at a construction phase. Protection of retained trees should be done through establishment of appropriate Tree Protection Zones (TPZ). Additionally, hoarding to be implemented around the project site to prevent unnecessary vegetation damage outside the project area. Any native flora species found within cleared land should be salvaged whenever possible.

#### **13.4.1 Selecting a Nursery, Salvaging of Native Plants, and Reinstatement**

##### **Selecting and Setting Up a Nursery**

Prior to site clearance, the Contractor should select an external nursery to house and maintain the salvaged saplings of native plants based on the stakeholder consultation (i.e., PUB, NParks). The Flora Specialist / Nursery Manager must have experience in transplanting sapling and cuttings from development sites while maintaining a high survival rate of saplings (>70%). The Flora Specialist/Nursery Manager should ensure the nursery has enough resources to handle at least 1000 number of saplings.

The nursery should be sealed with an irrigation system installed. If the nursery is located in areas with full sun exposure, the Flora Specialist / Nursery Manager must ensure there are measures in place to provide areas with indirect sunlight (e.g., shade cloth) where the light is between 60-90 Photosynthetic Active Radiation (PAR) and maintain at least 85% humidity (e.g., misting) depending on the growth requirements of saplings. The temperatures within the nursery should not exceed 30°C. The daily monitoring of the environmental conditions must be taken and documented as a record. Until the saplings are available for reinstatement purposes, the Flora Specialist / Nursery Manager must be responsible for sapling maintenance.

##### **Salvage and Transfer Native Plants to Nursery**

Before any planned or additional/ad-hoc site clearance, the Flora Specialist, who is preferably also an ISA-certified Arborist, shall identify, take photos of, and tag native tree, shrub, climber, fern, and palm saplings with local conservation status of LC, VU, EN and CR with unique tag numbers (Figure 13.7). The reason why LC species are also to be salvaged is because LC species tend to be fast growing and help to create conducive environment for growth of late-successional forest species. The tagged saplings should be in healthy and vigorous condition, free from dead wood, bruises, or

other root or branch injuries. They should also be free from insect and/or disease infestation. If any fruits of species with local conservation status of EN and CR are observed during salvaging, the Flora Specialist should collect the fruits and seeds and consult with Native Plant Centre personnel (located at Pasir Panjang Nursery) whether these fruits and seeds should be submitted to Native Plant Centre. If Native Plant Centre does not wish to collect the fertile materials, the fruits and seeds may be released outside project area such that they do not get destroyed by ground clearance.

Only saplings with a girth of 0.3 m or less are to be tagged and harvested. If the species is not known, samples should be sent to the Singapore Botanic Gardens Herbarium to confirm the species. The root ball size of saplings will be determined in consultation with the Arborist. The general rule of thumb of rootball size for transplanting is that for every 1 cm increase in tree stem diameter, there should be 8cm increase in root ball width. For example, a diameter of 5cm would require 40 cm root ball width. The height of root ball should range from 35–50 cm depending on the root condition. Tree pruning should be minimal, using appropriate sharp tools that are wiped clean before each cutting to remove dead branches and defective branches prior to being transplanted. The tree saplings should be watered well 24 hours prior to trenching if the ground is dry. The Flora Specialist / Nursery Manager should subsequently arrange personnel to salvage the tagged saplings.

There shall be sufficient support installed prior to trenching to ensure that trees do not move. Trenching should be performed manually around the outer edge of the designated root ball size. The root balls should be wrapped and secured with geotextile and polythene wraps. Tree saplings should then be lifted from the secured root ball and loaded onto the transport vehicle. Secure tree saplings firmly onto the transport vehicle and cover the sapling crown with tarp netting prior to transporting. Covering the tree crown with tarp nettings should help to reduce transpiration pull and protect the saplings from wind burn.

If a plant (e.g., climber) cannot be transplanted in full, cuttings of 1–1.2 m should be made and propagated. The cuttings shall be placed in a sealed plastic bag with water to maintain humidity. Epiphytes and ferns can be collected and tied to a fern slab or, in cases where the epiphyte or fern is attached to fallen branches or log, the epiphyte or ferns can be collected together with the branch or log. Transportation of transplanted epiphyte or ferns need to be done with proper care to reduce transportation shock to the saplings/plants. It would be acceptable to mount the epiphytic ferns to bark of a mature healthy tree (instead of being transported to nursery) if suitable trees can be found in vicinity of project site. It is important to ensure that epiphytic ferns are relocated to trees or areas (if the fern is attached to fallen branch or log) of similar microclimates with where such ferns were collected.

In the nursery, tree saplings should be unloaded from the transport vehicle, lifting the sapling from the secure root ball. The saplings should be placed into the bags or containers in the nursery site. After placing the tree sapling into the polythene bag, remove the geotextile/polythene wraps and backfill the space/gaps with clean approved soil mix. Staking support shall be provided as and when necessary to keep the plant upright. The saplings shall be placed with adequate spacing from one another in a semi-

shade shelter in temporary nursery sites. Cuttings from climbers shall be dipped into growth hormone and inserted into a container with a mixture of sand and soil. Individual saplings shall be salvaged from site, transported, and placed in the temporary nursery site within a day.

The Flora Specialist/Nursery Manager shall tally the total number of salvaged saplings and provide an inventory consisting at least the tag number, photo, and species of each salvaged sapling.



**Figure 13.7.** Example of a tagged native sapling



**Figure 13.8.** Example of plants in a nursery

### ***Growth and Maintenance of Salvaged Native Plants***

The newly harvested saplings should be maintained until the saplings have sufficiently established. The transplanted climbers and epiphytes are to be placed under shade until they have established and sprouting of young leaves occur. Once each tree has been

transplanted, daily maintenance activities should begin. The Flora Specialist/Nursery Manager should provide all material, equipment, and labours required for maintenance of salvaged saplings. Sufficient space between salvaged saplings should be provided such that they have room to grow in size. Table 13.6 outlines the maintenance regime.

To ensure good form and structure of the harvested saplings, it is recommended to carry out formative pruning on the harvested tree saplings in accordance with ANSI A-300 pruning standards (ISA). Manual weeding should be carried out regularly for the harvested saplings. All weeds should be removed before mulching. Regular mulching should be applied to all harvested saplings at the receiving site. The mulch used should be friable, odour-free high grade mature composts.

**Table 13.6.** Maintenance regimes for salvaged plants at nursery

| Regime        | Frequency                        | Material                                                                                                                                                                                                         |
|---------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Watering      | Daily, except for the rainy days | Non potable water<br>5L for trees <0.1m in girth and at least 30L for trees >0.3m in girth                                                                                                                       |
| Fertilization | Fortnightly                      | E.g., Fish kelp, humic acid, high EC molasses                                                                                                                                                                    |
| Pesticide     | As and when required             | Pesticide should be avoided whenever possible. Use organic certified solutions instead                                                                                                                           |
| Mulching      | Once a month                     | Mulch should have a minimum thickness of 20 mm, should be an approved friable, odour-free high grade mature compost or an approved mix, with C:N ratio between 12:1 and 25:1. Mulch pH must be between 5.5 to 7. |
| Weeding       | Once a week                      | Weeding should be done manually and before mulching                                                                                                                                                              |
| Pruning       | As and when required             | Clean cutters. Cutters must be wiped clean in between cuts.                                                                                                                                                      |

### **Site Reinstatement Works Using Salvaged Native Plants**

Three months after the transfer of salvaged saplings to the nursery, the Flora Specialist/Nursery Manager shall tally the remaining saplings and estimate the survival rate of each species as an effort to improve industry knowledge of survival rate of native species. With the final list of surviving saplings, the Flora Specialist/ Arborist shall inform the Contractor the remaining species and the quantity of salvaged saplings available for site reinstatement purposes, especially site reinstatement for manhole and sewerage alignment works east of the CCKWW plant and strengthen forest edges at working boundary of CCKWW extension areas.

Compared to conventional method of using cowgrass (*Axonopus compressus*) to turf back the project site after temporary site-clearance, using salvaged native-plants for reinstatement works bring three layers of ecological and biodiversity benefits.

- Firstly, the forests where sewerage alignment works take place are abandoned kampong habitat with low native plant diversity, therefore introducing the native trees salvaged from CCKWW extension area into the abandoned kampong will help enhance the forest by facilitating forest succession processes from

abandoned kampong to native-dominated forests, potentially leading to net positive environmental impact for the abandoned kampong.

- Secondly, the salvaged plants will be already of certain height, which will help rapidly close the forest canopy gap created by sewerage alignment works or ad-hoc site clearance and prevent invasion of non-native plants into more sensitive habitat of native-dominated secondary forests.
- Thirdly, by using salvaged native trees to strengthen forest edges and reinstate site after completion of construction works, it helps to preserve and disperse native flora germplasm rather than introducing foreign species which might deter conservation of native germplasms.

Overall, these measures should be able to mitigate and conserve certain degree of biodiversity loss and even enhance forest habitat by utilising salvaged native trees for reinstatement works.

After all the remaining salvaged plants had been planted on reinstatement sites, the following ISA Best Management Practices (BMP) are to be implemented, including watering for at least eight weeks to ensure survival and establishment.

The criteria of satisfactory saplings include (but are not limited to):

- Saplings are in good health and free from injury with good taper
- Saplings of tree species can be self-supporting and planted without staking (with the exception of climber and shrub)
- Saplings of excurrent species should have a defined central leader with apical bud intact
- Root crown must be at the surface of the rootball
- When shaking or handling the unsupported rootball at least 90% of the soil volume must remain intact
- Rootball is shaved or pruned if girdling root is observed within planting bag

#### **13.4.2 Assessment and Monitoring of Trees to be Protected**

Before any construction activity begins (including both planned and ad-hoc site clearance), the Arborist shall perform pre-construction tree assessment for trees at proposed development site. The Arborist should utilise Site Layout Plan and Topography Plan to determine extent of tree roots and tree crown affected by the development. Based on site observation, construction drawing and design, estimated work area boundary, the Arborist shall produce tree assessment report recording tree information such as site and tree photos, species, height, girth, crown spread, tree health, form, structure, site observations and assessment on whether the tree will be affected by proposed development footprint, and if so, what are the recommended mitigation measure to construction impact on trees. This tree assessment report will then serve as record of pre-development tree condition, and the Arborist will have to refer to this report as benchmark when performing monthly monitoring for trees.

Tree pruning such as crown cleaning, formative pruning, structural pruning, or crown thinning should be the first line of mitigation measures. Tree Support Systems such as Tree Guying and Root Anchoring must be considered if the tree is deemed to be

susceptible to wind throw.

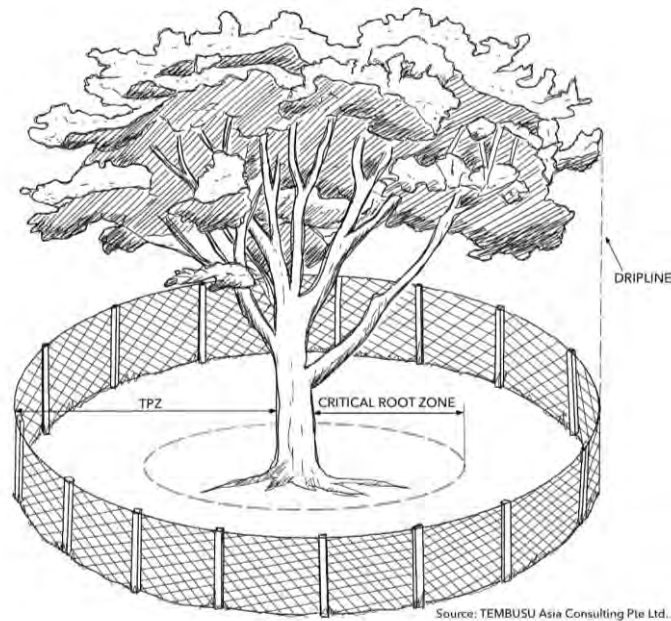
### 13.4.3 Tree Protection Zones

While the design of CCKWW extension area and sewerage alignment doesn't allow retention of large trees within most of the construction footprint, it is possible to retain large trees that fall either (1) inside a footprint that can afford flexibility in location or (2) outside of construction footprint but within areas with other construction needs. For example, manhole or equipment storage site can be shifted if a native mature tree was found within or at the border of such site. A TPZ is an area demarcated to protect individual trees, so as to minimize the impacts of construction activities (including root damage, mechanical trunk damage, damage due to soil compaction, etc) on the tree. NParks (2018) has included some guidelines on Tree Protection Zones in their *Guidelines on Greenery Provision and Tree Conservation for Developments*.

TPZ size varies depending on tree size (Table 13.7). In general, the remaining space should be sufficient for implementation of design and required infrastructure. Should the Contractor require ad-hoc space in the midst of construction, the area should be surveyed for native trees to be retained on site and native saplings to be salvaged to the nursery before any clearance. If any materials spill into TPZ, the spillage should be cleaned up immediately and the Arborist must be informed. The Contractor shall notify Arborist immediately if trees to be retained are damaged. If the tree suffers from substantial damage as determined by the Arborist, the Contractor should replace the tree of the same species and size.

**Table 13.7.** TPZ size required for different girth range

| Girth                                                 | Recommended TPZ (radius)                                    |
|-------------------------------------------------------|-------------------------------------------------------------|
| <0.5m                                                 | 1 m                                                         |
| >0.5m but less than 1m                                | 2 m                                                         |
| >1m but less than 1.5m                                | 3 m                                                         |
| >1.5m but less than 2m                                | 4 m                                                         |
| >2m                                                   | 5 m                                                         |
| Fig trees and Trees with Critically Endangered status | Prescribed individually by Arborist on a case-by-case basis |



**Figure 13.9.** Tree Protection Zone Diagram

The following specifies guidelines for construction activities within and outside the TPZ, extracted from NParks (2018).

#### Inside TPZ

- There must be no excavation, raising or lowering of soil level, compaction or any form of construction activities including temporary works within the hoarded area.
- Dumping of debris, excavated materials and/or storage of construction materials and equipment are not allowed within the TPZ.
- The demolition of drains, structures within the TPZ should be carried out manually and backfilled with Approved Soil Mixture (ASM) immediately.
- Trees are to be watered regularly if rainfall is inadequate.
- Trees are to be fertilised if soil tests or deficiency symptoms indicate they are nutrient stressed.

#### Outside TPZ

- If major roots are encountered during excavation, the applicant may like to seek advice from a Certified Arborist, as cutting of major roots may affect the stability of the tree. Where possible, alternative proposals should be explored to avoid the need to cut the roots.
- In cases where the trees are managed by NParks (e.g., trees within the park connector planting verge), or are required by NParks to be conserved (e.g., trees with girth >1.0m within TCA or vacant land), approval from NParks must be obtained before the major root can be cut. If approval is granted by NParks to cut the roots, this must be done with a clean cut using a chainsaw.
- All building debris and chemical wastes should not be burned or buried within green verges on the site.

#### **13.4.4 Tree Felling Within Forested Areas**

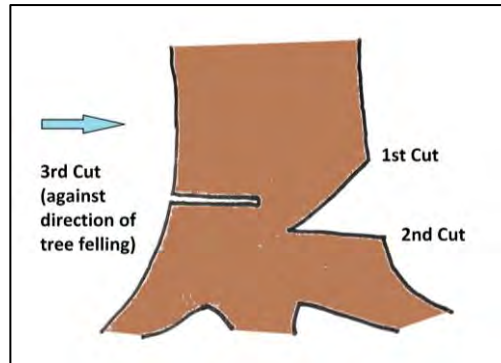
Before felling trees, the Arborist must ensure native saplings have already been tagged, identified, and transplanted to the nursery. The Arborist must also survey, identify, and confirm the trees to be protected in the surrounding area and establish a tree felling protection distance based on site condition and tree crown spread.

The trees to be felled shall be inspected for any fauna as per guidelines provided in wildlife management plan (Section 13.3). Such trees will be marked with red & white tape and no tree felling operation shall be carried out within 5 m from the said tree until further instruction.

Before commencing tree felling works, the supervisor, banksman, and foreman shall scout the area a final time to ensure that the tree felling protection zone is clear of human activities, while fauna specialist will inspect the tree felling protection zone if the site is free of wildlife activities. Once the areas are cleared, the banksman should signal the excavator operator to commence work for trees felling. The excavator operator should first clear off the shrubs and small trees (<5 m height) to create a clear line of sight for the whole area and to keep away blind spot areas which are blocked by small trees or tall shrubs. The excavator operator shall operate cautiously and fell all small trees and shrubs in a controlled manner, aware of the location of the other trees to be protected.

When opening is completed, the excavator operator should clear off small trees and shrubs along the path into the site so as to demark the area that they are supposed to work within. When the paths are cleared, the operator should then move inward to fell trees within the area. If the operator faces a tree with height between 5 and 7 m, they should clear off shrubs in the surrounding area so that the foreman can move closer to the tree. The recommended tree cutting method should be the notch cut (Figure 13.10). The foreman shall determine the direction of falling and ensure the tree does not land on any property, cause injury, or damage nearby Trees to be Protected. To avoid trees leaning to an unintended direction when performing a third cut, the excavator should assist to prevent fall back and guide the tree to fall into the intended direction. Once the tree has been felled, the tree cutter shall cut the tree trunk into shorter lengths for easier loading during clearing of the debris from site.

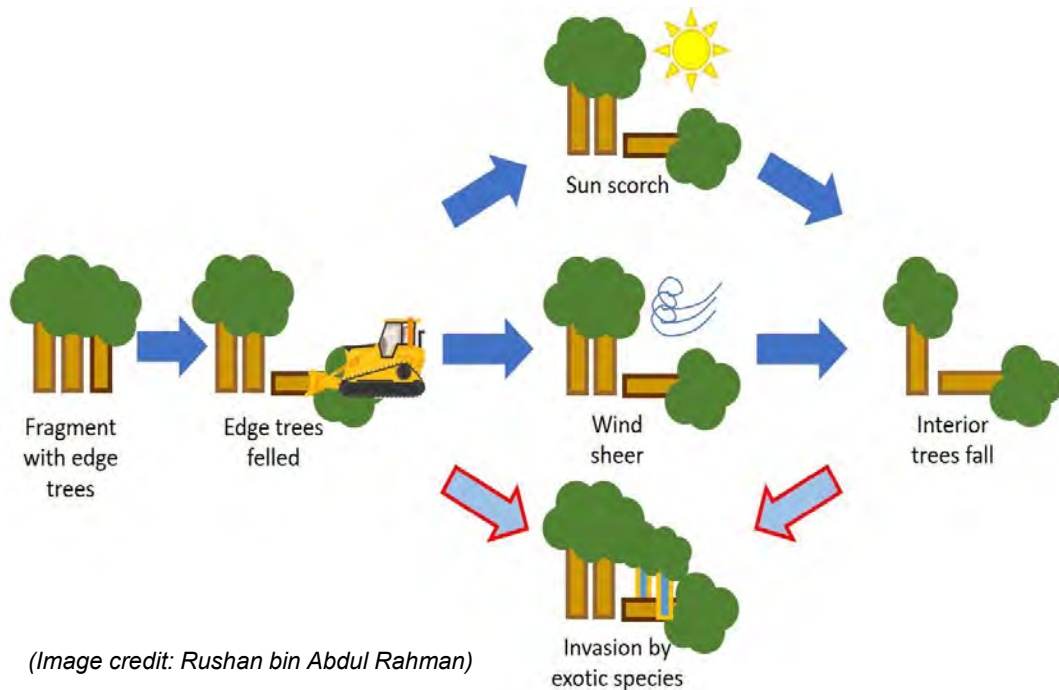
If the tree has a height greater than 7 m, the tree height has to be reduced by using a lorry crane to perform crown reduction first. Before the lorry crane can enter the site, the Contractor has to prepare proper access for the lorry crane to enter the site and access the tree location. The Contractor must ensure that the access ground is firm enough to allow the lorry crane to deploy its outrigger. When the access is ready, the Contractor will then mobilize the lorry crane to enter the site to reduce tree height to 7 m in order to adopt the notch cut method for trees less than 7 m in height.



**Figure 13.10.** Illustration of a notch cut

#### **13.4.5 Flora Protection Outside Working Boundaries**

Forest clearing tend to create gaps and forest fragments where non-native species can enter sensitive habitat such as native-dominated old secondary forest, and pre-dispose trees to various abiotic stress. One way to prevent such occurrence is to set up a screening buffer outside the building footprint. Buffers are the most efficient means of protecting forest fragments as they absorb and reduce impacts from surrounding development. As recommended, the Contractor should use salvaged plants to replant forest edges outside working boundary whenever a gap occurs due to a tree fall or ad-hoc site clearance to protect flora diversity and forest habitats outside working boundaries.



(Image credit: Rushan bin Abdul Rahman)

**Figure 13.11.** Illustration of edge effect

Figure 13.11 indicates the different edge effects for flora. Removal of edges can result in interior trees becoming damaged from sun scorch, weakened by wind shear and invaded by competing invasive plants. Damaged trees from sun scorch and trees weakened by wind shear can fall, exposing more interior habitat to invasion and damage.

### **13.5 Lighting Management Plan (LMP)**

A Lighting Management Plan (LMP) shall be developed as part of the Construction EMMP as recommended in section 4.6.1 and 9.3.1. This section provides items to be covered in the LMP.

#### **13.5.1 Purpose**

The purpose of LMP shall be:

- Provide guidance for construction lighting to reduce impacts of artificial lighting at night (ALAN) on the natural environment;
- Provide details on light utilisation on site, including the approximate location and specification of each lighting unit; and
- Recommend light mitigation measures that are to be implemented on the site.

#### **13.5.2 General Lighting Management Principles**

Based on the international practices, following four lighting management principles are to be taken into consideration while formulating the LMP during construction phase.

##### **1. Additional lighting is to be installed only when it is needed**

Given the proximity of the project site to forested areas, the ALAN are to be installed only when it is determined that potential safety hazards from the project activities can be mitigated with the use of ALAN.

##### **2. Leverage adaptive controls to minimise lighting usage for specific tasks**

In light of the schedule of project activities, ALAN usage must be planned in advance to minimise the lighting duration (i.e., minimum number of nights/hours required) and intensity required to complete the works in an area. Adaptive controls (e.g., automatic switches, motion sensors, timers) are to be deployed where appropriate and feasible to limit the ALAN's duration, intensity, and/or extent.

##### **3. Additional lighting is to be designed to reduce beam spread**

ALAN are to be directed inward the project site and downward so that only the objects or areas intended are lighted. The lightings are to be shielded and installed at the appropriate height to avoid any spills into the surroundings.

##### **4. Use appropriate lightings and equipment in the project site**

ALAN with long wavelength (defined as above 580 nm, e.g., amber/orange light) must be used wherever possible to reduce disturbance to sensitive animals. Usage of lightings with short wavelength (defined as below 500 nm, e.g., blue and ultra-violet light) has to be reduced with the sources filtered as much as feasible when used. Moreover, the construction equipment deployed in the project site must have non-reflective, dark-coloured surface as far as practicable.

#### **13.5.3 Lighting Control Measures**

The LMP shall include the following lighting control measures to be implanted at the project site.

- Spatial layout of lighting to be utilised in project site
- Designs of lighting utilised – including height and shielding design
- Specifications of lighting utilised – lighting colour, spectrum, and brightness level
- Additional measures in place in the event that white or non-shielded lighting is required
- Implementation monitoring of mitigation measures

## **13.6 Biodiversity Monitoring Requirements**

Specific monitoring requirements for biodiversity are detailed in this section. The other monitoring requirements for physical parameters are presented in Table 13.9.

### **13.6.1 Fauna Monitoring**

There should be regular checks to ensure that the implementation of mitigation measures for fauna protection are in place and effective in the mitigation of impacts. Sensitive biodiversity should also be monitored to ensure that their presence has not been adversely affected by the works. The following section outlines visual site inspection and aquatic monitoring recommendations.

#### Visual Site Inspection

Monthly inspections of hoarding surrounding the worksite:

- There should be no clearance of vegetation outside the hoarding boundary. Additionally, the hoarding should be implemented with no gaps between, to ensure that animals are not able to enter the site. Specifications for hoarding should also follow directives from NParks.

Monthly inspection of habitats:

- Besides forested areas immediately surrounding the hoarding, visual inspections should also be made of sensitive habitats, especially Stream 3. There should be no visible impacts to the stream, including, but not limited to, loss of vegetation, siltation, visible increased sedimentation or erosion, presence of oil, etc.

#### Aquatic Fauna Surveys

Due to the importance of Stream 3 as habitat for rare aquatic fauna, monthly aquatic fauna surveys are recommended. The surveys should target fish, decapod crustaceans, and molluscs. The points and methodology should closely follow the EIA, so that data can be comparable to baseline. Additionally, monthly odonates surveys along Stream 3 should also be carried out. The data collected should be compared on a monthly basis, which can provide an indication on any major changes in aquatic fauna diversity.

### **13.6.2 Flora Monitoring**

Due to the presence of many plant species of conservation value within and around the project site, there should be regular checks to ensure that the implementation of mitigation measures for flora and arboriculture protection are in place and effective in the mitigation of impacts. The following section outlines monitoring recommendation for flora protection.

For the monitoring of flora, the flora specialist and arborist should have access to the documents listed in Section 13.4.2.

#### Monitoring of Retained Trees

During the construction stage, the Arborist shall also conduct monthly inspections of trees to be protected. The monthly inspection report will cover 2 main aspects: current tree health and TPZ condition. Table below presents criteria to be reflected in the monthly inspection report. Photos of each tree are to be included in monthly inspection report. In addition, past or current maintenance activities will be reflected in the monthly inspection report (such as Last Tree Pruning Date) during the development period.

**Table 13.8.** Parameters to take note during monthly tree inspections

| Tree Health                                             | TPZ Condition                                        |
|---------------------------------------------------------|------------------------------------------------------|
| Foliage colour (Normal, Chlorotic, Necrotic)            | TPZ barriers installed/good condition                |
| Foliage density (Normal, Sparse)                        | Evidence of illegal encroachment                     |
| Leaf size (Normal, Small)                               | Evidence of damage to tree                           |
| Epicormic shoots (Yes, No)                              | Evidence of toxic splash                             |
| Twig dieback (Yes, No)                                  | Evidence of illegal compaction                       |
| Wound wood roll development (Good, Average, Poor, None) | Evidence of materials storage                        |
| Vigour (Good, Average, Poor)                            | Evidence of machinery, equipment and vehicle storage |
| Pests/Diseases (Yes, No)                                |                                                      |

#### Monitoring of Nursery Plants

Regular monitoring of salvaged plants in the nursery should be conducted on a monthly basis. Monitoring for tree health should follow the parameters indicated in Table 13.8. Additionally, it is necessary to ensure that the regime presented in Table 13.6 have been effectively implemented. Every three months, a tally of the survivability of the salvaged plants should be conducted.

### **13.6.3 Water Quality Monitoring**

Given the importance of good water quality in the maintenance of healthy aquatic habitats, in-situ and ex-situ surveys of water quality in Stream 3 should be conducted. The points, methods used, and parameters measured, should closely follow what was studied in this EIA, and monitoring should take place on a fortnightly basis during the construction phase. Water quality parameters should be compared to the baseline to ensure that there are no major changes in water quality.

## **13.7 Environmental Monitoring Plan**

Table 13.9 provides the overview of recommended environmental monitoring plan and EMMP measures for this project which is to be incorporated into the CEMMP by the Contractor. The environmental monitoring locations are to be finalised by EMMP team during CEMMP formulation in consultation with relevant stakeholders.

**Table 13.9.** Recommended Environmental Monitoring Plan for construction phase

| Monitoring Category                                                                                                                                                                                                                                                                                         | Impact                                          | Monitoring Parameters                                                                                                                                                                                                                                                                                     | Monitoring Method                                                                                                               | Location                                          | Standards / Criteria                                                                                                                                                                                                                  | Time / Duration / Frequency                                                                                                                                                 | Reporting                                                                                  | Implementation                                                                                                      | Supervision                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Biodiversity Monitoring</b> <ul style="list-style-type: none"> <li>Avoiding clearance of vegetation outside working boundaries</li> <li>Minimization of disturbance to sensitive species</li> <li>Minimization of disturbance to sensitive habitats</li> <li>Avoiding human wildlife conflict</li> </ul> | <u>On-site Visual and Compliance Monitoring</u> |                                                                                                                                                                                                                                                                                                           |                                                                                                                                 |                                                   |                                                                                                                                                                                                                                       |                                                                                                                                                                             |                                                                                            |                                                                                                                     |                                  |
|                                                                                                                                                                                                                                                                                                             | Habitat Loss and Degradation                    | <ul style="list-style-type: none"> <li>Hoarding to be erected prior to vegetation clearance to demarcate working boundaries</li> <li>Successful reinstatement of habitats upon work completion where required</li> <li>Retain vegetated buffer between construction areas and waterbodies</li> </ul>      | <ul style="list-style-type: none"> <li>Visual monitoring</li> <li>Compliance check</li> <li>Water quality monitoring</li> </ul> | Entire project site (near / within forested area) | <ul style="list-style-type: none"> <li>Proper installation of temporary hoarding/ barriers</li> <li>Absence of vegetation clearance outside working boundaries</li> </ul>                                                             | <ul style="list-style-type: none"> <li>Weekly during construction phase</li> </ul>                                                                                          | <ul style="list-style-type: none"> <li>Monthly Environmental Performance Report</li> </ul> | <ul style="list-style-type: none"> <li>Contractor / CEMMP In-charge</li> <li>Ecologist</li> <li>Arborist</li> </ul> | PUB/ Superintending Officer (SO) |
|                                                                                                                                                                                                                                                                                                             | Species Mortality                               | <ul style="list-style-type: none"> <li>Properly designated Tree Protection Zones (TPZ) prior to construction</li> <li>Visual inspection of trees and holes for nesting birds prior to felling</li> <li>Directional clearance of vegetation</li> <li>Translocation of aquatic fauna in Stream 2</li> </ul> | <ul style="list-style-type: none"> <li>Visual monitoring</li> <li>Compliance check</li> </ul>                                   | Entire project site (near / within forested area) | <ul style="list-style-type: none"> <li>Proper TPZ installation</li> <li>Retention of tree health</li> <li>Absence of mechanical damage on trees</li> <li>Absence of nesting birds</li> <li>Absence of large mammal species</li> </ul> | <ul style="list-style-type: none"> <li>Prior to vegetation clearance (for wildlife translocation and fauna inspection)</li> <li>Weekly during construction phase</li> </ul> | <ul style="list-style-type: none"> <li>Monthly Environmental Performance Report</li> </ul> | <ul style="list-style-type: none"> <li>Contractor / CEMMP In-charge</li> <li>Ecologist</li> <li>Arborist</li> </ul> | PUB/ SO                          |
|                                                                                                                                                                                                                                                                                                             | Human-wildlife Conflict                         | <ul style="list-style-type: none"> <li>Briefing to on-site workers on dos and don'ts, as well as notes on safety.</li> <li>Areas are to be demarcated for food consumption and storage.</li> </ul>                                                                                                        | <ul style="list-style-type: none"> <li>Visual monitoring</li> <li>Compliance check</li> </ul>                                   | Entire project site (near forested area)          | <ul style="list-style-type: none"> <li>No injuries due to wild animals</li> </ul>                                                                                                                                                     | <ul style="list-style-type: none"> <li>Weekly during construction phase</li> </ul>                                                                                          | <ul style="list-style-type: none"> <li>Monthly Environmental Performance Report</li> </ul> | <ul style="list-style-type: none"> <li>Contractor / CEMMP In-charge</li> <li>Ecologist</li> </ul>                   | PUB/ SO                          |
|                                                                                                                                                                                                                                                                                                             | <u>On-site Biodiversity Monitoring</u>          |                                                                                                                                                                                                                                                                                                           |                                                                                                                                 |                                                   |                                                                                                                                                                                                                                       |                                                                                                                                                                             |                                                                                            |                                                                                                                     |                                  |
|                                                                                                                                                                                                                                                                                                             | Disturbance to aquatic fauna                    | <ul style="list-style-type: none"> <li>To follow baseline surveys at Stream 3</li> </ul>                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>Aquatic fauna surveys</li> </ul>                                                         | Stream 3                                          | <ul style="list-style-type: none"> <li>To follow baseline surveys at Stream 3</li> </ul>                                                                                                                                              | <ul style="list-style-type: none"> <li>Monthly during construction phase</li> <li>Monthly for 3</li> </ul>                                                                  | <ul style="list-style-type: none"> <li>Monthly Environmental Performance Report</li> </ul> | <ul style="list-style-type: none"> <li>Contractor / CEMMP In-charge</li> <li>Ecologist</li> </ul>                   | PUB/ SO                          |

| Monitoring Category                                                                                                                                                                                                       | Impact                                                          | Monitoring Parameters                                                                                                                                                                                                                                               | Monitoring Method                                                                                                    | Location                                                                                                                         | Standards / Criteria                                                                                       | Time / Duration / Frequency                                                                                                             | Reporting                                                                                               | Implementation                                                                                            | Supervision |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------|
|                                                                                                                                                                                                                           |                                                                 |                                                                                                                                                                                                                                                                     |                                                                                                                      |                                                                                                                                  |                                                                                                            | consecutive months after construction is complete                                                                                       |                                                                                                         |                                                                                                           |             |
| <b>Noise Monitoring</b><br><br><ul style="list-style-type: none"> <li>Minimization of biodiversity disturbance due to construction noise</li> <li>Minimization of nuisances to human due to construction noise</li> </ul> | <u>On-site Visual and Compliance Monitoring</u>                 |                                                                                                                                                                                                                                                                     |                                                                                                                      |                                                                                                                                  |                                                                                                            |                                                                                                                                         |                                                                                                         |                                                                                                           |             |
|                                                                                                                                                                                                                           | Disturbance to biodiversity and human due to construction noise | <ul style="list-style-type: none"> <li>Noise barriers around construction work areas</li> <li>Utilization of quieter equipment and vehicles with low noise levels</li> <li>PPE use by construction personnel at all times while on the construction site</li> </ul> | <ul style="list-style-type: none"> <li>Visual monitoring</li> <li>Compliance check</li> </ul>                        | Entire project site                                                                                                              | Environmental Protection and Management (Control of Noise at Construction Sites) 2008                      | <ul style="list-style-type: none"> <li>Weekly during entire construction phase</li> </ul>                                               | <ul style="list-style-type: none"> <li>Monthly Environmental Performance Report</li> </ul>              | <ul style="list-style-type: none"> <li>Contractor / CEMMP In-charge</li> <li>ECO</li> </ul>               | PUB/ SO     |
|                                                                                                                                                                                                                           | <u>On-site Noise Monitoring</u>                                 |                                                                                                                                                                                                                                                                     |                                                                                                                      |                                                                                                                                  |                                                                                                            |                                                                                                                                         |                                                                                                         |                                                                                                           |             |
|                                                                                                                                                                                                                           | Noise generated from construction work                          | <ul style="list-style-type: none"> <li>Leq 12 hrs</li> <li>Leq 1 hr</li> <li>Leq 5 mins</li> </ul>                                                                                                                                                                  | Sound level meter                                                                                                    | <ul style="list-style-type: none"> <li>N1 (Near NTU)</li> <li>N2 (Near Dunbar walk)</li> <li>N3 (West of CCKWW plant)</li> </ul> | Environmental Protection and Management (Control of Noise at Construction Sites) 2008                      | <ul style="list-style-type: none"> <li>Continuous (24x7) boundary noise monitoring</li> <li>During entire construction phase</li> </ul> | <ul style="list-style-type: none"> <li>Monthly Environmental Performance Report</li> </ul>              | <ul style="list-style-type: none"> <li>Contractor / CEMMP In-charge</li> <li>ECO</li> </ul>               | PUB/ SO     |
| <b>Water Quality Monitoring</b><br><br><ul style="list-style-type: none"> <li>Minimization of impact to waterbodies due to contaminated site run-off</li> <li>Minimization of impact to</li> </ul>                        | <u>On-site Visual and Compliance Monitoring</u>                 |                                                                                                                                                                                                                                                                     |                                                                                                                      |                                                                                                                                  |                                                                                                            |                                                                                                                                         |                                                                                                         |                                                                                                           |             |
|                                                                                                                                                                                                                           | ECM non-compliance                                              | <ul style="list-style-type: none"> <li>Verify implementation of ECM Plan</li> <li>Perimeter cut-off drains, perimeter silt fence, silt traps, sedimentation basin and silt treatment system</li> </ul>                                                              | <ul style="list-style-type: none"> <li>Visual monitoring</li> <li>Compliance check</li> <li>ECM checklist</li> </ul> | Construction area with earthworks                                                                                                | ECM Plan designed by a Qualified Erosion Control Professional (QECP)                                       | <ul style="list-style-type: none"> <li>Daily compliance monitoring</li> <li>During entire construction phase</li> </ul>                 | <ul style="list-style-type: none"> <li>Monthly Environmental Performance Report</li> </ul>              | <ul style="list-style-type: none"> <li>Contractor / CEMMP In-charge</li> <li>ECO</li> <li>QECP</li> </ul> | PUB/ SO     |
|                                                                                                                                                                                                                           | <u>On-site Water Quality Monitoring</u>                         |                                                                                                                                                                                                                                                                     |                                                                                                                      |                                                                                                                                  |                                                                                                            |                                                                                                                                         |                                                                                                         |                                                                                                           |             |
|                                                                                                                                                                                                                           | ECM discharge (sediment runoff)                                 | Total Suspended Solids (TSS)                                                                                                                                                                                                                                        | Implementation of TSS monitor and CCTV                                                                               | Final ECM discharge points                                                                                                       | <ul style="list-style-type: none"> <li>Less than 50 mg/L for TSS</li> <li>Sewerage and Drainage</li> </ul> | <ul style="list-style-type: none"> <li>Real-time continuous during entire construction</li> </ul>                                       | <ul style="list-style-type: none"> <li>TSS report</li> <li>Monthly Environmental Performance</li> </ul> | <ul style="list-style-type: none"> <li>Contractor / CEMMP In-charge</li> <li>ECO</li> </ul>               | PUB/ SO     |

| Monitoring Category                                                                                                                                           | Impact                                                            | Monitoring Parameters                                                                                                                                                                                                                                                                         | Monitoring Method                                                                             | Location                                                                       | Standards / Criteria                                                                                                                    | Time / Duration / Frequency                                                                              | Reporting                                  | Implementation                                                                              | Supervision |
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| terrestrial habitats due to erosion of topsoil                                                                                                                |                                                                   |                                                                                                                                                                                                                                                                                               | including a Silty Imagery Detection System (SIDS)                                             |                                                                                | (Surface Water Drainage) Regulation 2007                                                                                                | phase                                                                                                    | Report                                     |                                                                                             |             |
|                                                                                                                                                               | Contamination of water resources through trade effluent discharge | <ul style="list-style-type: none"> <li>All parameters identified in EPM (Trade Effluent) Regulations for Controlled Watercourse</li> <li>Additional parameters: Aluminium, Conductivity, Turbidity, Total nitrogen, Total phosphorous, Total organic carbon, Ammonia, Enterococcus</li> </ul> | • Ex-situ monitoring                                                                          | • At every discharge outlet                                                    | • EPM (Trade Effluent) Regulations for Controlled Watercourse                                                                           | <ul style="list-style-type: none"> <li>Once a month</li> <li>During entire construction phase</li> </ul> | • Monthly Environmental Performance Report | <ul style="list-style-type: none"> <li>Contractor / CEMMP In-charge</li> <li>ECO</li> </ul> | PUB/ SO     |
|                                                                                                                                                               | Degradation of stream habitat (Construction stage monitoring)     | • All parameters identified in EPM (Trade Effluent) Regulations for Controlled Watercourse                                                                                                                                                                                                    | • In-situ and ex-situ monitoring                                                              | • Same locations as per baseline surveys at Stream 3 during construction phase | <ul style="list-style-type: none"> <li>EPM (Trade Effluent) Regulations for Controlled Watercourse</li> <li>Baseline results</li> </ul> | • Fortnightly during construction phase                                                                  | • Monthly Environmental Performance Report | <ul style="list-style-type: none"> <li>Contractor / CEMMP In-charge</li> <li>ECO</li> </ul> | PUB/ SO     |
|                                                                                                                                                               | Degradation of stream habitat (post-construction monitoring)      | • All parameters identified in EPM (Trade Effluent) Regulations for Controlled Watercourse                                                                                                                                                                                                    | • In-situ and ex-situ monitoring                                                              | • All streams                                                                  | <ul style="list-style-type: none"> <li>EPM (Trade Effluent) Regulations</li> <li>Baseline results</li> </ul>                            | • Monthly for 3 consecutive months after construction is complete                                        | • Monthly Environmental Performance Report | <ul style="list-style-type: none"> <li>Contractor / CEMMP In-charge</li> <li>ECO</li> </ul> | PUB/ SO     |
| <b>Air Quality Monitoring</b> <ul style="list-style-type: none"> <li>Minimization of human health &amp; biodiversity impacts due to dust pollution</li> </ul> | On-site Visual and Compliance Monitoring                          |                                                                                                                                                                                                                                                                                               |                                                                                               |                                                                                |                                                                                                                                         |                                                                                                          |                                            |                                                                                             |             |
|                                                                                                                                                               | Fugitive dust emissions                                           | <ul style="list-style-type: none"> <li>Verify implementation of dust suppression plan</li> <li>Watering to reduce dust emissions from exposed areas</li> <li>Washing bay</li> <li>Implementation of vehicular</li> </ul>                                                                      | <ul style="list-style-type: none"> <li>Visual monitoring</li> <li>Compliance check</li> </ul> | All construction areas                                                         | • Approved dust suppression plan                                                                                                        | • During entire construction phase                                                                       | • Monthly Environmental Performance Report | <ul style="list-style-type: none"> <li>Contractor / CEMMP In-charge</li> </ul>              | PUB/ SO     |

| Monitoring Category                                                                                                                              | Impact                                                    | Monitoring Parameters                                                                                                                                                                                                           | Monitoring Method                                                                             | Location                                                                                       | Standards / Criteria                                                                                                                                          | Time / Duration / Frequency                                                                                | Reporting                                                                                  | Implementation                                                                                   | Supervision |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------|
| <ul style="list-style-type: none"> <li>Minimization of human health impacts due to exhaust emissions</li> </ul>                                  |                                                           | <ul style="list-style-type: none"> <li>speed limit</li> <li>Covered stockpiles</li> <li>Use of PPE (face mask) by construction personnel</li> </ul>                                                                             |                                                                                               |                                                                                                |                                                                                                                                                               |                                                                                                            |                                                                                            |                                                                                                  |             |
|                                                                                                                                                  | Exhaust emission from construction machineries operations | <ul style="list-style-type: none"> <li>Maintenance frequency of vehicles and machineries</li> </ul>                                                                                                                             | <ul style="list-style-type: none"> <li>Visual monitoring</li> <li>Compliance check</li> </ul> | All construction areas                                                                         | <ul style="list-style-type: none"> <li>No visible exhaust plume, dark smoke etc.</li> </ul>                                                                   | <ul style="list-style-type: none"> <li>During entire construction phase</li> </ul>                         | <ul style="list-style-type: none"> <li>Monthly Environmental Performance Report</li> </ul> | <ul style="list-style-type: none"> <li>Contractor / CEMMP In-charge</li> </ul>                   | PUB/ SO     |
|                                                                                                                                                  | <u>On-site Air Quality Monitoring</u>                     |                                                                                                                                                                                                                                 |                                                                                               |                                                                                                |                                                                                                                                                               |                                                                                                            |                                                                                            |                                                                                                  |             |
|                                                                                                                                                  | Particulate matter emission from construction activities  | <ul style="list-style-type: none"> <li>PM<sub>2.5</sub> and PM<sub>10</sub></li> </ul>                                                                                                                                          | <ul style="list-style-type: none"> <li>Dust sampler</li> </ul>                                | <ul style="list-style-type: none"> <li>A1 (Near NTU)</li> <li>A2 (Near Dunbar walk)</li> </ul> | <ul style="list-style-type: none"> <li>Singapore Ambient Air Quality Targets</li> </ul>                                                                       | <ul style="list-style-type: none"> <li>24hr (1 day) continuous monitoring</li> <li>Once a Month</li> </ul> | <ul style="list-style-type: none"> <li>Monthly Environmental Performance Report</li> </ul> | <ul style="list-style-type: none"> <li>Contractor / CEMMP In-charge</li> </ul>                   | PUB/ SO     |
| <b>Light Impact Monitoring</b> <ul style="list-style-type: none"> <li>Minimization of light pollution impacts on ecological processes</li> </ul> | <u>On-site Visual and Compliance Monitoring</u>           |                                                                                                                                                                                                                                 |                                                                                               |                                                                                                |                                                                                                                                                               |                                                                                                            |                                                                                            |                                                                                                  |             |
|                                                                                                                                                  | Light pollution affecting sensitive species               | <ul style="list-style-type: none"> <li>Appropriate positioning of lights</li> <li>Scheduling of activity during nightworks</li> <li>Turn off all unnecessary lights outside working hours</li> </ul>                            | <ul style="list-style-type: none"> <li>Visual monitoring</li> <li>Compliance check</li> </ul> | Entire project site (near forested area)                                                       | <ul style="list-style-type: none"> <li>Absence of nightworks</li> <li>All artificial lights to be downward facing, turned away from forested areas</li> </ul> | <ul style="list-style-type: none"> <li>During entire construction phase</li> </ul>                         | <ul style="list-style-type: none"> <li>Monthly Environmental Performance Report</li> </ul> | <ul style="list-style-type: none"> <li>Contractor/ CEMMP In-charge</li> <li>Ecologist</li> </ul> | PUB/ SO     |
| <b>Vector Monitoring</b> <ul style="list-style-type: none"> <li>Minimizing the impacts due to increase in vector related diseases</li> </ul>     | <u>On-site Visual and Compliance Monitoring</u>           |                                                                                                                                                                                                                                 |                                                                                               |                                                                                                |                                                                                                                                                               |                                                                                                            |                                                                                            |                                                                                                  |             |
|                                                                                                                                                  | Increase in the Incidence of vectors & related diseases   | <ul style="list-style-type: none"> <li>Verify implementation of vector control management plan</li> <li>Engagement of NEA registered vector control operator</li> <li>Appointment of an in-house vector control team</li> </ul> | <ul style="list-style-type: none"> <li>Visual monitoring</li> <li>Compliance check</li> </ul> | Entire project site                                                                            | <ul style="list-style-type: none"> <li>Vector control management plan</li> <li>Control of Vectors and Pesticides Act, 2002</li> </ul>                         | <ul style="list-style-type: none"> <li>During entire construction phase</li> </ul>                         | <ul style="list-style-type: none"> <li>Monthly Environmental Performance Report</li> </ul> | <ul style="list-style-type: none"> <li>Contractor/ CEMMP In-charge</li> <li>ECO</li> </ul>       | PUB/ SO     |

| Monitoring Category                                                                                                                                                 | Impact                                                         | Monitoring Parameters                                                                                                                                                                                | Monitoring Method                                                                             | Location                                                                                                                      | Standards / Criteria                                                                                                      | Time / Duration / Frequency                                                        | Reporting                                                                                  | Implementation                                                                                           | Supervision |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------|
| <b>Waste Management Monitoring</b> <ul style="list-style-type: none"> <li>Minimizing the impacts due to improper disposal of hazardous and general waste</li> </ul> | <u>On-site Visual and Compliance Monitoring</u>                |                                                                                                                                                                                                      |                                                                                               |                                                                                                                               |                                                                                                                           |                                                                                    |                                                                                            |                                                                                                          |             |
|                                                                                                                                                                     | Improper disposal of hazardous waste lead to land pollution    | <ul style="list-style-type: none"> <li>Engagement of NEA licensed waste collector for hazardous waste</li> <li>Record of waste disposal</li> </ul>                                                   | <ul style="list-style-type: none"> <li>Visual monitoring</li> <li>Compliance check</li> </ul> | Entire project site                                                                                                           | <ul style="list-style-type: none"> <li>Environmental Public Health (Toxic Industrial Wastes) Regulation, 2000</li> </ul>  | <ul style="list-style-type: none"> <li>During entire construction phase</li> </ul> | <ul style="list-style-type: none"> <li>Monthly Environmental Performance Report</li> </ul> | <ul style="list-style-type: none"> <li>Contractor/ CEMMP In-charge</li> <li>ECO</li> <li>QECF</li> </ul> | PUB/ SO     |
|                                                                                                                                                                     | Improper disposal of construction waste lead to land pollution | <ul style="list-style-type: none"> <li>Verify implementation of solid waste management plan</li> <li>Engagement of NEA licensed general waste collector</li> <li>Record of waste disposal</li> </ul> | <ul style="list-style-type: none"> <li>Visual monitoring</li> <li>Compliance check</li> </ul> | <ul style="list-style-type: none"> <li>Construction waste storage location</li> <li>General waste storage location</li> </ul> | <ul style="list-style-type: none"> <li>Environmental Public Health (General Waste Collection) Regulation, 2000</li> </ul> | <ul style="list-style-type: none"> <li>During entire construction phase</li> </ul> | <ul style="list-style-type: none"> <li>Monthly Environmental Performance Report</li> </ul> | <ul style="list-style-type: none"> <li>Contractor/ CEMMP In-charge</li> <li>ECO</li> </ul>               | PUB/ SO     |

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